

Merged research report for the Surreal project

Definable Surreal Numbers and Omnific Integers

Ambient definability, reversible and bounded omnific coding,
hereditary ordinal definability, a transcendence dichotomy,
and the maximal initial core

Part I built from two manuscripts dated September 23, 2026;
Parts II and III from three manuscripts dated October 4, 2026

Abstract

A surreal number cannot be called definable without specifying a language, a universe, and permitted parameters. We develop ambient set-theoretic notions, keep them apart from definability in the ordered field and its exponential expansion, and give an algebraic theory of the resulting numbers. For a fixed parameter set, the ambiently definable surreals form a real closed field closed under the Conway monomial map, the exponential and the omnific floor; their omnific integers form an integer part whose fraction field is exactly that field, with a denominator chosen from the same parameters. Over a transitive model the definable field is an elementary exponential subfield. Ordinal-definable surreals are exactly the surreals of HOD.

Two reversible omnific codes compress every surreal: a sign code with leading term ω , zero constant term and coefficients in $\{1, 2, 3\}$, landing in the omnific interval $\omega < z < \omega + \omega^{1/2}$, and the monomial code $x \mapsto \omega^{(1+x/\sqrt{1+x^2})/2}$, landing between 0 and ω and localizing to every infinitely wide omnific interval. Consequently ordinal definability of all omnific integers in one such interval is equivalent to $V = \text{HOD}$, and the parameter-free version in a model is equivalent to pointwise definability. If $V \neq \text{HOD}$, the first interval contains an explicit Ord-indexed algebraically independent family over the HOD surreals. If ρ is the first ordinal of a transitive model M not definable from A , then ρ is an epsilon number and the definable surreals of birthday below ρ form the largest initial subclass of the definable field; it is again an elementary exponential subfield and the fraction field of its own omnific integers. Explicit normal forms show in both directions that definability of a whole series differs from definability of its terms. We also treat complexity-bounded escape ordinals, homogeneous forcing that adds no reals, relativization, the surcomplex pairs, language dependence, and the finite quotients of the definable omnific ring.

Part II (added 4 October 2026) lets the real parameter vary: the resulting field is captured by one real exactly when $\text{HOD}_{\mathbb{R}}$ satisfies Choice; closure under countable Hahn sums is equivalent to real-ordinal definability of countable ordinal sequences, holds in a Cohen model, fails after Namba forcing over L , and so is independent of ZFC; and the common fixed field of the strong automorphisms fixing all reals and ordinals is the field of normal forms over the rational span of the ordinals. Part III (added the same day) interprets set theory parameter-free in the field with the omnific predicate and the omega map, proves absoluteness of normal forms between inner models without a same-reals hypothesis, and characterizes definable sections of the real decoder by correctness of ω_1 in HOD.

Part I merges two independently written manuscripts and Part II two more; shared results are printed once with their sources named, and every source's limitations are kept. It is an AI-assisted draft: it has not been refereed, and nothing in it has been formalized in Lean.

Sources: manuscript 04 (the base text) and manuscript 07, both pinned to repository commit [9a385d3](#); their titles are in [section 1.4](#). The first non-HOD birthday is printed by citation of the universes report. The provenance of every section is recorded in [appendix A](#); non-claims are collected in [appendix B](#). Finite checks accompany the report and test finite identities only. Parts II and III (batch 93, manuscripts 02, 05 and 06) record their provenance in [sections 19.5](#) and [30.3](#).

Contents

I Ambient definability, reversible and bounded omnific coding, hereditary ordinal definability, a transcendence dichotomy, and the maximal initial core	5
1 The question, the results, and how this report was assembled	5
1.1 The question	5
1.2 Main results	5
1.3 What is prior work, and what is proposed here	6
1.4 How this report was assembled	7
1.5 Conventions and renamed symbols	8
1.6 Repository snapshot and dependency boundary	9
2 Foundations and canonical representations	9
2.1 Universes, models, and class notation	9
2.2 Signs, simplicity, and cuts	10
2.3 Normal forms and two different exponentials	10
2.4 Omnific integers and the correct floor formula	11
2.5 The imported exponential-field theorem	12
3 Which notion of definability?	12
3.1 Ambient definability with parameters	12
3.2 Ordinal parameters	13
3.3 Intrinsic ordered-field definability	13
3.4 Provable descriptions and computability	14
4 The field and integer part of ambiently definable numbers	14
4.1 Value group and residue field	17
5 The ordinal-definable surreal field is the HOD field	18
6 Reversible omnific codes	20
6.1 A sign code in a fixed omnific interval [04]	20
6.2 A monomial code below omega [07]	21
6.3 A uniformly definable bounded family with unbounded birthdays [07]	23
7 One interval detects the entire set-theoretic universe	23
8 From one missing number to proper-class transcendence	26
9 Uniform definability is not termwise definability	28
10 The maximal initial definable core	31
10.1 The first omitted ordinal is an epsilon number	31
10.2 The core theorem	32
10.3 What the core theorem does and does not say	33
11 The first birthday of a non-ordinal-definable surreal	34
11.1 A current set-theoretic restriction on the threshold	35
12 Complexity-bounded definability and escape ordinals	35
12.1 The bounded hierarchy [04]	36
12.2 Why the separate definitions cannot be collected uniformly	36
13 Forcing can change the theory without adding reals	37
14 Relativization and a surcomplex companion	39
14.1 An allowed set of ordinal information [04]	39

14.2 Canonical surcomplex pairs [04, 07]	40
15 Changing the language changes the answer	40
15.1 Adding the ordinary exponential still does not define omega [07]	40
15.2 Simplicity already defines omega [07]	41
15.3 The field with the omnific predicate: what the collection proves [merge]	41
16 Arithmetic of the definable omnific ring	42
17 Foundational boundaries, further questions, and formalization	44
17.1 From set models to an intended universe [07]	44
17.2 Questions	44
17.3 A realistic Lean decomposition	46
18 Conclusion	47
A Provenance, proof scope, and reproducibility	47
A.1 Where each section comes from	47
A.2 Claim ledger	48
A.3 Stale statements and corrections	49
A.4 What each source inspected	50
A.5 Files and checks	51
B Non-claims	51
B.1 Source 04	51
B.2 Source 07	53
B.3 The merge	54
C A small gallery of examples [07]	54
D Notation and reading guide	54
II Real parameters, countable assembly, and strong symmetries	56
19 Scope, contributions, and relationship to the repository	56
19.1 What is proved here	57
19.2 Proof and formalization status	57
19.3 What source 05 asks, and what it adds [05]	58
19.4 Relation to Part I and to neighbouring reports [write]	59
19.5 Provenance of this Part [write]	59
19.6 Notation in this Part [write]	61
19.7 Where source 05 is printed [write]	62
20 Foundations: sets, signs, series, and integers	62
20.1 The ambient convention	62
20.2 Normal forms and summability	63
20.3 Omnific integers and the sign of the infinitesimal tail	63
20.4 Source 05's conventions [05]	64
21 A parameter policy is part of a definition	65
21.1 External definitions without ordinal parameters	65
21.2 Ordinal and real-plus-ordinal definitions	66
21.3 Basic field and integer-part closure	66
21.4 Intrinsic ordered-field definability	67
21.5 Source 05's parameter dictionary [05]	67
21.6 Source 05's algebraic baseline: pointers and second routes [05]	69
22 Varying real parameters and the associated inner models	71
22.1 The real-parameter fields record exactly the parameter degrees	73

22.2	Choice is equivalent to stabilization at one real	73
22.3	Countable birthdays, spectra, and the first omitted birthday [05]	76
23	Countable information, Hahn sums, and countable cuts	78
23.1	Two further forms of the same principle	79
23.2	What a countable sum can carry	80
23.3	Source 05's separated-band multiplexer [05]	80
23.4	Further equivalent forms of countable assembly [05]	81
23.5	A canonical countable-assembly completion [05]	83
23.6	Certificate coding details [05]	85
24	A proper real-parameter field with exact countable closure	85
24.1	Source 05's treatment of the Cohen model [05]	87
25	A countable Hahn sum can defeat every real parameter	90
25.1	A negative example without a large-cardinal hypothesis	92
25.2	Source 05's Prikrý example [05]	93
26	The strong stabilizer of the reals and all ordinals	96
26.1	The rational span of the ordinals	96
26.2	Two automorphism tools	96
26.3	Moving every exponent outside the rational ordinal span	97
26.4	Consequences for intrinsic definability	98
26.5	The smaller field obtained from pure-field definitions	99
26.6	An independent family in one omnific interval	100
26.7	Naming the omega map	102
26.8	Remaining stabilizer and language questions	103
27	Arithmetic consequences and distinctions that remain	103
27.1	The omnific ring remembers the parameter hierarchy	103
27.2	A real parameter can encode long countable signs	104
27.3	Initiality and order topology	104
27.4	Examples by language	105
27.5	Consequences and common incorrect inferences [05]	105
28	A precise agenda for further research	107
28.1	The remaining stabilizer and language questions	107
28.2	The real-parameter hierarchy and countable information	107
28.3	Normal forms, parameter recovery, and integer arithmetic	108
28.4	Source 05's questions [05]	109
28.5	Further questions and research added at the write [write]	111
29	Formalization plan and reproducibility ledger	111
29.1	A safe sequence of formal developments	111
29.2	What was checked and what was imported	112
29.3	Package contents	113
29.4	Source 05's formalization plan and proof ledger [05]	113
III	Definable operations: parameter hierarchies, absolute normal forms, uniform coding, and an interpretation of set theory	115
30	Scope, conclusions, and relation to the repository	115
30.1	The existing starting point	116
30.2	Principal results in operational form	116
30.3	Provenance of this Part [write]	117
30.4	Notation in this Part [write]	117
30.5	Relation to Parts I and II and to neighbouring reports [write]	118

31 Foundations and the objects being defined	118
31.1 Set theory, models, and class notation	118
31.2 Normal forms, monomials, and omnific integers	119
32 A precise hierarchy of definable numbers	120
32.1 Ambient parameters and language parameters	120
32.2 Why ordinal definability is internally expressible	120
32.3 Elementary closure and the role of whole inputs	121
32.4 A reusable bounded code	122
33 Real parameters: individual descriptions and uniform coding	123
33.1 A single real suffices when all surreals are ROD	123
33.2 A uniform decoder of countable sign sequences	124
33.3 The corresponding theorem at every infinite cardinal	125
34 Canonical surreal operations are absolute without a same-reals hypothesis	126
34.1 The limit step for complete normal forms	128
34.2 A Levy-hierarchy refinement	130
34.3 Limits of the absoluteness theorem	132
35 What the choice of language permits	133
35.1 The ordered field alone	133
35.2 The ordinary exponential and the omnific predicate	133
35.3 Definability in the omnific ring itself	134
36 Definable support calculus	135
36.1 Recovering the coefficient at zero	135
36.2 Every coefficient and every truncation	137
36.3 A bounded lexicographic pairing operation	137
36.4 Powersets of supports and arbitrary relations	138
37 Interpreting set theory in bounded omnific monomials	139
37.1 Pointed well-founded extensional graphs	139
38 Definable Hahn summation of coded families	141
39 An intrinsically definable commuting family of derivations	143
40 Further consequences for parameters, fields, and operations	145
40.1 Coefficientwise operations and full product rings	145
40.2 The inner-model transcendence consequence	146
40.3 Exact preservation of definability levels by the bounded code	146
40.4 Elementary theories and elementary inclusions	147
40.5 An operation calculus has no universal self-evaluator	148
41 Size, first omissions, and the limits of collection	148
41.1 Two different first-omission ordinals	149
42 Further research questions and proposed program	150
42.1 Further questions and research added at the write [write]	152
43 Proof dependencies, contribution ledger, and verification status	153
43.1 What is proved here and what is imported	153
43.2 A concrete formalization sequence	154
References	155

[write] Since 4 October 2026 the report has three Parts: Part II (from [section 19](#); batch 93, manuscripts 02 and 05, labels `dsn:rp:` and `dsn:ca:`) treats real parameters, countable assembly and strong symmetries, and Part III (from [section 30](#); manuscript 06, labels `dsn:op:`) definable operations and an interpretation of set theory. Part I's numbering and labels are unchanged.

Part I

Ambient definability, reversible and bounded omnific coding, hereditary ordinal definability, a transcendence dichotomy, and the maximal initial core

1 The question, the results, and how this report was assembled

1.1 The question

The familiar phrase “definable real number” conceals several choices. A positive root of a rational polynomial is definable in an ordered field; a computable real can instead be defined by a statement about its algorithm; and a set-theoretic definition can refer to ordinals, models, and forcing extensions. These are different notions, not alternative notations for one fixed countable field. The user-specified reference on definable reals is a useful starting point [18]; it is not used as a proof source. The foundational analysis and the pointwise-definability results used below come from Hamkins–Linetsky–Reitz [12].

For surreals the distinction becomes essential, because the sign sequence representing a number is a set while the totality of all surreals is a proper class internally. The pure ordered field has no parameter-free definable infinite element, whereas the first infinite ordinal ω has a very short parameter-free definition in set theory. An omnific integer can simultaneously have elementary-looking normal-form coefficients and contain the information of an arbitrary set of ordinals. Our goal is to make these facts into a coherent theory rather than to announce an unspecified collection of “describable” numbers.

The central object is not a syntactic list of expressions and not a putative truth predicate inside the universe under discussion. Fix a model M of ZFC and an external collection A of allowed parameters. An element of M is *A-definable over M* when a standard first-order formula of set theory, using a finite tuple from A , uniquely specifies it in M (definition 3.1). We write D_A^M for the definable closure, K_A^M for its surreals and I_A^M for its omnific integers. When the universe V itself is meant, the same words are a metatheoretic scheme.

1.2 Main results

Let $\mathbf{No}$ be represented by ordinal-length sign sequences, and let $\mathbf{Oz}$ be the classical omnific-integer ring. Write

$$\mathcal{U} = \{z \in \mathbf{Oz} : \omega < z < \omega + \omega^{1/2}, \text{ct}(z) = 0\}. \quad (1.1)$$

The exponentiation $x \mapsto \omega^x$ throughout this report is the *Conway monomial map*, also written $\Omega(x)$; it is not $x \mapsto \exp(x \log \omega)$ (section 2.3).

The definable field [04, 07]. For every parameter family the definable surreals K_A form a real closed field closed under Ω , Ω^{-1} on monomials, $\exp$, $\log$ and the omnific floor; $I_A = K_A \cap \mathbf{Oz}$ is an integer part and $K_A = \text{Frac}(I_A)$ with a denominator chosen by one rule from the same parameters (theorem 4.1). Over a transitive model $K_A^M \prec \mathbf{No}^M$ in the exponential language (theorem 4.4). The ordinal-definable surreals are the surreal field of HOD (theorem 5.1).

Reversible compression [04, 07]. There is a parameter-free set-theoretically definable injection $\mathbf{C} : \mathbf{No} \rightarrow \mathcal{U}$ with a parameter-free definable inverse on its image, preserving the definability degree for every parameter family (theorem 6.2, source 04). The monomial code $Q(x) = \omega^{\tau(x)}$, $\tau(x) = \frac{1}{2}(1 + x/\sqrt{1+x^2})$, is a parameter-free definable order embedding of $\mathbf{No}$ into $\mathbf{Oz} \cap (0, \omega)$ with the same property; it localizes to every infinitely wide omnific interval, and there is a definable class function esc on the ordinals with values in $\mathbf{Oz} \cap (0, \omega)$ and unbounded birthdays (theorems 6.8 to 6.10, source 07). Neither code is asserted to be a ring homomorphism or to be definable in the pure ordered field.

One interval tests the universe [04, 07]. Every element of $\mathcal{U}$ is ordinal definable if and only if $V = \text{HOD}$, and the same holds for $\mathbf{Oz} \cap (0, \omega)$ and for the monomials ω^a with $0 < a < 1$. In the external semantics of a model $M \models \text{ZFC}$, every element of $\mathcal{U}^M$ is parameter-free definable in M if and only if M is pointwise definable (theorems 7.2 and 7.7). Thus a fixed leading asymptotic term, or a fixed numerical bound, does not bound set-theoretic information.

A transcendence dichotomy [04]. Either $\mathbf{No} = K_{\text{HOD}}$, or $\mathcal{U}$ contains an Ord-indexed algebraically independent family over K_{HOD} (theorem 8.3). The proof is explicit: starting with any $t \in (0, 1) \setminus K_{\text{HOD}}$, use

$$z_\alpha = \omega + \omega^{t\omega^{-(\alpha+1)}} \quad (\alpha \in \text{Ord}). \quad (1.2)$$

The obstruction to a polynomial relation is separation of normal-form supports into distinct additive cosets of K_{HOD} .

The maximal initial core [07]. Let M be a transitive model. If some ordinal of M is not A -definable and ρ is the least such ordinal, taken externally, then ρ is an epsilon number, and

$$\text{Core}_A^M = K_A^M \cap (\mathbf{No}_{<\rho})^M$$

is the largest initial subclass of K_A^M . It is an elementary exponential subfield of $\mathbf{No}^M$, and it is the fraction field of its own omnific integers, with the birthday-controlled denominator $\omega^{b(x)+1}$ (theorems 10.3 and 10.5).

Two uniformity gaps [04, 07]. A fixed definable support and individually definable coefficients need not determine a definable coefficient assignment (theorems 9.1 and 9.3); conversely a uniformly definable omnific normal form can have a nondefinable term (proposition 9.5 and theorem 9.6). The correct invariant is the entire canonical normal-form graph (proposition 4.9).

Further results. The first birthday of a non-ordinal-definable surreal is the first subset disagreement between HOD and V (theorem 11.2, the case $M = \text{HOD}$ of a theorem of the universes report); complexity-bounded definitions admit escape ordinals (theorem 12.3); no definable set contains its own definable hull (theorem 12.5); homogeneous forcing produces the dichotomy without adding reals (theorem 13.2); and the definable omnific ring has only the finite quotients $\mathbb{Z}/n\mathbb{Z}$ (theorem 16.2).

1.3 What is prior work, and what is proposed here

The sign construction, normal forms, the monomial map, real closedness, the exponential, and the omnific integer part are classical [1, 2, 3, 4, 6]. Quantifier elimination, o-minimality, ordinal definability, HOD , partial truth predicates, and Mostowski collapse are standard tools

[7, 8, 10, 12]. Van den Dries and Ehrlich supply the elementary exponential-field and epsilon-cutoff results [3, 4]; Bournez and Guilmant restate the corrected cutoff theorem [5]. We do not claim these tools, or their routine closure consequences, as new.

Hamkins proved in 2012 that the existence of a definable ordinal-indexed dense image in $\mathbf{No}$, and the density of the ordinal-definable surreals, are equivalent to $V = \text{HOD}$ [13]. This density theorem is prior work; the bounded omnific tests below strengthen the form of the hypothesis, not the underlying idea (theorem 7.5). Source 04 did not cite it; the merge adds the credit.

Chen, Hamkins, and Yang have announced that the surreal ordered field expanded by birthday order is bi-interpretable with set theory [17]. The collection credits that announcement in its birthday-cutoff report (`hset:thm:chy`). In particular, the general idea of encoding well-founded membership graphs by surreal signs is *not* a new contribution of this report. No theorem below needs an unverified version of that announced bi-interpretation.

The candidate-original contributions are: the particular reversible omnific codes with fixed leading data or fixed numerical bounds, and their one-interval tests for $V = \text{HOD}$ and for pointwise definability (04, 07); the explicit *proper-class transcendence dichotomy in the same interval* (04); the local coding, bounded birthday escape and bounded noncollection (07); the two explicit normal-form uniformity gaps (04, 07); and the maximal initial elementary exponential core with its birthday-controlled denominators (07). The birthday, complexity and forcing results are refinements and applications with less ambitious originality claims. Neither source’s literature search located these precise statements; neither search was exhaustive, and no theorem is described as settling a named open problem in the published literature. Mathematical correctness and bibliographic novelty are separate questions.

1.4 How this report was assembled

The report is built from two manuscripts, both dated 23 September 2026 and both pinned to repository commit `9a385d3957bdf9ea79f9a524751c90bd2c894` (35 commits before the placement commit `a4dcb91`); they are items 04 and 07 of batch 27 and keep those numbers here.

- **04:** *Definable Surreal Numbers and Omnific Integers: Ambient definability, reversible coding, hereditary ordinal definability, and a transcendence dichotomy.* It is the base text. It contributes the general set-model framework, the sign code and the two-term code, the one-interval tests (including the pointwise test for arbitrary set models), support-subfield amplification and the HOD dichotomy, localization, the termwise counterexample and the collector, the first non-HOD birthday, the Σ_n escape, forcing, relativization and the surcomplex companion.
- **07:** *Definable Surreal Numbers and Omnific Integers: Ambient truth, bounded coding, and the maximal initial core.* It works over an externally fixed transitive model throughout. It contributes the exponential elementarity of the definable field, the cut denominator rule, the valuation theorem, the monomial and local codes, the birthday escape and bounded noncollection, the bit code and the reverse gap, the maximal initial core and its denominators, Hamkins’s density theorem with the topology corollary, the language-dependence results, and the arithmetic of the definable omnific ring.

The source manuscripts are not shipped. Their code, recorded data and audits are, under the prefixes `04-hod-transcendence-` and `07-initial-core-`.

Why one report, and why 04 is the base. The two manuscripts have the same title and the same spine: the definable field and its integer part with same-parameter denominators, the

HOD identification, bounded tests for $V = \text{HOD}$ and for pointwise definability, the gap between whole normal forms and their terms, noncollection, and the definable closure of the pure field. Each has a large result the other lacks. On the shared theorems 04 has the weaker hypotheses: its pointwise test holds for arbitrary set models, including ill-founded ones, and its field theory is stated both for V as a scheme and for set models. 07's stronger conclusions (exponential elementarity, the core) are proved under 07's own transitive-model hypothesis, and they keep it here. The two manuscripts do not contradict each other or any report in the collection.

Printed once. Each of the following is printed once, crediting both sources: unique-output closure ([lemma 3.2](#)); the omnific floor ([proposition 2.3](#)); the definable closure of the pure field ([theorem 3.4](#)); the definable field, integer part and fractions ([theorem 4.1](#), with both denominator rules); closure under whole definable cuts and sums ([proposition 4.8](#)); the HOD identification and the normal-form criterion ([theorem 5.1](#) and [lemma 5.2](#)); the bounded $V = \text{HOD}$ test ([theorem 7.2](#)); the pointwise test ([theorem 7.7](#)); noncollection ([theorem 12.5](#)); the set-coding lemma ([lemma 7.1](#)); and the complexification ([proposition 14.2](#)).

Kept as second routes. 04's least-ordinal denominator and 07's cut denominator ([theorem 4.1](#)); 04's sign code and 07's bit code for the fixed-support phenomenon ([theorems 9.1](#) and [9.3](#)); 04's collector and 07's reverse gap for the converse phenomenon ([proposition 9.5](#) and [theorem 9.6](#)); 04's two-term code and 07's monomial code ([proposition 6.5](#) and [theorem 6.8](#)).

Cited, not reprinted. 04's first-birthday theorem is the case $M = \text{HOD}$, $N = V$ of `univ:prop:beta` in the universes report, with the same proof; it is printed by citation ([theorem 11.2](#)). 07's arithmetic section restricts results of the omnific Diophantine and quotient reports to definable fragments; the full-class antecedents are named at each statement ([section 16](#)).

Added by the merge , each tagged [merge] with a complete proof or an explicit citation:

- the fix of a gap in 04's proof that a proper K_{HOD} is not intrinsically definable ([proposition 5.4](#));
- the remark on o-minimality of $\mathbf{No}_{\text{exp}}^M$ used tacitly by 07 ([remark 2.5](#));
- the extension of 07's bounded conditions of the pointwise test to arbitrary set models ([theorem 7.7](#)) and of 07's bounded noncollection to arbitrary parameter families ([theorem 12.5](#));
- the answer from the collection to both sources' language questions ([proposition 15.3](#)), and the re-scoped [question 17.1](#);
- the relation to `odg:q:size` ([remark 16.6](#)).

[Appendix A.3](#) lists every stale statement corrected.

[write] **Added 4 October 2026, batch 93.** This subsection describes Part I. Parts II and III were added from three manuscripts of batch 93 (02 and 05, merged, and 06), each with its own provenance subsection ([sections 19.5](#) and [30.3](#)). Their notes in this Part are the dated notes after [remark 17.6](#) and in [appendix A](#); no statement, proof, number or label of this Part changed.

1.5 Conventions and renamed symbols

The two sources used several letters for different objects. The following convention is fixed before any use. Symbols of 04, the base, are kept where possible and 07's are renamed; the few local renamings in 04's text are listed as well.

- D_A^M is the definable closure of A in $(M, \in)$ (07's $\text{Def}_M(A)$); $K_A^M = D_A^M \cap \mathbf{No}^M$ are the definable surreals (07's $D_M(A)$); $I_A^M = K_A^M \cap \mathbf{Oz}^M$ (07's $I_M(A)$); O_A^M the definable ordinals; ρ_A^M the first omitted ordinal (07's $\rho_M(A)$).
- Core_A^M is 07's maximal initial core (07's $K_M(A)$). The letter K never denotes the core.
- Π is the purely infinite ideal (07's $\mathcal{J}$); $\Pi_A = K_A \cap \Pi$ (07's $\mathcal{J}_D$); $R_A = K_A \cap \mathbb{R}$ (07's R_D).
- $\mathbf{C}$ is 04's sign code, $\mathbf{D}$ its inverse; $\mathbf{Q}$ and $\mathbf{Q}_{a,b}$ are 07's monomial and local codes (07's C , $C_{a,b}$).
- $\mathbf{T}$ is 04's two-term code, built on the rational map h ; esc is 07's birthday escape (07's T). The rational map of 04's collector is renamed ζ , and the width of 07's local code is ℓ (07's h).
- $e_\beta = \omega^{-(\beta+1)}$ (04) and $\iota_\beta = 1/(\beta+2)$ (07's e_β).
- B_n is 04's set of Σ_n -definable surreals; 07's set of surreals of birthday at most α is written $\mathbf{No}_{\leq \alpha}$ (07's B_α), its subsets of ordinals $X \subseteq \alpha$ (07's B), and 04's ordinal code of a set is X_a (04's B).
- 04's decoder of sets from signs is Set (04's E); $E_\alpha(X)$ is 07's bit code. In the set-coding lemma 04's index ρ is written ϱ , and in the forcing lemma 04's set $A \subseteq \theta$ is written Y , since ρ and A are reserved.
- λ is a normal-form length and λ_H the first non-HOD birthday; an epsilon number is written ϵ , and an infinitesimal ε (07 used λ for epsilon numbers and for other ordinals).
- $\omega^x = \Omega(x)$ is the Conway map (07 wrote a bold ω for the surreal ω); $\underline{\alpha}$ is the surreal ordinal α when ordinal and surreal arithmetic might be confused.
- $\prec$ is elementary substructure; $\prec_s$ is the proper-prefix (simplicity) relation.

Other reports use some of these letters differently: the automorphisms report writes $\mathcal{U}_{\mathbf{Oz}}$ for a group, whereas $\mathcal{U}$ here is an interval of $\mathbf{Oz}$; the universes report writes $\beta(M, N)$ for the first new birthday, which is λ_H when $M = \text{HOD}$ and $N = V$.

1.6 Repository snapshot and dependency boundary

Both manuscripts inspected the repository at commit [9a385d3](#) through a GitHub connector: 04 read the catalogue and the introductions to the birthday-cutoff and omnific Diophantine reports [19, 20, 22]; 07 read the README, the catalogue and the detailed guides of the universes, birthday and omnific Diophantine reports [21]. The catalogue describes AI-assisted research drafts and separates written arguments from Lean coverage. This report follows that distinction. It imports no nonclassical Diophantine, quotient, automorphism, or bi-interpretation claim as a proof dependency; the relations to other reports stated below are credits and cross-references, checked at the placement commit. [Appendix A.4](#) records the limited audit scope of each source.

2 Foundations and canonical representations

2.1 Universes, models, and class notation

Our mathematical statements are made in ZFC, with proper-class notation understood schematically. Each individual surreal is a *set*. A class function used below is specified by one set-theoretic

formula; all its values and all sums at which it is evaluated are set-sized. An Ord-indexed independent family means a definable class assignment whose every finite subfamily is algebraically independent. It does not mean that the entire family is a set or that its class sum exists.

When a literal satisfaction relation is needed, fix a set model $M \models \text{ZFC}$ in an ambient metatheory, and interpret $\mathbf{No}^M$, $\mathbf{Oz}^M$, and the relevant definitions internally. This conditional framework does not assert that ZFC proves the existence of such an M . The model-theoretic statements explicitly identified as external apply to arbitrary set models. For nonstandard models, their “ordinals,” “finite sequences,” and surreal codes are internal objects, not necessarily actual well-founded sequences. Where a proof uses a least *externally* chosen ordinal (section 10) or external o-minimality (theorem 4.4), M is assumed transitive, as in source 07; then its ordinals and finite sequences are genuine, and a surreal of M is literally an external sign sequence, although M may have only some of the subsets of a given ordinal.

2.2 Signs, simplicity, and cuts

A canonical surreal is a function

$$s : \alpha \longrightarrow \{-, +\}, \quad \alpha \in \text{Ord}.$$

Its birthday is $b(s) = \alpha$. Comparison is by the first disagreement, with a missing sign ordered between $-$ and $+$. The ordinal α itself is represented by the all-plus sequence of length α , written $\underline{\alpha}$ when needed. The standard real embedding and the arithmetic operations are parameter-free definable in set theory [1, 2]. We write $y \prec_s x$ when y is a proper prefix of x , and $y \preceq_s x$ when equality is allowed.

A subclass $F \subseteq \mathbf{No}$ is *initial* if it contains every prefix of each of its elements. This is not the same as being an initial segment of the numerical order. The notation $\mathbf{No}_{<\lambda}$ always refers to birthdays, not numerical magnitude.

For sets L, R of surreal numbers with $L < R$, there is a unique simplest separator $\{L \mid R\}$. It is a prefix of every separator of the cut. Both the restriction that L, R be sets in the universe of construction and the word “simplest” are essential. These standard facts are part of the Conway–Gonshor simplicity theory; see also [9].

2.3 Normal forms and two different exponentials

Every nonzero surreal has a unique Conway normal form

$$x = \sum_{\xi < \lambda} r_\xi \omega^{g_\xi}, \quad r_\xi \in \mathbb{R} \setminus \{0\}, \quad g_\xi > g_\eta \quad (\xi < \eta < \lambda). \quad (2.1)$$

The length λ is an ordinal, and the support $\text{supp}(x) = \{g_\xi : \xi < \lambda\}$ is reverse well-ordered and set-sized. Conversely, any such coefficient/exponent sequence has a surreal sum. Addition and multiplication are Hahn addition and convolution. We use $\text{NF}(x)$ for the *whole indexed sequence* $\xi \mapsto (g_\xi, r_\xi)$, including its length, not merely the collection of its individual entries; for zero it is the empty function.

The uniqueness theorem supplies parameter-free set-theoretic operations $x \mapsto \text{NF}(x)$ and evaluation of valid normal forms. It supplies no algorithm for arbitrary infinite codes and no assertion of topological convergence of finite partial sums. The factorisation paper [6, Sections 1–2] is a convenient modern reference for the normal-form convention and the distinction between series fields and omnific subrings.

We write $\Omega(a) = \omega^a$ for the Conway omega map. It is an increasing isomorphism from the additive group of surreals to the multiplicative group of canonical monomials. Distinct

exponents give distinct Archimedean scales: if $a < b$, then $n\omega^a < \omega^b$ for every positive ordinary integer n . The inverse Ω^{-1} is defined on monomials. Gonshor's function $\exp$ is a different function, extending ordinary real exponentiation with base e . Throughout, $\mathcal{L}_{\exp}$ means the ordered-field language with this $\exp$, *not* with Ω , and $\mathcal{L}_{\text{or}}$ the ordered-field language. In general one must not replace ω^a by $\exp(a \log \omega)$. All coding constructions below use the Conway omega map.

Lemma 2.1 (Ordinal domination [04]). *Every set $S \subseteq \mathbf{No}$ is strictly bounded in absolute value by a surreal ordinal. Moreover, the least ordinal δ satisfying $|s| < \delta$ for every $s \in S$ is definable from S .*

Proof. A sign sequence of length α lies between the all-minus and all-plus sequences of length α , hence $|s| \leq \text{b}(s)$. Replacement gives the set of birthdays. An ordinal strictly exceeding all of them gives the required bound. The qualifying ordinals form a nonempty definable class, so it has a least member, uniquely specified from S . $\square$

2.4 Omnific integers and the correct floor formula

Definition 2.2. An *omnific integer* is a surreal whose normal form has no negative exponents and whose coefficient at exponent 0 is an ordinary integer. Thus

$$\mathbf{Oz} = \Pi \oplus \mathbb{Z}, \quad \Pi = \{x \in \mathbf{No} : \text{supp}(x) \subseteq \mathbf{No}_{>0}\}. \quad (2.2)$$

The empty support is permitted, so $0 \in \Pi$. The map $\text{ct} : \mathbf{Oz} \rightarrow \mathbb{Z}$ takes the coefficient at exponent 0.

This is the standard ring, not the ring generated by the ordinals. For example, $\sqrt{2}\omega$, $\omega^{1/2}$ and $\omega + 17$ are omnific, but $\omega + \pi$, $\omega + 1/2$ and ω^{-1} are not. A positive omnific integer may be much smaller than ω while exceeding every ordinary integer. The direct sum in (2.2) is additive, not a direct product of unital rings. Positive exponents stay positive under addition, so Π is an ideal and ct is a ring retraction.

Proposition 2.3 (Integer part [04, 07]). *The ring $\mathbf{Oz}$ is discretely ordered with least positive element 1. For every $x \in \mathbf{No}$ there is exactly one $z \in \mathbf{Oz}$ such that $z \leq x < z + 1$. This $z = \lfloor x \rfloor_{\mathbf{Oz}}$ is parameter-free definable from x in set theory, and $\lfloor x \rfloor_{\mathbf{Oz}} \preceq_s x$.*

Proof. Write, by splitting the normal form, $x = P + r + \varepsilon$, where $P \in \Pi$, $r \in \mathbb{R}$, and ε is infinitesimal. Put

$$\lfloor x \rfloor_{\mathbf{Oz}} = P + \begin{cases} r - 1, & r \in \mathbb{Z} \text{ and } \varepsilon < 0, \\ \lfloor r \rfloor, & \text{otherwise.} \end{cases} \quad (2.3)$$

A nonzero positive element of Π is larger than every real, since its leading exponent is positive. Hence every positive element of $\mathbf{Oz}$ is at least 1. Formula (2.3) directly verifies the two inequalities. If two omnific integers had the required property, their nonzero difference would have absolute value less than 1, contradicting discreteness. Both the normal-form formula and the unique integer-part property are set-theoretic definitions.

For the prefix assertion (07), use the standard simplicity characterization $z = \{z - 1 \mid z + 1\}$ for $z \in \mathbf{Oz}$. The inequalities place x strictly between $z - 1$ and $z + 1$, and the simplest separator is a prefix of every separator. $\square$

Remark 2.4. The correction when r is integral and $\varepsilon < 0$ is essential: $\lfloor \omega - \omega^{-1} \rfloor_{\mathbf{Oz}} = \omega - 1$, not ω . The proposition is classical [1, 6]; its proof is included to make subsequent definability closure entirely explicit. The floor and its correction are also treated in the omnific Diophantine report [22].

2.5 The imported exponential-field theorem

Source 07 uses the following classical input. The surreal exponential field is an elementary extension of the real exponential field, and it is o-minimal. If ϵ is an epsilon number, then $\mathbf{No}_{<\epsilon}$, with the induced exponential, is an elementary substructure of $\mathbf{No}$ and is closed under logarithms of positive elements. These are consequences of [3, Theorem 2.1 and Corollary 5.5]; the latter is stated even in the richer restricted-analytic exponential language. The original paper must be read with its erratum [4], as the birthday-cutoff report insists; Bournez and Guilmant prove that $\mathbf{No}_\epsilon$ is stable under exponential and logarithm exactly when ϵ is an epsilon number [5], which is the closure input used in [theorem 10.3](#), and elementarity then follows from model completeness of the real exponential field with logarithm, which is what [3, Corollary 5.5] packages. Source 07 cited the original paper without the erratum; the merge adds it. Our conclusions about definable fragments use only the countable language $\mathcal{L}_{\text{exp}}$.

Remark 2.5 (o-minimality inside a transitive model [merge]). Source 07 uses tacitly that $\mathbf{No}_{\text{exp}}^M$ is o-minimal as an external structure when M is transitive. This holds formula by formula. The o-minimality of the real exponential field and the elementary equivalence of the surreal and real exponential fields are theorems of ZFC, and by uniform finiteness each formula $\varphi(x, \bar{y})$ has a standard bound N_φ such that ZFC proves that every fibre $\varphi(\mathbf{No}_{\text{exp}}, \bar{b})$ is a union of at most N_φ points and intervals. Applied inside M , this statement is first order with a standard numeral, so it holds externally of $\mathbf{No}_{\text{exp}}^M$.

Lemma 2.6 (Classical birthday controls [07]). *If $r\omega^a$ is a nonzero term in the normal form of x , then*

$$b(a) \leq b(x), \quad b(r\omega^a) \leq b(x).$$

The normal-form length is at most $b(x)$, and for every surreal a ,

$$b(a) \leq b(\Omega(a)) \leq \omega^{b(a)},$$

where the power on the right is ordinal exponentiation.

These are the estimates of [3, Lemmas 4.1–4.2 and the proof of Proposition 4.7], imported as established results rather than reproved from the sign-expansion analysis. Independently, the sign order immediately gives $|a| \leq \underline{b(a)}$, because the all-plus and all-minus sequences are the extremal sequences of a fixed length. For the merge, the upper bound $b(\Omega(a)) \leq \omega^{b(a)}$ was re-derived from Gonshor’s sign expansion of ω^a ; the exponent bound $b(a) \leq b(x)$ was not re-checked, and [theorem 10.5](#) uses only its consequence $a \geq -\underline{b(x)}$.

3 Which notion of definability?

3.1 Ambient definability with parameters

Definition 3.1 ([04, 07]). Let $M \models \text{ZFC}$ be a set model, and let A be an external collection of allowed elements of M . An element $a \in M$ is *ambiently A -definable in M* if some standard first-order formula $\varphi(v, \bar{u})$ and some finite tuple $\bar{p}$ from A satisfy

$$M \models \forall v [\varphi(v, \bar{p}) \leftrightarrow v = a].$$

Write $D_A^M = \text{dcl}_{(M, \in)}(A)$, and define

$$K_A^M = D_A^M \cap \mathbf{No}^M, \quad I_A^M = D_A^M \cap \mathbf{Oz}^M, \quad O_A^M = D_A^M \cap \text{Ord}^M.$$

When $A = \emptyset$, say *parameter-free definable*. For a finite tuple p write K_p^M . Superscripts are omitted when the universe is fixed.

For a set model, D_A^M is an external set, definable using the external satisfaction relation of M . In particular,

$$|D_A^M| \leq \max(\aleph_0, |A|). \quad (3.1)$$

This estimate counts standard formulas and finite parameter tuples. It is external: it does not put into M a list of all the elements definable in M . Even though $\mathbf{No}^M$ and K_A^M are sets externally, $\mathbf{No}^M$ is a proper class internally, and K_A^M need not be represented by a set of M .

For the intended universe V , the same terminology is a metatheoretic scheme: for each particular formula we can ask whether it uniquely defines a given object. We do not add a first-order truth predicate for V , and do not assume that the full collection $D_\emptyset^V$ is an internally available set or class.

Lemma 3.2 (Unique-output closure [04, 07]). *Suppose a set-theoretic formula defines a unique output $F(\bar{x})$ from each input in its domain. If every component of $\bar{a}$ lies in D_A , then $F(\bar{a}) \in D_A$.*

Proof. Choose definitions for the finitely many components of $\bar{a}$. Existentially quantify those components, assert their defining formulas, and assert the graph formula for F . This is a single formula using only finitely many parameters from A , with a unique solution. $\square$

This elementary lemma is the mechanism behind all positive closure results in the report. It does not assert that $D_A^M \prec M$: general set-theoretic existential statements need not have a definable witness (07).

3.2 Ordinal parameters

Definition 3.3 ([04, 07]). A surreal or omnific integer is *ordinal definable* if its canonical set code is in OD. Define

$$K_{\text{HOD}} = \mathbf{No} \cap \text{OD}, \quad I_{\text{HOD}} = \mathbf{Oz} \cap \text{OD}.$$

The standard internal definition of OD says that an object is uniquely definable over some set structure $(V_\theta, \in)$ using finitely many ordinal parameters below θ . Reflection identifies this with the usual notion of definability in V from ordinal parameters. The extra ordinal θ is allowed; no truth predicate for V is required. Inside a model M the same definition gives $K_{\text{HOD}}^M = \mathbf{No}^M \cap \text{OD}^M$.

Both OD and HOD are definable classes of ZFC, unlike a uniform truth-based relation between arbitrary formulas and their unique solutions in V . HOD is the transitive class of sets whose transitive closures, including the sets themselves, consist of ordinal-definable sets; it is an inner model of ZFC with a canonical definable well-order [10, 15]. Every ordinal is ordinal definable, using itself as an allowed parameter. This alone shows why K_{HOD} need not be a countable collection.

3.3 Intrinsic ordered-field definability

For a specified language $\mathcal{L}$ on $\mathbf{No}^M$, *intrinsic definability* means $\text{dcl}_{(\mathbf{No}^M, \mathcal{L})}(P)$ for a set P of surreal parameters. The quantifiers then range over surreals, and only the symbols of $\mathcal{L}$ may be used. The three notions — ambient, ordinal and intrinsic — should never be conflated. The set-theoretic definition of ω is parameter-free, but ω is not definable without parameters in the pure ordered field, or even in its exponential reduct (proposition 15.1). Conversely, allowing every surreal as a parameter trivializes definability of individual surreals.

Theorem 3.4 (Classical language contrast [04, 07]). *In the pure ordered-field language, the parameter-free definable surreals are exactly the real algebraic numbers. Their intersection with $\mathbf{Oz}$ is exactly $\mathbb{Z}$. More generally, in a real closed surreal field F (for example $F = \mathbf{No}^M$) and for any external parameter set $P \subseteq F$,*

$$\mathrm{dcl}_F(P) = \mathrm{rcl}_F(\mathbb{Q}(P)),$$

the relative real algebraic closure of $\mathbb{Q}(P)$ in F .

Proof. Quantifier elimination for real closed fields reduces a definable singleton to a Boolean combination of polynomial sign conditions over the field generated by finitely many parameters [7, 8]. If a point is transcendental over that field, it is not a zero of any of the finitely many nonzero polynomials appearing in the formula, so their signs are constant on a small interval around it and the formula cannot isolate that point. Thus the point is algebraic over $\mathbb{Q}(P)$. Conversely, an algebraic element is specified as the k th real root of its polynomial, for a fixed finite k , with coefficients given by field terms in finitely many parameters. Real algebraic elements of $\mathbf{No}$ lie in its standard real subfield. A real is omnific exactly when it is an integer, by (2.2). $\square$

For the full proper-class field, the argument is a formula-by-formula application of the same quantifier-elimination theorem. In particular, ω is *ambiently* parameter-free definable, but is not parameter-free definable in the pure field. Thus “the parameter-free definable omnific integers” can mean just $\mathbb{Z}$ when definability is taken in the ordered field, but can include ω , $\omega^{1/2}$ and arbitrarily elaborate bounded codes when taken in set theory; both assertions are correct in their respective languages. Nor does adding the word “omnific” to our prose add a predicate to that language. The structure $(\mathbf{No}; +, -, \cdot, <, 0, 1, \mathbf{Oz})$ is a different structure; what the collection now proves about its definable closure is in [proposition 15.3](#).

3.4 Provable descriptions and computability

For a recursively axiomatized theory T , one can restrict to formulas φ for which $T \vdash \exists!x (\mathbf{No}(x) \wedge \varphi(x))$. These are *T -provably unique descriptions*, or provable names. Their interpretation in a model of T need not be invariant between different models. For example the formula which selects 0 if CH holds and 1 otherwise has a provably unique output in ZFC, but its value depends on the model (07). Transitivity, syntactic shortness, and provable uniqueness are separate issues from absoluteness; the statement “ T proves uniqueness” is not a replacement for choosing the semantic universe.

A genuine effective presentation of a surreal sign sequence gives an ambient definition when the presentation uniquely determines a well-founded sequence. The converse fails already for real numbers: the characteristic real of a noncomputable but arithmetically definable set is definable. We therefore do not identify definability with any one of the computable-surreal representations catalogued by the collection, or with a numerical approximation algorithm, and we provide no algorithm deciding whether an arbitrary formula uniquely defines a surreal. When a construction is called uniform, its defining set-theoretic formula is fixed; no internal truth predicate is thereby supplied.

4 The field and integer part of ambiently definable numbers

Unless stated otherwise the statements of this section hold for V as a scheme and for any set model M ; the superscript M refers to such a model, whose internal operations on surreals and normal forms are used. Put $\Pi_A = K_A \cap \Pi$ and $R_A = K_A \cap \mathbb{R}$.

Theorem 4.1 (Definable field and integer part [04, 07]). *The ambiently A -definable surreals K_A form a real closed field. It is closed under Ω , under Ω^{-1} on monomials, under $\exp$ and positive logarithms, and under the omnific floor. The ring $I_A = K_A \cap \mathbf{Oz}$ is a discretely ordered integer part of K_A , and*

$$K_A = \text{Frac}(I_A). \quad (4.1)$$

More precisely, every nonzero $x \in K_A$ can be written $x = p/q$ with $p, q \in \Pi_A$ and $q > 0$ a monomial, the pair (p, q) being chosen by one parameter-free definable rule from x ; two such rules are

$$q = \omega^\delta \quad (04), \quad q = \omega^b, \quad b = \{ \{0\} \cup \{-a : a \in \text{supp } x\} \mid \emptyset \} \quad (07), \quad (4.2)$$

where δ is the least positive ordinal with $\delta + g > 0$ for all $g \in \text{supp } x$. The field K_A is an elementary substructure of the ambient surreal ordered field in the pure ordered-field language.

Proof. The constants $0, 1$, addition, subtraction, multiplication, inverse away from zero, Ω , Ω^{-1} , $\exp$ and $\log$ are uniquely definable set-theoretic operations, so [lemma 3.2](#) proves the field axioms and the closure assertions by restriction. Every positive element has a unique positive square root in $\mathbf{No}$, so that square root belongs to K_A . For an odd-degree polynomial with coefficients in K_A , real closedness of $\mathbf{No}$ gives at least one root; its finite set of roots has a least member, uniquely definable from the coefficient tuple. That root lies in K_A . The square-root and odd-degree-root criterion proves real closedness. This is an application of the classical real-closed-field theorem, not a new proof of real closedness of $\mathbf{No}$.

The floor from [proposition 2.3](#) belongs to D_A whenever its input does. Hence every $x \in K_A$ has its omnific floor in I_A . Discreteness and the subring property restrict from $\mathbf{Oz}$.

For the fraction-field assertion, let $0 \neq x \in K_A$, and put $S = \text{supp}(x)$. The whole normal form and S are definable from x . By [lemma 2.1](#) there exists a positive ordinal δ with $\delta + g > 0$ for $g \in S$. Choose the *least* such positive ordinal (04). It is definable from x , so $\delta \in K_A$, and $q = \omega^\delta$ and $p = qx$ belong to K_A . Alternatively (07), form the set cut b in (4.2); then $b > 0$ and $b + a > 0$ for every $a \in S$, and b is definable from x . In either case the normal form of $p = qx$ is obtained by adding the exponent of q to every exponent of x , without changing the coefficients. Its exponents are strictly positive, so $p, q \in \Pi_A$, and $x = p/q$. The zero case is $0 = 0/1$. The reverse inclusion follows because K_A is a field. Finally, every inclusion of real closed fields is elementary by quantifier elimination [7]. $\square$

Remark 4.2. The proof does not suppose that each exponent of an A -definable normal form is A -definable. It only uses definability of the *entire support* to choose one bound. Confusing these two assertions would invalidate the fraction-field argument. The equality $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ and the support-clearing method are not new: they are `odg:thm:fractions` in the omnific Diophantine report [22]. The point here is that the denominator can be chosen canonically enough to preserve the *same* allowed parameters; merely invoking an unspecified denominator would not establish this conclusion.

Source 07 states the closure under Ω , $\exp$ and the floor, and the cut denominator, for a transitive model; the proofs use only [lemma 3.2](#), so they hold in the generality of 04's statement. The same applies to [proposition 4.9](#) and [theorem 4.10](#) below.

Lemma 4.3 (Definable selection in an o-minimal field [07]). *Let F be an o-minimal expansion of an ordered field. For each formula $\varphi(y, \bar{x})$ there is a parameter-free definable partial function $s_\varphi(\bar{x})$ whose domain is exactly*

$$\{\bar{a} : F \models \exists y \varphi(y, \bar{a})\}$$

and whose value satisfies φ . A substructure closed under these functions is elementary in F . Intersections of substructures closed under all of them are elementary as well.

Proof. A nonempty definable subset of the line is a finite union of points and intervals. Its leftmost component and its finite endpoints are definable from its parameters. Select its unique point if it is a point. For an interval choose its left endpoint if that endpoint belongs to it; otherwise use the midpoint when both endpoints are finite, the finite endpoint minus or plus 1 in the one-sided unbounded cases, and 0 for the whole line. The alternatives and endpoints are uniformly definable for the given formula. Thus this is a definable selection, not a choice made separately with new parameters for each fibre.

Apply the Tarski–Vaught criterion to the selected witness. One-coordinate selection suffices for all existential formulas; tuples can be handled successively. Closure of an intersection follows because the same single-valued selectors are used in every substructure. $\square$

Theorem 4.4 (Exponential elementarity [07]). *Let M be a transitive set model of ZFC. In the ordered exponential language,*

$$(K_A^M, +, \cdot, <, \exp) \prec (\mathbf{No}^M, +, \cdot, <, \exp).$$

In particular the same holds in the ordered-field reduct.

Proof. By section 2.5 and remark 2.5 the right-hand structure is o-minimal. Each fixed $\mathcal{L}_{\exp}$ -formula translates into a fixed formula of set theory: restrict its quantifiers to $\mathbf{No}$, and replace its function symbols by their definable graphs. Thus a selector of lemma 4.3, evaluated at a tuple from K_A^M , is an ambient uniquely definable output. It lies in K_A^M by lemma 3.2. The selection lemma gives elementarity. This translation is performed for each standard formula separately. The proof never combines all translations into a truth predicate inside M . $\square$

Remark 4.5. A language with a symbol for every restricted real analytic function can contain nondefinable real information in its symbols. The countability and ambient-definability statements here concern $\mathcal{L}_{\exp}$, not that arbitrary uncountable expansion (07).

Corollary 4.6 (Open induction [04]). *The nonnegative part of I_A satisfies induction for every quantifier-free formula in the ordered-ring language, with parameters from I_A .*

Proof. A quantifier-free formula with parameters in I_A defines a semialgebraic subset of the real closed field K_A , hence a finite union of intervals and points. Every nonempty intersection of such a set with $(I_A)_{\geq 0}$ has a least member. Indeed, the integer-part property supplies the first integer at or immediately above each finite left endpoint, with a one-unit adjustment for a strict endpoint; an interval unbounded to the left is treated by the lower bound 0. Take the least of the finitely many nonempty pieces. If an induction instance failed, the set of counterexamples would therefore have a least member. It is not 0, and its predecessor is nonnegative and satisfies the formula, contradicting the inductive step. This is the standard integer-part route to open induction [11, 6]. $\square$

For the full ring $\mathbf{Oz}$ open induction, its failure beyond open formulas, and the failure of Peano arithmetic are in the omnific Diophantine report (`odg:rem:foursquares`, `odg:thm:induction`); the corollary is the definable-fragment instance.

Related (batch 86, added 3 October 2026): open induction is sharp here. By proposition 16.1, I_A is a discretely ordered ring with division with remainder by every ordinary integer and with the ring retraction `ct` onto $\mathbb{Z}$. The discrete-initial-subgroups report (its source 03; AI-assisted, unrefereed, not formalized) shows that such a ring satisfies the single bounded induction instance of its dyadic standard-cut formula Std^{dy} only if it is $\mathbb{Z}$ (`isg:sc:thm:boundary`); so the nonnegative part of I_A satisfies open induction but not $\text{!}\Delta_0$, as for $\mathbf{Oz}$. (That report proves its theorems for set-sized rings and reads class applications as in `isg:sc:sec:classes`; the same reading applies to I_A .) The `polish-models-of-omnific-arithmetic` report proves

open induction by the route of the corollary above for two finite principal-part integer parts: $\mathbb{Z} \oplus \bigoplus_{q>0} \mathbb{R}X^q$, in a Borel presentation (`pma:bor:thm:arithmetic`), and its countable subring with real algebraic coefficients, inside a Polish ordered field (`pma:ser:prop:iopen`).

Proposition 4.7 (Finite ambiguity does not enlarge the hull [04]). *An element of $\mathbf{No}$ belonging to a finite ambiently A -definable set of surreals is itself ambiently A -definable. In particular, ambient model-theoretic algebraicity and definability coincide on the surreal sort. Every surreal algebraic over K_A already belongs to K_A .*

Proof. Order the finite candidate set by the surreal order. Each member is its k th member for a fixed finite k ; this property is expressible by a formula with the same parameters. For the final assertion, the roots of a nonzero polynomial over K_A form such a finite definable set. Equivalently, a real closed field has no proper algebraic ordered extension. $\square$

This finite-ordering argument is a particularly transparent case of model-theoretic algebraicity; broader set-theoretic questions about algebraicity and implicit definability are studied in Hamkins–Leahy [14]. It concerns a genuinely finite candidate set, not an internally nonstandard finite set viewed from outside a nonstandard model.

Proposition 4.8 (Whole-family closure [04, 07]). *If $L, R \in D_A$ are sets of surreals with $L < R$, their simplest separator belongs to K_A . If an entire set-coded, strongly summable family of surreals belongs to D_A — in particular a valid normal form whose whole indexed sequence is A -definable — its canonical sum belongs to K_A .*

Proof. In both cases the output is a unique parameter-free definable function of the input set codes; apply lemma 3.2. For a cut, both option sets must be jointly available as A -definable sets, not merely have individually A -definable elements. Strong summability is the normal-form condition that the union of supports is reverse well ordered and each exponent receives only finitely many nonzero contributions. $\square$

The hypothesis concerns the *whole input code*. An arbitrary set of individually definable entries need not satisfy it (section 9). This distinction is central, not a technical qualification at the edge of the theory.

Proposition 4.9 (The normal-form invariant [07]). *For every $x \in \mathbf{No}^M$,*

$$x \in K_A^M \iff \text{NF}(x) \in D_A^M.$$

If $x \in K_A^M$ and $\xi < \text{dom NF}(x)$ is an A -definable ordinal, then g_ξ and r_ξ are A -definable.

Proof. The normal-form map and its inverse are unique parameter-free definable constructions. Evaluation of a set-coded function at an input is also uniquely definable. Apply lemma 3.2 in both directions. $\square$

Without the definability of the index, the final assertion need not hold (proposition 9.5 and theorem 9.6). Without the definability of the *whole* graph, its converse need not hold either (theorems 9.1 and 9.3).

4.1 Value group and residue field

Define $v(x) = -g_0$ for nonzero x in (2.1), and $v(0) = \infty$. This is the normal-form valuation; it uses the negative of the largest Conway exponent. Let

$$\mathcal{O}_A = \{x \in K_A : |x| < n \text{ for some } n \in \mathbb{N}_{>0}\}, \quad \mathfrak{m}_A = \{x \in K_A : |x| < 1/n \text{ for every } n \in \mathbb{N}_{>0}\}.$$

Here the quantification over $\mathbb{N}$ is ambient set-theoretic, not a claim that $\mathbb{N}$ is intrinsically definable in an o-minimal field.

Theorem 4.10 (Value group and residue field [07]). *The value group of K_A is canonically the ordered additive group $(K_A, +)$: $v(K_A^\times) = K_A$. Standard part induces an isomorphism $\mathcal{O}_A/\mathfrak{m}_A \simeq R_A$. Every finite $x \in K_A$ has a unique decomposition $x = r + \varepsilon$ with $r \in R_A$ and $\varepsilon \in \mathfrak{m}_A$.*

Proof. The leading exponent is a unique definable function of a nonzero normal form, so $v(x) \in K_A$. Conversely $v(\omega^{-a}) = a$ for every $a \in K_A$, and the monomial lies in K_A .

For finite x , its real constant term is its standard part; this remains true when there is no constant term, in which case it is zero. This unique real is definable from x , hence belongs to R_A . The ordinary normal-form multiplication and addition laws show that standard part is a surjective ring homomorphism with kernel exactly the infinitesimals. Uniqueness of the decomposition follows because no nonzero real is infinitesimal. $\square$

This self-similarity of the value group should not be mistaken for a claim that K_A contains every Hahn series whose entries belong to K_A ; [section 9](#) gives explicit obstructions.

5 The ordinal-definable surreal field is the HOD field

The canonical sign representation makes hereditary definability much more transparent than it is for arbitrary sets. In this section V is the universe of construction; every statement can equally be read inside a model, with V , OD and HOD relativized.

Theorem 5.1 (HOD identification [04, 07]). *With canonical sign-sequence representatives,*

$$K_{\text{HOD}} = \mathbf{No} \cap \text{OD} = \mathbf{No} \cap \text{HOD} = \mathbf{No}^{\text{HOD}}, \quad I_{\text{HOD}} = \mathbf{Oz} \cap \text{OD} = \mathbf{Oz} \cap \text{HOD} = \mathbf{Oz}^{\text{HOD}}. \quad (5.1)$$

These are equalities of sign-sequence classes. The class K_{HOD} is an initial subclass of $\mathbf{No}$ and a real closed field that contains every ordinal and is closed under Ω , Ω^{-1} on monomials, $\exp$, positive logarithms and the omnific floor. The ring I_{HOD} is an integer part with $\text{Frac}(I_{\text{HOD}}) = K_{\text{HOD}}$.

Proof. Let $s : \alpha \rightarrow \{-, +\}$ be ordinal definable. Every element in its transitive closure, other than s itself, is built from ordinals and the two fixed sign symbols by finitely many fixed set constructions, so it is ordinal definable. Since s itself is ordinal definable, $s \in \text{HOD}$. The converse is part of the definition of HOD. HOD contains every ordinal and is transitive, so being an ordinal-length sign sequence is absolute, and its surreal sign sequences are exactly the ambient sign sequences that belong to HOD. This gives the carrier equalities for $\mathbf{No}$ in (5.1).

The closure assertions follow by substituting ordinal definitions into the unique-output constructions of [theorem 4.1](#); the ordinals used are permitted parameters. The floor and the least-ordinal denominator give the integer-part and fraction claims for $I_{\text{HOD}} = \mathbf{Oz} \cap \text{OD}$. Initiality: a prefix $x \upharpoonright \beta$ is definable from x and the ordinal β (07).

It remains to compare with the internal arithmetic of HOD. Comparison is computed from the same signs; canonical options are the same initial segments; and the recursive arithmetic on set-sized option closures is absolute between a transitive inner model and the ambient universe, so the ordered-field structure of $\mathbf{No}^{\text{HOD}}$ is the restriction of that of $\mathbf{No}$ (this part is `univ:prop:absolute` in the universes report, for any same-ordinal inner model). For the monomial map one may use its recursive cut with integer and dyadic multipliers, so no new reals enter the recursion (04); source 07 asserts the same compatibility for the normal form,

the exponential and the purely infinite/constant splitting. The universes report proves the arithmetic and cut parts only, so for these further operations the recursion argument just sketched is the source. For the omnific carriers, an OD surreal has an OD normal-form graph whose entries at ordinal indices are OD, hence the whole graph is in HOD (lemma 5.2) and its canonical interpretation agrees in the two universes; this gives $\mathbf{Oz} \cap \text{HOD} = \mathbf{Oz}^{\text{HOD}}$ (07). $\square$

When V is replaced by a transitive model M , theorem 4.4 with all ordinals of M as the external parameter set also shows that K_{HOD}^M is an elementary exponential subfield of $\mathbf{No}^M$ (07). This is stronger than merely being closed under exp.

Lemma 5.2 (Support of an ordinal-definable number [04, 07]). *If $x \in K_{\text{HOD}}$ has normal form (2.1), then for every $\xi < \lambda$,*

$$r_\xi \in \mathbb{R} \cap \text{HOD}, \quad g_\xi \in K_{\text{HOD}}.$$

In fact, $\text{NF}(x) \in \text{HOD}$, and

$$x \in K_{\text{HOD}} \iff \text{NF}(x) \in \text{HOD}. \quad (5.2)$$

The converse with only “every coefficient and exponent is OD” is false whenever there is a non-OD subset of an ordinal.

Proof. The entire normal form is uniquely definable from x , so is ordinal definable. An individual entry is uniquely definable from x and the ordinal index ξ . Its real coefficient is ordinal definable, hence hereditarily ordinal definable under the usual real coding. Its surreal exponent belongs to HOD by theorem 5.1. The length, indices, and the fixed tuple coding are also hereditarily ordinal definable. Hence the whole sequence belongs to HOD. Conversely, $\text{NF}(x) \in \text{HOD}$ implies that it is ordinal definable, so its unique surreal value is ordinal definable. The counterexamples are theorem 9.1 and $E_\alpha(X)$ of theorem 9.3 for a non-OD $X \subseteq \alpha$. $\square$

The last equivalence requires the whole sequence to lie in HOD. The strictly weaker condition “every coefficient and exponent lies in HOD” is insufficient, even with a HOD support and rational coefficients. In particular, K_{HOD} is not being identified with the unrestricted ambient Hahn field containing *all* externally assembled series with coefficients and exponents in HOD.

Corollary 5.3 (No algebraic defect [04]). *Every $x \in \mathbf{No} \setminus K_{\text{HOD}}$ is transcendental over K_{HOD} . The inclusion $K_{\text{HOD}} \subseteq \mathbf{No}$ is elementary in the ordered-field language, whether or not $V = \text{HOD}$.*

Proof. Apply proposition 4.7 with ordinal parameters, and then quantifier elimination. $\square$

Proposition 5.4 (The HOD field is not intrinsically definable when proper [04]; proof corrected [merge]). *If $K_{\text{HOD}} \neq \mathbf{No}$, no ordered-field formula, even with finitely many surreal parameters, defines K_{HOD} inside $\mathbf{No}$.*

Proof. It suffices to show that an infinite definable subfield F is the whole field; K_{HOD} is infinite. An infinite definable subset of a real closed field is a finite union of points and intervals, so F contains an open interval (a, b) . The endpoints need not lie in F , so choose $a' < b'$ inside (a, b) ; they belong to F , and so does $c = b' - a' > 0$. For $0 < y < c$ the point $a' + y$ lies in $(a', b') \subseteq F$, so $y \in F$; hence $(0, c) \subseteq F$. For any $x > 0$, the number $y = 1/(x + 2/c)$ lies in $(0, c)$, hence in F , and then $x = 1/y - 2/c \in F$. Negatives and zero follow, so F is the entire field. $\square$

Source 04 took c from “differences” of the interval without placing the endpoints in the subfield; the choice of a', b' inside the interval closes that gap. The same fact, that a proper real closed subfield has empty interior, is used in proposition 7.6.

6 Reversible omnific codes

6.1 A sign code in a fixed omnific interval [04]

For each ordinal β , viewed as a surreal ordinal, define

$$e_\beta = \omega^{-(\beta+1)}. \quad (6.1)$$

The e_β form a strictly decreasing class of positive infinitesimal surreals. In particular $0 < e_\beta \leq \omega^{-1} < 1/2$. Every e_β is ordinal definable. For a sign sequence $s : \alpha \rightarrow \{-, +\}$ put

$$c_s(\beta) = \begin{cases} 1, & s(\beta) = -, \\ 2, & s(\beta) = +. \end{cases}$$

Definition 6.1 (Self-delimiting omnific code [04]). Define

$$C(s) = \omega + \sum_{\beta < \alpha} c_s(\beta) \omega^{e_\beta} + 3\omega^{e_\alpha}, \quad \alpha = \text{b}(s). \quad (6.2)$$

The coefficient 3 is an end marker, not a third sign value.

Theorem 6.2 (Reversible fixed-leading-term coding [04]). *Formula (6.2) defines a parameter-free set-theoretic injection $C : \mathbf{No} \rightarrow \mathcal{U}$. Its image is set-theoretically definable, and there is a parameter-free definable inverse D on that image. For every model M and every allowed parameter family A ,*

$$s \in D_A^M \iff C^M(s) \in D_A^M. \quad (6.3)$$

In particular, s and $C(s)$ are mutually definable and have the same ordinal-definability status.

Proof. The exponents in (6.2) occur in strictly decreasing order: first 1, then e_β for $\beta \leq \alpha$. They form a set, because α is an ordinal. Each coefficient is nonzero and real, so the expression is a valid Conway normal form. Every exponent is positive. Thus the value is an omnific integer with constant term zero. Its leading term is exactly ω .

The tail $C(s) - \omega$ is positive. Its leading exponent is $e_0 = \omega^{-1}$, including the empty-sign case, in which the leading tail coefficient is 3. Since $e_0 < 1/2$, comparison of normal forms gives $C(s) - \omega < \omega^{1/2}$. Hence $C(s) \in \mathcal{U}$.

To decode, take the unique normal form of z . Verify that its leading term is ω , that its remaining exponents are exactly e_β for an initial ordinal segment $\beta \leq \alpha$, that the last coefficient is 3, and that every preceding coefficient is 1 or 2. The map $\beta \mapsto e_\beta$ is injective, so the end marker uniquely determines α . At each $\beta < \alpha$, recover $-$ from coefficient 1 and $+$ from coefficient 2. This produces exactly one sign sequence s and satisfies $D(C(s)) = s$.

These are set-theoretic definitions: the assertion that there exists an ordinal α and the specified indexed normal form is a first-order assertion about sets. It neither quantifies over class sequences nor needs a field-intrinsic way to recognize ω . The unique-output closure lemma applied to both maps proves (6.3). The same proof is a theorem internal to any model; external definability uses standard formulas for the two maps. $\square$

Example 6.3. The first three codes are

$$C(0) = \omega + 3\omega^{\omega^{-1}}, \quad C(1) = \omega + 2\omega^{\omega^{-1}} + 3\omega^{\omega^{-2}}, \quad C(-1) = \omega + \omega^{\omega^{-1}} + 3\omega^{\omega^{-2}}.$$

The code for 0 is not 0. The coding theorem makes no assertion of arithmetic compatibility.

Corollary 6.4 (Congruence and leading-data blindness [04]). *Every code has leading exponent 1, leading coefficient 1, constant term 0, and coefficients from $\{1, 2, 3\}$. Moreover, for every ordinary positive integer n ,*

$$\mathbf{C}(s) \in n\mathbf{Oz}.$$

These invariants therefore do not bound ambient definability complexity.

Proof. Division by an ordinary $n > 0$ scales the real coefficients and leaves all exponents positive, so the quotient is still omnific. The remaining assertions follow from the normal form. Since every surreal is recoverable from its code, the listed common invariants cannot separate definability degrees. $\square$

Proposition 6.5 (A complementary two-term code [04]). *There is also a parameter-free definable order isomorphism*

$$\mathbf{T} : \mathbf{No} \longrightarrow \mathcal{U}_2 := \{\omega + \omega^g : 0 < g < 1/2\} \subseteq \mathcal{U}.$$

Every value has exactly two nonzero normal-form terms, both with coefficient 1, and is mutually ambiently definable with its input.

Proof. The rationally defined map

$$h(x) = \frac{1}{4} \left(1 + \frac{x}{1 + |x|} \right)$$

is an increasing bijection from $\mathbf{No}$ onto $(0, 1/2)$. For its inverse, put $u = 4g - 1$ and use $x = u/(1 - |u|)$. Therefore $\mathbf{T}(x) = \omega + \omega^{h(x)}$ has the stated normal form, is strictly increasing, and has exactly the stated image. Extracting the lower exponent from the unique normal form and applying the inverse of h recovers x by a parameter-free set-theoretic operation. $\square$

The two codes have different purposes. The elementary two-term code allows an arbitrary surreal exponent to carry all the input information. The sign code $\mathbf{C}$ instead fixes its exponents in an ordinal-definable template, so that all information is stored in the assignment of a finite coefficient alphabet. It is the latter feature that proves the failure of termwise HOD closure (theorem 9.1).

Remark 6.6 (An interval is still a proper class [04]). Although $\mathcal{U}$ is bounded by two explicitly given surreals, it is a proper class in the ambient universe. Otherwise its subset $\text{ran}(\mathbf{C})$ would be a set and Replacement applied to $\mathbf{D}$ would make all of $\mathbf{No}$ a set. Ordered boundedness is not a size bound for the full surreal field.

6.2 A monomial code below omega [07]

Lemma 6.7 ([07]). *In every real closed ordered field F , the map*

$$\tau(x) = \frac{1}{2} \left(1 + \frac{x}{\sqrt{1 + x^2}} \right)$$

is an order isomorphism $F \rightarrow (0, 1)_F$. Its inverse is

$$\tau^{-1}(t) = \frac{2t - 1}{\sqrt{1 - (2t - 1)^2}}. \tag{6.4}$$

Both maps are parameter-free definable in the ordered-field language.

Proof. Since $1 + x^2 > x^2$, we have $|x| < \sqrt{1 + x^2}$, so $0 < \tau(x) < 1$. If $u \in (-1, 1)$, put $x = u/\sqrt{1 - u^2}$. Direct calculation gives

$$1 + x^2 = \frac{1}{1 - u^2}, \quad \frac{x}{\sqrt{1 + x^2}} = u.$$

This proves the inverse formula. To check monotonicity without analytic assumptions, the function $x/\sqrt{1 + x^2}$ preserves sign. For $0 \leq x < y$, comparison of its squares reduces to

$$x^2(1 + y^2) < y^2(1 + x^2),$$

which is equivalent to $x^2 < y^2$. The negative case follows by oddness, and opposite signs are immediate. $\square$

Theorem 6.8 (Bounded omnific coding [07]). *The function*

$$\mathbf{Q}(x) = \Omega(\tau(x)) = \omega^{\tau(x)} \tag{6.5}$$

is a parameter-free ambient-definable order embedding of $\mathbf{No}$ into $\mathbf{Oz} \cap (0, \omega)$. Its image is exactly $\{\omega^a : 0 < a < 1\}$, and its inverse there is $\tau^{-1} \circ \Omega^{-1}$. For every positive ordinary integer n ,

$$n < \mathbf{Q}(x) < \omega/n. \tag{6.6}$$

For any M and A as in definition 3.1,

$$x \in K_A^M \iff \mathbf{Q}(x) \in I_A^M. \tag{6.7}$$

More strongly, $\text{dcl}_{(M, \epsilon)}(x) = \text{dcl}_{(M, \epsilon)}(\mathbf{Q}(x))$.

Proof. The exponent $\tau(x)$ lies strictly between 0 and 1. A one-term normal form with a positive exponent is purely infinite, hence omnific. Comparison of the exponents 0, $\tau(x)$, and 1 gives (6.6). Both τ and Ω are increasing, so their composite is an order embedding. The displayed inverse follows from uniqueness of the monomial exponent and lemma 6.7. Each direction is a unique ambient definition, so lemma 3.2 gives (6.7). Substitution of these definitions also gives the equality of definable closures, with or without additional parameters. $\square$

This theorem does not assert that the interval is order-isomorphic to all of $\mathbf{No}$: the image consists only of selected monomials. Nor does it give an ordered-field-definable map; such a map into this discrete set would contradict o-minimality. Its use of the ambient omega map is indispensable. The code $\mathbf{Q}$ and 04's two-term code $\mathbf{T}$ rest on the same idea, an algebraic compression of $\mathbf{No}$ into an interval followed by the omega map; $\mathbf{Q}$ has a single monomial term and lands below ω , $\mathbf{T}$ has two terms and lands in $\mathcal{U}$.

Theorem 6.9 (Local omnific coding [07]). *Let $a < b$ be omnific integers. If $\ell = b - a$ is positive infinite and $\beta > 0$ is its leading normal-form exponent, then*

$$\mathbf{Q}_{a,b}(x) = a + \omega^{\beta\tau(x)} \tag{6.8}$$

is an order embedding $\mathbf{No} \rightarrow (a, b) \cap \mathbf{Oz}$, ambient-definable over a, b , with an inverse definable over the same parameters. Consequently x and $\mathbf{Q}_{a,b}(x)$ are interdefinable over a, b . If $b - a$ is finite, $(a, b) \cap \mathbf{Oz}$ is finite, so no injection from all surreals into it exists.

Proof. The leading coefficient of ℓ is positive. Since $0 < \beta\tau(x) < \beta$, the monomial in (6.8) is positive infinite but of strictly smaller Archimedean order than ℓ . It is therefore strictly between 0 and ℓ . The sum is omnific, and monotonicity follows from $\beta > 0$. For a point z in the image, recover $\gamma = \Omega^{-1}(z - a)$, then $\tau(x) = \gamma/\beta$, then x from (6.4). The leading exponent β is definable from a, b , so no additional parameter was introduced. If ℓ is finite it is an ordinary positive integer by (2.2). Any omnific difference strictly between 0 and ℓ is then one of $1, \dots, \ell - 1$, so the interval contains exactly those finitely many translates of a . $\square$

The relevant condition is infinite *width*, not infinite endpoints. For example $(\omega, \omega + 3)$ contains just two omnific integers, while $(\omega, 2\omega)$ can encode every surreal by the theorem. Compare [corollary 8.5](#), which places an independent family at every definable infinite scale.

6.3 A uniformly definable bounded family with unbounded birthdays [07]

An order embedding into a numerically bounded interval need not preserve birthdays. Nevertheless its definable inverse lets us force birthdays to escape every prescribed ordinal bound.

Theorem 6.10 (Canonical birthday escape [07]). *There is a parameter-free set-theoretically definable class function $\text{esc} : \text{Ord} \rightarrow \mathbf{Oz} \cap (0, \omega)$ such that $b(\text{esc}(\alpha)) > \alpha$ for every ordinal α . It requires no global choice and no truth predicate.*

Proof. For a fixed ordinal α , the class $\mathbf{No}_{\leq \alpha} = \{z \in \mathbf{No} : b(z) \leq \alpha\}$ is a set: its members are sign sequences of bounded ordinal length. By Separation and Replacement, the inverse image

$$Q_\alpha = \{Q^{-1}(z) : z \in \mathbf{No}_{\leq \alpha} \cap \text{ran}(Q)\}$$

is a set. Define

$$u_\alpha = \{Q_\alpha \mid \emptyset\}, \quad \text{esc}(\alpha) = Q(u_\alpha). \quad (6.9)$$

Every construction is uniquely and uniformly definable from α . Suppose $b(\text{esc}(\alpha)) \leq \alpha$. Then $\text{esc}(\alpha) \in \mathbf{No}_{\leq \alpha} \cap \text{ran}(Q)$, so its unique inverse u_α belongs to Q_α . But the defining cut makes u_α strictly greater than every member of Q_α , including itself. This contradiction proves the strict birthday bound. $\square$

Corollary 6.11 ([07]). *For every A -definable ordinal α of M , there is an A -definable omnific integer z with $0 < z < \omega$ and $b(z) > \alpha$. The ordinal-definable omnific integers in $(0, \omega)$ have birthdays unbounded in Ord in every universe of ZFC, independently of whether $V = \text{HOD}$.*

Proof. Apply [theorem 6.10](#) internally and [lemma 3.2](#). With ordinal parameters allowed, every $\text{esc}(\alpha)$ is ordinal-definable. $\square$

The last statement does not imply density of ordinal-definable surreals. Birthday cofinality of a bounded family and density in the numerical order are different properties; Hamkins's density theorem has the substantially stronger hypothesis discussed in [theorem 7.5](#). The escape esc is also different from 04's Σ_n escape of [theorem 12.3](#), which escapes a definability level rather than a birthday.

7 One interval detects the entire set-theoretic universe

We first spell out the standard set-coding step needed for the global consequences. This step is not an original bi-interpretation claim.

Lemma 7.1 (Every set is recoverable from some surreal [04, 07]). *There is a parameter-free partial set-theoretic function Set on surreal sign sequences such that every set is $\text{Set}(s)$ for some s . No parameter-free selection of such an s from the set being coded is asserted. In particular, if every surreal of a model M is parameter-free definable over M , then every set of M is.*

Proof. For a set a , Choice gives a bijection $f : \kappa \rightarrow \text{TC}(\{a\})$ for some ordinal κ . Transport membership to a well-founded extensional relation R on κ , and record the unique index ϱ with $f(\varrho) = a$.

Fix a parameter-free class pairing of ordinals $p : \text{Ord}^2 \rightarrow \text{Ord}$. One construction well-orders ordinal pairs first by their maximum and then lexicographically, and uses the order type of each predecessor set. Every predecessor collection is a set, so this defines a class bijection. (07 uses instead the injection $(\xi, \eta) \mapsto \kappa \cdot \xi + \eta$ with ordinal arithmetic.) Form the set of ordinals

$$X_a = \{p(0, \kappa), p(1, \varrho)\} \cup \{p(2, p(\xi, \eta)) : (\xi, \eta) \in R\}.$$

Let $\lambda = \sup\{\gamma + 1 : \gamma \in X_a\}$ and take the sign sequence of length λ whose positive positions are exactly X_a . From this sequence recover X_a , then κ, ϱ, R , and finally the value at ϱ of the unique Mostowski collapse. These operations are parameter-free and unique whenever the code is valid. For invalid codes leave Set undefined. The last sentence follows by [lemma 3.2](#) applied inside M . Mostowski collapse is standard [10]; the same pointed-graph mechanism is present in the announced surreal/set-theory dictionary [17]. $\square$

The use of Choice in this proof is existential. It does *not* produce a canonical definable well-ordering of every transitive closure from its underlying set. In particular, the lemma must not be strengthened to a definable injection of arbitrary sets into omnific integers without further assumptions (07).

Theorem 7.2 (The ordinal-definability test [04, 07]). *The following statements are equivalent in ZFC:*

- (i) $V = \text{HOD}$;
- (ii) *every surreal is ordinal definable*;
- (iii) *every omnific integer is ordinal definable*;
- (iv) *every member of $\mathcal{U}$ is ordinal definable* [04];
- (v) *every value of the code $\mathcal{C}$ is ordinal definable* [04];
- (vi) *every omnific integer in $(0, \omega)$ is ordinal definable* [07];
- (vii) *every monomial ω^a with $0 < a < 1$ is ordinal definable* [07].

Proof. The implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iv) $\Rightarrow$ (v) and (iii) $\Rightarrow$ (vi) $\Rightarrow$ (vii) are immediate from the inclusions. If (v) holds, recover each s from $\mathcal{C}(s)$ by [theorem 6.2](#); if (vii) holds, recover each x from $\mathcal{Q}(x)$ by [theorem 6.8](#). Either way every surreal is ordinal definable. For any set a , choose a surreal s with $\text{Set}(s) = a$ by [lemma 7.1](#). Its canonical sign code lies in HOD by [theorem 5.1](#). More explicitly, the associated ordinal set X_a lies in HOD, and HOD decodes the same well-founded extensional relation and its Mostowski collapse. Thus $a \in \text{HOD}$. Since a was arbitrary, $V = \text{HOD}$. $\square$

The equivalence of (i) and (ii) is in substance part of Hamkins’s 2012 answer ([theorem 7.5](#)); source 04 did not cite it. The bounded conditions (iv)–(vii) are the forms proposed by the two sources.

Corollary 7.3 (Two terms already detect HOD [04]). *The equivalent conditions in [theorem 7.2](#) remain equivalent if condition (iv) is replaced by “every member of $\mathcal{U}_2$ is ordinal definable.”*

Proof. Use the two mutually definable maps in [proposition 6.5](#) to recover every surreal from such a two-term omnific integer, and then apply [theorem 7.2](#). $\square$

Remark 7.4. There is no Choice-based definable selection hidden in the converse. We choose a code only to prove its existence; hypothesis (ii) says *every* such surreal code is ordinal definable. This is much weaker as a proof requirement than constructing a canonical code from an arbitrary set in a universe not satisfying $V = \text{HOD}$.

Theorem 7.5 (Hamkins, 2012; included for comparison [07]). *The ordinal-definable surreals are dense in $\mathbf{No}$ if and only if $V = \text{HOD}$. Hamkins further proves equivalence with a parameter-free definable map $\text{Ord} \rightarrow \mathbf{No}$ having dense image, and with definable surjectivity and bijectivity in that setting [13].*

Proof of the density equivalence. If x is a sign sequence which is not OD, let x^- and x^+ be its extensions by one minus sign and one plus sign. Every number strictly between x^- and x^+ extends x : an earlier disagreement, or termination before the length of x , puts it outside this interval. If such a number y were OD, then the prefix $y \upharpoonright b(x)$ would be OD using the ordinal parameter $b(x)$. That prefix is x , a contradiction. Thus density forces every surreal to be OD, and [theorem 7.2](#) yields $V = \text{HOD}$. The converse is immediate. $\square$

The proof just given is the prefix argument from Hamkins’s earlier result, not a new density proof claimed here. The bounded tests use the explicit omnific codes, whereas the density criterion is already known.

Proposition 7.6 (Specialization of the old-field theorem [07]). *If $V \neq \text{HOD}$, the subclass $\mathbf{No}^{\text{HOD}}$ is closed and nowhere dense in the fine order topology of $\mathbf{No}$.*

Proof. For every $x \notin \mathbf{No}^{\text{HOD}}$, the interval (x^-, x^+) in the previous proof contains no HOD surreal. Hence the complement is open, and $\mathbf{No}^{\text{HOD}}$ is closed.

For completeness, any proper real closed subfield $F \subsetneq \mathbf{No}$ has empty interior. Suppose an open interval were contained in F , and choose $c < d$ in that interval. For arbitrary $y \in \mathbf{No}$, $z = c + (d - c)\tau(y)$ lies between c and d , so $z \in F$. Then $(z - c)/(d - c) \in F$, and the algebraic inverse (6.4), using the positive square root in the real closed field F , implies $y \in F$. This would give $F = \mathbf{No}$. Apply this to the proper field $\mathbf{No}^{\text{HOD}}$ and combine empty interior with closedness. $\square$

The universes report proves the more general assertion for proper same-ordinal inner models, including its surcomplex analogue (`univ:thm:nowheredense`; source 07 cited it as “Theorem 14.3” of that report). [Proposition 7.6](#) is the case $M = \text{HOD}$ of that theorem and is not counted as a new result. It also illustrates the difference between fine surreal topology and the usual topology on the embedded real line.

Theorem 7.7 (The pointwise-definability test [04, 07]; items (v)–(vi) for arbitrary set models [merge]). *Let $M \models \text{ZFC}$ be any set model. In external semantics, the following are equivalent:*

- (i) M is pointwise definable;
- (ii) every member of $\mathbf{No}^M$ is parameter-free definable in M ;
- (iii) every member of $\mathbf{Oz}^M$ is parameter-free definable in M ;
- (iv) every member of $\mathcal{U}^M$ is parameter-free definable in M [04];
- (v) every omnific integer z of M with $0 < z < \omega$ is parameter-free definable in M [07];
- (vi) every monomial ω^a of M with $0 < a < 1$ is parameter-free definable in M [07].

Proof. The implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iv) and (iii) $\Rightarrow$ (v) $\Rightarrow$ (vi) are immediate. Each $s \in \mathbf{No}^M$ is recovered by a fixed standard formula from $\mathcal{C}^M(s) \in \mathcal{U}^M$, and each $x \in \mathbf{No}^M$ from the monomial $\mathcal{Q}^M(x)$, so (iv) and (vi) each imply (ii). Internally to M , every set has a code as in [lemma 7.1](#). For any fixed $a \in M$, choose such a code $s \in M$. A standard formula defines s by (ii), and composing it with the standard decoder formula defines a . This does not require M ’s internal relation to be externally well-founded: the collapse operation is the one that M proves exists uniquely. $\square$

Source 04 proved (i)–(iv) for arbitrary set models; source 07 proved (i)–(iii), (v) and (vi) for transitive M . The argument for (vi) $\Rightarrow$ (ii) uses only internal formulas, so 04’s proof covers 07’s conditions for arbitrary set models; this extension is the merge’s.

Pointwise definable models of ZFC exist whenever ZFC is consistent; Hamkins–Linetsky–Reitz also prove strong extension theorems and treat nonstandard as well as transitive models [12]. Their existence theorem is prior work, and no existence assumption is needed for the equivalence. The new feature of [theorem 7.7](#) is the restriction of the test to a single fixed-leading-term or bounded omnific interval. It also explains why an external counting argument cannot be converted into an internal proof that some member of $\mathbf{Oz}^M$ is undefinable: the formulas that define particular elements are not presented to M as a surjective internal list. In a pointwise definable model every set can be definable while the model still satisfies its own uncountability assertions.

8 From one missing number to proper-class transcendence

This section contains the strongest algebraic result, due to source 04. We first isolate a support argument that does not involve definability at all.

Lemma 8.1 (Disjoint support cosets [04]). *Let $K \subseteq \mathbf{No}$ be a subfield, and let $G \subseteq (\mathbf{No}, +)$ be an additive subgroup such that*

$$\text{supp}(a) \subseteq G \quad (0 \neq a \in K).$$

Suppose $(g_i)_{i \in J}$ is a family such that every nonzero finitely supported integer combination satisfies

$$\sum_i m_i g_i \notin G. \quad (8.1)$$

Then the family $(\omega^{g_i})_{i \in J}$ is algebraically independent over K . The assertion applies to a class family when the condition is read on each finite subfamily.

Proof. Take a nonzero polynomial in finitely many distinct variables,

$$P(Y_1, \dots, Y_n) = \sum_{\nu \in F} a_\nu Y_1^{\nu_1} \cdots Y_n^{\nu_n},$$

where $F \subseteq \mathbb{N}^n$ is finite and the displayed coefficients are nonzero. Since $\omega^u \omega^v = \omega^{u+v}$, the normal-form support of the term at multi-index ν after substitution is

$$\text{supp}(a_\nu) + \sum_{j=1}^n \nu_j g_j \subseteq G + \sum_{j=1}^n \nu_j g_j. \quad (8.2)$$

For $\nu \neq \mu$, these cosets are disjoint. Otherwise subtracting an exponent in their intersection would give $\sum_j (\nu_j - \mu_j) g_j \in G$, contrary to (8.1). Every displayed term is nonzero, and its coefficients cannot be cancelled by any other term. The finite sum is therefore nonzero. This proves the required absence of polynomial relations. No infinite polynomial, class sum, or interchange of summations is used. $\square$

Related (batch 89, added 4 October 2026): the case $K = \mathbb{R}$, $G = \{0\}$ is proved again, without this lemma, as Theorem 43.1 (`hset:kw:thm:double`) of `foundations-and-computation/birthday-cutoffs-and-hereditary-sets`, Part VII (batch-89 manuscript 04; AI-assisted, unrefereed, not formalized): for an injection $j : X \rightarrow \mathbf{No}$ the numbers $t_x = \omega^{\omega^{j(x)}}$ are algebraically independent over $\mathbb{R}$ and are positive infinite omnific integers, in ZF. That

part uses it to show that embedding the free rings $\mathbb{Z}[T_X]$ into $\mathbf{Oz}$, or the free fields $\mathbb{R}(T_X)$ into $\mathbf{No}$, for every set X is exactly the first Kinna–Wagner principle (its Theorem 43.5, `hset:kw:thm:equivalences`); its Remark 40.4 records the overlap with this lemma.

Related (batch 90, added 4 October 2026): a second case, with irrational coefficients in the exponents, is Lemma 3.2 (`csf:lem:independence`) of `foundations-and-computation/cantor-families-of-surreal-subfields` (batch-90 manuscript 04; AI-assisted, unrefereed, not formalized): take $K = k$, the field of real algebraic numbers, $G = \{0\}$ and $g_{q,b} = b\omega^q$ for $q \in \mathbb{Q}$ and $b \in \{1\} \cup \{\sqrt{p} : p \text{ prime}\}$; nonzero integer combinations of the $g_{q,b}$ are nonzero by the $\mathbb{Q}$ -linear independence of square roots of squarefree integers (its Lemma 3.1, `csf:lem:radicals`). That report proves the lemma directly and uses the monomials $\omega^{b\omega^q}$ as a transcendence basis of a countable real closed field whose relative real closures K_A form a closed Cantor family of subfields on which embeddability is a complete analytic quasiorder (its Theorem 7.4, `csf:thm:universality`); a dated note after its Lemma 3.2 credits this lemma.

Theorem 8.2 (Support-subfield amplification [04]). *Let K be a proper subfield of $\mathbf{No}$ satisfying*

$$\omega \in K, \quad e_\alpha \in K \ (\alpha \in \text{Ord}), \quad \text{supp}(a) \subseteq K \ (0 \neq a \in K), \quad (8.3)$$

where $e_\alpha = \omega^{-(\alpha+1)}$. For any $t \in (0, 1) \setminus K$, the class family

$$y_\alpha = \omega^{te_\alpha}, \quad z_\alpha = \omega + y_\alpha \quad (\alpha \in \text{Ord}) \quad (8.4)$$

is algebraically independent over K , both for the y family and for the z family. Each z_α lies in $\mathcal{U}$ and has exactly two nonzero normal-form terms, both with coefficient 1. Such a t exists for every proper K satisfying (8.3).

Proof. First, the family (e_α) is linearly independent over $\mathbb{Q}$. Indeed, a finite rational linear combination is already a normal form with distinct exponents $-(\alpha+1)$; after deleting zero coefficients it cannot vanish. Hence for distinct $\alpha_1, \dots, \alpha_n$ and integers m_j not all zero,

$$c = \sum_{j=1}^n m_j e_{\alpha_j} \in K \setminus \{0\}.$$

If $\sum_j m_j t e_{\alpha_j} = tc$ belonged to K , division by c would give $t \in K$, a contradiction. Thus the exponents te_α satisfy (8.1) for the additive subgroup $G = (K, +)$. Lemma 8.1 proves algebraic independence of the y_α .

Translation by $\omega \in K$ is a polynomial automorphism on each finite tuple of variables. Therefore a nonzero relation among the $z_\alpha = \omega + y_\alpha$ would give a nonzero relation among the y_α , which is impossible.

Finally,

$$0 < te_\alpha < e_\alpha \leq \omega^{-1} < 1/2.$$

Consequently ω^{te_α} is a positive, purely infinite omnific monomial strictly below $\omega^{1/2}$. The exponents 1 and te_α are distinct and positive, so the asserted two-term normal form is exact and $z_\alpha \in \mathcal{U}$.

For existence, take $x \in \mathbf{No} \setminus K$. Then $u = |x| > 0$ is not in K , since otherwise $x = u$ or $x = -u$ would be in K . Set $t = u/(1+u)$. It lies strictly between 0 and 1. If $t \in K$, then $u = t/(1-t) \in K$, again a contradiction. $\square$

Theorem 8.3 (The HOD-surreal transcendence dichotomy [04]). *Exactly one of the following occurs:*

- (i) $V = \text{HOD}$ and $\mathbf{No} = K_{\text{HOD}}$;

(ii) $V \neq \text{HOD}$, and for any $t \in (0, 1) \setminus K_{\text{HOD}}$, the family

$$\left(\omega + \omega^{t\omega^{-(\alpha+1)}} \right)_{\alpha \in \text{Ord}}$$

is algebraically independent over K_{HOD} and is contained in $\mathcal{U}$.

In case (ii), for every cardinal κ there is an algebraically independent subset of $\mathcal{U}$ of cardinality κ over K_{HOD} . In particular, the transcendence defect has no set-sized cardinal bound.

Proof. [Theorem 7.2](#) identifies $V \neq \text{HOD}$ with $K_{\text{HOD}} \neq \mathbf{No}$. The field K_{HOD} contains ω and every e_α because these are ordinal definable. Its support property is [lemma 5.2](#). Apply [theorem 8.2](#). Restricting the class family to $\alpha < \kappa$ gives a set family by Replacement. Algebraic independence implies distinctness, hence that set has cardinality κ . $\square$

Remark 8.4 (What “proper-class transcendence” means [04]). The theorem gives a concrete class injection with finite polynomial independence, definable from one parameter t . It does not choose a transcendence basis for $\mathbf{No}$, invoke Global Choice, or assign an ordinary cardinal to a proper class. Each individual z_α is transcendental over K_{HOD} , but this pointwise assertion alone would be much weaker than the simultaneous algebraic independence proved above. In a set model, the assertions have their internal cardinal and ordinal meaning, not an assertion that its externally set-sized carrier is literally a proper class in the metatheory.

Corollary 8.5 (Localization at other definable scales [04]). *Assume $V \neq \text{HOD}$. For every $p \in I_{\text{HOD}}$, every positive $\gamma \in K_{\text{HOD}}$, and every $t \in (0, 1) \setminus K_{\text{HOD}}$, the family*

$$w_\alpha = p + \omega^{\gamma t e_\alpha} \quad (\alpha \in \text{Ord}) \tag{8.5}$$

is algebraically independent over K_{HOD} and lies in $(p, p + \omega^\gamma) \cap \mathbf{Oz}$.

Proof. For every nonzero finite integer combination c of the e_α , we have $c \in K_{\text{HOD}} \setminus \{0\}$. If $\gamma t c \in K_{\text{HOD}}$, division by γc would put t in K_{HOD} . The coset proof therefore applies to the exponents $\gamma t e_\alpha$. Translation by $p \in K_{\text{HOD}}$ preserves independence. The exponents are positive and less than γ , so the added monomials are omnific and lie strictly between 0 and ω^γ . $\square$

The widths in this corollary are infinite monomial widths. It would be false to replace them indiscriminately by positive real widths: an interval of width less than 1 contains at most one omnific integer. The result is localization at an arbitrary definable *infinite scale*, not density of omnific integers in $\mathbf{No}$.

9 Uniform definability is not termwise definability

There are two different pitfalls. Individually definable terms need not form a definable sequence; conversely, a definable sequence need not have definable entries at arbitrary ordinal indices. Each source gives an explicit example of each; both pairs are kept, since the mechanisms differ.

Theorem 9.1 (Definable support and terms do not suffice [04]). *Assume $V \neq \text{HOD}$. There is a valid normal form q satisfying all of the following:*

- (i) $q \in \mathcal{U} \setminus I_{\text{HOD}}$;
- (ii) *its support, as a whole set, belongs to HOD ;*
- (iii) *every coefficient is in $\{1, 2, 3\}$, and every individual monomial term belongs to K_{HOD} ;*
- (iv) *the whole indexed coefficient/exponent sequence does not belong to HOD .*

Thus K_{HOD} is not closed under arbitrary ambient Hahn-summable families of its elements, even when the support belongs to HOD .

Proof. Choose $s \notin K_{\text{HOD}}$ by [theorem 7.2](#), and let $\alpha = \text{b}(s)$ and $q = \text{C}(s)$. The reversible code gives $q \notin I_{\text{HOD}}$. Its support is the set

$$\{1\} \cup \{e_\beta : \beta \leq \alpha\},$$

which is definable from the ordinal α and whose members all belong to HOD . Hence the support itself belongs to HOD . Every individual coefficient is one of three fixed rationals, and every individual exponent is ordinal definable, so every individual term is ordinal definable. If the whole normal form belonged to HOD , [lemma 5.2](#) would give $q \in K_{\text{HOD}}$, a contradiction. The summable family here is simply the family of its distinct monomial terms, so no additional summability issue occurs. $\square$

Remark 9.2. In this example even the *set of coefficient values* is a finite HOD set. What fails is the definability of which coefficient is assigned to which exponent. Saying “the coefficients are rational” erases precisely the information carrying the original sign sequence.

Theorem 9.3 (Uniform subset coding by omnific normal forms [07]). *For an ordinal α and $X \subseteq \alpha$, put $\iota_\beta = 1/(\underline{\beta} + 2)$ and*

$$E_\alpha(X) = \omega^{3/4} + \sum_{\beta < \alpha} (1 + \chi_X(\beta)) \omega^{\iota_\beta}. \quad (9.1)$$

Then this is a well-defined omnific integer satisfying

$$\omega^{3/4} \leq E_\alpha(X) < 2\omega^{3/4} < \omega, \quad (9.2)$$

with strictness in the first inequality when $\alpha > 0$. The pair (α, X) and $E_\alpha(X)$ are mutually ambient-definable without additional parameters. In particular,

$$E_\alpha(X) \in K_A^M \iff \alpha, X \in D_A^M,$$

and the analogous equivalence holds with ordinal definability in place of A -definability.

Proof. The ordinal embedding is strictly increasing, so the exponents ι_β are strictly decreasing and positive. Their largest possible value is $1/2$, below the marker exponent $3/4$. Thus (9.1) is already a decreasing normal form, all of whose exponents are positive. It is omnific. Its leading term has coefficient 1 at exponent $3/4$, so its lower terms are strictly smaller in Archimedean order; this proves (9.2).

Starting from (α, X) , Replacement and the normal-form construction define the number uniquely. Starting from the number, take its canonical normal form, remove the first, marker term, and take the order type of the remaining decreasing support. This is α , including the empty case. The coefficient at the β th remaining place is 2 precisely when $\beta \in X$. Hence X is recovered uniquely. The definability equivalences follow from composition. $\square$

The marker in (9.1) is necessary for this simple uniform empty-case decoder and bound. No cancellation can obscure a bit: the exponents are distinct and the coefficients are never zero. The bit code and 04’s sign code C exploit the same phenomenon; C is self-delimiting through its coefficient alphabet $\{1, 2, 3\}$, while E_α uses a marker term and lands below ω .

Corollary 9.4 (Individually definable entries are insufficient [07]). *Suppose $X \subseteq \omega$ belongs to M but is not A -definable. Then $E_\omega(X)$ is not A -definable, although its support is parameter-free definable and every one of its real coefficients and exponents is individually parameter-free definable. If X is not ordinal-definable, the entire number is not ordinal-definable either.*

Proof. The support is the fixed set $\{3/4\} \cup \{1/(n+2) : n \in \mathbb{N}\}$. Its entries are rational, and each coefficient is either 1 or 2. Nondefinability of the sum follows from the decoder in [theorem 9.3](#). $\square$

There is no contradiction in saying that every coefficient is definable while the coefficient assignment is not. The statement “the coefficient at n is 1” or “it is 2” may require a nonuniform choice of which true statement to use. The missing information is exactly the subset X . For arbitrary $X \subseteq \alpha$, all exponents in (9.1) are ordinal-definable and the support is definable from α ; a non-OD X still gives a non-OD sum. Thus even allowing ordinal parameters does not turn pointwise definability of data into uniform definability.

Proposition 9.5 (A definable integer can encode nondefinable entries [04]). *There is a parameter-free ambiently definable $q_* \in \mathcal{U}$ whose normal-form coefficients canonically encode every member of $\mathbb{R} \cap \text{HOD}$. In any universe or fixed transitive model in which some member of $\mathbb{R} \cap \text{HOD}$ is not parameter-free definable, q_* has a normal-form coefficient that is not parameter-free definable. The last hypothesis is essential and is not asserted in every model.*

Proof. The set $R_H = \mathbb{R} \cap \text{HOD}$ is parameter-free definable. Restrict the canonical set-like HOD well-order to it, and let $(r_\xi)_{\xi < \lambda}$ be the unique increasing enumeration. The entire enumeration and its ordinal order type λ are parameter-free definable. Define the injective real function

$$\zeta(r) = \frac{3}{2} + \frac{r}{2(1+|r|)} \in (1, 2).$$

Its inverse on $(1, 2)$ is parameter-free definable: if $u = 2\zeta(r) - 3$, then $r = u/(1 - |u|)$. Form

$$q_* = \omega + \sum_{\xi < \lambda} \zeta(r_\xi) \omega^{e_\xi}. \quad (9.3)$$

This is a valid normal form and is parameter-free definable from the whole enumeration. The set R_H is nonempty, and the leading tail exponent is $e_0 < 1/2$, so $q_* \in \mathcal{U}$. Every $r \in R_H$ occurs in exactly one coefficient, after applying ζ . If that r is not parameter-free definable, neither is $\zeta(r)$, since the displayed inverse would otherwise define r . $\square$

This does not contradict unique-output closure. Extracting the coefficient at position ξ uses ξ as an additional parameter; a parameter-free definition of q_* need not define every such ordinal position. Ordinal definability allows that parameter, which is why all coefficients of an ordinal-definable surreal do belong to HOD.

Theorem 9.6 (The reverse uniformity gap [07]). *Suppose $\mu \in \text{Ord}^M$ is A -definable and there is $\beta < \mu$ which is not A -definable. Then*

$$U_\mu = \sum_{\gamma < \mu} \omega^{\iota_\gamma}, \quad \iota_\gamma = \frac{1}{\underline{\gamma} + 2}, \quad (9.4)$$

belongs to $I_A^M \cap (0, \omega)$, but its exponent ι_β and its corresponding monomial are not A -definable.

Proof. The entire normal-form family in (9.4) is uniformly defined from μ . Hence so is its sum, which is a positive omnific integer with leading exponent $1/2$. If ι_β were A -definable, taking its reciprocal and subtracting 2 would make the surreal ordinal $\underline{\beta}$, and therefore the set-theoretic ordinal β , A -definable. If the monomial were A -definable, applying Ω^{-1} would give the same contradiction. $\square$

This theorem is conditional on the stated ordinal configuration. One must not assume that an ordinal which is uncountable *inside* a countable transitive model necessarily has an externally undefinable predecessor; pointwise definable models are a reminder that it need not. [Section 10](#) gives an exact criterion for when the reverse gap can occur. The collector (04) obtains a nondefinable coefficient from real information in HOD; the reverse gap (07) obtains a nondefinable exponent from ordinal information. The two are different mechanisms.

A compact lesson for proposed definitions [07]. Defining a “definable surreal” to mean “a normal form with definable coefficients and exponents” produces an unstable notion unless a representation and a uniformity condition are specified. [Theorems 9.1](#) and [9.3](#) show that it can admit numbers with an undefinable information sequence. [Proposition 9.5](#) and [theorem 9.6](#) show that it can reject uniformly definable numbers. Using the entire canonical graph, as in [proposition 4.9](#), avoids both errors and agrees exactly with ambient definability.

10 The maximal initial definable core

Throughout this section, which is source 07’s, M is an *externally fixed transitive* set model of ZFC and $A \subseteq M$; the proofs use a least externally omitted ordinal, which need not exist for a nonstandard model. We write $K = K_A^M$, $I = I_A^M$ and $O = O_A^M = \text{Ord}^M \cap D_A^M$. The next elementary observation links the simplicity tree to definability of ordinals.

Proposition 10.1 (Initiality criterion [07]). *The subclass K_A^M is initial in $\mathbf{No}^M$ if and only if O_A^M is downward closed in Ord^M .*

Proof. If K_A^M is initial and $\alpha \in O_A^M$, the all-plus surreal α is in K_A^M . Each β for $\beta < \alpha$ is a prefix, hence definable. Its birthday is β , so $\beta \in O_A^M$. Conversely, let $x \in K_A^M$. Its birthday α is definable from x , so $\alpha \in O_A^M$. If $\beta < \alpha$, downward closure makes β A -definable. The prefix $x \restriction \beta$ is uniquely definable from x, β and therefore belongs to K_A^M . $\square$

10.1 The first omitted ordinal is an epsilon number

Assume $O_A^M \neq \text{Ord}^M$. Since M is transitive, the external nonempty set $\text{Ord}^M \setminus O_A^M$ has a least element. Denote it by

$$\rho_A^M = \min(\text{Ord}^M \setminus O_A^M). \quad (10.1)$$

This is a metatheoretic definition. It does not say that the minimum is defined from A in M .

Lemma 10.2 (Cantor-normal-form closure [07]). *The ordinal $\rho = \rho_A^M$ satisfies $\omega^\rho = \rho$, where the power is ordinal exponentiation. In particular $\rho > \omega$.*

Proof. The ordinals 0 and all finite ordinals are parameter-free definable. Suppose a nonzero ρ were not a fixed point of ordinal exponentiation with base ω . Its finite Cantor normal form would be

$$\rho = \omega^{\gamma_0} n_0 + \cdots + \omega^{\gamma_{k-1}} n_{k-1}, \quad \gamma_0 > \cdots > \gamma_{k-1}, \quad n_j \in \mathbb{N}_{>0},$$

with every $\gamma_j < \rho$. Indeed, the leading exponent is at most ρ ; equality would force $\rho = \omega^\rho$, and would leave no further term or coefficient greater than one. All exponents in this finite expression are A -definable by minimality of ρ . Ordinal exponentiation, finite ordinal sums, and finite coefficients are uniquely definable operations. The finite expression would therefore make ρ A -definable, a contradiction. The least epsilon number is greater than ω (and is itself parameter-free definable), so the final assertion follows. $\square$

The argument is ordinary ordinal definability, not a new theorem about Cantor normal form. Its role is to identify exactly the cutoff at which the classical epsilon-field theorem becomes available.

10.2 The core theorem

Theorem 10.3 (Maximal initial elementary exponential core [07]). *Let M be transitive, and suppose $\rho = \rho_A^M$ exists. Define*

$$\text{Core}_A^M = K_A^M \cap (\mathbf{No}_{<\rho})^M. \quad (10.2)$$

Then:

- (i) Core_A^M is initial, and every initial subclass of K_A^M is contained in it.
- (ii) Its surreal ordinals are exactly the ordinals below ρ .
- (iii) It is an elementary exponential subfield of $\mathbf{No}^M$, hence real closed and closed under positive logarithms.
- (iv) $\text{Core}_A^M \cap \mathbf{Oz}^M$ is an integer part of Core_A^M .
- (v) $\text{Core}_A^M \cap \mathbb{R}^M = K_A^M \cap \mathbb{R}^M$.

If $O_A^M = \text{Ord}^M$, the same initiality and elementary-field conclusions hold with $\text{Core}_A^M = K_A^M$.

Proof. Let $x \in \text{Core}_A^M$, and put $\alpha = \text{b}(x) < \rho$. Every $\beta < \alpha$ is A -definable by the definition of ρ . Hence $x \restriction \beta$ is A -definable and has birthday below ρ . This proves initiality.

No A -definable surreal has birthday exactly ρ , since the birthday is definable from the number. If an initial subclass $S \subseteq K_A^M$ contained x with birthday greater than ρ , it would also contain $x \restriction \rho$. The birthday of that prefix is ρ , giving the same contradiction. Therefore every such S is contained in (10.2). This proves (i), and (ii) follows because every ordinal below ρ is definable and the all-plus sequence has that ordinal as birthday.

By lemma 10.2, ρ is an epsilon number. Thus the classical cutoff theorem (section 2.5) gives

$$(\mathbf{No}_{<\rho})_{\text{exp}}^M \prec \mathbf{No}_{\text{exp}}^M.$$

Theorem 4.4 gives $(K_A^M)_{\text{exp}} \prec \mathbf{No}_{\text{exp}}^M$. Fix the parameter-free definable selection functions of lemma 4.3 for the common ambient exponential field. Each elementary substructure is closed under them: for parameters in it, the uniquely defined selected value belongs to it by elementarity. Their intersection is therefore closed under the same selectors. By lemma 4.3, the intersection in (10.2) is elementary. This proves (iii). This step uses common definable selectors; it is not an unsupported assertion that arbitrary intersections of elementary submodels are elementary.

For (iv), the omnific floor of $x \in \text{Core}_A^M$ belongs to K_A^M by theorem 4.1 and is a prefix of x by proposition 2.3. Initiality therefore places the floor in Core_A^M . Finally every embedded real has birthday at most $\omega < \rho$, giving (v).

If all ordinals of M are A -definable, proposition 10.1 makes K_A^M initial, and theorem 4.4 gives the remaining field assertion. $\square$

Corollary 10.4 (Parameter monotonicity [07]). *If $A \subseteq A' \subseteq M$, then $\text{Core}_A^M \subseteq \text{Core}_{A'}^M$, interpreting the core as in theorem 10.3 in the no-omission case. When both first omitted ordinals exist, $\rho_A^M \leq \rho_{A'}^M$.*

Proof. The initial class Core_A^M is contained in $K_{A'}^M$, so the maximality of $\text{Core}_{A'}^M$ gives the inclusion. Every ordinal below ρ_A^M is also A' -definable, giving the ordinal inequality. $\square$

Theorem 10.5 (Birthday-controlled omnific denominators [07]). *Let $\epsilon \in \text{Ord}^M$ be any epsilon number, and put*

$$K_{A, < \epsilon} = K_A^M \cap (\mathbf{No}_{< \epsilon})^M, \quad I_{A, < \epsilon} = I_A^M \cap (\mathbf{No}_{< \epsilon})^M.$$

Then $\text{Frac}(I_{A, < \epsilon}) = K_{A, < \epsilon}$. For nonzero $x \in K_{A, < \epsilon}$ with $\alpha = \text{b}(x)$, the explicit choice

$$q = \omega^{\alpha+1}, \quad p = qx \tag{10.3}$$

gives $p, q \in \Pi^M \cap K_{A, < \epsilon}$ and $x = p/q$. The rule is definable from x alone, without ϵ as an additional parameter. In particular,

$$\text{Frac}(\text{Core}_A^M \cap \mathbf{Oz}^M) = \text{Core}_A^M.$$

Proof. For each exponent a of x , lemma 2.6 gives $\text{b}(a) \leq \alpha$, hence $a \geq -\alpha$. Therefore

$$a + \alpha + 1 \geq -\alpha + (\alpha + 1) = 1 > 0.$$

All exponents of p are positive, so p and q are purely infinite. The number q is the surreal ordinal corresponding to the ordinal $\omega^{\alpha+1}$. As ϵ is an epsilon number and $\alpha < \epsilon$, we have $\alpha + 1 < \epsilon$ and $\omega^{\alpha+1} < \epsilon$. Thus $\text{b}(q) < \epsilon$. The epsilon-cutoff field is closed under multiplication, so $p = qx$ also has birthday below ϵ . Both p and q are definable from x , and hence belong to K_A^M . The zero case is $0/1$. The core assertion follows by taking $\epsilon = \rho_A^M$ when that ordinal exists, and by theorem 4.1 in the no-omission case. $\square$

Corollary 10.6 (Normal-form heredity and valuation of the core [07]). *The field $\text{Core} = \text{Core}_A^M$ is closed under Ω and its inverse on monomials. Every exponent, coefficient, and monomial term in the normal form of an element of Core belongs to Core . Its normal-form value group is $(\text{Core}, +)$, and its residue field is $R_A = K_A^M \cap \mathbb{R}^M$.*

Proof. Suppose first that $\rho = \rho_A^M$ exists. The bound $\text{b}(\Omega(a)) \leq \omega^{\text{b}(a)} < \rho$ for $a \in \text{Core}$, together with ambient definability, proves closure under Ω . The opposite birthday inequality gives closure under its inverse on monomials.

If $x \in \text{Core}$, its normal-form length is at most $\text{b}(x) < \rho$. Every index in that length is therefore A -definable. Proposition 4.9 makes each coefficient and exponent definable. Their birthdays, and those of the corresponding terms, are below ρ by lemma 2.6; coefficients are real and hence have birthday at most ω . Thus they lie in Core . In the no-omission case, every ordinal index is definable and the same reasoning requires no birthday cutoff. Now repeat the proof of theorem 4.10: leading exponents stay in Core , $\omega^{-a} \in \text{Core}$ realizes every value $a \in \text{Core}$, and standard part stays in the common real subfield from theorem 10.3(v). $\square$

10.3 What the core theorem does and does not say

The full ambient-definable field need not be initial: a definable high ordinal can have an undefinable predecessor. Nevertheless it always has the canonical initial elementary field just constructed. The reverse normal-form gap of theorem 9.6 occurs exactly when O_A^M is not downward closed, equivalently when K_A^M is not initial.

The ordinal ρ is not an additional allowed parameter in the definition of K_A^M . It indexes an externally described intersection. Calling it “the first undefinable ordinal” in the metatheory does not turn it into a definition inside M . This is precisely the distinction that prevents a Berry-style contradiction.

The fraction-field conclusion required the additional birthday argument of [theorem 10.5](#); it was not automatic from initiality or real closedness. In general an initial real closed subfield need not be the fraction field of its omnific integer part: the real algebraic subfield has omnific integer part $\mathbb{Z}$. Here it is the epsilon cutoff and the definability of the ordinal denominator that remove this obstruction.

The ordinal ρ_A^M and 04's threshold λ_H ([section 11](#)) are different invariants: ρ is the first ordinal not definable from A , and λ_H is the first birthday of a non-ordinal-definable surreal. With ordinal parameters every ordinal is definable, so the core of the ordinal-definable field is the whole field K_{HOD} , which is initial ([theorem 5.1](#)).

Proposition 10.7 (Failure of finite-parameter continuity [07]). *Suppose M contains an ordinal μ which is uncountable in the external metatheory. Let $A = \{\beta : \beta \leq \mu\} \subseteq M$. Then*

$$\bigcup_{A_0 \subseteq A \text{ finite}} \text{Core}_{A_0}^M \subsetneq \text{Core}_A^M.$$

Proof. For finite A_0 , only countably many ordinals are externally A_0 -definable. Some ordinal below μ is therefore omitted, so $\rho_{A_0}^M < \mu$. The surreal ordinal $\underline{\mu}$ is not in $\text{Core}_{A_0}^M$, since its birthday exceeds that cutoff. On the other hand every ordinal at most μ is itself an allowed parameter in A . Thus the first omitted ordinal for A , if there is one, is greater than μ , and $\underline{\mu} \in \text{Core}_A^M$. Parameter monotonicity gives the inclusion, and this element proves it strict. $\square$

This is a conditional statement about externally uncountable transitive models. It does not apply merely because M calls an ordinal uncountable. It shows that, although each individual definition uses finitely many parameters, requiring *all* prefixes to remain definable is a genuinely non-finitary condition.

11 The first birthday of a non-ordinal-definable surreal

When $V \neq \text{HOD}$, there is a genuine definable first place where ordinal-definable signs cease to exhaust the construction. It is not the birthday of a canonically selected nondefinable number.

Definition 11.1 ([04]). Assuming $V \neq \text{HOD}$, define

$$\lambda_H = \min\{\alpha \in \text{Ord} : \mathcal{P}(\alpha) \not\subseteq \text{HOD}\}. \quad (11.1)$$

Existence follows from the set-of-ordinals coding in [lemma 7.1](#). All terms in this definition are first-order definable classes or set constructions.

HOD is a parameter-free definable transitive inner class of V containing every ordinal and satisfying ZFC, so the pair $M = \text{HOD}$, $N = V$ satisfies the standing hypothesis (H) of the universes report, and $\lambda_H = \beta(\text{HOD}, V)$ in that report's notation.

Theorem 11.2 (Exact birthday threshold [04]; the case $M = \text{HOD}$ of `univ:prop:beta`). *Assume $V \neq \text{HOD}$. Then*

$$\lambda_H = \min\{\text{b}(s) : s \in \mathbf{No} \setminus K_{\text{HOD}}\}. \quad (11.2)$$

The ordinal λ_H is an infinite cardinal in both HOD and V . It is itself a parameter-free definable ordinal, and therefore a parameter-free definable omnific integer.

Proof. The equality (11.2) and the two cardinality assertions are `univ:prop:beta` of the universes report [21] (from that report's source 08), applied with $M = \text{HOD}$ and $N = V$ and

combined with $K_{\text{HOD}} = \mathbf{No}^{\text{HOD}}$ (theorem 5.1); that report also proves $\delta(\text{HOD}, V) \leq \text{cf}(\lambda_H)$ for its omitted-cut invariant δ . Its proof is the argument source 04 gives: signs of length α and subsets of α are interdefinable from α ; a HOD bijection with a smaller ordinal would pull a new subset back; and a V -bijection with a smaller cardinal would give a HOD code of a well-order of type λ_H . The birthday-cutoff report proves the equality and the inner-model cardinality for transitive models $M \subsetneq N$ of ZFC with the same ordinals as `hset:thm:firstbirthday`, which covers $\text{HOD}^N \subseteq N$ for a transitive set model N . The remaining assertion is 04's: (11.1) is a parameter-free unique definition of λ_H , since HOD is parameter-free definable, and ordinal surreals are omnific. $\square$

Corollary 11.3 (No definable witness selection [04]). *The set of non-ordinal-definable signs of length λ_H is a nonempty parameter-free definable set, but none of its members is ordinal definable. In particular, there is no ordinal-definable selection of one of those signs, or of one of their omnific codes.*

Proof. The set is defined using the definable predicate HOD and the parameter-free ordinal λ_H . Its nonemptiness follows from the definition of that ordinal. A selected ordinal-definable member would contradict the defining condition. An ordinal-definable code would yield such a member by the decoder. $\square$

11.1 A current set-theoretic restriction on the threshold

The threshold is the first cardinal where V and HOD disagree about subsets. It therefore inherits restrictions already proved in set theory; these are *not* new surreal theorems in substance. Benhamou, Cummings, Goldberg, Hayut, and Poveda prove that a singular strong-limit cardinal of uncountable cofinality cannot be the first such disagreement, and, from a measurable cardinal, obtain a model whose first disagreement is $\aleph_\omega$ [16]. Hence their first theorem gives:

Corollary 11.4 (Translation of a cited preprint theorem [04]). *If λ_H is singular and strong limit, then $\text{cf}(\lambda_H) = \omega$.*

Proof. An infinite singular cardinal has infinite cofinality. Uncountable cofinality is excluded by the cited compactness theorem, so its cofinality must be ω . $\square$

The preprint was submitted on August 25, 2026. It is important not to replace the cardinal conclusion of theorem 11.2 by an unsupported regular-cardinal claim: the cited relative consistency result explicitly permits a singular first threshold. Source 04 credited only “the older repository discussion of the first new birthday across universes” in the birthday-cutoff report; the exact theorem is `univ:prop:beta`, and the same statement for transitive same-ordinal models is `hset:thm:firstbirthday`.

12 Complexity-bounded definability and escape ordinals

The unrestricted phrase “all parameter-free definable numbers” is external. A fixed syntactic complexity bound, by contrast, admits an internal treatment. This gives a precise version of the familiar intuition that definitions of bounded complexity can be diagonalized against without creating a paradox.

12.1 The bounded hierarchy [04]

Fix a standard natural number $n \geq 1$. Use a standard syntactic coding of formulas in the Lévy hierarchy, with padding so that Σ_n formulas are also admitted at every higher level. There is a partial truth formula $\text{Sat}_{\Sigma_n}(e, x)$ for this fixed fragment. The existence of each fixed-fragment truth definition is standard; it does not provide one truth formula for all values of n simultaneously [10, 12].

Definition 12.1 ([04]). Let S_n be the collection of unique solutions of parameter-free Σ_n formulas, and put $B_n = S_n \cap \mathbf{No}$. More explicitly, $x \in S_n$ means that some code e of a formula with one free variable satisfies

$$\text{Sat}_{\Sigma_n}(e, x) \wedge \forall y(\text{Sat}_{\Sigma_n}(e, y) \longrightarrow y = x).$$

Each formula is required to have a unique solution among all sets, not just among the surreals.

Lemma 12.2 ([04]). *For every fixed standard n , both S_n and B_n are parameter-free definable sets of ZFC.*

Proof. For a formula code having a unique solution, send the code to that solution; send every other code to a fixed default set. The fixed-fragment truth formula defines this total function on a subset of ω , so Replacement gives its range as a set. Separation removes the default value when necessary and restricts to actual unique solutions. A further Separation restricts to surreal codes. All definitions use the fixed standard integer n only through its finite numeral, so there are no free parameters. $\square$

Theorem 12.3 (Definability-complexity escape [04]). *For every fixed standard $n \geq 1$, let*

$$\delta_n = \sup\{b(x) + 1 : x \in B_n\}, \quad d_n = \text{the all-plus surreal of length } \delta_n. \quad (12.1)$$

Then d_n is a parameter-free definable omnific integer, but $d_n \notin B_n$. There is a standard $m > n$ with $d_n \in B_m$. Consequently the hierarchy (B_n) does not eventually stabilize.

Proof. Replacement gives the set of birthdays and the ordinal supremum in (12.1). The all-plus sequence is then uniquely defined, so d_n is parameter-free definable and is an ordinal surreal. If $d_n \in B_n$, the definition of the supremum would imply

$$b(d_n) + 1 = \delta_n + 1 \leq \delta_n,$$

which is impossible. On the other hand, the parameter-free formula just given for d_n is a particular finite formula. After conversion to prenex form and adding dummy quantifiers, it belongs to some Σ_m fragment. Thus $d_n \in B_m$. By nesting and its failure to lie in B_n , we must have $m > n$. $\square$

Remark 12.4 (No unproved complexity estimate [04]). The theorem does not assert $m = n + 1$, a closed expression for δ_n , or a computable method that outputs the full transfinite sign sequence of d_n . The syntactic complexity of the normal-form, birthday, and sign-coding formulas depends on coding conventions. Only existence of a finite higher level is used. Nor is the sequence of escape ordinals asserted to be cofinal in all ordinals.

12.2 Why the separate definitions cannot be collected uniformly

Theorem 12.5 (Noncollection [04, 07]; bounded clause for arbitrary A [merge]). *Let M be a model or the universe, and A an allowed parameter family.*

- (i) *There is no ambiently A -definable set $S \subseteq \mathbf{No}$ such that $K_A \subseteq S$. In particular, no ambiently A -definable map $\omega \rightarrow \mathbf{No}$ has range containing all of K_A [04, 07].*
- (ii) *No ambiently A -definable set contains every A -definable omnific integer in $(0, \omega)$ [07].*

Proof. (i) If such an S were given, the simplest strict upper bound $u = \{S \mid \emptyset\}$ is uniquely definable from S , so $u \in K_A$, but $u > S$, so $u \notin S$ (07). Source 04 uses instead the all-plus surreal of the least ordinal strictly above all birthdays in S , whose birthday exceeds that of every member of S . For a definable map on ω , Replacement gives an A -definable range set, reducing to the first assertion.

(ii) Restrict a putative set S to $\text{ran}(\mathbf{Q})$ and apply $\mathbf{Q}^{-1}$ pointwise. Separation and Replacement produce an A -definable set of surreals $S' = \{\mathbf{Q}^{-1}(z) : z \in S \cap \text{ran}(\mathbf{Q})\}$. Every $x \in K_A$ has $\mathbf{Q}(x) \in I_A \cap (0, \omega)$ by [theorem 6.8](#), so the assumed property of S would give $x \in S'$. This contradicts (i). $\square$

Source 07 stated the theorem for a finite parameter tuple p of a transitive M ; the proof uses only that the sets involved are defined from finitely many allowed parameters, so it holds for any A . Thus external countability does not even give an internally definable list of the definable omnific integers in this bounded interval. The theorem concerns *sets* of M , not a universal assertion that K_A cannot be a definable class: in a pointwise definable model it is simply $\mathbf{No}^M$, which is a definable class (07).

Corollary 12.6 (Failure of a uniform bounded-truth collector [04]). *In an ω -standard model of ZFC there is no single parameter-free formula that, for each $n \geq 1$, uniquely defines the set B_n above, uniformly in n .*

Proof. A uniform formula would, by Replacement, give a parameter-free set $B = \bigcup_{n \geq 1} B_n$. Every parameter-free definable surreal is uniquely defined by a finite formula and hence belongs to some B_n . Thus B would contain $K_\emptyset$, contrary to [theorem 12.5](#). The ω -standardness hypothesis ensures that internal natural indices are precisely the standard finite levels discussed here. $\square$

This is the exact point at which the false “least undefinable surreal” argument tries to use more truth than it has. Separate formulas defining each fixed B_n do not constitute one internal formula defining the sequence of all these sets. The argument also does not claim that the membership class of definable elements can never accidentally be definable: in a pointwise definable model it is simply the entire universe. What fails there is an internal surjective set-sized list of those elements, not the external fact that every one of them has some definition.

13 Forcing can change the theory without adding reals

We now give a concrete source of models exhibiting the dichotomy. The forcing facts are standard, but proofs of the relevant homogeneity and closure steps are included. The result connects ambient definability of real numbers to genuinely transfinite surreal information without conflating the two.

Lemma 13.1 (Homogeneity over a constructible ground [04]). *Let W be a transitive model of ZFC satisfying $V = L$, and let G be generic for an ordinal-definable weakly homogeneous forcing $\mathbb{P} \in W$. Then, in $W[G]$,*

$$\text{HOD}^{W[G]} = W.$$

The assertion has the usual relative-model interpretation; it does not assume that a generic filter over the actual universe is a set in that universe.

Proof. First let $Y \subseteq \theta$ be ordinal definable in $W[G]$. Membership in Y is expressed there by one formula $\psi(\beta, \bar{\alpha})$ with a finite ordinal tuple. For each $\beta < \theta$, weak homogeneity implies that the top condition decides $\psi(\check{\beta}, \check{\bar{\alpha}})$. Indeed, if one condition forced the sentence and another forced its negation, an automorphism could make their images compatible, while fixing all check names for ordinals. Compatible conditions cannot force contradictory sentences. Thus

$$Y = \{\beta < \theta : 1_{\mathbb{P}} \Vdash \psi(\check{\beta}, \check{\bar{\alpha}})\} \in W$$

by the definability of the forcing relation.

Every HOD set is coded within HOD by a set of ordinals: use its canonical HOD well-order on a transitive closure and the pointed membership-relation code. Such a code is itself ordinal definable, so the previous paragraph places it in W . The absolute Mostowski collapse in W then recovers the original set. This proves $\text{HOD}^{W[G]} \subseteq W$. Conversely, forcing does not change the ordinals or the constructible hierarchy, so $L^{W[G]} = W$. Every constructible set is ordinal definable in $W[G]$ using its position in the canonical constructible well-order, and so are all members of its transitive closure. Therefore $W \subseteq \text{HOD}^{W[G]}$. These are the standard homogeneity/HOD arguments [10]. $\square$

Theorem 13.2 (Prescribed regular first birthday [04]). *Let $W \models \text{ZFC} + V = L$ be transitive, and let κ be an infinite regular cardinal of W . Force with*

$$\text{Add}(\kappa, 1)^W = \{p : p \text{ is a partial map } \kappa \rightarrow 2, |\text{dom}(p)|^W < \kappa\},$$

ordered by extension, and let $V = W[G]$. Then

- (i) $\text{HOD}^V = W$ and $K_{\text{HOD}}^V = \mathbf{No}^W$ as ordered fields;
- (ii) the least birthday of a surreal outside K_{HOD}^V is exactly $\lambda_H^V = \kappa$;
- (iii) $\mathcal{U}^V$ contains the explicit ordinal-indexed independent family of [theorem 8.3](#) over the old field $\mathbf{No}^W$;
- (iv) if $\kappa > \omega$, then $\mathbb{R}^V = \mathbb{R}^W$, and every real of V is ordinal definable in V , despite (ii) and (iii).

Proof. For two conditions p, q , flip the bits on the set of common-domain coordinates at which they disagree. This is a ground-model automorphism sending p to a condition compatible with q . Thus the forcing is weakly homogeneous and ordinal definable, so [lemma 13.1](#) applies. The field identification follows from [theorem 5.1](#) and the absoluteness of canonical surreal arithmetic.

Regularity of κ makes the union of a descending chain of length less than κ a condition. Given a name for a sequence of ordinals of length $\mu < \kappa$, recursively strengthen a condition to decide each coordinate, taking lower bounds at limit stages and at the end. The resulting ground-model sequence is forced to equal the name. Hence the forcing adds no subsets of any $\mu < \kappa$. For $\kappa = \omega$, the corresponding assertion about finite subsets is immediate. This closure also prevents a collapse of the cardinal κ by a shorter sequence.

The union $g = \bigcup G$ is a total subset of κ , and is new: for every ground-model $a \subseteq \kappa$, the set of conditions already disagreeing with a at some coordinate is dense. Since $\text{HOD}^V = W$, the first non-HOD subset therefore occurs exactly at κ . [Theorem 11.2](#) proves (ii). The new sign sequence with positive positions g lies outside K_{HOD}^V ; normalize its absolute value as in [theorem 8.2](#) to obtain the parameter t for (iii).

When $\kappa > \omega$, the no-new-short-sequences conclusion in particular gives no new subsets of ω , hence no new reals. All old reals belong to $W = \text{HOD}^V$, which proves (iv). $\square$

Example 13.3 (All reals ordinal definable, but many omnific integers not [04]). Take $\kappa = \omega_1^W$ in [theorem 13.2](#). In the extension, every real is ordinal definable; every surreal of birthday less than ω_1^W is in the HOD-surreal field; and a non-HOD surreal first appears at birthday ω_1^W . Nevertheless a single fixed interval of purely infinite omnific integers already contains independent sets of every internal cardinality over the old surreal field. This is a strict separation between the real question and the surreal question, not a consequence of adding an undefinable real.

Example 13.4 (A Cohen real [07]). Let a transitive ground universe satisfying $V = L$ be extended by a Cohen real c . In the extension $\text{HOD} = L$ (this is [lemma 13.1](#) for Cohen forcing, which is ordinal-definable and weakly homogeneous; the standard argument concerns OD subsets of ordinals and does *not* assert that homogeneous forcing adds no new OD sets of arbitrary kind). Consequently

$$z_c = \omega^{3/4} + \sum_{n < \omega} (1 + \chi_c(n)) \omega^{1/(n+2)}$$

is a non-OD omnific integer in $(0, \omega)$. The displayed support is parameter-free definable and every coefficient is either 1 or 2. This realizes [corollary 9.4](#) concretely, relative to the existence of the indicated ground model and generic extension. More generally, if an extension adds a non-OD subset X of an ordinal α , [theorem 9.3](#) creates a non-OD omnific integer below ω , even when that forcing adds no real numbers. This is an immediate conditional application of the code, not a claim that every forcing extension has the same HOD.

Remark 13.5 (What is, and is not, absolute [04]). The forcing theorem compares old sign sequences and their arithmetic. It does not assert that every old parameter-free definition keeps its truth or uniqueness in an extension. An ambient definition can mention the continuum, a generic object, or the set-theoretic universe itself. Its truth can change even when the particular sign sequence and all arithmetic performed on old inputs stay unchanged. Nor is the relation $V = \text{HOD}$ preserved by arbitrary forcing.

14 Relativization and a surcomplex companion

14.1 An allowed set of ordinal information [04]

Let p be a fixed set of ordinals. Write $\text{OD}(p)$ for definitions using p together with finitely many ordinals, and $\text{HOD}(p)$ for hereditary membership in this class. Here the parameter is the *single set* p , not permission to use its arbitrary elements as unrelated parameters. Since p is a set of ordinals, it belongs to $\text{HOD}(p)$; this is an inner model of ZFC with its corresponding canonical definable well-order [10].

Proposition 14.1 (Relativized theory [04]). *Put $K_p = \mathbf{No} \cap \text{OD}(p)$ and $I_p = \mathbf{Oz} \cap \text{OD}(p)$. Then*

$$K_p = \mathbf{No} \cap \text{HOD}(p) = \mathbf{No}^{\text{HOD}(p)}, \quad K_p = \text{Frac}(I_p).$$

The field K_p is real closed and I_p is an integer part. Every member of $\mathcal{U}$ is in $\text{OD}(p)$ if and only if $V = \text{HOD}(p)$. If $V \neq \text{HOD}(p)$, the family

$$\omega + \omega^{t\omega^{-(\alpha+1)}} \quad (\alpha \in \text{Ord})$$

is algebraically independent over K_p for every $t \in (0, 1) \setminus K_p$.

Proof. The hereditary sign-code argument only uses that ordinals and the fixed sign tags are hereditarily definable; permitting p does not change it. Unique-output closure gives the field, floor, and fraction claims. The reversible code uses no extra parameter, while the

membership-graph collapse works inside $\text{HOD}(p)$ because that inner model contains p . Finally every normal-form entry of an $\text{OD}(p)$ surreal is definable from p and ordinal parameters, so its support lies in K_p . The support-subfield amplification proof applies without change. $\square$

This relativization describes precisely how adding allowed information changes the theory. It is different from allowing *all* surreal parameters, which makes every individual surreal trivially definable by equality to itself.

14.2 Canonical surcomplex pairs [04, 07]

For completeness, define the surcomplex field by pairs $\mathbf{No}(i) = \mathbf{No} \times \mathbf{No}$ with $i = (0, 1)$ and the usual quadratic extension operations. This fixed coordinate representation names a real axis and an imaginary unit.

Proposition 14.2 (Definable surcomplex closure [04, 07]). *In the canonical pair representation, the ambiently A -definable surcomplex numbers are exactly $K_A(i)$, and they form an algebraically closed field. The ordinal-definable ones are exactly $K_{\text{HOD}}(i)$. If $V \neq \text{HOD}$, the real omnific family in [theorem 8.3](#) remains algebraically independent over $K_{\text{HOD}}(i)$ inside $\mathbf{No}(i)$ [04].*

Proof. Pairing and the two coordinate projections are parameter-free set-theoretic operations. Thus a pair is A -definable precisely when both coordinates are. The quadratic extension of a real closed field by i is algebraically closed. For the independence assertion, write a purported complex-coefficient polynomial relation as $P_0(\bar{z}) + iP_1(\bar{z}) = 0$, with polynomials over K_{HOD} . At real inputs both values are real, so both are zero; a nonzero relation would contradict real algebraic independence. $\square$

There is no conflict with complex conjugation or with the failure of a pure field to choose an imaginary unit. The proposition uses *ambient canonical coordinates* with their distinguished orientation. An abstract algebraically closed-field language without those coordinates has a different definable closure, just as the abstract real field does; coordinate-based definitions are not silently transferred to that reduct.

15 Changing the language changes the answer

The pure ordered field is [theorem 3.4](#).

15.1 Adding the ordinary exponential still does not define omega [07]

Proposition 15.1 ([07]). *In the exponential-field reduct,*

$$\text{dcl}_{\mathbf{No}_{\text{exp}}^M}(\emptyset) \subseteq \mathbb{R}^M.$$

In particular ω is not parameter-free definable in that reduct.

Proof. The embedded real exponential field is elementary in the surreal exponential field by the classical theorem recalled in [section 2.5](#). A parameter-free formula having a unique solution in the larger structure must have a solution in the elementary substructure, and uniqueness makes these the same solution. Thus the solution is real. $\square$

We make no attempt to identify the exact parameter-free definable closure in the real exponential field. In particular, this argument supplies no conclusion about whether any particular familiar transcendental real beyond those directly named by the language is definable there.

15.2 Simplicity already defines omega [07]

Proposition 15.2 ([07]). *In the structure $(\mathbf{No}^M; <, \prec_s)$, the ordinal subclass and the element ω are parameter-free definable.*

Proof. The condition

$$\text{OrdSign}(x) : \quad \forall y (y \prec_s x \longrightarrow y < x)$$

holds exactly for all-plus sequences. It holds for an ordinal because each proper prefix is smaller. If x has a minus sign, the prefix ending just before that sign is greater than x , so the condition fails. On this definable ordinal subclass, the numerical order is the usual ordinal order. A nonzero ordinal is a limit exactly when it has no greatest ordinal predecessor; this is first-order expressible by

$$x > 0 \quad \wedge \quad \forall y (\text{OrdSign}(y) \wedge y < x \longrightarrow \exists z (\text{OrdSign}(z) \wedge y < z < x)).$$

The least such ordinal is ω . □

The formula OrdSign is not new: the birthday-cutoff report uses the same formula, “every proper prefix of x lies below x ”, in the proof of `hset:prop:nondef`, which shows that neither birthday nor prefix simplicity is definable in the ordered-field reduct or any o-minimal expansion. Source 07 did not credit it. Much stronger interpretations become possible after adding birthday structure: the birthday-cutoff report develops birthday-enriched interpretations and bi-interpretations [20], and Chen–Hamkins–Yang’s related work has been publicly announced by Hamkins [17]. None of those interpretation results is needed to prove the coding and core theorems above, and none is asserted here as a new result.

15.3 The field with the omnific predicate: what the collection proves [merge]

Both sources asked what becomes definable in $\mathfrak{S} = (\mathbf{No}; +, -, \cdot, <, 0, 1, \mathbf{Oz})$, the field with only the omnific predicate added (04’s Question 1, 07’s question on intermediate languages). At their pin the collection did not address this. It now contains results that largely answer it; we record them with proofs from the cited theorems.

Proposition 15.3 (The pair $(\mathbf{No}, \mathbf{Oz})$ [merge]).

- (i) $\text{dcl}_{\mathfrak{S}}(\emptyset) \subseteq \mathbb{R}$. In particular no infinite or infinitesimal surreal, and so neither ω nor any code value of [section 6](#), is parameter-free definable in $\mathfrak{S}$.
- (ii) $\mathbb{R}$ and $\mathbb{Z}$ are parameter-free definable subsets of $\mathfrak{S}$. Hence every real that is parameter-free definable in $(\mathbb{R}; +, \cdot, <, \mathbb{Z})$ belongs to $\text{dcl}_{\mathfrak{S}}(\emptyset)$; since that structure interprets second-order arithmetic, these include every real whose binary expansion is parameter-free definable in second-order arithmetic.
- (iii) For every set $P \subseteq \mathbf{No}$, neither the class of Conway monomials nor the graph of $x \mapsto \omega^x$ is definable in $\mathfrak{S}$ with parameters from P .

Proof. (i) Every field automorphism of $\mathbf{No}$ preserves the order, since the positive elements are the nonzero squares; so a field automorphism preserving $\mathbf{Oz}$ is an automorphism of $\mathfrak{S}$. Fix a formula $\varphi(v)$ with a unique solution x in $\mathfrak{S}$. By induction on the subformulas of this fixed formula, every class automorphism σ of $\mathfrak{S}$ preserves φ , so $\sigma(x) = x$. By `opa:thm:fixed` of the automorphisms report [24], the common fixed field of all field automorphisms of $\mathbf{No}$ preserving $\mathbf{Oz}$ is $\mathbb{R}$. Hence $x \in \mathbb{R}$. (The dilation S_2 of `odg:def:ex:dilation`, $\sum_g a_g \omega^g \mapsto \sum_g a_g \omega^{2g}$, already suffices: it is a field automorphism preserving $\mathbf{Oz}$, and it fixes x only if $2 \text{supp } x = \text{supp } x$;

for a reverse well-ordered set this forces $\text{supp } x \subseteq \{0\}$, since a nonzero member g would give the increasing sequence $g, 2g, 4g, \dots$ or $g, g/2, g/4, \dots$ inside the support.)

(ii) By `odg:thm:realrecovery` and `odg:thm:cor:internal` of the omnific Diophantine report [22], the pure ring **Oz** parameter-free interprets **No** as its fraction field and defines in it the subfield $\mathbb{R}$ by a formula whose quantifiers range over **Oz**. In $\mathfrak{S}$ the map $(a, b) \mapsto a/b$ on **Oz** $\times$ (**Oz** $\setminus \{0\}$) is definable, so relativizing that formula to **Oz** defines $\mathbb{R} \subseteq \mathbf{No}$ without parameters. Then $\mathbb{Z} = \mathbb{R} \cap \mathbf{Oz}$. A parameter-free definition in $(\mathbb{R}; +, \cdot, <, \mathbb{Z})$ relativizes to one in $\mathfrak{S}$. That structure defines $\mathbb{N}$ and codes subsets of $\mathbb{N}$ by binary expansions of reals, which gives the last clause. (The pure ring **Oz** already defines $\mathbb{Z}$ and $\mathbb{N}$, `odg:thm:cor:arithmetic`.)

(iii) This is `opa:thm:parameters` of the automorphisms report. $\square$

So among ambiently definable surreals only reals can become parameter-free definable after adding the omnific predicate, and the codes of section 6, which recognize ordinal-indexed monomials and normal-form coefficients through set theory, have no counterpart there. The exact identification of $\text{dcl}_{\mathfrak{S}}(\emptyset)$ inside $\mathbb{R}$ is not known; it is squeezed between the reals definable in $(\mathbb{R}; +, \cdot, <, \mathbb{Z})$ and $\mathbb{R}$. The expansion by the omega map is not addressed by these results. Question 17.1 is re-scoped accordingly.

16 Arithmetic of the definable omnific ring

These results of source 07 apply to $I = I_A$ and, with the same proofs, to the OD omnific class. Their full-class antecedents are in the omnific Diophantine and quotient reports [22, 23], named at each statement below; the arguments are included to verify that the required division and support operations preserve ambient definability. They are restricted duplicates, not new results.

Proposition 16.1 (The constant-term retraction [07]). *There is a split exact sequence of additive groups and a ring retraction*

$$0 \longrightarrow \Pi_A \longrightarrow I_A \xrightarrow{\text{ct}} \mathbb{Z} \longrightarrow 0, \quad I_A = \Pi_A \oplus \mathbb{Z}.$$

The ideal Π_A is divisible as an additive group. For every positive ordinary integer n ,

$$nI_A = \Pi_A \oplus n\mathbb{Z}, \quad I_A/nI_A \simeq \mathbb{Z}/n\mathbb{Z}. \quad (16.1)$$

Furthermore $\Pi_A^2 = \Pi_A$.

Proof. The purely infinite part and the constant coefficient are uniquely definable from the normal form, so both remain in K_A . Products of terms with positive exponents still have positive exponents; therefore constant term is a ring homomorphism and Π_A is an ideal. For $p \in \Pi_A$, division by n changes coefficients but not positive support, so $p/n \in \Pi_A$. This proves divisibility and (16.1).

For the final assertion take nonzero $p \in \Pi_A$ with exponent set $S \subseteq \mathbf{No}_{>0}$. The simplest separator $\delta = \{0 \mid S\}$ is positive and strictly below every exponent of p . It is definable from p . Factor $p = \omega^\delta(p\omega^{-\delta})$. Both factors have strictly positive normal-form exponents and belong to K_A , so both are in Π_A . Thus $\Pi_A \subseteq \Pi_A^2$; the reverse containment follows because it is an ideal. The zero case is automatic. $\square$

The full-class statement is `odg:thm:finitequotients` ($n\mathbf{Oz} = \Pi + n\mathbb{Z}$, $\mathbf{Oz}/n\mathbf{Oz} \cong \mathbb{Z}/n\mathbb{Z}$).

Theorem 16.2 (All finite quotients [07]). *Every unital ring homomorphism from I_A to a finite ring factors through $\text{ct} : I_A \rightarrow \mathbb{Z}$. Every finite-index ideal of I_A is nI_A for a unique positive integer n . Consequently the profinite completion of I_A is canonically $\widehat{I_A} \simeq \widehat{\mathbb{Z}}$. Also ct is the unique unital ring homomorphism $I_A \rightarrow \mathbb{Z}$.*

Proof. Let $f : I_A \rightarrow R$ and suppose the additive group of R has finite exponent m . For $p \in \Pi_A$ choose $q \in \Pi_A$ with $p = mq$. Then $f(p) = mf(q) = 0$. Thus f factors through $I_A/\Pi_A \simeq \mathbb{Z}$. If an ideal J has finite index, apply this argument to $I_A \rightarrow I_A/J$ to get $\Pi_A \subseteq J$. The image of J in $\mathbb{Z}$ is $n\mathbb{Z}$ for a unique $n \geq 1$. Hence $J = \Pi_A \oplus n\mathbb{Z} = nI_A$. The compatible quotient identifications in (16.1) identify the inverse system of all finite quotients with that of $\mathbb{Z}$. For a homomorphism $f : I_A \rightarrow \mathbb{Z}$, the image of each $p \in \Pi_A$ is divisible by every positive integer because p is. The only such integer is zero. Again f factors through ct , and unitality forces its restriction to $\mathbb{Z}$ to be the identity. $\square$

For the full ring these are `odg:thm:finitequotients`, `odg:cor:charideals` and `odg:eq:profinite` of the omnific Diophantine report, and the much stronger universal theorem `osq:thm:universal` of the quotient report, which factors every homomorphism into a set-sized ring through ct . Related (batch 34): the surcomplex-automorphism report proves the finite-quotient statement, and the transfer of solvability of integer polynomial systems through ct , uniformly for its integer-part rings R_F over set-sized real closed fields F , with $\mathbf{Oz} = R_{\mathbb{R}}$ and 2^c pairwise nonisomorphic rings among them (`saut:gr:prop:finite`, `saut:gr:prop:dio`); it cites the theorem above for the rings I_A .

Proposition 16.3 (A size-sensitive nontransfer [07]; already in the quotient report). *For a transitive set model M and finite A , the externally countable ring I_A^M admits a homomorphism to a set-sized ring which does not factor through constant term.*

Proof. Take the identity map $I_A^M \rightarrow I_A^M$. Its target is a set externally, and is countable by (3.1). It cannot factor through constant term because $\omega \in I_A^M$, $\omega \neq 0$, and $\text{ct}(\omega) = 0$. $\square$

There is no contradiction with the full-class theorem. Its source is the entire proper class of omnific integers in the same universe in which target size is measured. Here the source is a fragment, and the external set I_A^M need not be an internal set of M . The finite-target theorem survives because its proof uses divisibility rather than a proper-class size obstruction. Source 07 presented this “external size-boundary warning” as supplied by it; it is already in the quotient report, in the paragraph “The universe-relative reading” of `osq:sub:foundations` (from that report’s manuscripts 04, 10 and 11), which observes that a set-sized fragment viewed in a larger universe has its own identity map as a set-sized target.

Proposition 16.4 (Non-Noetherian behaviour [07]). *The ring I_A is not Noetherian. In fact it has the strict ascending chain*

$$(\omega) \subsetneq (\omega^{1/2}) \subsetneq (\omega^{1/3}) \subsetneq \dots$$

Proof. Every displayed monomial is parameter-free ambient-definable. Since $\omega^{1/n} = \omega^{1/(n+1)}\omega^{1/n-1/(n+1)}$ and the second factor is omnific, the ideals are increasing. Equality would put $\omega^{1/(n+1)}/\omega^{1/n} = \omega^{-1/(n(n+1))}$ in I_A . This is a nonzero infinitesimal, hence is not omnific. Therefore each inclusion is strict. $\square$

The full-ring chain $(\omega) \subsetneq (\omega^{1/2}) \subsetneq (\omega^{1/4}) \subsetneq \dots$ is `odg:prop:notnormal`.

Proposition 16.5 (Elementary Diophantine transfer [07]). *A finite system of polynomial equations with ordinary integer coefficients has a solution in I_A if and only if it has a solution in $\mathbb{Z}$. This equivalence need not persist when nonvanishing conditions are added.*

Proof. Apply the ring retraction ct coordinatewise to any solution in I_A . Conversely every integer solution already belongs to I_A . For the failure with nonvanishing, the equation $X^2 = 2Y^2$ has the nonzero omnific solution $X = \sqrt{2}\omega$, $Y = \omega$, both parameter-free ambient-definable, whereas the only ordinary integer solution is $(0, 0)$. $\square$

The full-ring statements are `odg:thm:transfer` and `odg:def:thm:ideal`, where $\exists y x^2 = 2y^2$ defines Π in **Oz**. In particular the nonnegative part of I_A is not a model of Peano arithmetic: PA proves that the displayed equation has no nonzero solution. This is distinct from open induction ([corollary 4.6](#)), the familiar weaker theory associated with integer parts of real closed fields [[11](#)]; no stronger induction assertion is being inferred from finite-quotient agreement with $\mathbb{Z}$.

Remark 16.6 (Relation to `odg:q:size [merge]`). The omnific Diophantine report asks (`odg:q:size`) which set-sized exponent groups or universe-relative constructions admit analogues of the common-monomial and set-sized quotient theorems. The definable fragments K_A and I_A supply one instance: same-parameter per-element monomial clearing holds ([theorems 4.1](#) and [10.5](#)); finite quotients remain $\mathbb{Z}/n\mathbb{Z}$ ([theorem 16.2](#)); and the set-sized quotient theorem fails through the identity map ([proposition 16.3](#), already observed in the quotient report). This is partial information only. The question is about general set-sized exponent groups and constructions, and it stays open.

17 Foundational boundaries, further questions, and formalization

17.1 From set models to an intended universe [07]

The set-model convention is a clean semantic setting, not a theorem of ZFC that such a model exists. All theorems quantified over such M are conditional statements. The set-theoretic coding constructions themselves, including $\mathbb{C}$, $\mathbb{T}$, $\mathbb{Q}$, $\mathbb{Q}_{a,b}$, $E_\alpha(X)$ and esc , are ordinary ZFC-definable class constructions and do not require the existence of a set model.

For an intended universe V , one may discuss parameter-free definability externally, or work in a stronger metatheory supplying the relevant truth apparatus. What one may not do is count all true definitions of V using an unmentioned truth predicate and then invoke Replacement *inside that same* V . The OD/HOD part avoids this difficulty because OD and HOD have internal first-order definitions. The noncollection theorem is formulated precisely so that no hidden internal list of definitions is used.

For nonstandard models of ZFC, [definition 3.1](#) still uses standard external formulas, and the closure-under-definitions principle continues to hold; so do the pointwise test and the coding theorems, whose proofs are internal. However, the ordinal and sign arguments about a *least externally omitted ordinal* need not transfer unchanged: the model's ordinals may not be externally well founded. [Theorem 10.3](#) and its corollaries are not claimed for such models. The pointwise-definability literature treats nonstandard models as well as transitive cases [[12](#)]; these distinctions should be retained when importing its results.

17.2 Questions

The questions below are further projects, not omitted hypotheses or unfinished steps in the proofs above. They are not asserted to be previously published open problems.

Question 17.1 (Languages between fields and set theory [04, 07]; re-scoped). Determine the parameter-free definable closure of $\mathfrak{S} = (\mathbf{No}; +, -, \cdot, <, 0, 1, \mathbf{Oz})$ inside $\mathbb{R}$, or of well-specified

set-sized models of its first-order theory. How much of ambient definability is captured by an expansion naming the omega map, which bounded coding maps become definable there, and what are its intrinsic definable closures?

Source 04 asked the first part in the form “which ambiently definable surreals become intrinsically definable after adding only the omnific predicate”; source 07 asked about $(\mathbf{No}; <, +, \cdot, \mathbf{Oz})$ and the omega-map expansion. By [proposition 15.3](#) (from the automorphisms and omnific Diophantine reports) only reals can, all reals definable in $(\mathbb{R}; +, \cdot, <, \mathbb{Z})$ do, and the monomials are not definable with set parameters; the exact real closure and the omega-map expansion remain open. The pure ordered field defines only real algebraic elements without parameters ([theorem 3.4](#)); order plus simplicity already defines ω ([proposition 15.2](#)); birthday-enriched interpretation results supply a much stronger endpoint, but do not settle every intermediate language by themselves.

Question 17.2 (Birthday cost of reversible omnific coding [04]). Obtain sharp upper and lower bounds for $b(C(s))$ in terms of $b(s)$, and compare alternative reversible codes with the same leading term and coefficient alphabet. Can a useful optimality statement be proved for a specified class of such codes?

The construction controls support order type and the leading scale, not the exact surreal birthday of its nested monomials. No unstated birthday estimate is used in the independence theorem. [Lemma 2.6](#) bounds $b(\omega^a)$ by $\omega^{b(a)}$, which is a starting point for upper bounds.

Question 17.3 (Intermediate support fields [04]). Classify naturally occurring proper subfields satisfying (8.3), especially those arising from relative definability or canonical inner models. Determine when stronger structural invariants than the independent family can be read from the quotient additive group $\mathbf{No}/K$.

The support-subfield theorem already gives one invariant uniformly: a single element outside such a field produces an ordinal-indexed independent family in one fixed omnific interval. Instances are K_{HOD} and $K_p = \mathbf{No} \cap \text{OD}(p)$ ([proposition 14.1](#)). For a proper same-ordinal inner model M as in the universes report, $\mathbf{No}^M$ would satisfy (8.3) if the omega map and normal forms were absolute between M and V ; `univ:prop:absolute` proves absoluteness of the arithmetic and of cuts only, so this instance is not claimed from that report. For inner models with the same reals it is now supplied by the large-cardinal report ([remark 17.6](#)).

Question 17.4 (Possible first omitted ordinals [07]). For which transitive M and parameter sets A can a specified epsilon number occur as ρ_A^M ? What additional closure constraints follow from definable ordinal functions beyond finite Cantor normal form?

There are immediate further constraints: O_A^M is closed under every parameter-free definable ordinal-valued function on definable inputs. For example, if an ordinal index is definable, its indexed epsilon number is definable. Determining the exact resulting external cut is a set-theoretic problem, not a matter of adding a few field operations.

Question 17.5 (Effective syntactic costs [07]). After fixing a proof system and a precise coding of formulas, how efficiently can one translate definitions through Q , E_α , and the denominator rule? Which translations preserve a chosen level of the set-theoretic formula hierarchy?

Interdefinability is proved here by fixed formula substitution. It does not automatically preserve quantifier complexity, and no optimal description-length theorem is asserted. This question links the present semantic theory to the collection’s separate work on computable surreals without identifying the two notions.

Remark 17.6 (The same-reals instance, from the large-cardinal report). The report `docs/foundations-and-computation/large-cardinal-embeddings-and-normal-forms` proves, in its absoluteness lemma (`lce:lem:absolute`, from three manuscripts of batch 30), that for transitive inner classes $M \subseteq V$ of ZFC with the same ordinals and the same reals the inclusion $\mathbf{No}^M \subseteq \mathbf{No}$ preserves the field operations, the omega map and canonical normal forms. Its `lce:rem:dsnsupport` draws the consequence for [question 17.3](#): if moreover $M \neq V$, then $K = \mathbf{No}^M$ satisfies (8.3) and is proper (a transitive inner model of ZFC containing every set of ordinals is V), so [theorem 8.2](#) applies to it. The targets of elementary embeddings $j : V \rightarrow M$ with a critical point, such as the ultrapower by a normal measure, are instances. Inner models that lack some real, such as L when $\mathbb{R} \not\subseteq L$, or HOD when it lacks a real, are not covered, and the classification asked in [question 17.3](#) stays open. This remark, and the sentence after [question 17.3](#) that points here, were added when the large-cardinal report was written; nothing else in this report changed.

[write] **Status of the questions after batch 93, 4 October 2026.** Parts [II](#) and [III](#) bear on four of these questions; all stay open as re-scoped here. [Question 17.1](#). Its omega-map half is answered at the level of an interpretation: in $(\mathbf{No}; +, -, \cdot, <, 0, 1, \mathbf{Oz}, \Omega)$ every normal-form coefficient and truncation is parameter-free definable ([theorem 36.3](#)), and $(V, \in)$ has a parameter-free quotient interpretation with representatives in the bounded monomial interval ([theorem 37.1](#)); source 02's [proposition 26.11](#) is the case of the unary code. The definable closures of that expansion, of $\mathfrak{S}$ and of the language with the omega map alone, and a bi-interpretation, remain open ([questions 42.1](#) and [42.2](#) and the questions of [section 28](#)). [Question 17.3](#). The instance this remark leaves open is settled: for every transitive inner model $M \neq V$ of ZFC, with or without all reals, $\mathbf{No}^M$ satisfies (8.3), because canonical operations and normal forms are absolute without a same-reals hypothesis ([theorem 34.1](#) and [corollary 34.4](#)), and [theorem 8.2](#) applies ([corollary 40.2](#)); so do L and HOD when they lack reals. The real-parameter field $K_{\mathbb{R}}$ is a further instance when $V \neq \text{HOD}_{\mathbb{R}}$ ([remark 22.10](#)). The classification stays open ([question 42.11](#)). [Question 17.5](#). Answered in part: the canonical operations, normal-form evaluation, $\mathbf{Oz}$ and the floor have Δ_1 graphs over ZFC, and x and $\mathbf{Q}(x)$ have the same Σ_n , Π_n and Δ_n singleton status with an additive description-length bound ([theorem 34.7](#), [proposition 34.9](#), and [corollary 40.3](#)); optimality and the other codes remain open. [Question 17.2](#) is refined by [question 42.7](#) and by the birthday-cost questions of [section 28](#), and [question 17.4](#) is extended by [question 42.11](#).

17.3 A realistic Lean decomposition

A formalization should separate the general Hahn-field mathematics from satisfaction and internal models (04). The shortest algebraic route is to formalize [lemma 8.1](#) for set-sized Hahn series and finite polynomials first: coefficient supports in different cosets cannot cancel. Next formalize the finite linear independence of monomials, the normalized-parameter argument, and the finite form of [theorem 8.2](#). These steps require no model-theoretic truth predicates or proper-class objects.

The set-theoretic track then needs a precise universe convention, canonical sign codes, unique normal forms, and definable class maps. HOD identification requires a model-of-ZFC interface, with standard external formulas kept distinct from internal formula codes. The statements about B_n should be separate theorems for each fixed fragment, not a falsely uniform truth evaluator. Source 07 proposes three interfaces: a sign/normal-form interface with definable canonical operations; an external definability-closure interface for set structures; and a model-theoretic selection interface for o-minimal exponential fields. The bounded code uses only the first two, while the core theorem additionally uses the published epsilon-cutoff theorem and the intersection argument. The repository documents sign, Hahn, size, and normal-form interfaces, but neither source nor this report maps the new statements to checked declarations, claims that the required interfaces already exist in Lean, or claims to have built or extended the Lean code [\[19\]](#).

18 Conclusion

There is no single language-independent class of “the definable surreals.” Once the ambient universe and permitted parameters are fixed, however, the basic algebra is robust: definable surreals form a real closed field, exponential and elementary over a transitive model, and definable omnific integers are an integer part with exactly that fraction field. Ordinal parameters reveal a particularly natural inner field, the canonical surreal field of HOD.

The strongest results are not counting observations. Explicit normal-form maps compress every surreal definability degree into a fixed-leading-term omnific interval, or into single monomials below ω . If HOD fails to contain the universe, one missing surreal generates an ordinal-indexed algebraically independent family in that same interval. The failure of the definable field to be initial is controlled exactly by the first omitted ordinal, and the resulting maximal initial fragment is still an elementary exponential field and the fraction field of its omnific integers. Uniform coefficient assignments, rather than small coefficient alphabets, numerical size or ordinary congruences, carry the missing information. These conclusions coexist with, rather than evade, the traditional cautions about truth, parameters, HOD, and pointwise definable models.

A Provenance, proof scope, and reproducibility

[write] **4 October 2026.** This appendix concerns Part [I](#). The provenance, files and checks of Parts [II](#) and [III](#) are in [sections 19.5 and 30.3](#) and in the README; their notation is in [sections 19.6 and 30.4](#).

A.1 Where each section comes from

Table 1: Provenance of the sections.

Section	Sources
1 Introduction	04’s introduction and 07’s scope section, merged; provenance and conventions added.
2 Foundations	04 §2 with 07 §2 (simplicity, initial classes, two exponentials, floor prefix, imported cutoff theorem and birthday controls); remark 2.5 merge.
3 Definability	04 §3 with 07 §3; field dcl printed once.
4 Field and integer part	04 §4 with 07 §4 (selection, $\mathcal{L}_{\text{exp}}$ elementarity, cut denominator, valuation, normal-form invariant).
5 HOD	04 §5 with 07 §8.1; proof of proposition 5.4 corrected.
6 Codes	04 §6 and 07 §§5.1–5.3.
7 Universe tests	04 §7 with 07 §§5.4, 8.2, 8.3.
8 Transcendence	04 §8.
9 Uniformity	04 §9 with 07 §6.
10 Core	07 §7.
11 First birthday	04 §10, printed by citation of the universes report.
12 Complexity	04 §11 with 07 §5.5.
13 Forcing	04 §12 with 07 §8.4.
14 Relativization	04 §13 with 07 §9.4.
15 Languages	07 §§9.2–9.3; proposition 15.3 merge.
16 Arithmetic	07 §10 with credits added.

Section	Sources
17 Questions	04 §14 and 07 §11.
Appendices	04's appendices and 07's audit section and gallery.

A.2 Claim ledger

Table 2: Status of the statement families.

Statement family	Source	Status	Dependency and qualification
Signs, normal forms, real closedness, omega map, exponential, floor	04, 07	Classical	Conway, Gonshor, van den Dries–Ehrlich (with erratum), L’Innocente–Mantova; elementary proofs supplied where useful.
Exponential elementarity and epsilon cutoffs	07	Published input	van den Dries–Ehrlich, erratum, Bournez–Guilmant. Not new.
Ambient field and integer-part closure	04, 07	Standard-method consequence	Unique-output closure, ordered finite root selection, and a uniformly chosen denominator. No priority claim for the closure principle; $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ is <code>odg:thm:fractions</code> .
$K_A^M \prec \mathbf{No}^M$ in $\mathcal{L}_{\text{exp}}$	07	Application	Fixed-formula translation and o-minimal definable selection; transitive M .
Global birthday/set-theory interpretation	04, 07	Prior announced work	Chen–Hamkins–Yang; not used as a black-box dependency and not claimed.
HOD/sign-code identification	04, 07	Foundational application	Hereditariness of a sign code, inner-model recursion, canonical normal forms. No claim to have introduced HOD-valued surreal fields.
Fixed-interval sign code	04	Proposed contribution	Theorem 6.2 ; search did not locate this precise code.
Bounded and local monomial codes	07	Proposed contribution	Theorems 6.8 and 6.9 , with two-way parameter preservation.
One-interval universe tests	04, 07	Proposed consequences	Theorems 7.2 and 7.7 ; standard pointed set coding followed by the omnific codes. Density form is Hamkins 2012; pointwise model existence is HLR.
Support-subfield amplification and HOD dichotomy	04	Principal proposed contribution	Theorems 8.2 and 8.3 ; finite support-coset proof. No transcendence-basis claim.

Statement family	Source	Status	Dependency and qualification
Two uniformity gaps	04, 07	Proposed explicit formulations	Theorems 9.1, 9.3 and 9.6 and proposition 9.5 ; the general distinction between pointwise and uniform definability is standard.
Maximal initial elementary core	07	Principal proposed contribution	Theorems 10.3 and 10.5 and corollary 10.6 ; birthday estimates and cutoffs are classical inputs; transitive M .
Bounded birthday escape and noncollection	07 (04)	Proposed consequences	Theorems 6.10 and 12.5 .
First non-HOD birthday	04	Duplicate, printed by citation	<code>univ:prop:beta</code> with $M = \text{HOD}$; also <code>hset:thm:firstbirthday</code> ; the compactness restriction is [16].
Bounded-complexity escape	04	Standard-method refinement	Fixed-fragment truth and Replacement; no all-level truth predicate and no asserted one-step complexity bound.
Forcing without new reals	04, 07	Application	Standard homogeneous closed forcing over L , combined with the independent-family theorem.
Density iff $V = \text{HOD}$	07	Hamkins 2012	Credited in theorem 7.5 .
Old-field closed nowhere denseness	07	Specialization	<code>univ:thm:nowheredense</code> .
Simplicity defines ω	07	Elementary	The formula is that of <code>hset:prop:nondef</code> .
Pair (No , Oz)	merge	From the collection	<code>opa:thm:fixed</code> , <code>opa:thm:parameters</code> , <code>odg:def:thm:realrecovery</code> .
Finite quotients, constant term, non-Noetherian, equations, size boundary	07	Restricted duplicates	<code>odg:thm:finitequotients</code> , <code>odg:prop:notnormal</code> , <code>odg:thm:transfer</code> , <code>osq:thm:universal</code> , <code>osq:sub:foundations</code> .
Finite verification	04, 07	Regression tests only	Finite symbolic codes, decoders, formal support-coset and floor checks. They prove no infinite or set-theoretic theorem.

A.3 Stale statements and corrections

1. “*All 51 reports.*” 04’s inspection note said that it was not a review of “all 51 reports”. At the pin the collection had 51 reports; at the placement commit it has 53 written reports, and this report was placed together with two others.

2. *First new birthday*. 04 credited only an “older repository discussion” in the birthday-cutoff report. The exact theorem, with the same proof, is `univ:prop:beta`; the birthday-cutoff report has it for same-ordinal models as `hset:thm:firstbirthday`. 04’s theorem is printed by citation.
3. *Hamkins 2012*. 04 did not cite Hamkins’s density/ $V = \text{HOD}$ answer, which is close to its equivalence of (i) and (ii) in [theorem 7.2](#); the credit is added.
4. *Size boundary*. 07 described its size-boundary warning as “supplied here”. It is in the quotient report (`osq:sub:foundations`).
5. *OrdSign*. 07’s formula defining the ordinals from order and prefix is the one in the proof of `hset:prop:nondet`; credit added.
6. *Cutoff theorem*. 07 cited van den Dries–Ehrlich without the erratum; the erratum and Bournez–Guilmant are added.
7. *Universes report*. 07’s “Theorem 14.3 in the report” is `univ:thm:nowheredense`. 07’s statement that the universes report “develops this compatibility explicitly” (for normal forms and the exponential) is only partly accurate: `univ:prop:absolute` covers the arithmetic and cuts.
8. *Restricted duplicates*. 07’s arithmetic results are restricted versions of `odg:thm:finite` quotients, `odg:prop:notnormal`, `odg:thm:transfer`, `odg:def:thm:ideal` and the open-induction discussion `odg:rem:foursquares`; the labels are now named.
9. *Omnific-predicate questions*. Neither source knew the automorphisms report or the reconstruction section of the omnific Diophantine report; their language questions are re-scoped ([proposition 15.3](#) and [question 17.1](#)).
10. *Proof gap*. 04’s proof that a proper K_{HOD} is not intrinsically definable took the interval endpoints in the subfield ([proposition 5.4](#)).
11. *Bibliography*. 07 gave Gonshor’s book as Lecture Note Series 124; it is number 110, as 04 has it.
12. *Delivery files*. Both sources described their own archives (`.tex`, PDF, README). Those files are not shipped here; the shipped files are listed in the README.

A.4 What each source inspected

Source 04. The repository was accessed through its GitHub connector at the pin. Inspection covered its README, the report catalogue, the introduction and provenance section of the birthday-cutoff/hereditary-set report, and the introduction and convention section of the omnific-Diophantine report. It was not a line-by-line audit of all reports or their preserved source manuscripts. No repository files were edited, no commit was made, and no Lean build was performed. The literature comparison used the supplied definable-real reference, primary works and author announcements on pointwise definability, HOD, surreal arithmetic, omnific normal forms, and quantifier elimination; the recent first-HOD-disagreement preprint was checked as well. The cited works and the catalogue did not reveal the fixed-interval code or the support-subfield amplification. This limited negative search evidence cannot establish publication priority.

Source 07. The repository was inspected at the pin through its connected GitHub interface: the README, the catalogue, and the detailed guides for the universes, birthday and omnific

Diophantine reports, whose cross-references identified the quotient report. These guides identify the statements, provenance and proof-review boundaries of the earlier manuscripts; they are not an exhaustive independent review of every proof or file. The literature search, on 23 September 2026, combined “definable surreal”, “ordinal definable surreal”, “surreal HOD”, “definable omnific”, “first undefinable ordinal surreal” and “definable surreal initial ordinal” with the named classical papers. The most relevant precedent located was Hamkins’s 2012 density answer; the epsilon-cutoff statement was checked against Corollary 5.5 of the original paper. No matching statement of the coding/core package was located; that supports describing these as proposed new formulations, and does not certify priority or exclude an equivalent result under different terminology.

The merge. Every cross-report label cited here was checked in the collection at the placement commit. The Bournez–Guilmant abstract was checked for the epsilon-number characterization.

A.5 Files and checks

The source is self-contained apart from ordinary LaTeX packages: its bibliography is included below, and no external graphics or bibliography database is required. The two finite scripts shipped with the report are exact regression checks. Neither decides definability, approximates HOD, or replaces ω by a large real number: that numerical substitution would destroy the exact non-Archimedean ordering relevant to the proofs.

Source 04’s checks. The script represents the nested exponents e_n by formal tags. Its recorded run passed 8,191 finite code round trips through sign length 12, 11 malformed-code rejections, 3,121 exact rational normalizations of the two-term map h with 3,120 strict order comparisons and 2 rejected endpoints, 2,157 formal infinitesimal floor cases, and 500 finite support-coset polynomial checks with 55,105 noncancelled formal support terms. A deliberately dependent exponent example correctly produced a vanishing polynomial, checking that the independence hypothesis is not silently omitted.

Source 07’s checks. The script checks finite normal-form bit encodings, their decoders and marker bounds, rational instances of the compression/inverse identities, and floor-correction examples. Its recorded run passed 8,191 bit codes (exhaustive for lengths 0 through 12), 4 malformed-code rejections, 555 rational compression/inverse pairs, 554 adjacent monotonicity comparisons, 1,161 floor-correction cases and 24 support-shift cases. It verifies no transfinite, set-theoretic satisfaction, HOD, epsilon-cutoff, elementarity, or core statement.

Both scripts were rerun for this merge on copies and reproduced their recorded results up to line endings.

B Non-claims

Every limitation stated by either source is kept here; most also appear at the relevant statement.

B.1 Source 04

N4.1 The report is not refereed, not formalized in Lean, and makes no mapping to Lean declarations; no Lean build was performed.

- N4.2 The coding and transcendence results are proposed as original on the strength of a limited, non-exhaustive search; priority is not established, and no theorem settles a named open problem in the published literature.
- N4.3 $\mathbf{C}$ is not a ring homomorphism, is not definable in the pure ordered field, and has no arithmetic compatibility (the code of 0 is not 0).
- N4.4 “Ord-indexed independent” means only that finite subfamilies are independent. No transcendence basis, no Global Choice, no class sum, and no ordinary cardinal of a proper class is asserted; in a set model the assertions have their internal meaning.
- N4.5 $D_{\mathcal{O}}^V$ is not an internal set or class; there is no truth predicate for V .
- N4.6 For nonstandard models, codes are internal objects; the finite-ambiguity argument concerns genuinely finite sets.
- N4.7 The definable closure of $(\mathbf{No}; +, -, \cdot, <, 0, 1, \mathbf{Oz})$ is not classified (see [proposition 15.3](#) for what the collection now proves).
- N4.8 Definability is not identified with computability or with the computable-surreal representations; normal-form uniqueness supplies no algorithm for infinite codes and no topological convergence of partial sums.
- N4.9 $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ is not new; the contribution is the uniformity of the denominator; no priority is claimed for the general closure principle.
- N4.10 Termwise HOD membership is insufficient; K_{HOD} is not the Hahn field of HOD entries.
- N4.11 The widths in the localization are infinite monomial widths; a real width below 1 contains at most one omnific integer; no density is claimed.
- N4.12 The B_n escape asserts no $m = n + 1$, no closed form, no algorithm, and no cofinality of the d_n ; the sequence of the B_n is not uniformly definable.
- N4.13 The BCGHP corollary is a translation of a cited theorem, not a new surreal theorem; λ_H is not claimed to be regular.
- N4.14 Forcing: old definitions need not keep their truth or uniqueness; $V = \text{HOD}$ is not preserved by arbitrary forcing; a generic filter over the actual universe is not assumed to be a set in it.
- N4.15 The surcomplex results use canonical coordinates; the abstract algebraically closed-field language behaves differently.
- N4.16 The questions are not published open problems and not omitted hypotheses.
- N4.17 The finite checks are regression tests only; ω is never replaced by a real, and no infinite or set-theoretic theorem is checked.
- N4.18 The Chen–Hamkins–Yang announcement is not a black-box dependency; the set-coding mechanism is not new.
- N4.19 No nonclassical Diophantine, quotient, automorphism or bi-interpretation claim of the repository is a proof dependency.
- N4.20 The collector has a nondefinable coefficient only under the stated hypothesis that some member of $\mathbb{R} \cap \text{HOD}$ is not parameter-free definable.
- N4.21 No birthday estimate for codes is claimed or used.

- N4.22 The existence of pointwise definable models is prior work (HLR).
- N4.23 No claim to have introduced HOD-valued surreal fields; the first-birthday threshold is the standard first subset disagreement.
- N4.24 The two-term code is an elementary observation, not a major novelty claim.

B.2 Source 07

- N7.1 The main setting is an externally fixed transitive set model; its existence is not a ZFC theorem, and theorems quantified over it are conditional.
- N7.2 The core theorem and its corollaries are not claimed for nonstandard models.
- N7.3 D_A^M is not claimed to be elementary in M ; K_A^M need not be represented by a set of M .
- N7.4 Provable uniqueness does not give absoluteness (the CH example).
- N7.5 There is no algorithm deciding definability; definability is not identified with computability.
- N7.6 The $\mathcal{L}_{\text{exp}}$ results do not cover the restricted-analytic language or other uncountable languages.
- N7.7 $\mathbf{Q}$ is not onto an interval, is not ordered-field-definable, and uses Ω , not $\exp$.
- N7.8 The birthday escape does not give density.
- N7.9 The set-coding lemma gives no canonical definable injection of sets into $\mathbf{Oz}$; its use of Choice is existential.
- N7.10 Noncollection concerns sets of M , not classes.
- N7.11 The reverse gap is conditional; an internally uncountable ordinal need not have an undefinable predecessor.
- N7.12 ρ is external, not a parameter (the Berry distinction).
- N7.13 An initial real closed subfield need not be the fraction field of its omnific part (real algebraic numbers with $\mathbb{Z}$).
- N7.14 The finite-parameter discontinuity needs an externally uncountable ordinal.
- N7.15 Hamkins 2012, the old-field topology result, pointwise-model existence, birthday bi-interpretations, full-class denominator clearing and full-class finite quotients are credited, not claimed.
- N7.16 There is no exact definable closure for the real exponential field and no claim about particular transcendental reals.
- N7.17 The complexification depends on the chosen orientation and is not transferred to the pure algebraically closed-field language.
- N7.18 The full-class quotient theorem does not transfer to fragments; no Peano arithmetic and no stronger induction is inferred from finite-quotient agreement.
- N7.19 The questions are not published open problems.
- N7.20 The finite checks verify no transfinite, HOD, epsilon-cutoff, elementarity or core statement.
- N7.21 No Lean formalization; priority is not certified; novelty and correctness are separate, and many arguments are short combinations of established theorems.

- N7.22 Interdefinability by formula substitution need not preserve quantifier complexity; no optimal description-length theorem is asserted.
- N7.23 The Cohen example is relative to the existence of the ground model; the homogeneity argument does not assert that no new OD sets of arbitrary kind are added, nor that every forcing extension has the same HOD.
- N7.24 The $\mathcal{L}_{\text{exp}}$ elementarity is proved formula by formula, with no truth predicate in M .
- N7.25 The self-similar value group is not a claim that K_A contains every Hahn series with entries in K_A .

B.3 The merge

- NM.1 [Proposition 15.3](#) does not identify $\text{dcl}_{\mathfrak{E}}(\emptyset)$; the omega-map expansion is not addressed.
- NM.2 `odg:q:size` is not answered; [remark 16.6](#) gives one instance.
- NM.3 Support-subfield amplification is not claimed for general same-ordinal inner models ([question 17.3](#)).
- NM.4 The agreement of HOD's internal omega map, exponential and normal forms with the ambient ones rests on the sources' recursion argument, not on the universes report, which covers arithmetic and cuts.

C A small gallery of examples [07]

Input or datum	Constructed number	What it illustrates
$x = 0$	$Q(0) = \omega^{1/2}$	A definable bounded omnific code
$x = 3/4$	$Q(3/4) = \omega^{4/5}$	Exact algebraic compression
$x = -3/4$	$Q(-3/4) = \omega^{1/5}$	Order preservation
$\alpha = 0, X = \emptyset$	$E_0(X) = \omega^{3/4}$	Marker and empty decoder
$\alpha = 2, X = \{1\}$	$\omega^{3/4} + \omega^{1/2} + 2\omega^{1/3}$	Finite bit assignment
$x = \omega - \omega^{-1}$	$\lfloor x \rfloor_{\mathbf{OZ}} = \omega - 1$	Negative infinitesimal correction
$(a, b) = (\omega, 2\omega)$	$Q_{a,b}(x) = \omega + \omega^{\tau(x)}$	Coding in an infinite-width interval
$s = 1$	$C(1) = \omega + 2\omega^{\omega^{-1}} + 3\omega^{\omega^{-2}}$	The sign code (04)

For the second row, $\sqrt{1 + (3/4)^2} = 5/4$, so $\tau(3/4) = (1 + 3/5)/2 = 4/5$. Every identity is an exact field identity, not a numerical approximation. The examples involving ω are normal forms with symbolic exponents and are not evaluated by substituting a large real number for ω .

D Notation and reading guide

Table 3: Notation.

Notation	Meaning
$\mathbf{No}, \mathbf{Oz}, \Pi$	Canonical surreal field, classical omnific-integer ring, purely infinite ideal (07's $\mathcal{J}$).
$b(s), \prec_s$	Ordinal length of the canonical sign sequence; proper prefix.
$\omega^x = \Omega(x)$	Conway monomial map; not $\exp(x \log \omega)$.
$\underline{\alpha}$	The surreal all-plus ordinal α .
$\text{NF}(x), \text{supp}(x)$	Whole indexed normal form; its exponent support.
M, A	A set model of ZFC (transitive where stated) and an external allowed parameter family.
D_A^M	External definable closure of A in $(M, \in)$ (07's $\text{Def}_M(A)$).
K_A, I_A, O_A	Ambiently A -definable surreals, omnific integers, ordinals (07's $D_M(A), I_M(A), O_M(A)$).
Π_A, R_A	$K_A \cap \Pi$ and $K_A \cap \mathbb{R}$.
$K_{\text{HOD}}, I_{\text{HOD}}$	Ordinal-definable surreals and omnific integers; $K_{\text{HOD}} = \mathbf{No}^{\text{HOD}}$.
$\mathcal{U}, \mathcal{U}_2$	$\{z \in \mathbf{Oz} : \omega < z < \omega + \omega^{1/2}, \text{ct}(z) = 0\}$ and its two-term part.
e_α	$\omega^{-(\alpha+1)}$, a positive infinitesimal surreal.
ι_β	$1/(\underline{\beta} + 2)$ (07's e_β).
$\mathbf{C}, \mathbf{D}$	04's sign code and its partial inverse.
$\mathbf{T}, h$	04's two-term code and its rational compression.
$\tau, \mathbf{Q}, \mathbf{Q}_{a,b}$	07's algebraic compression, monomial code and local code (07's $C, C_{a,b}$).
$\text{esc}(\alpha)$	07's bounded code with birthday above α (07's T).
$E_\alpha(X)$	07's bit code of $X \subseteq \alpha$.
Set, X_a	04's decoder of sets from signs, and the ordinal code of a (04's E, B).
ζ, q_*	04's collector map and collector.
$\rho_A^M, \text{Core}_A^M$	First ordinal of M not A -definable (external), and the maximal initial core (07's $K_M(A)$).
ϵ, ε	An epsilon number; an infinitesimal.
λ_H	First ordinal with a non-HOD subset; the first birthday of a non-OD surreal.
B_n, d_n	Fixed- Σ_n uniquely definable surreals, and the definable ordinal surreal escaping that level.
$\mathbf{No}_{\leq \alpha}$	Surreals of birthday at most α (07's B_α).
$\mathfrak{S}$	$(\mathbf{No}; +, -, \cdot, <, 0, 1, \mathbf{Oz})$.
ct, st, v	Constant term, finite standard part, normal-form valuation.

For the main new algebraic argument alone, read [sections 2, 5 and 8](#). For the definability concept and its model-theoretic limitations, read [sections 3, 7 and 12](#). For the initial core, read [sections 4 and 10](#). The codes in [section 6](#) connect the routes.

Part II

Real parameters, countable assembly, and strong symmetries

Real parameters, HOD, countable assembly, Hahn summation, forcing, and the strong stabilizer of the reals and the ordinals

Merged from two research manuscripts prepared for Vladimir Reshetnikov, both dated 4 October 2026

Abstract. Definability of a surreal number depends both on its language and on its allowed parameters. This Part distinguishes external parameter-free definability, real-only definability, ordinal definability, definability from a fixed real and ordinals, and definability from an arbitrarily varying real and ordinals. The last regime yields the initial real closed field $K_{\mathbb{R}} = \bigcup_{r \in \mathbb{R}} \mathbf{No}^{\text{HOD}(r)}$ and its omnific integer part $I_{\mathbb{R}}$. This union is captured by one real parameter exactly when $\text{HOD}_{\mathbb{R}}$ satisfies Choice, and every surreal is definable from a real and ordinals only if one real suffices for the whole universe. Countable assembly of ordinal information, closure of $K_{\mathbb{R}}$ under countable Hahn sums, interpolation of countable cuts, closure under countable sequences of definitions, and $\text{ROD} = \text{COD}$ are all equivalent; the test can be made with one explicit coefficient-1 omnific integer below ω , and the surreals definable from countable ordinal sequences are the least ambiently stable, countably Hahn-closed enlargement. Cohen forcing gives a proper such field with an exactly ω_1 -long cofinal parameter hierarchy; a Namba extension over L makes countable closure fail with no large-cardinal hypothesis, so it is independent of ZFC if ZFC is consistent; a Prikry extension gives a cardinal-preserving failure from a measurable cardinal. Finally, the common fixed field of the strong automorphisms fixing every real and every ordinal is exactly the field of normal forms with exponents in the rational span of the ordinals, which answers the strong form of a question of the self-embeddings report.

[write] **Added 4 October 2026, batch 93.** This Part merges manuscripts 02 and 05 of batch 93 of ProveIt’s incoming-reports intake (sources and pins in [section 19.5](#)). Like Part I it is AI-assisted, unrefereed and not formalized. Manuscript 02, *Definable Surreal Numbers and Omnific Integers: Real Parameters, Countable Closure, and Strong Symmetries*, is the base text and is printed in full: **its Section k is Section $k+18$ here and its statement or equation $k.m$ is $(k+18).m$** , so its Theorem 5.1 is [theorem 23.1](#), its Corollary 4.5 is [corollary 22.5](#) and its Theorem 7.6 is [theorem 25.6](#). Manuscript 05, *Definable Surreal Numbers and Omnific Integers: Real parameters, ordinal information, and countable assembly*, proves the same central theorems by its own routes and adds the canonical completion theorem, parameter spectra and a covering criterion. Its material follows source 02’s in the section it belongs to, marked [05], so source 02’s numbers are undisturbed; a theorem proved by both is printed once, with both sources named and the different proof kept as a marked second route. [Table 5](#) locates every numbered item of source 05. Source numbers 02, 05 and 06 in Parts II and III are batch-93 manuscript numbers; Part I’s sources 04 and 07 are batch-27 manuscripts. Notation changed from the sources is listed in [section 19.6](#); apart from those changes, the dated [write] notes and the editorial changes listed in [section 19.5](#), no statement, proof, number or non-claim of source 02 was changed. Throughout, “this paper” in source 02’s text and “this article” in source 05’s text mean the respective manuscript.

19 Scope, contributions, and relationship to the repository

There is no single satisfactory meaning of “a definable surreal number” until three choices have been made: the ambient universe, the language, and the parameter policy. For example, ω has an elementary parameter-free description in set theory, but cannot be defined from any real

parameters in the pure ordered-field structure of the surreals. Allowing ordinal parameters avoids one obstruction caused by Tarski’s undefinability theorem, but it does not make every surreal definable in every universe. Allowing arbitrary real parameters includes every real, yet can still leave a proper class of surreal numbers outside the resulting field.

The present paper is an extension of the existing research collection, not a replacement attribution for it. The ProveIt snapshot used here is `3c25fbb5515748040decb853139ed39543885b90`, observed on 4 October 2026. The directly relevant predecessor is *Definable Surreal Numbers and Omnific Integers* (Part I). It already develops fixed-parameter definable fields, their omnific integer parts, the identification with HOD, reversible bounded coding, transcendence amplification, forcing examples, and several intrinsic-language obstructions. The reports on universes, automorphisms, and self-embeddings provide additional antecedents [21, 24, 25]. The foundational facts used below also have classical sources [1, 2, 3, 28].

19.1 What is proved here

The principal extensions established in this paper are as follows.

1. A classification of the fields K_r and rings I_r obtained from one real parameter and arbitrary finite ordinal information. Their inclusion order is exactly ordinal-definability reducibility on real parameters. The field obtained by allowing the real parameter to vary is captured by one real precisely when $\text{HOD}_{\mathbb{R}}$ satisfies Choice.
2. An equivalence between countable ordinal information, external countable-sequence closure, countable Hahn summation, and countable-cut interpolation. This is stronger than merely placing countably many individual numbers in a common fixed-real field.
3. An explicit Cohen model in which the parameter fields have an exactly ω_1 -long cofinal chain, all countable cuts are filled, and a cut with $\aleph_1$ endpoints is omitted. A Namba model gives the opposite countable-closure behavior without a large-cardinal assumption; a Prikry model adds a cardinal-preserving example under the stated measurable-cardinal assumption.
4. An exact answer to the *strong* version of the question `sse:sr:q:stabilizers` in the self-embeddings report [25]: the common fixed field of the strong automorphisms fixing every real and every ordinal is the full normal-form field over the rational span of the ordinals. The answer remains the same if preservation of **Oz** is imposed as well.

These are proved mathematical statements with their hypotheses displayed. The claim of extension is relative to the inspected repository snapshot; it is not a certification of priority over the entire literature. In particular, the general Hahn-lifting and support-orbit techniques are antecedents, not discoveries of this paper. The unconstrained, possibly nonstrong stabilizer problem remains open here.

19.2 Proof and formalization status

Every new theorem has an ordinary mathematical proof below. The main arguments were independently checked for parameter, model, and support issues during preparation. No Lean or Rocq verification of the new theorems is claimed. The source repository itself distinguishes unformalized research documents from its trusted theorem boundary. Compilation of this paper checks its presentation, not its mathematics. The small validation script shipped with the paper checks only explicit rational inequalities used in one coding construction.

19.3 What source 05 asks, and what it adds [05]

Source 05 organizes the subject around one concrete question:

If every term of a countable summable surreal family is definable from some real and finitely many ordinals, must its sum admit such a definition?

The answer is not an unconditional yes. Source 05 proves an exact criterion and exhibits both a proper field satisfying it and an explicit relative-model counterexample; the counterexample uses only coefficient-1 monomials and produces an omnific integer smaller than ω . It also stresses the same three choices as source 02, with the example of ω , which is parameter-free definable in set theory and hence as a canonical sign sequence, but is not parameter-free definable in the pure ordered field, whose parameter-free definable elements are the real algebraic numbers. Naming the set $\mathbb{R}$ as one ambient parameter adds nothing, since $\mathbb{R}$ is parameter-free definable; allowing an arbitrary member of $\mathbb{R}$ is very different; allowing ordinal parameters as well is a third, stronger policy; and allowing a *countable sequence of ordinals* as one parameter is not the same as allowing a finite tuple of ordinals.

What source 05 found in the repository. Source 05 read Part I at its pin as already establishing the definable real closed field and its omnific fraction ring, the identification with $\mathbf{No}^{\text{HOD}}$, fixed-set-of-ordinals relativizations, reversible bounded omnific codes, whole-family versus termwise definability gaps, and noncollection results, and it presents none of these as new. The real-parameter special case follows by taking the fixed set of ordinals of [proposition 14.1](#) to be a real; what needs additional analysis is the *union over all real parameters*, above all its external countable assembly properties. Source 05 used Part I as a research predecessor, not as a substitute for proofs or as an independently certified library. Its description of this report’s README as “a merged, 55-page report from two manuscripts dated 23 September 2026” whose formalization mapping is “marked pending” was true at its pin; [\[write\]](#) the report had 56 pages from batch 90 on and now has the three Parts listed in [section 19.5](#); the formalization mapping is still pending.

Relation to the published literature. Canonical sign sequences, Conway normal forms and the omnific integer part are classical; useful modern accounts are [\[37, 6\]](#), alongside [\[1, 2\]](#). Definability from countable ordinal sequences is classical, not a new notion: Solovay uses it explicitly in [\[38, III.2.2–III.2.6\]](#). In the Levy–Solovay extension, countable sequences of real–ordinal definable sets are real–ordinal definable; a recent explicit formulation is [\[39, Proposition 2.2\(i\), version 1\]](#). These facts prevent the sources from claiming that countable assembly itself was previously unnoticed. Source 05’s contribution is the *surreal and omnific equivalence*, its explicit separated-band code, the precise minimal-closure statement, and the detailed Cohen/Prikry comparison. The forcing ingredients are standard, but their role is proved rather than hidden behind a general appeal to homogeneity. Source 05’s literature search was selective; priority for these particular combinations remains unverified.

Source 05’s status statement. Complete mathematical arguments are supplied for the stated new bridges and applications. These are proposed research results, not refereed or Lean-verified theorems. Classical set-theoretic ingredients and earlier repository results are credited separately. No claim of historical priority, a resolved named classical conjecture, or an optimal consistency bound is made.

Source 05’s guide, in this Part’s numbering. The multiplexer [theorem 23.5](#) and the further forms of the assembly theorem in [theorem 23.9](#) identify the exact obstruction to countable

sums; the completion [theorem 23.15](#) supplies the canonical enlargement when the obstruction is present; [theorem 24.1](#) with source 05's ring form [\(24.2\)](#) describes the real-parameter field in a Cohen product, and [theorem 24.9](#) makes its properness, countable closure and first omitted birthday explicit; [theorem 25.7](#) produces source 05's negative example without adding reals.

19.4 Relation to Part I and to neighbouring reports [\[write\]](#)

The following antecedents in the collection were checked at the write (4 October 2026) and are credited here; the sources credit some of them themselves, and the others were written before or after their pins.

- [Proposition 22.1](#) combines the classical inner model $\text{HOD}(\mathbb{R})$ with the relativized theory of [proposition 14.1](#) (Part I) for p a real; the fixed-real fields K_r are that proposition's K_p .
- [Section 25](#) strengthens to real-ordinal definability the ordinal-definable gaps of Part I: [theorem 9.1](#), [corollary 9.4](#) and the Cohen real of [example 13.4](#) already separate termwise from uniform definability for OD; the ROD versions are new here.
- The Prikry gap of [section 25](#) is related to the omitted (ω, ω) gap of the old field in Prikry extensions, `univ:cor:prikry` (iii), (v) of [foundations-and-computation/sur real-fields-across-universes](#) [\[21\]](#); Namba forcing under CH is already used there (`univ:gs:thm:nambainput`, imported from [\[36, Section 7.6\]](#) in its classical tree presentation, and `univ:gs:cor:namba`). [Theorem 25.6](#) uses the Laver-style presentation and is new.
- In [section 26](#), [corollary 26.6](#) without the predicate for Ord, and the nondefinability assertions after [theorem 26.10](#) without that predicate, follow already (per finite tuple, and more sharply) from `opa:sc:cor:innerbound`, `opa:thm:parameters` and `opa:sc:thm:undef` (ii), (iv) of [surreal/omnific-preserving-automorphisms](#) [\[24\]](#); see the dated notes in [section 26](#). With a predicate for Ord they are new. [theorem 26.5](#) answers the strong form of `sse:sr:q:stabilizers` of [surreal/surreal-self-embeddings](#) [\[25\]](#); the nonstrong remainder is [question 28.1](#).
- [Proposition 27.1](#) restricts [theorem 16.2](#) and [proposition 16.5](#) (Part I, themselves restrictions of results of the omnific Diophantine and set-sized-quotient reports [\[22, 23\]](#)); the topological proposition of [section 27](#) re-proves the mechanism of `univ:thm:nowheredense` for the ZF inner model $\text{HOD}_{\mathbb{R}}$.
- Part III (source 06) proves [corollary 22.5](#) independently (its Theorem 4.2, printed here once, with source 06's second proof and its further clauses in [remark 22.7](#)), and its interpretation theorem [theorem 37.1](#) goes far beyond [proposition 26.11](#).

No statement of this Part is formalized. Its placement beside the collection's formal developments confers no formal status; no Lean or Rocq declaration of ProveIt states any of its theorems.

19.5 Provenance of this Part [\[write\]](#)

This Part was built from two manuscripts of batch 93, delivered in the archives `definable_surreals.zip` and `definable_surreals_real_parameters.zip` (arrival commit `2faa3b37a`) and placed in commit `47a77daba`.

Source	Title and date	Pin	Printed here
02	<i>Definable Surreal Numbers and Omnific Integers: Real Parameters, Countable Closure, and Strong Symmetries</i> , 4 October 2026	3c25fbb55	Sections 19–29 in full (base); labels <code>dsn:rp:</code>
05	<i>Definable Surreal Numbers and Omnific Integers: Real parameters, ordinal information, and countable assembly</i> , 4 October 2026	a9ab9a698	inserted by subject, table 5 ; labels <code>dsn:ca:</code>

The full pins, both commits of 4 October 2026 and ancestors of the placement commit, are

3c25fbb5515748040decb853139ed39543885b90 (source 02; “Verify batch-90 file placement and route five later research arrivals”), a9ab9a69872c2fb7464097af32fd7b76c75cb030 (source 05; “Audit eight arithmetic compiler reports”).

The two manuscripts are not editions of one text: their pins differ, they share under one per cent of their eight-word sequences, and neither cites the other. They prove the same central theorems independently. The merge made the following choices.

1. *Base.* Source 02 is the base: its negative direction needs only the consistency of ZFC (Namba forcing over L , [theorem 25.6](#)) where source 05 needs a measurable cardinal, and it proves the larger set of statements on the shared spine.
2. *One copy of each shared theorem.* The countable-assembly equivalence (source 02’s Theorem 5.1, source 05’s Theorem 7.2) is printed as [theorem 23.1](#); source 05’s additional equivalent clauses are [theorem 23.9](#) with source 05’s proof as a second route. The Cohen classification (02’s Theorem 6.1, 05’s Theorem 9.3) is [theorem 24.1](#), with 05’s ring form and proof as a second route; 05’s Theorem 9.4 is printed as [theorem 24.9](#), naming which clauses are source 02’s. The global test (05’s Proposition 5.4) is printed for its new invariant $\lambda_{\mathbb{R}}$ only.
3. *Two codes.* Source 02’s rational-slot code Z_f (in the proof of [theorem 23.1](#)) and source 05’s multiplexer $\mathcal{Z}(f)$ ([theorem 23.5](#)) are the same construction with different band constants, $[q_n, \frac{5}{4}q_n)$ and $[q_n, \frac{3}{2}q_n)$ for $q_n = 2^{-n-1}$; both are kept.
4. *Prikry.* Source 02’s [theorem 25.4](#) and source 05’s Theorem 10.2 prove the same phenomenon with different codes; source 05’s version is [theorem 25.7](#), with its graph-coding proof that the Prikry sequence is not real-ordinal definable as a second route to source 02’s order-type argument.
5. *Background.* Source 05’s re-proofs of Part I background (its Sections 3–4) are printed once: as pointers to Part I and to source 02, with source 05’s proofs kept where they take a different route ([section 21.6](#)).
6. *Superseded statements.* Source 05’s Section 10.3 and its Question 13.1 present a measurable cardinal as the known upper bound for failure of countable assembly; source 02’s [theorem 25.6](#) needs only $\text{Con}(\text{ZFC})$. Both texts are kept with dated notes. Source 05’s Question 13.7 is answered negatively by [corollary 22.5](#).
7. *Questions.* Questions asked by both sources are printed once, with both discussions; source 05’s other questions follow source 02’s in [section 28](#), with the questions added at the write.

Editorial changes, besides the notation of [section 19.6](#): the two abstracts were merged into the abstract above; citations of the predecessor report became references into Part I, and

citations of other reports of the collection name their labels; the two bibliographies were merged into the report's ([3], [28], [13], [8] and [6] are the sources' VDE, DoraisHamkins, HamkinsDense, vdD and LM); source 02's two labels on its Section 8 became one; source 05's cover page, contents and build appendix were dropped, their content kept in this subsection and in the README.

Files. The report directory ships, from source 02, `code/08-real-parameters-finite_code_check.py` (exact rational checks of the slot inequalities and the inverse formula of Z_f ; no recorded output was delivered) and `code/08-real-parameters-build.sh`, which builds the undelivered `definable_surreals.tex` and is kept as delivered, not for use; from source 05, `code/09-countable-assembly-verify_code.py` with its recorded output `data/09-countable-assembly-verification.json` (20,508 inverse checks, 206 sequence checks, 1,000 band checks and four invalid-input rejections), `code/09-countable-assembly-Makefile` and the audit `09-countable-assembly-PROOF_AUDIT.md`. The delivered files are byte-identical to the delivery and still use delivery names; the README gives rerun instructions on a copy. Both scripts check finite rational specializations only; they do not implement infinite ordinals, satisfaction, HOD or forcing, and they are not proofs of any theorem here. The manuscripts' sources, PDFs and delivery READMEs and source 05's checksum manifest are not shipped and survive in the arrival commit's archives.

19.6 Notation in this Part [write]

Symbols of Part I keep their meaning; in particular $\mathcal{U}$ is Part I's code interval (1.1), $\mathbf{C}$ is Part I's sign code, $\mathfrak{S} = (\mathbf{No}; +, -, \cdot, <, 0, 1, \mathbf{Oz})$, and $e_\beta = \omega^{-(\beta+1)}$ as in (8.3). Symbols defined inside a proof are local to it. The sources' symbols were changed as follows.

Source symbol	Here	Reason
02: $\mathcal{U} = \bigcup_r \text{HOD}(r)$	$H_{\mathcal{U}}$	$\mathcal{U}$ is Part I's code interval $\mathcal{U}$, in which source 02's own examples land
02: $\mathbb{C}_\omega^\mathbb{R}$ (assembly principle); 05: $\mathbf{A}_\omega$	$\mathbf{A}_\omega$	$\mathbf{C}$ is Part I's sign code; source 02's principle (23.1) (countable sets of ordinals) and source 05's (countable ordinal sequences) are equivalent
02: $\text{OD}_\mathbb{R}$; 05: ROD	ROD	one name for one class
05: $K_{\text{ro}}, I_{\text{ro}}, H_{\text{ro}}, \lambda_{\text{ro}}$	$K_\mathbb{R}, I_\mathbb{R}, \text{HOD}_\mathbb{R}, \lambda_\mathbb{R}$	source 02's names; the classes are the same
05: $K_{\text{od}} = \mathbf{No} \cap \text{OD}, I_{\text{od}}$	$K_{\text{HOD}}, I_{\text{HOD}}$	Part I's names; equal by theorem 5.1
05: $\mathcal{Z}(f)$ (macro <code>\Code</code>)	$\mathcal{Z}(f)$	the macro name <code>\Code</code> is Part I's sign code $\mathbf{C}$; the glyph $\mathcal{Z}$ is unchanged; it differs from source 02's Z_f
05: $e_n(a)$ (band exponents)	$b_n(a)$	e_β is Part I's $\omega^{-(\beta+1)}$
02, section 23: e_n (slot exponents)	g_n	the same
02, section 25: $e_n = \Omega(-(\kappa_n + 1))$	e_{κ_n}	this is Part I's e_β at $\beta = \kappa_n$; no change of meaning
02, section 26: H, G, K	$\Gamma_\mathbb{Z}, \Gamma_\mathbb{Q}, \mathbb{R}(\text{Ord})$	H_r , Part I's K (theorem 8.2) and G (lemma 8.1) mean other things
02: $\mathcal{F}_G, I_G$	$\mathcal{F}_\Gamma, I_\Gamma$	follows the previous line
02: $\ell(x)$ (leading exponent)	$\text{lead}(x)$	Part I's ℓ is a width
02: $E = \sum_n \omega^{-n}/n!$	$\mathcal{E}$	Part I's $E_\alpha(X)$ is a bit code
02, section 26: τ, τ_s, S_τ (additive automorphisms)	ψ, ψ_s, S_ψ	τ is the bounded-code exponent of (6.4), also in source 02's section 22
02: $e_\alpha = \omega^{-\alpha}$ (theorem 26.10)	m_α	differs from Part I's $e_\alpha = \omega^{-(\alpha+1)}$ by the index shift
02: $B = \{\Omega(t) : 0 < t < 1\}$; 06: U	$\mathcal{Q}$	the range of the bounded code $\mathbf{Q}$ (theorem 6.8); B and U are used otherwise

Source symbol	Here	Reason
02: $\mathfrak{P} = (\mathbf{No}; +, \cdot, <, \mathbf{Oz})$	$\mathfrak{S}$	Part I's name
02: bd, tc, rcl, $\sqsubset \neq$; 05: b	b, TC, rcl, $\prec_s$	Part I's typography

Tempting false readings: $H_{\cup}$ is a union of inner models, not an inner model in general ([theorem 22.3](#)); $K_{\mathbb{R}}$ is the union of the fixed-real fields K_r , not $\mathbf{No} \cap \text{OD}(\mathbb{R})$ for the single set $\mathbb{R}$, which is just K_{HOD} ([proposition 21.2](#)); the field $\mathcal{F}_{\Gamma}$ is a fixed field of automorphisms, not a definable closure ([proposition 26.12](#)); and ω^a is the Conway map $\Omega(a)$, not $\exp(a \log \omega)$.

19.7 Where source 05 is printed [write]

Table 5: Source 05's numbered items and their places in this report.

Source 05	Here
Definitions 2.1–2.2	section 20.4 (definition 20.2)
Proposition 3.1, Lemma 3.2	proposition 21.2 , lemma 21.3
Proposition 3.3	(21.2) and theorem 3.4
Lemma 4.1, Theorem 4.2,	second routes in section 21.6 to lemma 3.2 , theorem 4.1 ,
Lemma 4.4, Remark 4.5,	proposition 14.1 , lemma 5.2 and theorem 4.10 of Part I and to
Theorem 4.6, Proposition 4.7,	propositions 21.1 and 22.1
Section 4.3	
Proposition 4.3	(20.3) and proposition 2.3 , with source 05's proof in section 21.6
Proposition 5.1	proposition 22.1 and the paragraph after it
Theorem 5.2	theorem 22.8
Proposition 5.3	theorem 22.2 , last clause
Section 5.1, Proposition 6.1	section 22.3 (spectra; the bounded code theorem 6.8)
Proposition 5.4	proposition 22.9 , with corollary 22.5
Theorem 6.2, Remark 6.3,	theorem 23.5 , the remark after it, example 23.7
Example 6.4	
Definition 7.1, Theorem 7.2	section 23.4 : theorem 23.1 with theorem 23.9
Corollary 7.3, Remark 7.4,	corollary 23.10 , the remark after it, proposition 23.12
Proposition 7.5	
Lemma 8.1, Definition 8.2,	lemma 23.13 , definition 23.14 , theorem 23.15 and the two
Theorem 8.3, Corollary 8.4,	statements after it
Remark 8.5	
Lemmas 9.1–9.2	lemmas 24.7 and 24.8
Theorem 9.3	theorem 24.1 ; source 05's form and proof in section 24.1
Theorem 9.4, Remark 9.5	theorem 24.9 and the remark after it
Lemma 10.1	lemmas 25.2 and 25.3 ; source 05's proof in section 25.2
Theorem 10.2, Remark 10.3	theorem 25.7 and the remark after it
Sections 10.3, 11, 12, 14	section 25.2 (with a dated note), section 27.5 , section 29.4 , section 27.5
Questions 13.1–13.10	section 28.4
Appendices A and B	section 23.6 ; section 29.4

20 Foundations: sets, signs, series, and integers

20.1 The ambient convention

Unless otherwise stated, work in an ambient universe V satisfying ZFC. Each surreal is a *set*, although the collection $\mathbf{No}$ is a proper class. We use the canonical sign-sequence representation

$$x : \alpha \longrightarrow \{-, +\}, \quad \alpha \in \text{Ord}.$$

The ordinal $\alpha = \text{b}(x)$ is its birthday. Prefix inclusion is simplicity. Numerical order is lexicographic with $- < \text{termination} < +$. The canonical ordinal α is the all-plus sequence of length α . We identify an ordinal with this canonical surreal when performing surreal field arithmetic; ordinal addition and surreal addition must otherwise be distinguished.

An inner model in this paper is a transitive definable class containing all ordinals and satisfying the indicated set theory. For an inner model M , $\mathbf{No}^M$ means its canonical sign sequences. In particular

$$\mathbf{No}^M = \mathbf{No} \cap M$$

as carriers, and their field operations agree with the ambient ones. For set models we explicitly say when satisfaction and cardinality are external. Forcing constructions can be read over transitive models or through the usual forcing-theoretic relative-consistency interpretation; we do not infer the existence of a transitive model from ZFC.

Statements involving a fixed first-order formula on $\mathbf{No}$ are translated into set theory by relativizing quantifiers to its definable carrier. Statements about all formulas are metatheorems or schemes unless a set-sized satisfaction predicate is supplied. Statements about all class automorphisms may be read in a class metatheory. The separating automorphisms constructed below are definable from set parameters, so their use does not require a class-sized choice function or a proper-class basis.

20.2 Normal forms and summability

Write $\Omega(x) = \omega^x$ for Conway's canonical omega map. Every surreal has a unique normal form

$$x = \sum_{\xi < \lambda} r_\xi \omega^{g_\xi}, \quad r_\xi \in \mathbb{R} \setminus \{0\}, \quad g_\xi > g_\eta \quad (\xi < \eta < \lambda). \quad (20.1)$$

The exponent support is a set and is reverse well ordered: every nonempty subset has a greatest element. Conversely, every such set-sized normal form represents a surreal. The map satisfies

$$\Omega(a + b) = \Omega(a)\Omega(b), \quad a < b \implies \Omega(a) \ll \Omega(b),$$

where $u \ll v$ for positive quantities means $nu < v$ for every positive ordinary integer n . We also write $u \prec v$ if $|u| \ll |v|$. The omega map is not the ordinary surreal exponential. In particular, $\Omega(1) = \omega$, whereas $\exp(1)$ is the ordinary real e .

A family $(x_i)_{i \in J}$ is *Hahn summable* (also called strongly summable) if the union of its exponent supports is reverse well ordered and each exponent occurs in only finitely many members. Its sum is defined coefficient by coefficient. This is an algebraic summation operation, not a claim of sequential convergence in the order topology. All uses of summability below have set-sized index sets.

The canonical normal form, the omega map and its inverse on monomials, and the simplest-number operation on set-sized cuts are uniquely definable set-theoretic constructions. These facts follow from the standard sign and normal-form theory [2]; the relevant absoluteness and definability issues are discussed in the predecessor (Part I). We give the hereditary membership argument whenever normal forms are used to transfer information between parameter regimes.

20.3 Omnific integers and the sign of the infinitesimal tail

An omnific integer is a surreal whose normal form has no negative exponents and whose coefficient at exponent zero is an ordinary integer:

$$\mathbf{Oz} = \{p + n : \text{supp}(p) \subseteq \mathbf{No}_{>0}, n \in \mathbb{Z}\}. \quad (20.2)$$

The coefficients of positive-exponent terms may be arbitrary reals. Thus $\sqrt{2}\omega$, $\omega^{1/2}$, and $\omega^{1/\omega}$ are omnific integers. This does not make them ordinary integers or ordinal numbers. The ring $\mathbf{Oz}$ is discretely ordered with least positive element 1.

Decompose a surreal uniquely as $x = p + r + \varepsilon$, where p is purely infinite, $r \in \mathbb{R}$, and ε is infinitesimal. Its unique omnific floor is

$$\lfloor x \rfloor_{\mathbf{Oz}} = \begin{cases} p + r - 1, & r \in \mathbb{Z} \text{ and } \varepsilon < 0, \\ p + \lfloor r \rfloor, & \text{otherwise.} \end{cases} \quad (20.3)$$

It is characterized by $z \leq x < z + 1$. The first case matters: for example, $\lfloor \omega - \omega^{-1} \rfloor_{\mathbf{Oz}} = \omega - 1$, not ω . Uniqueness also makes the floor definable in the ordered field with the predicate $\mathbf{Oz}$.

20.4 Source 05's conventions [05]

Source 05 works with the same conventions; its formulations add the following points.

The universe convention. We reason in ZFC about canonical set codes. A surreal is a sign function $s : \alpha \rightarrow \{-, +\}$ for an ordinal α ; its birthday is $b(s) = \alpha$. The class of all such functions is $\mathbf{No}$. No surreal is itself a proper class. The notation $\mathbf{No}$ is a definable proper class, not an extra set to which unrestricted power-set operations may be applied.

A claim about arbitrary ambient formulas is read as a scheme, one claim for each fixed standard formula. Equivalently, one may fix externally a transitive set model $M \models \text{ZFC}$ and use genuine external satisfaction for $(M, \in)$. Then $\mathbf{No}^M$ is a set in the metatheory, while M regards its surreal numbers as a proper class. The existence of such an M is not asserted by ZFC; model-based results are conditional on the specified model. All forcing statements use this usual relative-model interpretation.

Unless marked otherwise, ordinals, reals, countability and definability are those of the ambient universe currently under discussion. In a forcing section that universe is the extension, not the ground model. We restrict the main model comparisons to transitive models; we make no claim about a least externally omitted ordinal in an ill-founded model.

Order, simplicity, and two exponentials. The order on signs is lexicographic, with a missing sign between $-$ and $+$. An ordinal α is represented by the all-plus sequence of length α . The simplicity relation is proper initial-segment inclusion of sign sequences. A subclass is *initial* when it contains every proper sign prefix of every one of its members.

The Conway map

$$\Omega(a) = \omega^a$$

is the canonical monomial map. It satisfies $\Omega(a + b) = \Omega(a)\Omega(b)$ and is strictly increasing. In particular, $0 < a < 1$ gives $1 < \omega^a < \omega$. Throughout, ω^a means this map; it is *not* shorthand for $\exp(a \log \omega)$ using Gonshor's exponential. Both constructions are ambiently definable, but they play different roles [2, 37].

When ordinals occur in a displayed *field* expression such as $\alpha/(1+\alpha)$, they are first embedded into $\mathbf{No}$ and the operations are surreal field operations. Ordinal-index manipulations, such as a well-order type or the domain of a sign sequence, remain ordinary ordinal operations. This convention is important in the code below.

Normal forms and omnific integers. Every surreal has a unique Conway normal form

$$x = \sum_{i < \lambda} r_i \omega^{a_i}, \quad r_i \in \mathbb{R} \setminus \{0\}, \quad a_i > a_j \quad (i < j), \quad (20.4)$$

where λ is an ordinal and the support is a set. Conversely, a set-sized reverse-well-ordered support with nonzero real coefficients gives a surreal. We use $\text{NF}(x)$ for the entire indexed graph $\langle (a_i, r_i) : i < \lambda \rangle$, not for a collection of individually specified entries. These are standard normal-form facts [1, 2, 6].

Definition 20.1. The class **Oz** of omnific integers consists of the normal forms with no negative exponent and with integer coefficient at exponent 0. A missing constant term has coefficient 0. For a parameter policy $\mathcal{P}$, a $\mathcal{P}$ -definable omnific integer is an omnific integer whose canonical surreal code is $\mathcal{P}$ -definable.

Thus $\sqrt{2}\omega$, $\omega^{1/2}$ and $\omega^{1/2} + \omega^{1/4} + \dots$ are omnific integers; $1/2$ and ω^{-1} are not. An omnific integer need not have ordinary integer coefficients at positive exponents. Nor does the interval $(0, \omega)$ contain only the ordinary positive integers. The integer-part theorem for **Oz** is classical [6]; we give its useful explicit formula in [proposition 2.3](#).

What a countable sum means.

Definition 20.2. A family $(x_j)_{j \in J}$, with J a set, is *Hahn-summable* if $\bigcup_{j \in J} \text{supp}(x_j)$ is reverse well ordered and each exponent occurs in the support of only finitely many x_j . Its sum is obtained by adding the finitely many coefficients at each exponent and removing zeros.

This is a formal, canonical summation operation, not convergence of partial sums in the order topology. In particular a reverse-well-ordered sequence of distinct monomials is summable. Countable support does not imply countable birthday: its exponents may encode very large ordinals. All sums in the main theorems are set-sized, and no sum over all ordinals is taken.

The normal-form and summation constructions are fixed definable operations on canonical set codes. Consequently a definable *whole summable family* always has a definable sum with the same parameters. The difficult issue is whether pointwise definability yields such a whole-family definition.

21 A parameter policy is part of a definition

21.1 External definitions without ordinal parameters

For a set model $M \models \text{ZFC}$ and an external family $A \subseteq M$, let D_A^M consist of objects uniquely defined in $(M, \in)$ by a standard first-order formula using a finite tuple from A . Write

$$D_{A, \mathbf{No}}^M = D_A^M \cap \mathbf{No}^M, \quad D_{A, \mathbf{Oz}}^M = D_A^M \cap \mathbf{Oz}^M.$$

Taking $A = \emptyset$ gives parameter-free definability. Taking a fixed real r as the only parameter, or taking $A = \mathbb{R}^M$, gives two different real-only notions. There are at most $\max(\aleph_0, |A|)$ such objects externally. The external enumeration of definitions need not belong to M and does not provide an internal enumeration or an internal birthday bound. Pointwise definable models are compatible with this observation [12].

In the intended universe, the corresponding phrase “definable without parameters in V ” is a semantic notion. It must not be replaced by an unrestricted internally definable truth predicate. Tarski’s obstruction applies here, as it also does to real-only definability without ordinal parameters [28]. A fixed complexity level or a fixed set structure V_θ has a satisfaction predicate; the entire universe does not have one in its own language.

21.2 Ordinal and real-plus-ordinal definitions

For a real r , define $\text{OD}(r)$ internally by rank-local satisfaction: an object x belongs to $\text{OD}(r)$ when there are an ordinal θ , a formula code, and finitely many ordinals below θ such that $x, r \in V_\theta$ and x is the unique object selected by that formula in $(V_\theta, \in)$ using those parameters and r . Reflection identifies this with semantic definability in V from r and finitely many ordinals. Conversely, a rank-local definition is an ambient definition because its rank and formula code may be included among the ordinal parameters. This construction uses only satisfaction for a set structure.

Let OD denote the case with no additional real, and define

$$\text{ROD} = \bigcup_{r \in \mathbb{R}} \text{OD}(r), \quad \text{HOD}_{\mathbb{R}} = \{x : \text{TC}(\{x\}) \subseteq \text{ROD}\}. \quad (21.1)$$

We use $\text{HOD}_{\mathbb{R}}$ for what is often written $\text{HOD}(\mathbb{R})$ in the real-plus-ordinal sense. A finite tuple of real parameters can be joined into a single real. The predicates in (21.1) are first-order definable in set theory. The standard general theory of $\text{OD}(X)$ provides this framework [28]; no uniform truth predicate for V is being asserted.

For a fixed real let

$$H_r = \text{HOD}(r) = \{x : \text{TC}(\{x\}) \subseteq \text{OD}(r)\}, \quad K_r = \mathbf{No} \cap \text{OD}(r), \quad I_r = \mathbf{Oz} \cap \text{OD}(r).$$

Also set $K_{\mathbb{R}} = \mathbf{No} \cap \text{ROD}$ and $I_{\mathbb{R}} = \mathbf{Oz} \cap \text{ROD}$. These definitions make explicit the difference between *a fixed real* and *a real that may depend on the number being defined*.

Giving the whole set $\mathbb{R}$ as one additional ambient parameter does not have the same effect as allowing its individual members as parameters. The set $\mathbb{R}$ is already parameter-free definable in V , so giving that set alone adds no ambient definability. Similarly, naming all ordinals as separate constants allows only finitely many of them in any one formula; it does not supply a countable sequence of ordinal parameters.

Regime	Allowed information in one definition	Formal status
Parameter-free	One finite formula	External semantics, or a fixed-model notion
Real-only	A finite tuple of reals	External semantics without a rank parameter
Ordinal-definable	Finitely many ordinals	Internal rank-satisfaction predicate OD
Fixed-real ordinal-definable	One specified r and finitely many ordinals	$\text{OD}(r)$ and H_r
Varying-real ordinal-definable	Some real r , depending on the object, and finitely many ordinals	ROD and $\text{HOD}_{\mathbb{R}}$
Countable ordinal information	A countable ordinal sequence and finitely many ordinals	$\text{OD}(\text{Ord}^\omega)$; usually stronger

21.3 Basic field and integer-part closure

The next proposition is background from the predecessor, with proof included to make the later arguments independent of informal closure claims (Part I).

Proposition 21.1 (Same-parameter closure). *In any of the ambient finite-parameter regimes, the definable surreals form an ordered real closed subfield. They are closed under every uniquely specified parameter-free ambient operation whose inputs belong to the regime. In particular*

this includes the omega map, its inverse on monomials, normal forms, the omnific floor, and summation when the whole family is available as a definable set. Their omnific integers form an integer part, and the field is their fraction field.

Proof. Combine the finitely many input definitions by substitution into the graph of the desired operation. This gives addition, multiplication, negation, reciprocal, and the other uniquely specified outputs without new parameters. A polynomial over the resulting field has its real roots in a finite numerical order. Its j th root, if it exists, is therefore uniquely definable from its coefficients and the ordinary integer j . Positive square roots and roots of odd-degree polynomials follow, giving real closure. The floor characterization gives the integer part.

For the fraction-field assertion take $x \neq 0$ with exponent support S . Let γ be the simplest surreal strictly greater than $\{0\} \cup \{-g : g \in S\}$. It is definable from x . Put $b = \omega^\gamma$ and $a = xb$. Every exponent of a and b is positive, so a, b are purely infinite omnific integers, $b \neq 0$, and $x = a/b$. The case $x = 0$ uses $0/1$. This construction preserves the original parameter information. $\square$

The words “whole family” cannot be dropped. The main countable theorem below measures exactly when that additional information can be assembled from individual real-plus-ordinal definitions.

21.4 Intrinsic ordered-field definability

Fix the language $L_{\text{of}} = \{0, 1, +, -, \cdot, <\}$. Quantifier elimination for real closed fields gives

$$\text{dcl}_{L_{\text{of}}}^{\text{No}}(A) = \{x \in \mathbf{No} : x \text{ is algebraic over } \mathbb{Q}(A)\} = \text{rcl}_{\mathbf{No}} \mathbb{Q}(A). \quad (21.2)$$

For a proper parameter class the right side uses only finite parameter subsets; this is the same convention as on the left. To see the nontrivial inclusion directly, a formula reduces to finitely many polynomial sign conditions. At a point transcendental over its coefficient field all nonzero polynomials in that finite list are nonzero, and their signs persist on an interval. The formula cannot isolate that point. Conversely, a root is isolated by its numerical position among the finitely many roots.

Consequently

$$\text{dcl}_{L_{\text{of}}}^{\text{No}}(\emptyset) = \mathbb{R}_{\text{alg}}, \quad \text{dcl}_{L_{\text{of}}}^{\text{No}}(\mathbb{R}) = \mathbb{R}.$$

With no parameters, the omnific intersection is $\mathbb{Z}$. These intrinsic conclusions do not conflict with ambient definitions of infinite surreals. Later we strengthen the intrinsic obstruction even after every real and ordinal is named and the omnific predicate is added.

21.5 Source 05’s parameter dictionary [05]

For a fixed structure $(M, \in)$ and a family $A \subseteq M$ of allowed parameters, let D_A^M be the external class of elements uniquely specified in M by a standard first-order formula and a finite tuple from A . Define

$$K_A^M = D_A^M \cap \mathbf{No}^M, \quad I_A^M = D_A^M \cap \mathbf{Oz}^M.$$

For the intended universe the same notation is interpreted schematically. Parameter-free definability means $A = \emptyset$. A fixed parameter r does not authorize arbitrary elements of r as unrelated parameters; any extraction must itself be definable. For real parameters, natural indices are harmless because each natural number is parameter-free definable.

Notation	Allowed information in an ambient definition	Surreal field
$D_\emptyset$	No parameters	$K_\emptyset$
RD	Finitely many real parameters, no ordinal parameters	K_{rd}
OD	Finitely many ordinals	K_{HOD}
OD(r)	One fixed real r , and finitely many ordinals	K_r
ROD	An arbitrary real, and finitely many ordinals	$K_{\mathbb{R}}$
COD	One countable sequence of ordinals	K_{co}

A finite real tuple, or a given countable sequence of reals, can be encoded by one real using a fixed pairing function on ω . We freely replace real numbers by their standard subset-of- ω codes, which preserves the corresponding definability policies. Thus

$$\text{ROD} = \bigcup_{r \in \mathbb{R}} \text{OD}(r), \quad K_{\mathbb{R}} = \bigcup_{r \in \mathbb{R}} K_r, \quad I_{\mathbb{R}} = \bigcup_{r \in \mathbb{R}} I_r. \quad (21.3)$$

Only finitely many ordinals are permitted in each ROD definition. A countable sequence of ordinal parameters cannot silently be substituted for that finite tuple.

Source 05's table writes K_{od} , K_{ro} for K_{HOD} , $K_{\mathbb{R}}$ (section 19.6). The class COD and its fields K_{co} , I_{co} are source 02's OD(Ord^ω) of corollary 23.3.

Proposition 21.2. *Adding $\mathbb{R}$ itself as a single set parameter to the ambient language $(V, \in)$ does not enlarge parameter-free definability. Adding $\mathbb{R}$ to the permitted ordinal parameters does not enlarge OD. In contrast, allowing arbitrary individual real parameters includes every real.*

Proof. Replace each occurrence of the parameter $\mathbb{R}$ by its parameter-free set-theoretic definition. Conversely a given real r is uniquely defined from the parameter r by $x = r$. $\square$

This proposition is about ambient set theory. Naming a real predicate in a weaker structure need not be a definitional extension of that weaker structure. The language must not be changed unnoticed.

Rank-local certificates, without a truth predicate for V . For ordinal-allowing policies there is a uniform internal definition. A certificate for $x \in \text{OD}(r)$ consists of an ordinal θ , a finite formula code e , and a finite ordinal tuple $\bar{\alpha}$, with all parameters and x in V_θ , such that

$$(V_\theta, \in) \models "x \text{ is the unique } y \text{ satisfying } \varphi_e(y, r, \bar{\alpha})". \quad (21.4)$$

Satisfaction for *set structures* is a single definable relation. The certificate therefore makes sense uniformly as e varies.

Lemma 21.3 (Certificate equivalence). *A set is globally ordinal definable from r if and only if it has a certificate of the form (21.4). Consequently OD(r), ROD, and COD are internally definable classes, with the displayed parameter policy respected.*

Proof. Given a fixed global definition with a unique output, reflection for that formula and its uniqueness assertion supplies a sufficiently large V_θ containing the parameters and the output. Conversely the unique output in (21.4) is globally definable from $r, \theta, \bar{\alpha}$ and the finite numeral coding e . The extra ordinal θ is permitted. For COD, place θ , the syntax code, and any finite auxiliary ordinals at the beginning of the countable ordinal parameter sequence. Existentially quantifying over these certificates gives the uniform internal predicates. $\square$

The same argument does *not* supply a uniform predicate for global parameter-free definability, or for global real-only definability: the rank parameter θ cannot simply be introduced under those policies. External counting of formulas likewise does not provide an internal enumeration of all definable objects. Pointwise definable models give an important warning against a naive “least undefinable object” argument [12]. This distinction will be enforced in the countable-assembly proof.

Intrinsic definability in the field reduct. Source 05’s Proposition 3.3 is (21.2) above (theorem 3.4 in Part I), proved by the same quantifier-elimination argument [8].

For ordinal parameters in the field language, the answer is similarly the relative real closure of the field generated by those parameters. It is not, merely by virtue of those parameter names, the ambient HOD-surreal field. Surreal sign coding and set-theoretic membership

21.6 Source 05’s algebraic baseline: pointers and second routes [05]

The principal statements of source 05’s Section 4 are already in Part I and in proposition 21.1 above, and are printed there. Source 05 includes proofs to isolate exactly which parameter and collection rules they use; where its route differs, the statement is recalled without a number and its proof is kept.

Source 05’s Lemma 4.1 (= lemma 3.2). *If $x_1, \dots, x_m$ are definable under a parameter policy closed under finite combination of parameters, and F is a fixed parameter-free ambiently definable partial function defined on this tuple, then $F(x_1, \dots, x_m)$ is definable under the same policy.*

Second proof (source 05). Conjoin the finitely many definitions of the inputs, quantify over those inputs, and apply the fixed graph formula for F . Input uniqueness and output uniqueness give a unique result. This is a formula-by-formula argument, not an internal quantification over global truth. $\square$

Source 05’s Theorem 4.2 (theorem 4.1 with proposition 21.1; the closure under exp and log is part of theorem 4.1). *Each field associated with a policy in the table of section 21.5 is a real closed subfield of $\mathbf{No}$. It is closed under the Conway map, its inverse on monomials, and the canonical surreal exponential and logarithm on their domains. If K is such a field and $I = K \cap \mathbf{Oz}$, then I is an integer part of K and*

$$K = \text{Frac}(I). \quad (21.5)$$

These assertions also apply to a fixed arbitrary allowed parameter family A , interpreted as in section 21.5.

Second proof (source 05). The field operations and their domain restrictions are canonical ambiently definable maps, so lemma 3.2 applies. Every positive element has a uniquely specified positive square root. Every odd-degree polynomial has a real root in $\mathbf{No}$, and one can specify its least real root, or more generally its k th real root in increasing order. Hence K is real closed. The named maps are canonical, so unique-output closure applies to them as well. The floor and fraction assertions are proved explicitly below. Only finite input tuples are used here. $\square$

Source 05’s Proposition 4.3 is the floor formula (20.3) (proposition 2.3); its proof:

Proof. If c is not an integer, its distances from the adjacent integers are positive reals and dominate the infinitesimal ε . If c is an integer, a negative infinitesimal places x just below $x_{>0} + c$, accounting for the first case. The stated inequalities follow in all cases. A nonzero omnific integer has absolute value at least 1: its leading positive exponent makes it infinite, or its normal form is a nonzero ordinary integer. Thus two candidates for z cannot differ by a nonzero omnific integer of absolute value less than 1. $\square$

Source 05's Lemma 4.4 (a denominator; compare (4.2) and the proof of proposition 21.1). *For every nonzero surreal x there is a parameter-free definable choice $q(x) \in \mathbf{Oz}_{>0}$ such that $xq(x) \in \mathbf{Oz}$. In particular if $x \in K$, both $q(x)$ and $xq(x)$ belong to I .*

Second proof (source 05). Let $S = \text{supp}(x)$. Choose the least ordinal $\delta > 0$ such that $\delta > -a$ for every $a \in S$, with the comparison made in **No**. This ordinal exists: each surreal is below some ordinal; Replacement on the set S and an ordinal supremum give one common upper bound. The least such ordinal is definable from x , without adding δ as an independent parameter. Set $q(x) = \omega^\delta$. Multiplication shifts every exponent of x to $a + \delta > 0$, so $xq(x)$ has zero constant term and no negative exponent. Both factors are omnific. This proves (21.5), with $x = 0$ handled by $0/1$. $\square$

Remark 21.4. For a proper-class ring, (21.5) means elementwise representation as a quotient with nonzero denominator; it does not require forming a set of all equivalence classes. In an external set-model interpretation, it is the usual fraction-field assertion for the resulting set-sized ring. Likewise the real-closed-field assertion is a scheme of finite algebraic conditions. Quantifier elimination also yields ordered-field elementarity of these inclusions in the corresponding semantic setting.

For a fixed real r , use $\text{HOD}(r)$ for the sets hereditarily in $\text{OD}(r)$, including the object itself in its transitive closure. Because r is a set of natural numbers, $r \in \text{HOD}(r)$; this is the standard relative HOD inner model of ZFC [10]. The superscript $\mathbf{No}^N$ will mean the canonical sign sequences belonging to a transitive inner model N , not a quotient by arbitrary game forms.

Source 05's Theorem 4.6 (= proposition 14.1 for a real; compare theorem 5.1 and proposition 22.1). *For a fixed real r ,*

$$K_r = \mathbf{No} \cap \text{OD}(r) = \mathbf{No} \cap \text{HOD}(r) = \mathbf{No}^{\text{HOD}(r)}, \quad I_r = \mathbf{Oz} \cap K_r. \quad (21.6)$$

The same holds with no real parameter, replacing $\text{OD}(r)$ by OD . Moreover K_r and K_{HOD} are initial in the sign-sequence order.

Second proof (source 05). A canonical sign sequence is a set of ordered pairs whose first coordinates are ordinals and whose second coordinates are fixed finite tags. Every member of the transitive closure of such a graph other than the graph itself is hereditarily ordinal definable: it is built from ordinals and finite sets by canonical finite operations. If the graph is $\text{OD}(r)$, it is therefore hereditarily $\text{OD}(r)$. The reverse inclusion is immediate. Membership in the canonical surreal class is absolute for transitive inner models, since a specified domain ordinal and a given function graph have the same meaning there. This proves the set-code identification.

For any $\beta < b(x)$, the prefix $x \upharpoonright \beta$ is definable from x and β . The ordinal β is allowed. Thus every prefix of an $\text{OD}(r)$ surreal is again $\text{OD}(r)$, proving initiality. $\square$

The equality here is first an equality of canonical sign-code classes, with ambient order and operations on the left. The later arguments do not require an unproved blanket absoluteness

theorem for all analytic operations between arbitrary inner models with different real lines. Ambient definability and the preceding hereditary argument suffice.

Source 05’s Proposition 4.7 (= the whole-normal-form paragraph after [proposition 22.1](#); compare [lemma 5.2](#)). *A surreal x belongs to K_r if and only if its entire normal-form graph belongs to $\text{HOD}(r)$. In particular every coefficient and every exponent of a member of K_r is $\text{OD}(r)$.*

Second proof (source 05). From x one defines $\text{NF}(x)$ and, using the allowed ordinal index i , each entry (a_i, r_i) . Each exponent is a sign sequence and each coefficient is a real, so the preceding hereditary argument applies to them. The graph, its indexed pairs, and all their transitive members are consequently $\text{OD}(r)$. Conversely the graph is itself $\text{OD}(r)$ if it is in $\text{HOD}(r)$, and its canonical evaluation uniquely defines x . $\square$

This proposition concerns the *whole graph*. Its converse cannot be weakened to “every entry is individually $\text{OD}(r)$ ”: the infinite assignment of entries to indices can carry information absent from r and every finite tuple of ordinal parameters. The strongest example in this article will have every entry already OD .

The natural valuation. For $x \neq 0$ with leading exponent a_0 , put $v(x) = -a_0$. On any of the fields K above the value group is $(K, +)$: the leading exponent is definable from x , and every value $a \in K$ is realized by ω^{-a} . The residue field is $K \cap \mathbb{R}$. Indeed, for a finite surreal the real constant term is its unique real standard part, and the kernel is the infinitesimal ideal. These are direct normal-form arguments, also recorded in (Part I). For $K_{\mathbb{R}}$ and K_{co} the residue field is the full ambient $\mathbb{R}$. Having all real residues will not imply that the whole field is **No**.

This is [theorem 4.10](#) of Part I for the fields of this Part.

22 Varying real parameters and the associated inner models

Throughout this section the ambient universe satisfies ZFC. We represent real parameters by subsets of ω ; the standard conversions to ordinary real numbers and to their canonical surreal representatives are definable. Allowing an arbitrary finite tuple of real parameters is equivalent to allowing one real, by interleaving binary codes. Define

$$\text{ROD} = \bigcup_{r \in \mathbb{R}} \text{OD}(r), \quad H_r = \text{HOD}(r),$$

and distinguish the two classes

$$\text{HOD}_{\mathbb{R}} = \{x : \text{tc}(\{x\}) \subseteq \text{ROD}\}, \quad H_{\cup} = \bigcup_{r \in \mathbb{R}} H_r. \quad (22.1)$$

The first is hereditary definability using individual real and ordinal parameters. The second is the union of the classes obtained by fixing one real parameter at a time. The distinction matters even though their canonical surreal numbers coincide. These definitions use the standard rank-satisfaction presentation of ordinal definability, so all four displayed classes are first-order definable. They do not require a truth predicate for V ; compare the general $\text{OD}(X)$ framework in [28, Section 4].

Proposition 22.1 (The hereditary model and its surreal numbers). *$\text{HOD}_{\mathbb{R}}$ is a transitive inner model of ZF containing all ordinals, all ambient reals, and the ambient set $\mathbb{R}$ itself. If*

$$K_r = \mathbf{No} \cap H_r, \quad I_r = \mathbf{Oz} \cap H_r, \quad K_{\mathbb{R}} = \mathbf{No} \cap \text{ROD}, \quad I_{\mathbb{R}} = \mathbf{Oz} \cap \text{ROD},$$

then

$$K_{\mathbb{R}} = \mathbf{No} \cap \text{HOD}_{\mathbb{R}} = \mathbf{No} \cap H_{\cup} = \bigcup_{r \in \mathbb{R}} K_r = \mathbf{No}^{\text{HOD}_{\mathbb{R}}}, \quad (22.2)$$

$$I_{\mathbb{R}} = \mathbf{Oz} \cap \text{HOD}_{\mathbb{R}} = \mathbf{Oz} \cap H_{\cup} = \bigcup_{r \in \mathbb{R}} I_r. \quad (22.3)$$

Here the last term of (22.2) uses the canonical sign-sequence construction inside $\text{HOD}_{\mathbb{R}}$.

Proof. Hereditary membership immediately makes $\text{HOD}_{\mathbb{R}}$ transitive. Every ordinal is ordinal definable; every real is definable from itself. The set $\mathbb{R}$ is parameter-free definable, and its hereditary members are reals and the elementary sets used to represent them. Thus all these objects belong to $\text{HOD}_{\mathbb{R}}$.

We give the inner-model argument to identify where choice is, and is not, used. A finite tuple of ROD parameters can be replaced by one real and finitely many ordinals. Consequently a uniquely defined output from such parameters is ROD. Pairing and union therefore preserve $\text{HOD}_{\mathbb{R}}$. Given $a, \bar{p} \in \text{HOD}_{\mathbb{R}}$, a set obtained by separation using a formula relativized to $\text{HOD}_{\mathbb{R}}$ is definable in V from $a, \bar{p}$, since $\text{HOD}_{\mathbb{R}}$ itself is definable. It is ROD, and its hereditary members already lie in $\text{HOD}_{\mathbb{R}}$. For replacement, apply replacement in V to the relativized functional formula. The resulting range is again ROD and has hereditary members in $\text{HOD}_{\mathbb{R}}$. Its internal powerset of a is

$$\{b \subseteq a : b \in \text{HOD}_{\mathbb{R}}\}.$$

This set exists by separation in V , is ROD from a , and is hereditary ROD. Infinity is supplied by ω , and extensionality and foundation are inherited by the transitive class. Thus the class satisfies ZF. This proof does not establish choice in $\text{HOD}_{\mathbb{R}}$.

If a canonical sign sequence s is $\text{OD}(r)$, every other member of $\text{tc}(\{s\})$ is ordinal definable: it is an ordinal, a finite sign tag, or one of the finite sets used to encode an ordered pair. Hence $s \in H_r$. Conversely, membership in $\text{HOD}_{\mathbb{R}}$ implies that s itself is ROD. This proves the asserted intersections and unions; the omnific assertion is their restriction to $\mathbf{Oz}$. Finally the property of being a function from an ordinal to the fixed two-element sign set is absolute for the transitive same-ordinal model $\text{HOD}_{\mathbb{R}}$. Its canonical surreal class is therefore precisely $\mathbf{No} \cap \text{HOD}_{\mathbb{R}}$. $\square$

By [proposition 21.1](#) and the fixed-real theory, each K_r is a real closed field and I_r is its canonical omnific integer part. The unions in (22.2)–(22.3) are directed: any finitely many real parameters can be combined into one real. Consequently $K_{\mathbb{R}}$ is a real closed field containing all ambient reals, and $I_{\mathbb{R}}$ is its omnific integer part. For example, the coefficients of a given polynomial all lie in one K_r , which supplies the needed positive square roots and odd-degree roots. This reasoning does not assume that $\text{HOD}_{\mathbb{R}}$ satisfies choice.

Every K_r and $K_{\mathbb{R}}$ is initial in the sign tree: if x is in the field, its prefix of any ordinal length is definable from x and that ordinal. All these fields contain every canonical ordinal. By quantifier elimination for real closed fields they are elementary subfields of $\mathbf{No}$ in the ordered-field language, interpreted formula by formula. They also satisfy the same-parameter fraction-field property of [proposition 21.1](#).

There is a useful whole-normal-form version. If $x \in K_r$, its complete normal form is $\text{OD}(r)$; each coefficient and exponent is definable from x and its ordinal index, hence belongs to H_r . The hereditary graph coding then puts the entire normal form in H_r . Conversely a whole normal form in H_r has an $\text{OD}(r)$ value. Therefore $x \in K_r$ if and only if its whole normal-form code belongs to H_r . The corresponding statement for $K_{\mathbb{R}}$ is obtained by allowing r to vary. Entrywise membership alone is not sufficient.

22.1 The real-parameter fields record exactly the parameter degrees

Define $r \leq_{\text{OD}} s$ to mean $r \in \text{OD}(s)$. All occurrences of relative HOD in the next theorem refer to the same ambient universe, rather than to iterated computations inside its various inner models.

Theorem 22.2 (Exact inclusion classification). *For real parameters r, s , the following are equivalent:*

$$r \leq_{\text{OD}} s \iff H_r \subseteq H_s \iff K_r \subseteq K_s \iff I_r \subseteq I_s. \quad (22.4)$$

The field $K_{r \oplus s}$ is the least member of this family containing both K_r and K_s , and similarly for H_r and I_r . Every countable collection of members of $H_{\cup}$ is contained in some H_t . In particular, every countable collection of members of $K_{\mathbb{R}}$ is contained in some K_t .

Proof. If r is $\text{OD}(s)$, substitute a definition of r into every definition from r and ordinal parameters. This gives $\text{OD}(r) \subseteq \text{OD}(s)$, and the hereditary condition gives $H_r \subseteq H_s$. Restriction gives the field and integer-ring inclusions.

Conversely, the canonical surreal representative of r lies in K_r . If $K_r \subseteq K_s$, its inverse real code is $\text{OD}(s)$. For the ring inclusion use the injective, parameter-free definable map

$$\mathbb{R} \longrightarrow \mathbf{Oz}, \quad a \longmapsto a\omega.$$

For $a \neq 0$ its image has one positive normal-form exponent and zero constant term; for $a = 0$ the assertion is immediate. The inverse on the image is division by ω . Thus $I_r \subseteq I_s$ also implies that r is $\text{OD}(s)$.

The two coordinates of $r \oplus s$ are definable from their interleaved code, so its class contains the two given classes. If t bounds both parameters in the preorder, the pair and its interleaving are $\text{OD}(t)$. This proves the least-upper-bound statement. It concerns least upper bounds within this family; no equality with a field-theoretic compositum is asserted.

Given $x_n \in H_{\cup}$, ordinary choice selects reals r_n such that $x_n \in H_{r_n}$. Interleave the sequence of real codes into a single real t . Each r_n is definable from t and the integer n , so $H_{r_n} \subseteq H_t$. Hence $x_n \in H_t$ for every n . Apply (22.2) for the surreal version. $\square$

The countable conclusion concerns the individual values. It does not say that the sequence $n \mapsto x_n$, or its range as a set, belongs to H_t . The ordinal information selecting those values can fail to be recoverable from a real. The Prikry example below realizes this failure even when all the values are ordinal definable.

22.2 Choice is equivalent to stabilization at one real

Theorem 22.3 (Stabilization theorem). *The following conditions are equivalent:*

1. $K_{\mathbb{R}} = K_r$ for some real r ;
2. $I_{\mathbb{R}} = I_r$ for some real r ;
3. $\mathbb{R} \subseteq H_r$ for some real r ;
4. $\text{HOD}_{\mathbb{R}} = H_r$ for some real r ;
5. $\text{HOD}_{\mathbb{R}} \models \text{AC}$;
6. $\mathbb{R} \in H_{\cup}$;
7. $H_{\cup} = \text{HOD}_{\mathbb{R}}$;

8. $H_\cup$ has an internal powerset of ω ;

9. $H_\cup$ is an inner model of ZF.

If these conditions fail, the family of fields $\{K_r : r \in \mathbb{R}\}$ has no countable cofinal subfamily under inclusion.

Proof. Suppose first that $\text{HOD}_{\mathbb{R}}$ satisfies choice. Since it contains the actual set $\mathbb{R}$, it contains a wellordering W of that set. Its internal isomorphism to an ordinal is still an isomorphism in V , so W really wellorders all ambient reals. There is a real s such that W is $\text{OD}(s)$. For each real a , its ordinal index in W uniquely defines a from W . Substitution shows that a is $\text{OD}(s)$. Since a real is hereditary $\text{OD}(s)$ whenever it is $\text{OD}(s)$, we obtain $\mathbb{R} \subseteq H_s$. This proves (5) $\Rightarrow$ (3).

If (3) holds, every allowed real parameter can be replaced by a definition from r and ordinals. Thus $\text{ROD} = \text{OD}(r)$, and taking hereditary membership yields (4). Condition (4) implies (5), since the fixed-real class H_r is an inner model of ZFC. It also implies (1), (2), (7) and (9). Conversely (1) implies (3), since $K_{\mathbb{R}}$ contains all reals; (2) implies it by the coding $a \mapsto a\omega$ from Theorem 22.2.

The class $H_\cup$ is transitive and contains every ambient subset of ω . If it has an internal powerset of ω , that set must therefore be the actual $\mathbb{R}$. Hence (9) $\Rightarrow$ (8) $\Rightarrow$ (6). If $\mathbb{R} \in H_r \subseteq H_\cup$, transitivity of H_r implies (3). Finally (7) implies (6), since $\text{HOD}_{\mathbb{R}}$ contains $\mathbb{R}$. These implications prove all equivalences.

For the last assertion, suppose a countable subfamily were cofinal. Theorem 22.2 supplies a single real t whose class contains every member of that subfamily. Cofinality then implies that it contains every K_r , giving $K_t = K_{\mathbb{R}}$, contrary to failure of stabilization. $\square$

Thus the failure of choice in $\text{HOD}_{\mathbb{R}}$ is visible in a concrete field question: there is no single real parameter whose definable surreal field includes all the others. In that case the union $H_\cup$ contains every real individually but omits their collection as a set and fails even powerset at ω .

Proposition 22.4 (The internally hereditarily wellorderable part). *One has*

$$H_\cup = \{x \in \text{HOD}_{\mathbb{R}} : \text{HOD}_{\mathbb{R}} \models \text{“tc}(\{x\}) \text{ is wellorderable”}\}.$$

Proof. If $x \in H_r$, its transitive closure belongs to H_r , and choice in H_r supplies a wellordering there. The same witness belongs to $\text{HOD}_{\mathbb{R}}$.

Conversely suppose $\text{HOD}_{\mathbb{R}}$ contains a bijection $e : \lambda \rightarrow T$, where $T = \text{tc}(\{x\})$ and λ is an ordinal. Inside $\text{HOD}_{\mathbb{R}}$ form

$$E = \{(\xi, \eta) \in \lambda^2 : e(\xi) \in e(\eta)\},$$

and choose the ordinal η_x with $e(\eta_x) = x$. The relation E is ROD , so it is $\text{OD}(s)$ for some real s . All the other members of its transitive closure are ordinal coded; hence $E \in H_s$. Its well-foundedness and extensionality hold in H_s , which therefore computes its Mostowski collapse. By uniqueness of well-founded collapse, this is the same collapse as in V , and its value at η_x is x . Thus $x \in H_s \subseteq H_\cup$. $\square$

Corollary 22.5 (A single real suffices if every surreal is ROD). *Over ambient ZFC,*

$$K_{\mathbb{R}} = \mathbf{No} \iff I_{\mathbb{R}} = \mathbf{Oz} \iff V = \text{HOD}_{\mathbb{R}} \iff \exists r \in \mathbb{R} V = \text{HOD}(r).$$

Proof. If every surreal is ROD , the characteristic sign sequence of each subset of an ordinal puts that subset in $\text{HOD}_{\mathbb{R}}$. For an arbitrary set x , choice in V supplies an ordinal code for the membership graph of $\text{tc}(\{x\})$. This code belongs to $\text{HOD}_{\mathbb{R}}$, and its well-founded collapse there

recovers x . Thus $V = \text{HOD}_{\mathbb{R}}$. The stabilization theorem then applies, because this inner model has the ambient axiom of choice, and yields one real r with $V = H_r$. The reverse implications are immediate.

To recover all surreals from all omnific integers, use the parameter-free injective map $x \mapsto \Omega(\Omega(x))$. Its outer exponent is positive, so its image consists of purely infinite omnific integers. Its inverse on its image is uniquely definable. Alternatively the bounded reversible coding from the existing framework gives the same implication. $\square$

Corollary 22.6 (A bounded omnific test for the entire universe). *The equivalent global conditions of [corollary 22.5](#) hold if and only if every omnific integer z with $1 < z < \omega$ belongs to ROD.*

Proof. Use the predecessor's reversible bounded monomial code, whose parameter behavior we spell out:

$$\tau(x) = \frac{1}{2} \left(1 + \frac{x}{\sqrt{1+x^2}} \right), \quad Q(x) = \Omega(\tau(x)).$$

The function τ bijects **No** with $(0, 1)$, and $Q(x)$ is a purely infinite omnific integer between 1 and ω . If $t = \Omega^{-1}(Q(x))$, then

$$x = \frac{2t - 1}{2\sqrt{t(1-t)}}.$$

Both directions are uniquely definable without new parameters. Thus ROD membership of every such omnific integer implies ROD membership of every surreal, and the global corollary applies. The bounded code itself is antecedent material from (Part I). $\square$

Remark 22.7 (Source 06's form of [corollary 22.5](#) [06]). Source 06 (Part III) proves [corollary 22.5](#) independently, as its Theorem 4.2, which is printed only here ([remark 33.2](#) in Part III points back to this remark). Its formulation: over ZFC the following are equivalent: (1) $V = \text{HOD}_{\mathbb{R}}$; (2) every set is ROD; (3) there is a real r with $V = H_r$; (4) every surreal is ROD; (5) every omnific integer is ROD; (6) every monomial ω^a with $0 < a < 1$ is ROD. Consequently, if every surreal has some real-ordinal definition, then one real r works for all surreals and all omnific integers simultaneously. Clause (6) strengthens the preceding corollary, since these monomials form the subclass $\mathcal{Q}$ of **Oz** $\cap (1, \omega)$.

Second proof (source 06). The equivalence of (1) and (2) follows directly from hereditary membership. To prove (2) $\Rightarrow$ (3), choose in V a well-order W of $\mathbb{R}$. By (2), W is definable from some real r and ordinal parameters. Every real s is then $\text{OD}(r)$: use, in addition, its ordinal position in W . Given any set x , a ROD definition of x uses a real s , which can now be replaced by its $\text{OD}(r)$ definition. Thus every set is $\text{OD}(r)$ and hence hereditarily so. This proves $V = H_r$. Conversely (3) implies (2).

Clearly (2) implies (4), (4) implies (5), and (5) implies (6). For (6) $\Rightarrow$ (4), use the parameter-free reversible monomial coding $x \mapsto \omega^{\tau(x)}$, where

$$\tau(x) = \frac{1}{2} \left(1 + \frac{x}{\sqrt{1+x^2}} \right) \in (0, 1).$$

The canonical omega-map inverse and $\tau^{-1}(t) = (2t - 1)/\sqrt{1 - (2t - 1)^2}$ recover x . Finally, any set x has, by Choice, a code consisting of a pointed well-founded extensional relation on an ordinal. Encode that relation as a set of ordinals and then as a surreal sign sequence. If that surreal is ROD, the unique Mostowski collapse and its marked element are ROD, so x is ROD. This proves (4) $\Rightarrow$ (2). $\square$

The same proof works relative to a fixed set of ordinals p : if every set is definable from p , some real depending on that set, and ordinal parameters, then $V = \text{HOD}(p, r)$ for one real r .

It is not enough merely to know that every real is available as a parameter; the hypothesis concerns all sets, or equivalently the full surreal or bounded omnific coding class. Nor does the theorem identify $\text{HOD}_{\mathbb{R}}$ with a fixed H_r when $\text{HOD}_{\mathbb{R}}$ is a proper inner model.

22.3 Countable birthdays, spectra, and the first omitted birthday [05]

Source 05's Proposition 5.1 (the basic structure of $K_{\mathbb{R}}$) is the paragraph after [proposition 22.1](#); its proof joins finitely many real parameters into one and applies the fixed-real results, initiality following either from the same argument or by taking a union of the initial classes K_r .

Theorem 22.8 (All countable-birthday surreals need only a real).

$$\mathbf{No}_{<\omega_1} \subseteq K_{\text{rd}} \subseteq K_{\mathbb{R}}. \quad (22.5)$$

The first inclusion needs no ordinal parameters.

Proof. If $x : \alpha \rightarrow \{-, +\}$ and α is countable, choose a well-order on a subset of ω of type α , together with a label on each point giving the sign of its image under the unique order isomorphism. One real codes the domain, relation, and labels. From that real alone, well-founded collapse recovers the order type, its order isomorphism, and hence the canonical sign sequence x . Finite and empty domains are covered by the same tagged coding convention. $\square$

This theorem concerns countable *sign length*, not countable normal-form support. The distinction is decisive in [theorem 25.7](#). This is also source 06's [remark 33.3](#) (Part III), where it follows from the surjectivity of the real decoder, and source 02 states it in [section 27](#).

Source 05's Proposition 5.3 is the last clause of [theorem 22.2](#), with the same proof; source 05 adds that the witness selection can be performed in ZFC using the internally definable certificate relation of [lemma 21.3](#), Collection, and Choice.

It is tempting, but wrong, to conclude that a countable summable family in $K_{\mathbb{R}}$ has its sum in this common K_r . Individual definitions may require a non- $\text{OD}(r)$ sequence of ordinal parameters. Joining the real parameters has not joined that ordinal information.

Parameter spectra. For a set x , define its real-parameter spectrum by

$$\text{Spec}(x) = \{r \in \mathbb{R} : x \in \text{OD}(r)\}.$$

This is a definable subset of $\mathbb{R}$. On reals, put $r \preceq s$ when $r \in \text{OD}(s)$, equivalently $r \in \text{HOD}(s)$. It is a preorder. Formula substitution shows that $r \preceq s$ implies $K_r \subseteq K_s$, so $\text{Spec}(x)$ is upward closed in this preorder. The join of a countable sequence of reals is an upper bound for all its entries. No assertion about the existence of a *least* real in a spectrum is implicit in this observation.

Finite algebraic operations may lose information. For instance, $\text{Spec}(x) \cap \text{Spec}(y) \subseteq \text{Spec}(x + y)$, but equality need not hold: $x + (-x) = 0$. Reversible coding, in contrast, will preserve the entire spectrum exactly.

Source 05's Proposition 6.1 (the bounded code Q of [theorem 6.8](#), with spectra). *Put*

$$\tau(x) = \frac{1}{2} \left(1 + \frac{x}{\sqrt{1+x^2}} \right), \quad Q(x) = \omega^{\tau(x)}. \quad (22.6)$$

Then Q is a parameter-free ambiently definable bijection from $\mathbf{No}$ onto the monomials ω^a with $0 < a < 1$, a subclass of $\mathbf{Oz} \cap (0, \omega)$. Its inverse is parameter-free definable. It preserves every parameter policy in this article, and $\text{Spec}(Q(x)) = \text{Spec}(x)$.

Second proof (source 05). The positive square root of $1 + x^2$ is greater than $|x|$, so $0 < \tau(x) < 1$. Given a monomial $z = \omega^a$ with $0 < a < 1$, recover a by the inverse Conway map. Put $y = 2a - 1$, so $-1 < y < 1$, and recover

$$x = \frac{y}{\sqrt{1 - y^2}}.$$

Direct field algebra verifies that these formulas are inverse. Unique-output closure gives parameter preservation in both directions. $\square$

This is the monomial code in (Part I), not a new code invented here. It is not a ring homomorphism and is not definable in the pure ordered-field language. It shows, for example, that the global conditions in [proposition 22.9](#) are equivalent to the much smaller looking assertion that every omnific integer in $(0, \omega)$ is ROD. A bounded omnific interval can retain arbitrary ambient definability information.

Bounded testing and the first omitted birthday. The hereditary real-ordinal class will be denoted

$$\text{HOD}_{\mathbb{R}} = \{x : \text{TC}(\{x\}) \subseteq \text{ROD}\}.$$

We use it as a definable class, without assuming that it has a canonical well-order or satisfies Choice. For the next statement no inner-model axiom for $\text{HOD}_{\mathbb{R}}$ is needed.

Proposition 22.9 (First omitted real-ordinal birthday [05]). *If $K_{\mathbb{R}} \neq \mathbf{No}$, the least birthday of a missing surreal is*

$$\lambda_{\mathbb{R}} = \min\{\alpha : \exists X \subseteq \alpha \ (X \notin \text{ROD})\} \geq \omega_1. \quad (22.7)$$

Source 05 proves this together with the equivalence of: every surreal is ROD; every subset of every ordinal is ROD; every set is ROD; and $V = \text{HOD}_{\mathbb{R}}$, which are part of [corollary 22.5](#).

Proof. A subset of an ordinal and its labelled sign sequence are interdefinable using that ordinal. This gives the surreal/subset equivalence. By Choice, any set a has a pointed well-founded extensional membership code on an ordinal domain: well-order its transitive closure and transport membership to that domain. A fixed ordinal coding of pairs turns this into a subset of an ordinal, with the distinguished point and domain included as finite ordinal parameters. If the code is ROD, its Mostowski collapse defines a under the same policy. No canonical choice of the code is needed, since the hypothesis covers all such codes. The hereditary equivalence is immediate. The same sign coding identifies the two least ordinals in (22.7); the lower bound is [theorem 22.8](#). $\square$

The predicate in (22.7) is internally definable, so the least ordinal exists whenever there is a witness. This is not a “least undefinable object” paradox: the least *birthday* is definable, while it need not pick a unique missing sign sequence. We do not claim that $\lambda_{\mathbb{R}}$ must be a cardinal in every universe; [theorem 24.9](#) computes it in a specific model.

Remark 22.10 (The real-parameter field is a support field [write]). The field $K_{\mathbb{R}}$ satisfies the hypothesis (8.3) of the support-subfield amplification theorem [theorem 8.2](#), so if $V \neq \text{HOD}_{\mathbb{R}}$, then for every $t \in (0, 1) \setminus K_{\mathbb{R}}$ the family $z_{\alpha} = \omega + \omega^{te_{\alpha}}$ ($\alpha \in \text{Ord}$) of (8.4) lies in $\mathcal{U}$ and is algebraically independent over $K_{\mathbb{R}}$. This was observed at the write (4 October 2026); neither source states it.

Proof. $K_{\mathbb{R}}$ is a subfield of $\mathbf{No}$ ([proposition 22.1](#) and the paragraph after it). The surreal ω is parameter-free definable, and $e_{\alpha} = \omega^{-(\alpha+1)}$ is definable from the ordinal α , so both lie in $K_{\text{HOD}} \subseteq K_{\mathbb{R}}$. If $0 \neq a \in K_{\mathbb{R}}$, say $a \in K_r$, then every exponent of a is definable from a and its

ordinal index in the normal form (the whole-normal-form paragraph after [proposition 22.1](#)), hence lies in $K_r \subseteq K_{\mathbb{R}}$; so $\text{supp}(a) \subseteq K_{\mathbb{R}}$. Finally $K_{\mathbb{R}} \neq \mathbf{No}$ exactly when $V \neq \text{HOD}_{\mathbb{R}}$, by [corollary 22.5](#), and then a $t \in (0, 1) \setminus K_{\mathbb{R}}$ exists: if $x \notin K_{\mathbb{R}}$ then $\tau(x) \notin K_{\mathbb{R}}$, because $K_{\mathbb{R}}$ is real closed and $x = \tau^{-1}(\tau(x))$ is given by the algebraic formula (6.4). [Theorem 8.2](#) applies. $\square$

23 Countable information, Hahn sums, and countable cuts

The preceding pointwise amalgamation theorem does not supply the sequence of definitions as one object. We now characterize exactly the missing condition. Define the first-order principle

$$\mathbf{A}_{\omega} : \quad \text{every countable set of ordinals belongs to ROD.} \quad (23.1)$$

Here and throughout this section, countability is measured in V . A countable sequence f of ordinals can be coded by the countable set $\{\lambda \cdot n + f(n) : n < \omega\}$ for an ordinal λ exceeding all $f(n)$. Conversely, a countable set is the range of a countable ordinal sequence. Thus $\mathbf{A}_{\omega}$ is equivalent to condition (1) below. Ordinal multiplication in this coding sentence is set-theoretic ordinal multiplication, not surreal multiplication.

Work in an ambient universe of ZFC. Write $\text{OD}(r)$ for ordinal definability with one additional real parameter r , and put

$$K_r = \mathbf{No} \cap \text{OD}(r), \quad K_{\mathbb{R}} = \bigcup_{r \in \mathbb{R}} K_r, \quad I_{\mathbb{R}} = \mathbf{Oz} \cap K_{\mathbb{R}}.$$

Here $\text{HOD}_{\mathbb{R}}$ means hereditary definability allowing an arbitrary real and finitely many ordinals at each object. It is not shorthand for one fixed-real HOD .

Theorem 23.1 (Countable assembly criterion). *The following conditions are equivalent.*

1. *Every function $f : \omega \rightarrow \text{Ord}$ is ordinal definable from a real.*
2. *$\text{HOD}_{\mathbb{R}}$ is closed under the ambient countable sequences of its elements.*
3. *Every ambient countable sequence of elements of $K_{\mathbb{R}}$ has an entire sequence code ordinal definable from a real.*
4. *$K_{\mathbb{R}}$ is closed under arbitrary countable strongly summable families of its elements.*
5. *Every countable strongly summable family of ordinal-definable purely infinite omnific integers has its sum in $I_{\mathbb{R}}$.*
6. *Whenever $A, B \subseteq K_{\mathbb{R}}$ are countable sets and $A < B$, some $y \in K_{\mathbb{R}}$ satisfies $A < y < B$.*

In (6), countability is computed in the ambient universe. Thus (6) is the ambient η_1 interpolation property.

Proof. Assume (1), and let $(x_n)_{n < \omega}$ be a sequence of elements of $\text{HOD}_{\mathbb{R}}$. For each n , choose a rank-local definition of x_n : a real r_n , a rank θ_n , a finite tuple of ordinal parameters, and a formula code that singles out x_n inside V_{θ_n} . Collection and ambient Choice permit these countably many choices. Join the reals and the natural-number formula codes into one real. The ordinal witnesses form a single countable ordinal sequence, which, by (1), is itself ordinal definable from a real. Join the two real parameters. Set-sized satisfaction then gives a uniform definition of the whole sequence $(x_n)_{n < \omega}$. Its transitive closure contains only the x_n , their hereditary descendants, and the elementary sequence coding apparatus. Consequently the sequence is in $\text{HOD}_{\mathbb{R}}$. This proves (2).

Condition (2) implies (3), since a surreal sign sequence is in $K_{\mathbb{R}}$ precisely when it belongs to $\text{HOD}_{\mathbb{R}}$. Condition (3) implies (4), because the strongly summable family is now available as

one set-coded input and Hahn summation is a parameter-free, uniquely determined operation. Condition (4) implies (5), because the sum of a strongly summable family of purely infinite surreals is purely infinite (or zero).

We prove (5) $\Rightarrow$ (1) by an explicit code. Given $f : \omega \rightarrow \text{Ord}$, regard each ordinal $f(n)$ as its canonical surreal, and set

$$q_n = 2^{-(n+1)}, \quad g_n = q_n + \frac{q_n}{4} \frac{f(n)}{1 + f(n)}, \quad t_n = \omega^{g_n}.$$

Since $0 \leq f(n)/(1 + f(n)) < 1$,

$$q_n \leq g_n < \frac{5}{4}q_n, \quad g_{n+1} < \frac{5}{8}q_n < g_n.$$

The family $(t_n)_{n < \omega}$ is strongly summable; each t_n is an ordinal-definable purely infinite omnific integer. Its sum

$$Z_f = \sum_{n < \omega} \omega^{g_n} \tag{23.2}$$

belongs to $(0, \omega) \cap \mathbf{Oz}$, with the displayed expression as its exact normal form. From its n th exponent recover

$$u_n = 4(g_n/q_n - 1), \quad f(n) = \frac{u_n}{1 - u_n}.$$

The canonical ordinal embedding has a definable inverse on its range. Thus Z_f and f are interdefinable without additional parameters. Condition (5) gives (1).

Condition (3) implies (6): enumerate the two countable sets, join their whole sequence codes, and take the unique simplest surreal between them. This construction is definable from that set-coded cut.

For (6) $\Rightarrow$ (1), use the same t_n and put $s_n = \sum_{k < n} t_k$. Set

$$A_f = \{s_{n+1} : n < \omega\}, \quad B_f = \{s_n + 2t_n : n < \omega\}.$$

Every endpoint is ordinal definable using only finitely many entries of f . The full sum Z_f witnesses $A_f < B_f$. By (6), choose $y \in K_{\mathbb{R}}$ between these sets. For every n ,

$$0 < y - s_{n+1} < 2t_{n+1} \prec t_n.$$

Hence the first $n + 1$ normal-form terms of y are exactly $t_0, \dots, t_n$. Its first ω terms therefore recover all the g_n and then f . Any additional normal-form tail occurs below every g_n and does not affect decoding. Thus f is ordinal definable from a real, proving (1). $\square$

Remark 23.2. Even without the equivalent conditions, every countable set of real parameters has a join r , so every countable collection of individual $K_{\mathbb{R}}$ elements is contained pointwise in one K_r . This does not in general put the whole chosen sequence in $\text{HOD}(r)$: the ordinal witnesses may carry additional countable information. The theorem identifies precisely the extra assembly principle.

23.1 Two further forms of the same principle

Corollary 23.3. *Each condition of [theorem 23.1](#) is also equivalent to each of:*

1. The transitive union $H_{\cup} = \bigcup_r H_r$ is closed under all ambient ω -sequences of its elements.
2. $\text{HOD}_{\mathbb{R}} = \text{HOD}(\text{Ord}^{\omega})$, where the right side permits a countable sequence of ordinals and finitely many ordinals as parameters.

Under these conditions, $\text{HOD}_{\mathbb{R}}$ satisfies *Dependent Choice*.

Proof. Assume A_ω and $x_n \in H_\cup$. Choose reals r_n with $x_n \in H_{r_n}$ and join them into r . Every x_n and all its hereditary descendants then belong to H_r . As in the theorem, use A_ω to assemble the ordinal records of the definitions and join their real witness to r . The whole graph is now $\text{OD}(t)$ for one real t , and its descendants belong to H_t . Hence the graph belongs to $H_t \subseteq H_\cup$. Conversely, ordinal sequences are sequences of elements of $H_\cup$, so closure implies their real-plus-ordinal definability.

For the second equivalence, reals are countable ordinal sequences, so $\text{ROD} \subseteq \text{OD}(\text{Ord}^\omega)$. Under A_ω every allowed ordinal-sequence parameter on the right is itself in ROD , and substitution gives $\text{OD}(\text{Ord}^\omega) = \text{ROD}$ and the hereditary equality. Conversely each ordinal-sequence graph is hereditarily definable from itself, so that equality makes every such graph ROD .

Finally, let a nonempty set and a serial relation on it belong to $\text{HOD}_{\mathbb{R}}$. In the ambient universe choose a dependent sequence, starting at any specified member. All its entries lie in $\text{HOD}_{\mathbb{R}}$, and the countable-sequence closure just proved puts the whole sequence in $\text{HOD}_{\mathbb{R}}$. This proves *Dependent Choice* there. We assert no converse from *Dependent Choice* alone. $\square$

Remark 23.4 (No automatic countable least join). Interleaving a specified countable sequence of real parameters gives an upper bound for their ordinal-definability degrees, but need not give their least upper bound. In a universe with a non-OD real, its sequence of bits consists of individually parameter-free definable bit values; their interleaving can nevertheless encode that non-OD real. Finite joins and countable sequence assembly have different logical content.

23.2 What a countable sum can carry

The code Z_f in (23.2) is uniformly interdefinable with f , without any additional parameters. Its individual monomials require only one ordinal each. This is a precise explanation for why a countable sum can leave a definability class: the sum can recover the entire ordering and association of infinitely many ordinal choices. The failure is not caused by divergent series; the family is Hahn summable with an explicitly verified support.

The interpolation implication is also exact. It does not require the interpolant to equal the sum. Every possible interpolant has the same first ω normal-form terms and therefore the same coded information. Allowing an arbitrary smaller tail cannot hide the code.

23.3 Source 05's separated-band multiplexer [05]

Source 05 builds the code of [theorem 23.1](#) as a separate theorem, with dyadic bands $[2^{-n-1}, 3 \cdot 2^{-n-2})$ in place of source 02's slots $[q_n, \frac{5}{4}q_n)$; it also records the parameter spectra of [section 22.3](#).

The general code does not by itself expose a sum of individually ordinal-definable terms. We now give a version tailored to that task. For a nonnegative surreal a set

$$u(a) = \frac{a}{1+a}, \quad e_n(a) = 2^{-n-1} + 2^{-n-2}u(a) \quad (n < \omega). \quad (23.3)$$

For $f : \omega \rightarrow \text{Ord}$, interpreting $f(n)$ as a surreal ordinal, put

$$\boxed{\mathcal{Z}(f) = \sum_{n < \omega} \omega^{e_n(f(n))}.} \quad (23.4)$$

No monotonicity assumption is made on f .

Theorem 23.5 (Countable ordinal multiplexer). *For every $f : \omega \rightarrow \text{Ord}$ the expression (23.4) is a well-defined omnific integer in $(0, \omega)$. Its normal form has support of order type ω in decreasing order and all coefficients are 1. Every individual monomial in the display is OD. The whole set f and the whole surreal $\mathcal{Z}(f)$ are interdefinable by fixed parameter-free ambient formulas. In particular*

$$\mathcal{Z}(f) \in K_{\mathbb{R}} \iff f \in \text{ROD}, \quad \text{Spec}(\mathcal{Z}(f)) = \text{Spec}(f). \quad (23.5)$$

Proof. For $a \geq 0$ one has $0 \leq u(a) < 1$, whence

$$2^{-n-1} \leq e_n(a) < 3 \cdot 2^{-n-2}. \quad (23.6)$$

The entire band for index $n + 1$ lies below the band for n , since

$$3 \cdot 2^{-n-3} < 2^{-n-1}.$$

Thus the sequence $e_n(f(n))$ is strictly decreasing regardless of the values or repetitions in f . Any nonempty subset of this support has a largest element, namely the one with the least natural index. The normal form is therefore valid and Hahn-summable. All its exponents are positive, so it is omnific with zero constant term. Its leading exponent is less than $3/4 < 1$; comparison of leading terms gives $0 < \mathcal{Z}(f) < \omega$.

Fixing n , the term $\omega^{e_n(f(n))}$ is definable from the ordinal $f(n)$ and the natural numeral n . This proves individual OD definability without asserting a uniform OD assignment of these ordinals as n varies.

Conversely, the unique normal form of $z = \mathcal{Z}(f)$ supplies its n th exponent a_n . Set

$$v_n = 2^{n+2}(a_n - 2^{-n-1}), \quad f(n) = \frac{v_n}{1 - v_n}. \quad (23.7)$$

The result is the all-plus surreal corresponding to the original ordinal, whose domain recovers that ordinal as a set. Normal-form uniqueness and the disjoint bands make this decoder unambiguous. Replacement over $n < \omega$ recovers the whole graph of f . Encoding and decoding are fixed definable operations, so each direction preserves all permitted parameters. $\square$

Remark 23.6 (Why this is stronger than small coefficients). The construction requires no nondefinable real coefficients, no uncountable support, and no freedom to permute the support after it is chosen. The entire possible failure of ROD definability is the countable sequence of ordinal information hidden *inside the exponents*. The real leading parts of their bands are fixed in advance. Even a repeated or oscillating ordinal sequence is faithfully encoded.

Example 23.7 (A finite check of the formulas). For the initial values $f(n) = n$, the first four exponents are

$$e_0(0) = \frac{1}{2}, \quad e_1(1) = \frac{5}{16}, \quad e_2(2) = \frac{1}{6}, \quad e_3(3) = \frac{11}{128}.$$

Substituting each one into (23.7) recovers 0, 1, 2, 3 exactly. The supplied verification script tests the rational formula and band separation on many finite examples. It does not implement infinite ordinal arithmetic, decide definability, or prove the forcing results.

23.4 Further equivalent forms of countable assembly [05]

Definition 23.8. The *real-ordinal assembly principle* A_ω is the assertion

$$\text{every function } f : \omega \rightarrow \text{Ord} \text{ belongs to ROD.}$$

Quantification is over set functions: every such function has an ordinal bound on its range. This is an ordinary first-order assertion of ZFC, using the certificate predicate for ROD.

The principle does not say that every ordinal is ROD; that is automatic. It says that the entire *assignment of ordinals to natural indices* can be described using one real and only finitely many ordinal parameters. The principle A_ω is equivalent to source 02's (23.1), as explained there; the two sources state it for sequences and for sets respectively.

Theorem 23.9 (Countable-assembly equivalence: source 05's further forms [05]). *Over ZFC the conditions of theorem 23.1 are equivalent to each of the following.*

- (iii) *Whenever $(x_n)_{n < \omega}$ is a sequence of sets with each $x_n \in \text{ROD}$, the whole sequence is ROD.*
- (v) *For every $f : \omega \rightarrow \text{Ord}$, the particular omnific integer $\mathcal{Z}(f)$ in (23.4) belongs to $I_{\mathbb{R}}$.*
- (vi) $\text{ROD} = \text{COD}$.
- (vii) $K_{\mathbb{R}} = K_{\text{co}}$.

Source 05's clauses (i) A_ω , (ii) every countable set of ordinals is ROD, and (iv) every countable Hahn-summable sequence of elements of $K_{\mathbb{R}}$ has its sum in $K_{\mathbb{R}}$, are theorem 23.1 (1), (23.1) and theorem 23.1 (4). In particular, testing countable sums of the very special individually OD monomials in (23.4) is already sufficient to test arbitrary countable Hahn-sum closure of the whole field $K_{\mathbb{R}}$. Clause (vi) is also source 02's corollary 23.3 (2).

Second proof of theorem 23.1 with these clauses (source 05). We prove the implications in several explicit steps.

$(i) \Rightarrow (iii)$: *assembling definitions, not merely real parameters.* Let $x_n \in \text{ROD}$. For each n select a rank-local certificate consisting of a real r_n , a rank ordinal θ_n , a formula code e_n , and a finite ordinal tuple $\bar{\alpha}_n$ uniquely specifying x_n in V_{θ_n} . Selection is legitimate: the certificate relation is internally definable, Collection first bounds a set of witnesses for the countably many inputs, and Choice selects one for each index.

Encode the sequence (r_n) , the natural formula codes, and the lengths of the finite tuples by a real r_* . Flatten all rank ordinals θ_n and all entries of $\bar{\alpha}_n$ into one sequence $g : \omega \rightarrow \text{Ord}$, using the tuple lengths to delimit the blocks and padding unused positions by 0. The exact pairing convention is fixed once and for all. By (i), g is $\text{OD}(s)$ for some real s and a finite ordinal tuple $\bar{\beta}$. Join s and r_* into a real t .

From $t, \bar{\beta}$ recover g , then all the rank-local certificates. Uniform satisfaction for the set structures V_{θ_n} recovers each unique x_n using one formula, uniformly in n . Replacement therefore yields the whole sequence, which is definable from $t, \bar{\beta}$. This uses no satisfaction predicate for the full universe. The rank sequence is precisely the information that (i) allows us to compress.

$(iii) \Rightarrow (iv)$. Apply (iii) to the sequence of canonical surreal codes. The whole family is ROD, and its Hahn sum is a fixed uniquely definable operation on that family, by definition 20.2 and lemma 3.2.

$(iv) \Rightarrow (v) \Rightarrow (i)$. Each term in $\mathcal{Z}(f)$ is OD and hence in $K_{\mathbb{R}}$, and the family is Hahn-summable. Thus (iv) implies (v). The decoder (23.7) then recovers f from $\mathcal{Z}(f)$, giving (i). This is the crucial reduction from unrestricted sums to a very narrow omnific test family.

$(i) \Leftrightarrow (ii)$. Under (i), a countable set of ordinals has an enumeration whose graph is ROD, so its range is ROD; the empty set is immediate. Conversely assume (ii), and let $B = \text{ran}(f)$. The increasing enumeration of B has a countable ordinal domain δ . A real can code a well-order on a subset of ω of type δ , together with the index in that well-order corresponding to each $f(n)$. Decoding this real and the ROD set B recovers f using one real and the finite ordinal parameters in a definition of B . This gives (i).

$(i) \Leftrightarrow (vi)$. A real is a countable sequence of bits, so a real and any finite ordinal tuple can be joined into one countable ordinal sequence. Hence $\text{ROD} \subseteq \text{COD}$ always. Under (i), any countable ordinal parameter sequence is itself ROD, and substituting its definition shows

$\text{COD} \subseteq \text{ROD}$. Conversely any $f : \omega \rightarrow \text{Ord}$ is COD from the parameter f itself, so (vi) implies (i).

Finally (vi) implies (vii). If (vii) holds, then for every f its code $\mathcal{Z}(f)$ belongs to K_{co} by definition and therefore to $K_{\mathbb{R}}$; being omnific, it belongs to $I_{\mathbb{R}}$. Thus (vii) implies (v), completing the equivalence. $\square$

Corollary 23.10 (The integer ring detects the same obstruction). *The conditions of [theorem 23.9](#) are also equivalent to $I_{\mathbb{R}} = I_{\text{co}}$, and to closure of $I_{\mathbb{R}}$ under countable Hahn-summable families of its elements.*

Proof. A Hahn-summable family of omnific integers has no negative exponents, and only finitely many nonzero contributions at exponent 0. Its constant coefficient is therefore an integer, so its sum is omnific. Field closure implies ring closure. Ring closure implies condition (v), since the test monomials are omnific. Equality of the integer rings similarly implies (v), and field equality implies equality of the rings by intersection with $\mathbf{Oz}$. $\square$

Remark 23.11 (Whole-family closure remains unconditional). Even when A_{ω} fails, a summable family whose *whole graph* is ROD has a ROD sum. Failure concerns pointwise membership only. Thus the theorem does not contradict standard closure under a specified definable transfinite construction. It identifies when an unspecified countable choice of definitions can be turned into one specified construction.

A useful covering criterion.

Proposition 23.12 (Countable ordinal covering). *Suppose every countable set of ordinals is contained in a countable ordinal-definable set of ordinals. Then A_{ω} holds, hence $K_{\mathbb{R}}$ and $I_{\mathbb{R}}$ are countably Hahn-sum closed.*

Proof. Let $f : \omega \rightarrow \text{Ord}$, and choose a countable OD set B containing its range. As in the proof of (ii) $\Rightarrow$ (i), a real codes the countable order type of B together with the indices of the values $f(n)$. The increasing enumeration of B is definable from B . Combining the real code with an ordinal definition of B gives an ROD definition of f . $\square$

The hypothesis is a sufficient *covering* principle, not the assertion that a single real codes all countable ordinal sequences. Different input sequences may require different reals, ordinal parameters, and ordinal-definable covers.

23.5 A canonical countable-assembly completion [05]

The class COD is the classical class of sets definable from countable ordinal sequences; compare [38, III.2.2]. It is useful here because countably many such parameter sequences can themselves be flattened into one. Write

$$K_{\text{co}} = \mathbf{No} \cap \text{COD}, \quad I_{\text{co}} = \mathbf{Oz} \cap \text{COD}.$$

This is an enlargement of $K_{\mathbb{R}}$, not a change in the ambient arithmetic.

Lemma 23.13 (Countable-ordinal assembly). *A countable sequence of COD sets is itself COD.*

Proof. Choose rank-local definition certificates for its entries, with countable ordinal parameter sequences. Flatten the countable matrix of parameter ordinals, the rank ordinals, and the natural syntax codes into one countable ordinal sequence. This single parameter recovers all the certificates. Uniform set satisfaction and Replacement recover the whole input sequence.

The witness selection is justified by Collection and Choice exactly as in [theorem 23.9](#). This is the familiar ordinal-sequence assembly mechanism underlying [38, III.2.6]. $\square$

Definition 23.14 (Ambient stability). A subclass $H \subseteq \mathbf{No}$ is *ambiently stable* if, for every fixed parameter-free set-theoretic formula defining a partial surreal-valued function on a finite tuple, applying that function to inputs from H gives an element of H whenever the output is unique and exists. This is a scheme, not a putative uniform truth predicate.

This is deliberately stronger than closure under the field operations alone. Without this qualification the minimality statement below would not follow: an arbitrary ambient definition cannot be recovered by field arithmetic merely because its ordinal parameter has been encoded in a monomial.

Theorem 23.15 (Canonical ambient-stable Hahn completion). *The class K_{co} is the least subclass of $\mathbf{No}$ which contains $K_{\mathbb{R}}$, is ambiently stable, and is closed under countable Hahn sums. More explicitly, any subclass H with these three properties contains K_{co} . Moreover K_{co} is initial and real closed, $K_{\text{co}} = \text{Frac}(I_{\text{co}})$, and I_{co} is countably Hahn-sum closed.*

Proof. The finite-parameter combination property for COD, [lemmas 3.2](#) and [23.13](#), and definability of summation prove ambient stability and countable Hahn-sum closure. The algebraic properties follow from [theorem 4.1](#); initiality follows by adjoining an ordinal prefix length to the countable ordinal parameter. The integer-ring assertion follows as in [corollary 23.10](#).

For minimality, let $x \in K_{\text{co}}$ and choose $f : \omega \rightarrow \text{Ord}$ and a fixed formula $\varphi(y, f)$ which uniquely defines the surreal x . Every term of $\mathcal{Z}(f)$ is in K_{HOD} and hence in $K_{\mathbb{R}} \subseteq H$. Countable Hahn-sum closure places $\mathcal{Z}(f)$ in H .

There is a fixed parameter-free partial function on surreals which first checks whether its input is in the range of the code, decodes f by [\(23.7\)](#), and returns the unique surreal satisfying $\varphi(y, f)$, when there is such a unique surreal. Its graph is a single set-theoretic formula with φ substituted literally. It may equivalently be totalized by returning 0 on other inputs. Ambient stability now gives $x \in H$. No additional parameter and no uniform evaluator for all formulas has been introduced. $\square$

Corollary 23.16 (The completion has no further countable-assembly stages). *Applying the same ambient-stability and countable-Hahn-closure operation to K_{co} produces K_{co} again. In a universe satisfying $\mathbf{A}_\omega$, $K_{\mathbb{R}}$ is already this completion; in the Prikry model below the inclusion $K_{\mathbb{R}} \subsetneq K_{\text{co}}$ is strict.*

Remark 23.17 (What the word completion does not mean). This is a minimality statement for a specified closure scheme. It is not a Dedekind completion, an order-topological completion, a completion of a metric space, or a claim that K_{co} is closed under every uncountable summable family. It also does not assert that the field operations and countable sums *alone* generate K_{co} .

Remark 23.18 (The completion needs only the ordinals [write]). In [theorem 23.15](#) the hypothesis “contains $K_{\mathbb{R}}$ ” can be weakened to “contains every ordinal”: K_{co} is the least subclass of $\mathbf{No}$ that contains every ordinal, is ambiently stable, and is closed under countable Hahn sums. Hence such a subclass contains all of $K_{\mathbb{R}}$, and in particular every real. This was observed at the write (4 October 2026); the proof is source 05’s.

Proof. K_{co} contains every ordinal, so by [theorem 23.15](#) it has the three properties. Conversely let H have them, and let $x \in K_{\text{co}}$ be the unique y with $\varphi(y, f)$, where $f : \omega \rightarrow \text{Ord}$. For each fixed n the map $\alpha \mapsto \omega^{b_n(\alpha)}$ is given by one parameter-free formula (the numeral n is written into it), so ambient stability applied to the input $f(n) \in \text{Ord} \subseteq H$ puts the n th term of $\mathcal{Z}(f)$ in H . Countable Hahn-sum closure puts $\mathcal{Z}(f)$ in H , and the fixed decoding map of the proof of [theorem 23.15](#) puts x in H . Only the ordinals, not all of $K_{\mathbb{R}}$, were used. $\square$

23.6 Certificate coding details [05]

This is source 05's Appendix A.

Here is one completely definite version of the bookkeeping used in [theorem 23.9](#). Fix a recursive pairing bijection $\pi : \omega^2 \rightarrow \omega$. A real r_* has disjoint tagged columns coding the bits of each r_n , the formula code e_n , and the tuple length ℓ_n . Define an ordinal sequence g by

$$g(\pi(n, 0)) = \theta_n, \quad g(\pi(n, j+1)) = \begin{cases} \alpha_{n,j}, & j < \ell_n, \\ 0, & j \geq \ell_n. \end{cases}$$

The natural-number information in r_* tells the decoder which zeroes are genuine entries and which are padding. Once g and r_* are available, every local formula, rank, and finite ordinal tuple is uniformly available.

The statement that x_n is the unique output is checked using $\text{Sat}(V_{\theta_n}, e_n, x_n, r_n, \bar{\alpha}_n)$. The outer operation is one fixed formula concerning satisfaction in a set, not an evaluation of φ_{e_n} in V . The only global formula that is substituted in the proof is the single selected definition of g from a real and finitely many ordinals. This is why no uniform global truth predicate is needed.

For COD, use an additional pairing layer to store the countable ordinal sequence parameter for each x_n and interleave the ranks, syntax codes, and tuple lengths as further ordinal entries. All finite syntactic data are natural numbers and therefore ordinals. One ω -sequence still suffices. This makes explicit that the flattening does not require an ordinal-indexed family of real parameters or a class choice principle.

24 A proper real-parameter field with exact countable closure

The following construction uses standard Cohen forcing and its homogeneity. Its purpose is to compute the full real-parameter field hierarchy and its countable behavior, rather than to claim a new forcing notion. For background on the forcing machinery see [\[29, 30\]](#).

Let $W \models \text{ZFC} + V = L$, let λ be an uncountable cardinal of W , and force with

$$\mathbb{P} = \text{Fn}(\lambda \times \omega, 2, <\omega).$$

Write $V = W[G]$, $c_\xi(n) = (\bigcup G)(\xi, n)$, and for ground-model countable $S \subseteq \lambda$ let $W_S = W[G \restriction (S \times \omega)]$. Using a canonical constructible enumeration of S , code this countable coordinate generic by a real r_S , so $W_S = L[r_S]$.

Theorem 24.1 (Exact atlas and its inclusion order). *In V ,*

$$K_{r_S} = \mathbf{No}^{W_S}, \quad K_{\mathbb{R}} = \bigcup_{S \in ([\lambda]^{\leq \omega})^W} \mathbf{No}^{W_S}.$$

Furthermore,

$$K_{r_S} \subseteq K_{r_T} \iff S \subseteq T.$$

All equalities concern canonical sign sequences and ambiently absolute surreal operations.

Proof. The forcing is ccc. A nice name for a real uses countably many maximal antichains, each countable, and thus only a ground-model countable set of coordinates. Consequently every real $r \in V$ belongs to some W_S .

Factor the forcing over W_S into the remaining Cohen coordinates. This tail is weakly homogeneous. If a set of ordinals is definable in V from $r \in W_S$ and ordinals, homogeneity decides each membership assertion at the top condition: compatible images of conditions

prevent two contrary decisions about check parameters. The forcing relation in W_S therefore reconstructs that set there. The sign code of a surreal is a set of ordinals up to fixed definable coding. It follows that $K_r \subseteq \mathbf{No}^{W_S}$.

For $r = r_S$ this gives one inclusion in the first equality. Conversely, the canonical $L[r_S]$ well-order shows that every element of W_S is ordinal definable in V from r_S . The first equality follows, and the real-name support argument proves the union formula.

If $S \subseteq T$, inclusion of the intermediate models gives field inclusion. If $\xi \in S \setminus T$, the real c_ξ belongs to $\mathbf{No}^{W_S}$ but is Cohen generic over W_T and is not in W_T . Thus field inclusion fails. $\square$

Corollary 24.2 (The ω_1 chain). *For $\lambda = \omega_1^W = \omega_1^V$, let r_δ code the coordinates below δ . Then*

$$K_{r_\delta} = \mathbf{No}^{L[G_{<\delta}]}, \quad K_{r_\delta} \subsetneq K_{r_{\delta+1}}, \quad K_{\mathbb{R}} = \bigcup_{\delta < \omega_1} K_{r_\delta}.$$

The preorder of real-parameter fields, ordered by inclusion, has cofinality exactly ω_1 . It has no largest field.

Proof. Every ground countable set of coordinates is bounded in ω_1 , giving the cofinal chain, and c_δ witnesses strictness. A countable cofinal family of real parameters would have a real join; its field would then contain every field in the family and so be largest, contradicting strict extension by a fresh Cohen coordinate. $\square$

Proposition 24.3 (The general cofinality invariant). *Let $\mathcal{D}$ be the preorder of the fields K_r in the Cohen extension above. Then*

$$\text{cof}(\mathcal{D}) = \text{cof}([\lambda]^{\leq \omega})^W, \subseteq).$$

The right-hand cofinality has the same value in W and V . In particular it is λ when $W = L$ and λ is regular uncountable.

Proof. The canonical fields form a cofinal copy of the displayed support poset. Conversely, for each member K_r of a cofinal family choose a countable ground support S_r containing a name for r . Any canonical field K_{r_T} is contained in some K_r , hence in $K_{r_{S_r}}$, and therefore $T \subseteq S_r$. Thus the chosen supports are cofinal.

To see preservation, let a name enumerate a new cofinal family in the ground support poset using μ indices. A ccc maximal antichain deciding each value yields only countably many possible ground values per index. The ground union of all these possibilities has size at most $\mu \cdot \aleph_0$ and is cofinal; otherwise a condition forcing cofinality would contradict a missed ground support. The reverse inequality holds because an old cofinal family remains cofinal in the unchanged ground poset. Finally, in L , regular uncountable λ satisfies $\lambda^{\aleph_0} = \lambda$ by GCH. The support poset has cardinality λ , and any cofinal family must cover λ by its countable members, so its cofinality is λ . $\square$

Theorem 24.4 (Countable closure and its exact first failure). *In the Cohen extension with $\lambda = \omega_1$,*

1. *Every countable sequence of ordinals is ordinal definable from a real, and all the countable assembly conditions above hold.*
2. *Every surreal of countable birthday belongs to $K_{\mathbb{R}}$.*
3. *The least birthday of a surreal outside $K_{\mathbb{R}}$ is exactly ω_1 .*
4. *The order $K_{\mathbb{R}}$ fills every external countable cut but omits a cut with $\aleph_1$ endpoints (the η_1 but not η_2 property).*

Proof. For a name $\dot{f}$ of an ordinal sequence, choose for each n a maximal antichain deciding $\dot{f}(n)$. The ccc makes their total coordinate support countable, so $f \in L[G_S] = L[r_S]$ and hence $f \in \text{OD}(r_S)$. This proves (1).

For (2), any countable sign sequence can be encoded by a real coding a countable well-order of its domain and the sign assignment. The Mostowski collapse decodes the actual ordinal domain. Such a surreal is definable from that real, even without additional ordinal parameters.

For (3), let

$$A = \{\xi < \omega_1 : c_\xi(0) = 1\}.$$

This set is in no W_S with S ground countable. Indeed, for any $B \subseteq \omega_1$ in W_S , the tail forcing has dense conditions disagreeing with B at a fresh coordinate outside S . Hence A is not real-ordinal definable by the atlas theorem. Its sign sequence $s_A : \omega_1 \rightarrow \{-, +\}$ is likewise outside $K_{\mathbb{R}}$, and has birthday ω_1 .

The η_1 assertion follows from countable assembly. To disprove η_2 , take the canonical cut of s_A : a proper prefix $s_A \upharpoonright \alpha$ goes on the left when $s_A(\alpha) = +$ and on the right when $s_A(\alpha) = -$. Every prefix has countable birthday and lies in $K_{\mathbb{R}}$. Any interpolant must extend the entire sign sequence s_A ; a first disagreement, or an early end, violates the corresponding cut inequality. But $K_{\mathbb{R}}$ is initial: a prefix of a real-ordinal definable sign sequence is definable using the same real and the prefix length as an additional ordinal. Thus an interpolant there would put s_A there, a contradiction. This is an unfilled cut of size ω_1 . $\square$

Remark 24.5 (Do not extend the atlas equality to all hereditary sets). The full set $\mathbb{R}^V$ belongs to $\text{HOD}_{\mathbb{R}}^V$: it is definable without parameters and each of its elements is definable using itself. Yet no countable coordinate model W_S contains $\mathbb{R}^V$ as a set, since it would then contain every real in that set by transitivity. Thus the surreal atlas does not imply $\text{HOD}_{\mathbb{R}} = \bigcup_S W_S$.

Remark 24.6 (Which cofinality is being computed). The cofinality in [corollary 24.2](#) concerns the *inclusion preorder of parameter fields*. It is not the order cofinality of the surreal field: every ordinal already belongs to every K_r . The strict chain and the absence of a largest parameter field imply, by the Choice criterion, that $\text{HOD}_{\mathbb{R}}$ fails Choice in this example. Nevertheless $\mathbf{A}_\omega$ holds, so $\text{HOD}_{\mathbb{R}}$ is closed under ambient countable sequences and satisfies Dependent Choice. Thus countable coherence and capture by one real are independent features of this construction.

24.1 Source 05's treatment of the Cohen model [05]

Source 05 works in the same extension, written with κ for source 02's λ and J for source 02's S : We now construct a nontrivial positive model. Let W be a transitive model of ZFC with $W \models V = L$. Let κ be an uncountable cardinal of W , let

$$\mathbb{P} = \text{Add}(\omega, \kappa)^W = \text{Fn}(\kappa \times \omega, 2, < \omega)^W,$$

and let G be W -generic. Set $V = W[G]$. For a ground-model countable $J \subseteq \kappa$, put

$$G_J = G \upharpoonright (J \times \omega), \quad W_J = W[G_J].$$

We identify the generic filter with its union when restricting its coordinate function. All $\text{OD}(r)$ and ROD classes in this section are computed in V .

Lemma 24.7 (Countable support and ordinal covering). *The Cohen product is ccc. Every real of V belongs to some W_J with $J \in W$ countable in W . Every countable set of ordinals of V is contained in a countable set of ordinals belonging to W .*

Proof. For ccc, apply the delta-system lemma to the finite domains of an uncountable family of conditions, and thin further so all conditions agree on their common finite root. Any two are compatible, so the family is not an antichain.

For a name for a real, choose for each natural index a maximal antichain deciding its bit. Each antichain is countable in W . The union of the finite coordinate supports appearing in these countably many antichains is a countable $J \in W$. The decisions, and hence the real, are determined by G_J .

For a name for $f : \omega \rightarrow \text{Ord}$, work below a condition forcing that it is such a function and choose a maximal antichain deciding each $f(n)$. The union B of all ordinal values decided by these antichains is a countable set in W , and $\text{ran}(f) \subseteq B$ in the generic extension. Enumerating an arbitrary countable set gives the final assertion. This proof also explains the usual preservation of ω_1 by ccc forcing [10]. $\square$

Lemma 24.8 (Homogeneity and ordinal subsets). *Let N be a transitive ground model and $P \in N$ a forcing such that any two conditions have stronger conditions with isomorphic forcing cones, with the induced name isomorphism fixing canonical names for ground-model sets. If $X \subseteq \theta$ in an extension $N[H]$ is definable there from finitely many ground-model parameters, then $X \in N$.*

Proof. For each $\xi < \theta$, write membership as a sentence $\psi(\check{\xi}, \check{a})$ using the given ground parameters. If one condition forced this sentence and another forced its negation, strengthen to isomorphic cones. Transport of names preserves the ground parameters, giving opposite decisions at corresponding tops, a contradiction. Hence the top condition decides each membership sentence. The definability theorem for forcing gives the ground-model set

$$X = \{\xi < \theta : 1_P \Vdash \psi(\check{\xi}, \check{a})\}.$$

This is the standard homogeneity argument, stated only for subsets of a specified ground ordinal. It does not assert that no new definable sets of arbitrary kind can appear. $\square$

For a Cohen product, finite bit flips make any two conditions compatible after an automorphism, and fix all ground check names. Thus the lemma applies also to the remaining-coordinate product over an intermediate model W_J .

Source 05's Theorem 9.3 (= **theorem 24.1**, with the ring equality). *In the extension just described,*

$$K_{\mathbb{R}}^V = \bigcup_{J \in [\kappa]^{\leq \omega} \cap W} \mathbf{No}^{W_J}, \quad (24.1)$$

$$I_{\mathbb{R}}^V = \mathbf{Oz}^V \cap \bigcup_{J \in [\kappa]^{\leq \omega} \cap W} \mathbf{No}^{W_J}. \quad (24.2)$$

Here $[\kappa]^{\leq \omega} \cap W$ means the sets countable in W . For each such J , choose a real g_J coding G_J using a constructible enumeration of J . Then

$$K_{g_J}^V = \mathbf{No}^{W_J}. \quad (24.3)$$

The equalities are equalities of canonical sign-code subclasses, with ambient surreal order and arithmetic. In particular their right-hand sides are subfields under that arithmetic.

Second proof (source 05). Let $x \in K_{\mathbb{R}}^V$ and choose a real r with $x \in \text{OD}^V(r)$. By lemma 24.7, $r \in W_J$ for some ground-model countable J . Factor the forcing into its J and $\kappa \setminus J$ coordinates. Over W_J the remaining factor is a weakly homogeneous Cohen product. Let $\alpha = b(x)$ and

let $A \subseteq \alpha$ be the positive-sign positions of x . The set A is definable in V from r and finitely many ordinals. These parameters all belong to W_J . By [lemma 24.8](#), $A \in W_J$, and hence its sign sequence x belongs to W_J . This proves the forward inclusion.

Conversely, choose a constructible enumeration of the ground-model countable set J . With this enumeration, a real g_J codes the whole array G_J , and

$$W_J = L[g_J]$$

with the constructible hierarchy taken up to the common ordinals of the models. The constructible enumeration and J itself are already in L . Every member of $L[g_J]$ is ordinal definable from g_J in V : specify a stage of the relative constructible hierarchy and a position in its canonical g_J -definable well-order. The hierarchy $L[g_J]$ has the same meaning in the intermediate and final models. Consequently every canonical surreal in W_J is in $K_{g_J}^V$, proving the reverse inclusion and [\(24.3\)](#). Intersecting with the ambient omnific predicate gives [\(24.2\)](#). $\square$

The formulation of the ring equality avoids using additional normal-form absoluteness as a hidden premise. The field property of each displayed subclass follows directly from its equality with the ambient definable field $K_{g_J}^V$.

Theorem 24.9 (Properness, closure, and the first omitted birthday). *In this Cohen extension the following all hold:*

- (i) $K_{\text{HOD}}^V = \mathbf{No}^W$ and $K_{\text{HOD}}^V \subsetneq K_{\mathbb{R}}^V \subsetneq \mathbf{No}^V$.
- (ii) $\mathbf{A}_\omega$ holds. Thus $K_{\mathbb{R}}^V = K_{\text{co}}^V$ and both the field and its omnific integer part are countably Hahn-sum closed.
- (iii) No single real r satisfies $K_r^V = K_{\mathbb{R}}^V$.
- (iv) $\lambda_{\mathbb{R}}^V = \omega_1^V$.
- (v) There is a non-ROD omnific integer in $(0, \omega)$.

Proof. Homogeneity with ordinal parameters gives $K_{\text{HOD}}^V \subseteq \mathbf{No}^W$, and every constructible set is ordinal definable in V , giving equality. A coordinate Cohen real c_i is not in W but is real-ordinal definable from itself. This makes the first inclusion in (i) strict.

For (ii), the countable ordinal cover from [lemma 24.7](#) lies in $W = L$, and every constructible set is ordinal definable in V . Thus [proposition 23.12](#) applies.

For properness, let $\eta = \omega_1^W = \omega_1^V$ and encode the generic array on $\eta \times \omega$ as a subset A of η , using the ordinal bijection $(\xi, n) \mapsto \omega \cdot \xi + n$. The multiplication in this sentence is ordinal multiplication. If A were ROD, the support argument in [theorem 24.1](#) would put it in some W_J for countable $J \in W$. Choose $i < \eta$ outside J . The coordinate real c_i is definable from A and i , so it would be in W_J . But c_i is Cohen generic over W_J and is not in W_J , a contradiction. Thus the sign sequence of length η with positive set A is a surreal outside $K_{\mathbb{R}}^V$, proving the second strict inclusion in (i). Its birthday is exactly η , and [theorem 22.8](#) supplies the matching lower bound, proving (iv).

For (iii), given r , choose countable J with $r \in W_J$. The forward support argument gives $K_r^V \subseteq \mathbf{No}^{W_J}$. Choose any $i \in \kappa \setminus J$; then $c_i \in K_{\mathbb{R}}^V$ but $c_i \notin K_r^V$. Finally apply the reversible bounded code Q of [theorem 6.8](#) to the missing sign sequence to obtain (v). $\square$

Clauses (ii), (iii) and (iv) are also source 02's: they are [theorem 24.4](#) (1), [corollary 24.2](#) with the remark on which cofinality is computed, and [theorem 24.4](#) (3); source 05's proofs above are second routes, through the covering criterion [proposition 23.12](#) for (ii). Clauses (i) and (v) are source 05's alone.

Remark 24.10 (The countable-coordinate boundary). If only countably many Cohen coordinates are added over L , the entire generic array is coded by one real g , and $V = L[g]$. In that case every set is $\text{OD}(g)$, so $K_g = K_{\mathbb{R}} = \mathbf{No}$. Uncountably many coordinates are what make the positive example both countably closed and proper. In the uncountable product, [theorem 22.2](#) is sharp in the sense that countable families of real parameters have a common real bound, while the full family of coordinate reals does not.

25 A countable Hahn sum can defeat every real parameter

We now construct a precise obstruction to arbitrary countable summation. The large-cardinal assumption in this example is explicit: let W be a transitive ground model of ZFC with a measurable cardinal κ and a normal nonprincipal κ -complete ultrafilter D on κ . Force over W with ordinary Prikry forcing $\mathbb{P}_D$, obtaining $V = W[G]$. Its conditions are pairs (s, A) , where s is a finite increasing sequence of ordinals below κ , $A \in D$, and every element of A lies above s . An extension appends finitely many elements from the old upper set and shrinks that upper set. A direct extension, written $\leq^*$, leaves the stem s fixed.

The forcing ingredients are classical results of Prikry [\[31\]](#); we include their proofs in the form needed here. For a detailed author-written exposition see [\[30, Section 9.2\]](#).

Lemma 25.1 (The Prikry property and preservation of reals). *Every forcing statement is decided by a direct extension of any given condition. Consequently $\mathbb{P}_D$ adds no bounded subset of κ , and W and V have the same reals.*

Proof. First normality gives the following finite-set homogeneity fact: if $c : [\kappa]^{<\omega} \rightarrow \mu$ with $\mu < \kappa$, some $H \in D$ makes c constant on $[H]^n$ separately for each $n < \omega$. For completeness prove the fixed-length statement by induction on n . The case $n = 1$ follows from κ -completeness of the ultrafilter. For the successor step, for each $a \in [\kappa]^n$ choose a color i_a whose set B_a of possible last coordinates above a belongs to D . For each $\eta < \kappa$ intersect the fewer than κ sets B_a with $a \subseteq \eta + 1$, and take the diagonal intersection of these intersections. Normality makes the result D -large. If a lies in it and β is a later point of it, then $\beta \in B_a$. The induction hypothesis applied to $a \mapsto i_a$ now gives homogeneity for length $n + 1$. Intersecting the countably many fixed-length homogeneous sets proves the stated fact.

Fix $p = (s, A)$ and a statement φ . Color a finite tuple $t \in [A]^{<\omega}$ by 1 if some $B \in D$ makes $(s \smallfrown t, B)$ a condition forcing φ , and by 0 otherwise. Choose $H \subseteq A$ in D homogeneous at every length, and put $q = (s, H)$. If q did not decide φ , it would have extensions q_0, q_1 forcing opposite answers. Extend the shorter stem until their stems have equal length. Their appended tuples t_0, t_1 belong to the same $[H]^n$, and hence have the same color. The tuple of the extension forcing φ has color 1. The other tuple therefore has an upper-set witness forcing φ as well. That witness and the extension forcing $\neg\varphi$ have the same stem and compatible upper sets, a contradiction. Thus q decides φ .

Suppose a name $\dot{X}$ is forced to be a subset of $\lambda < \kappa$. For each $\xi < \lambda$ use the property just proved to find a direct extension, with a common fixed stem, deciding $\xi \in \dot{X}$. Intersect their upper sets. By κ -completeness the result is a condition deciding every bit, hence forcing $\dot{X}$ to equal a ground-model set. Such conditions are dense, so no bounded subset of κ is added. In particular no subset of ω is added. $\square$

Lemma 25.2 (An explicit isomorphism between tail cones). *Prikry forcing is cone homogeneous. As a result, its largest condition decides every sentence whose parameters are canonical names for ground-model sets.*

Proof. Given $p = (s, A)$ and $q = (t, B)$, remove from $A \cap B$ a bounded initial segment containing both stems, leaving $D_0 \in D$. Set $p' = (s, D_0)$ and $q' = (t, D_0)$. The map

$$(s \frown u, E) \mapsto (t \frown u, E)$$

is an order isomorphism from $\mathbb{P}_D/p'$ onto $\mathbb{P}_D/q'$. Both cones allow exactly the same additional finite tuples u from D_0 and exactly the same remaining upper sets E . The inverse replaces t by s .

An isomorphism of forcing notions preserves forcing and takes each canonical name for a ground-model set to the same canonical name over the other cone. If two conditions forced opposite answers to a sentence with only such parameters, strengthen them to the two isomorphic cones above. Invariance of forcing would give the opposite answers at corresponding largest conditions, which is impossible. The density of deciding conditions now shows that the largest condition decides the sentence. $\square$

Lemma 25.3 (Real-ordinal-definable ordinal sets remain in the ground). *Every ROD subset of an ordinal in $V = W[G]$ belongs to W .*

Proof. Suppose $X \subseteq \eta$ is uniquely defined in V by $\varphi(X, r, \bar{\alpha})$, where r is a real and $\bar{\alpha}$ is a finite tuple of ordinals. By Lemma 25.1, $r \in W$; all the ordinal parameters belong to W as well. For each $\xi < \eta$, let $\theta(\xi, r, \bar{\alpha})$ assert that the uniquely φ -defined object contains ξ . By Lemma 25.2 the largest condition decides this statement. The forcing theorem and separation in W produce

$$X_0 = \{\xi < \eta : 1_{\mathbb{P}_D} \Vdash \theta(\check{\xi}, \check{r}, \check{\bar{\alpha}})\} \in W.$$

The truth lemma gives $X = X_0$. This argument does not require $\mathbb{P}_D$ itself to be ordinal definable: it is an ordinary ground-model parameter in the construction of X_0 . $\square$

Theorem 25.4 (Failure of countable Hahn closure in a Prikry extension). *In $V = W[G]$ there is a countable summable sequence $(y_n)_{n < \omega}$ of ordinal-definable omnific integers such that*

$$z = \omega + \sum_{n < \omega} y_n$$

is not ROD. Moreover z is an omnific integer with zero constant term and

$$\omega < z < \omega + \Omega(1/2).$$

Thus all summands belong to the single field $\mathbf{No} \cap \text{HOD}^V$, while their Hahn sum does not belong to $K_{\mathbb{R}}^V$.

Proof. The union of the generic stems is an increasing cofinal sequence $C = \{\kappa_n : n < \omega\}$ in κ . The sets of conditions requiring any specified finite stem length, and requiring a point above any specified $\beta < \kappa$, are dense. This proves both countability and cofinality. Since κ is regular in W , $C \notin W$. Lemma 25.3 therefore gives $C \notin \text{ROD}^V$.

Put

$$e_{\kappa_n} = \Omega(-(\kappa_n + 1)), \quad y_n = \Omega(e_{\kappa_n}).$$

Each y_n is definable from the ordinal κ_n and is a monomial with positive exponent, so it is an ordinal-definable omnific integer. The exponents e_n are strictly decreasing, making the family Hahn-summable. They all satisfy $0 < e_n \leq \Omega(-1) < 1/2$. The sum $q = \sum_n y_n$ is therefore purely infinite and positive, with leading exponent less than $1/2$. Normal-form comparison gives $0 < q < \Omega(1/2)$, so $z = \omega + q$ has the asserted location and constant term.

Its normal form recovers the generic set by the fixed formula

$$C = \{\alpha \in \text{Ord} : \Omega(-(\alpha + 1)) \in \text{supp}_{\text{exp}}(z - \omega)\}.$$

The support and this inverse decoding are uniquely definable set operations. If z were ROD, then so would be C , a contradiction. All the decoding takes place in V ; no additional absoluteness theorem about normal forms between W and V is needed. $\square$

The theorem is a relative-consistency construction from a measurable cardinal. It does not assert failure of countable closure in every universe. In particular, $V = L$ makes every set ordinal definable. Its role is to show exactly why a countable family of values can fit inside one definable surreal field without its complete description, or its Hahn sum, fitting inside any field obtained from real and ordinal parameters.

Corollary 25.5 (A missing countable cut). *In the same Prikry extension there are countable sets of ordinal-definable surreal endpoints $A < B$ with no interpolant in $K_{\mathbb{R}}$. Thus the pointwise real-parameter amalgamation theorem does not imply even external countable-cut interpolation.*

Proof. Apply the finite-prefix cut construction in [theorem 23.1](#) to the increasing enumeration of the new cofinal set C . Every interpolant would recover that ordinal sequence, contradicting [lemma 25.3](#). $\square$

25.1 A negative example without a large-cardinal hypothesis

The measurable cardinal in the Prikry construction is useful for its cardinal preservation, but is unnecessary for the failure of countable assembly. A suitable Namba extension gives that failure from the consistency of ZFC alone. We specify the forcing presentation because two versions commonly called Namba forcing are not forcing equivalent.

Let $W \models \text{ZFC} + V = L$ and let $\kappa = (\omega_2)^W$. Use the version $\mathbb{N}$ whose conditions are trees $T \subseteq \kappa^{<\omega}$ with a stem s_T , with a single path below that stem, such that *every* node at or above the stem has κ immediate successors in T . The order is inclusion. This is the Laver-style presentation. The classical no-new-reals theorem under CH applies to this version: [\[33\]](#) explicitly distinguishes the two presentations and states that the preservation proof applies to both; [\[34, Theorem 23 and the comparison table on p. 4\]](#) also records the result for this presentation. We import this classical preservation theorem and prove the additional properties used here.

Theorem 25.6 (Stabilization with failure of countable assembly). *Let $G \subseteq \mathbb{N}$ be W -generic and $V = W[G]$. Then*

$$\text{HOD}_{\mathbb{R}}^V = \text{HOD}^V = L^V = W, \quad K_{\mathbb{R}}^V = \text{No}^W.$$

Every real-parameter field K_r^V equals this same field. Nevertheless there is a countable set of ordinals that is not ROD in V . Consequently countable Hahn closure, and the equivalent countable assembly and ambient η_1 properties, fail. In particular, choice in $\text{HOD}_{\mathbb{R}}$ and stabilization of the real-parameter fields do not imply these countable-closure properties.

Proof. We first verify cone homogeneity for exactly the specified forcing. Given two conditions T, S , consider only their nodes at or above their respective stems. Each is an abstract rooted tree of height ω in which every node has exactly κ children. By choosing a bijection between the successor sets at each pair of corresponding nodes and recursing through the finite levels, construct a rooted-tree isomorphism F between them in W .

For a condition $R \subseteq T$, map its nodes above s_T by F and restore the fixed proper prefixes of s_S . The result is a condition below S : its stem is the image of s_R , and the number of children

of every node above that stem remains κ . This construction preserves inclusion and has the analogous inverse induced by F^{-1} . Thus it is an isomorphism

$$\mathbb{N}/T \cong \mathbb{N}/S.$$

In particular the forcing is cone homogeneous, with the canonical names for all ground-model sets fixed under the induced name maps.

CH holds in W , so the cited Namba theorem gives $\mathbb{R}^V = \mathbb{R}^W$. The argument of Lemma 25.3, whose only inputs were cone homogeneity and preservation of reals, now shows that every ROD subset of an ordinal in V belongs to W .

All reals of V are constructible and hence ordinal definable in V . It follows by substitution that $\text{ROD}^V = \text{OD}^V$ and $\text{HOD}_{\mathbb{R}}^V = \text{HOD}^V$. To see that this class is contained in W , take any $x \in \text{HOD}^V$. Choose inside that inner model supplies an ordinal code for the membership graph of $\text{tc}(\{x\})$. The code is an ordinal-definable set of ordinals, hence belongs to W by the preceding homogeneity argument. Its well-founded collapse in W recovers x . Conversely every constructible set is ordinal definable in V by its position in the canonical constructible wellorder. Thus $\text{HOD}^V = W = L^V$. The canonical sign-sequence identification gives the equality of surreal fields; because each real is already ordinal definable, adding any one real parameter does not enlarge the field.

Namba forcing adds a cofinal function $g : \omega \rightarrow \kappa$. The dense sets requiring longer stems and a stem value above a specified ordinal prove this directly. Its range A is a countable unbounded subset of κ in V . It cannot belong to W : a fixed subset of an ordinal has the same order type in transitive universes, while any countable ordinal remains below $\omega_1^W = \omega_1^V$; a ground-model countable cofinal subset would contradict regularity of κ in W . Equivalently, take the increasing cofinal sequence of successive new maxima extracted from g ; its order type is ω , making this contradiction immediate. Therefore A is not ROD, by the proved ground-model containment.

Apply the countable assembly criterion. An explicit failed Hahn sum is obtained, as before, from the monomials

$$\Omega(\Omega(-(\alpha + 1))) \quad (\alpha \in A).$$

They are individually ordinal-definable omnific integers, and their sum recovers A from its exponent support. Adding ω places the missing omnific integer in the same bounded cell as in Theorem 25.4. $\square$

Since $V = L$ itself satisfies the countable assembly criterion and the Namba extension just constructed violates it, that criterion is independent of ZFC provided ZFC is consistent. The same is therefore true of arbitrary countable Hahn closure for the real-ordinal-definable surreal field. The Prikry construction supplies the complementary illustration in which the relevant ground-model cardinal is preserved.

[write] On the imported Namba input, 4 October 2026. Source 02's only imported input in theorem 25.6 is that this presentation of Namba forcing adds no reals under CH. The write could not confirm a textbook statement for this presentation; the references and what is missing are recorded in question 28.17. The cone homogeneity, the computation $\text{HOD}_{\mathbb{R}}^V = \text{HOD}^V = L^V = W$ and the nondefinability argument are proved above.

25.2 Source 05's Prikry example [05]

The positive Cohen model does not establish a theorem of ZFC that $\mathbf{A}_\omega$ always holds. A forcing that adds a new countable ordinal sequence while preserving all reals supplies the opposite behavior, provided its homogeneity prevents the new sequence from becoming ordinal definable from an old real.

Let $W \models \text{ZFC}$ be transitive and let $U \in W$ be a normal, nonprincipal, κ -complete ultrafilter on a measurable cardinal κ . Ordinary Prikry forcing P_U has conditions (s, A) , where s is a finite increasing sequence from κ , $A \in U$, and all members of A lie above the last member of s . An extension appends finitely many members of A to the stem and shrinks the remaining tail. A *direct extension* does not change the stem. Let $C = \langle \kappa_n : n < \omega \rangle$ be the generic Prikry sequence and $V = W[C]$.

The precise imported forcing input. The classical Prikry property says that every statement can be decided by a direct extension. For a normal measure, ordinary Prikry forcing adds an increasing cofinal ω -sequence in κ while adding no bounded subsets of κ . A modern proof and discussion appear in [32]. We use this standard theorem, not a general claim about all Prikry-type or tree-Prikry forcings.

For clarity, the no-new-bounded-subsets consequence follows directly from the Prikry property and κ -completeness. Given a name for a subset of $\gamma < \kappa$, decide its membership statements by direct extensions with a fixed stem and intersect their U -tails. The resulting tail still belongs to U and gives a condition deciding the entire subset to be a ground-model set. This argument works below every condition. In particular

$$\mathbb{R}^{W[C]} = \mathbb{R}^W. \quad (25.1)$$

The cofinality of C follows from the dense requirements to extend the stem beyond each fixed $\xi < \kappa$. Source 02 proves this input instead (lemma 25.1).

Source 05's Lemma 10.1 (= lemmas 25.2 and 25.3; source 05's ultrafilter is written U). *The forcing P_U satisfies the cone-isomorphism hypothesis of lemma 24.8. Consequently, every ROD subset of an ordinal in $V = W[C]$ belongs to W .*

Second proof (source 05). Given $p = (s, A)$ and $q = (t, B)$, choose $\gamma < \kappa$ at least as large as every entry of either finite stem, and put

$$D = (A \cap B) \setminus (\gamma + 1).$$

Then $D \in U$, and (s, D) and (t, D) strengthen the original conditions. Every condition in the first cone has the form $(s \smallfrown v, E)$ with the appended finite sequence v drawn from D , and the second cone has the corresponding condition $(t \smallfrown v, E)$. Replacing s by t and leaving v, E unchanged is an order isomorphism. When v is empty, every allowed tail is above both stems by the choice of D ; when it is nonempty, both stems have the same last appended point. Thus the construction works even if the original stem lengths differ.

The recursively induced isomorphism of names takes each canonical ground-model name to its counterpart. Apply lemma 24.8. A ROD definition in V uses a real from W by (25.1) and finitely many ordinals, also from W . All its parameters are therefore ground-model parameters, proving the last assertion. No assumption that U itself is ordinal definable is required for this containment argument. $\square$

Source 05's version of theorem 25.4 uses its multiplexer and proves that the Prikry sequence is not real-ordinal definable by coding its graph, a second route to source 02's argument through order types.

Theorem 25.7 (A non-ROD countable omnific sum). *In the ordinary Prikry extension above, let $f(n) = \kappa_n$. Then*

$$z_C = \sum_{n < \omega} \omega^{2^{-n-1} + 2^{-n-2} \frac{\kappa_n}{1 + \kappa_n}} \in \mathbf{Oz}^V \cap (0, \omega) \quad (25.2)$$

is not ROD in V . Every individual summand is OD in V , the family is Hahn-summable, and every coefficient in its normal form is 1. Therefore

$$\neg A_\omega, \quad K_{\mathbb{R}}^V \subsetneq K_{\text{co}}^V, \quad I_{\mathbb{R}}^V \subsetneq I_{\text{co}}^V.$$

In particular $K_{\mathbb{R}}^V$ and $I_{\mathbb{R}}^V$ need not be closed under countable Hahn sums, even of elements of the common ordinal-definable subfield and subring.

Proof. The function f cannot belong to W : if it did, its domain ω and cofinality in κ would be absolute, contradicting regularity of κ in W . Code its graph by

$$X_f = \{\kappa \cdot n + f(n) : n < \omega\} \subseteq \kappa \cdot \omega,$$

using ordinal multiplication and addition. The coding and decoding are definable from the ordinal κ . If f were ROD, so would be X_f . By [lemma 25.3](#), $X_f \in W$, and decoding inside W would give $f \in W$, a contradiction. Thus $f \notin \text{ROD}$. This argument uses the graph, not an assumption that every ground-model set which is countable in the extension was countable in the ground.

Apply [theorem 23.5](#) to this f . It proves every assertion about the normal form, summability, bounds, and individual definability in (25.2). If z_C were ROD, its fixed decoder would recover the whole f as a ROD set, contrary to the preceding paragraph. On the other hand f is itself an allowed countable ordinal parameter, so $z_C \in I_{\text{co}}$. The strict inclusions and failure of A_ω follow. $\square$

Remark 25.8 (What the negative result really separates). The failure is not caused by a new undefinable real: there are no new reals at all. It is not caused by choosing different real parameters for different summands: every summand is already OD. Nor is it caused by unsummability or a noncanonical limit convention. What is missing is a real–ordinal definition of the sequence $\langle \kappa_n : n < \omega \rangle$. A countable sequence of very large ordinals need not be encodable by one real and finitely many ordinal parameters.

The exact relative-consistency conclusion. The Cohen construction is available over a constructible transitive model of ZFC and gives a proper $K_{\mathbb{R}}$ satisfying countable Hahn-sum closure. The Prikry construction is available over a transitive model with a measurable cardinal and gives failure of that closure. Thus, under the stated model-existence hypotheses, the two behaviors occur in models of ZFC. In the usual forcing-relative-consistency reading, ZFC supplies the positive direction, while a measurable cardinal is an upper bound for the consistency strength of the negative direction.

No lower bound matching measurability is proved. In particular we do not claim equiconsistency with a measurable cardinal, and we do not claim that a measurable cardinal is necessary for failure of A_ω . Ordinary Prikry forcing was chosen because both no-new-reals and the needed cone isomorphisms have transparent proofs. Replacing it by a weaker-consistency construction is a separate research problem.

[write] **Superseded, 4 October 2026, batch 93.** The last two paragraphs are source 05's, written without knowledge of source 02. [Theorem 25.6](#) is exactly the weaker-consistency construction they call a separate research problem: from a transitive model of $\text{ZFC} + V = L$, which exists if any transitive model of ZFC does, Namba forcing gives failure of A_ω with no new reals and with the needed cone homogeneity proved. In the forcing relative-consistency reading, failure of A_ω is therefore consistent relative to ZFC alone, and since it is a statement of ZFC its consistency strength is exactly that of ZFC. The measurable cardinal remains what source 05 says it is: a sufficient hypothesis for this particular, cardinal-preserving construction. Nothing in source 05's paragraphs is wrong; they claim no lower bound. See also [question 28.12](#).

26 The strong stabilizer of the reals and all ordinals

Naming all real and ordinal parameters is still much weaker than allowing arbitrary set-theoretic definitions. This section computes an exact invariant that makes the difference visible. Arithmetic on surreal exponents is surreal arithmetic; addition of ordinal surreals is their natural sum. We work in ZFC, using canonical sign sequences. Statements about class automorphisms can equivalently be read in a conservative class theory with set choice.

For a subclass $B \subseteq \mathbf{No}$, put

$$\text{Ser}(B) = \{x \in \mathbf{No} : \text{supp}(x) \subseteq B\},$$

where $\text{supp}(x)$ is the set of exponents in the Conway normal form of x . The empty support represents zero. We use the classical facts that each support is a set, reverse well ordered by the surreal order, and that an ordinal's Conway normal form is its finite Cantor normal form [2]. Define

$$\Gamma_{\mathbb{Z}} = \text{span}_{\mathbb{Z}}(\text{Ord}), \quad \Gamma_{\mathbb{Q}} = \text{span}_{\mathbb{Q}}(\text{Ord}), \quad \mathcal{F}_{\Gamma} = \text{Ser}(\Gamma_{\mathbb{Q}}). \quad (26.1)$$

The spans use finite sums, so no class sum occurs.

A field automorphism is called *strong* here if it and its inverse transport every set-indexed summable family to a summable family and commute with its Conway sum. Every Hahn automorphism constructed below has this property. The exact theorem concerns this explicitly specified class; it does not assume that every field automorphism is strong.

26.1 The rational span of the ordinals

Lemma 26.1. *The elements of $\Gamma_{\mathbb{Q}}$ are precisely the finite normal forms*

$$g = \sum_{j=1}^m q_j \omega^{\gamma_j}, \quad q_j \in \mathbb{Q}, \quad \gamma_1 > \cdots > \gamma_m, \quad \gamma_j \in \text{Ord}.$$

For $\Gamma_{\mathbb{Z}}$, replace rational coefficients by integer coefficients. Hence

$$\Gamma_{\mathbb{Q}} \cap \mathbb{R} = \mathbb{Q}, \quad \Gamma_{\mathbb{Q}} \cap \{\text{infinitesimal surreals}\} = \{0\}.$$

Moreover, $\text{Ser}(\text{Ord})$ consists of finite normal forms with real coefficients and ordinal exponents.

Proof. Expand each ordinal in a finite rational linear combination into Cantor normal form and collect equal exponents. Conversely, ω^{γ} is an ordinal when γ is an ordinal, so every displayed expression belongs to $\Gamma_{\mathbb{Q}}$. The integer assertion is identical.

A nonzero such normal form with leading exponent zero is a nonzero rational. If its leading exponent is positive, it is infinite in absolute value. These are the only possibilities, proving the intersections. Finally, a reverse well-ordered subset of the ordinals is finite: an infinite set of ordinals has a strictly increasing sequence of its first ω elements. This proves the last assertion. $\square$

26.2 Two automorphism tools

The Hahn-lift construction and invariant-support method below already occur in the repository's automorphism and self-embedding reports [24, 25]. We give the details needed for the new fixed-field calculation. In particular, the set-parameter span-of-supports result `opa:thm:hull` and elementary shears in [24] are antecedents. The issue settled here is the stabilizer of the proper class of all canonical ordinals, which remains explicitly open in [25] at the pinned snapshot.

Lemma 26.2 (Hahn lifts). *Let $\psi : (\mathbf{No}, +, <) \rightarrow (\mathbf{No}, +, <)$ be an additive order automorphism. Then*

$$S_\psi \left(\sum_{g \in S} r_g \omega^g \right) = \sum_{g \in S} r_g \omega^{\psi(g)} \quad (26.2)$$

is a strong ordered-field automorphism of $\mathbf{No}$, fixes $\mathbb{R}$ pointwise, and preserves $\mathbf{Oz}$ in both directions. If ψ fixes every ordinal, then so does S_ψ .

Proof. An increasing bijection transports a reverse well-ordered set to a reverse well-ordered set. Reindexing therefore defines the map. It preserves addition, the largest exponent where normal forms differ, and the coefficient there; hence it preserves order. Inverses reindex by the inverse exponent maps.

For S_ψ , additivity of ψ gives $\omega^{\psi(g+h)} = \omega^{\psi(g)} \omega^{\psi(h)}$, so normal-form convolution for multiplication is preserved. Reindexing preserves the reverse-well-order and local finiteness conditions for summability in both directions; coefficientwise addition proves strongness.

An additive order automorphism fixes zero and preserves positive exponents. Thus S_ψ fixes real constant series and preserves the normal-form conditions for $\mathbf{Oz}$: all nonconstant exponents positive, constant term an integer. Finally, an ordinal has finite normal form with ordinal exponents and natural coefficients, so an exponent map fixing ordinals fixes this normal form. $\square$

Lemma 26.3 (A moving support point prevents a fixed series). *If an increasing bijection h of a linearly ordered class satisfies $h(S) = S$ for a reverse well-ordered set S , it fixes every member of S . Consequently*

$$S_\psi(x) = x \implies \text{supp}(x) \subseteq \text{Fix}(\psi).$$

Proof. If $h(s) > s$, then $s, h(s), h^2(s), \dots$ is a strictly increasing sequence in S . If $h(s) < s$, use inverse iterates. Both contradict reverse well ordering. Equality of reindexed normal forms implies invariance of their supports, proving the consequences. $\square$

26.3 Moving every exponent outside the rational ordinal span

Theorem 26.4 (The additive stabilizer). *The common fixed subgroup of all additive order automorphisms of $\mathbf{No}$ that fix Ord pointwise is exactly $\Gamma_{\mathbb{Q}}$. For every $e \notin \Gamma_{\mathbb{Q}}$ there is such an automorphism ψ , definable from set parameters, that moves e .*

Proof. An additive automorphism of a divisible torsion-free group is $\mathbb{Q}$ -linear, so every automorphism in question fixes $\Gamma_{\mathbb{Q}}$.

Choose once and for all a $\mathbb{Q}$ -linear map $d : \mathbb{R} \rightarrow \mathbb{R}$ whose kernel is exactly $\mathbb{Q}$. Such a map is obtained by extending $\{1\}$ to a Hamel basis, writing $\mathbb{R} = \mathbb{Q} \oplus W$, and taking the projection onto W . This is the explicit use of set choice; the single map d is a set parameter.

For each $s \in \mathbf{No}$, let $c_s(x)$ be the coefficient at exponent s , zero if absent, and define

$$d_s = \begin{cases} d, & s \in \text{Ord}, \\ \text{id}_{\mathbb{R}}, & s \notin \text{Ord}. \end{cases}$$

Put

$$N_s(x) = d_s(c_s(x))\omega^{s-1}, \quad \psi_s(x) = x + N_s(x). \quad (26.3)$$

Coefficient extraction is additive. Since $s-1 < s$, the coefficient at s in $N_s(x)$ vanishes. Thus $N_s^2 = 0$ and $\psi_s^{-1} = 1 - N_s$.

If $N_s(x) \neq 0$, then x has leading exponent at least s , whereas the added term has exponent $s - 1 < s$. The leading sign is unchanged. Thus ψ_s preserves signs and, by additivity, order. An ordinal has coefficient zero at a nonordinal exponent and a natural-number coefficient at an ordinal exponent. Both are killed by the corresponding d_s , so every ψ_s fixes all ordinals.

Finally, an element fixed by every ψ_s has no nonordinal support exponent and has rational coefficients at every ordinal support exponent. Its support is finite by Lemma 26.1, so it belongs to $\Gamma_{\mathbb{Q}}$. Thus some ψ_s moves each $e \notin \Gamma_{\mathbb{Q}}$. The entire family is given by one class formula with the one set parameter d ; no basis of the proper-class group **No** is used. $\square$

Theorem 26.5 (Exact strong fixed field). *The common fixed field of all strong field automorphisms of **No** fixing $\mathbb{R} \cup \text{Ord}$ pointwise is*

$$\text{Fix}_{\text{strong}}(\mathbb{R} \cup \text{Ord}) = \mathcal{F}_{\Gamma} = \{x \in \mathbf{No} : \text{supp}(x) \subseteq \text{span}_{\mathbb{Q}}(\text{Ord})\}. \quad (26.4)$$

*The same equality holds if these automorphisms are additionally required to preserve **Oz**.*

Proof. Let σ be any strong field automorphism fixing the reals and ordinals. A field automorphism of a real closed field preserves its order, since positive elements are the nonzero squares. For $g = \sum_{j=1}^m q_j \alpha_j \in \Gamma_{\mathbb{Q}}$, one has

$$\omega^g = \prod_{j=1}^m (\omega^{\alpha_j})^{q_j}.$$

Each ω^{α_j} is an ordinal, hence fixed. Positive rational powers are unique in an ordered field, so ω^g is fixed. Real coefficients are fixed as well. Strongness then fixes every normal-form sum supported on $\Gamma_{\mathbb{Q}}$. This proves the lower inclusion even without an assumption about **Oz**.

If $x \notin \mathcal{F}_{\Gamma}$, choose $e \in \text{supp}(x) \setminus \Gamma_{\mathbb{Q}}$. The additive stabilizer theorem gives ψ fixing every ordinal but moving e . Its Hahn lift S_{ψ} is strong, fixes $\mathbb{R} \cup \text{Ord}$, and preserves **Oz**. If it fixed x , the support-orbit lemma would force $\psi(e) = e$, a contradiction. Thus the upper inclusion already follows from the explicitly constructed **Oz**-preserving automorphisms. $\square$

This exactly answers the *strong* version of the $\mathbb{R} \cup \text{Ord}$ -stabilizer question recorded as **sse:sr:q:stabilizers** in the pinned self-embedding report [25]. The lower inclusion uses preservation of arbitrary summable families; the theorem does not compute the common fixed field without that hypothesis.

No collection of classes is formed as a set. The lower inclusion is a theorem about each class automorphism with the stated properties. For the upper inclusion, each $x \notin \mathcal{F}_{\Gamma}$ has an explicit automorphism defined from set parameters. These statements suffice for applications to each fixed first-order formula.

26.4 Consequences for intrinsic definability

Let $\mathfrak{S}$ be the surreal ordered field with predicate **Oz**. We may also add predicates for $\mathbb{R}$ and Ord ; the next assertion holds for either language. A permitted parameter class means that each formula may use a *finite tuple* from that class.

Corollary 26.6. *For every $A \subseteq \mathbb{R} \cup \text{Ord}$,*

$$\text{dcl}_{\mathfrak{S}}(A) \subseteq \mathcal{F}_{\Gamma}.$$

This includes permitting all real and ordinal parameters simultaneously.

Proof. The automorphisms in the upper-inclusion proof fix A , preserve $\mathbf{Oz}$, and fix the real and ordinal subclasses pointwise. They are automorphisms of either expansion. A uniquely defined element is fixed by every such automorphism. For a proper-class structure, invariance is proved by induction on the subformulas of the particular formula; no uniform truth predicate is needed. $\square$

[write] **Antecedent, 4 October 2026.** For a finite tuple of real and ordinal parameters, and in the language with $\mathbf{Oz}$ and a constant for every real but without predicates for Ord, this corollary follows already, and more sharply, from `opa:sc:cor:innerbound` of `surreal/omnific-preserving-automorphisms`: an element definable over a set A there is supported in the rational span of the exponents of the members of A , and for $A \subseteq \mathbb{R} \cup \text{Ord}$ that span is a finite-dimensional rational subspace of $\Gamma_{\mathbb{Q}}$. Source 02 does not cite it. The versions with a predicate for Ord, which elementary shears need not preserve, are new here.

Proposition 26.7. *The class $\mathcal{F}_{\Gamma}$ is a real closed subfield of $\mathbf{No}$. Its intersection $I_{\Gamma} = \mathcal{F}_{\Gamma} \cap \mathbf{Oz}$ is an integer part, and $\mathcal{F}_{\Gamma} = \text{Frac}(I_{\Gamma})$.*

Proof. It is a subfield by its fixed-field characterization. Every automorphism fixing the coefficients of a polynomial preserves its finite ordered list of real roots, hence each individual root. Positive square roots and roots of odd-degree polynomials exist in $\mathbf{No}$, and therefore belong to the fixed field. This gives real closedness.

The omnific floor is preserved by every order automorphism preserving $\mathbf{Oz}$, since it is the unique $z \in \mathbf{Oz}$ with $z \leq x < z + 1$. The strong $\mathbf{Oz}$ -preserving fixed-field characterization gives the integer-part assertion.

For $0 \neq x \in \mathcal{F}_{\Gamma}$, choose an ordinal δ with $\delta + g > 0$ for every $g \in \text{supp}(x)$. Such an ordinal exists because the support is a set and ordinals are cofinal in $\mathbf{No}$. Then $b = \omega^{\delta}$ is an ordinal, and $a = bx$ has only positive exponents. Thus $a, b \in I_{\Gamma}$ and $x = a/b$. $\square$

26.5 The smaller field obtained from pure-field definitions

Write $F_{\text{alg}} = \text{rcl}_{\mathbf{No}}(\mathbb{R}(\text{Ord}))$. Quantifier elimination for real closed fields gives the classical identity

$$\text{dcl}_{(\mathbf{No}; +, \cdot, <)}(\mathbb{R} \cup \text{Ord}) = F_{\text{alg}},$$

already recorded in the definability report (Part I). It uses finite parameters even when the permitted parameter class is proper. The next calculation makes its growth restrictions exact.

For $x \neq 0$, write $\text{lead}(x) = \max \text{supp}(x)$ for its leading exponent. One has $\text{lead}(xy) = \text{lead}(x) + \text{lead}(y)$, and a finite sum with a unique largest leading exponent is nonzero.

Proposition 26.8 (Exact leading-exponent group). *One has*

$$\text{lead}(\mathbb{R}(\text{Ord})^{\times}) = \Gamma_{\mathbb{Z}}, \quad \text{lead}(F_{\text{alg}}^{\times}) = \Gamma_{\mathbb{Q}}.$$

Every positive infinite $x \in F_{\text{alg}}$ therefore satisfies $x > \omega^{1/n}$ for some positive integer n .

Proof. The ring $\mathbb{R}[\text{Ord}]$ consists of finite normal forms with real coefficients and ordinal exponents. Finite products add ordinal exponents by natural addition, which still gives ordinals; conversely each ω^{α} , $\alpha \in \text{Ord}$, is a generator. A nonzero quotient in $\mathbb{R}(\text{Ord})$ consequently has leading exponent $\alpha - \beta \in \Gamma_{\mathbb{Z}}$. Every member of $\Gamma_{\mathbb{Z}}$ is a difference of two ordinals, after collecting positive and negative summands, and $\omega^{\alpha-\beta} = \omega^{\alpha}/\omega^{\beta} \in \mathbb{R}(\text{Ord})$.

For $0 \neq x \in F_{\text{alg}}$, choose a polynomial relation $\sum_i a_i x^i = 0$ over $\mathbb{R}(\text{Ord})$, with nonzero displayed coefficients. The largest of $\text{lead}(a_i) + i \text{lead}(x)$ must occur at least twice. For suitable $i \neq j$,

$$\text{lead}(x) = \frac{\text{lead}(a_j) - \text{lead}(a_i)}{i - j} \in \Gamma_{\mathbb{Q}}.$$

Conversely, for $g \in \Gamma_{\mathbb{Q}}$, choose a positive integer n with $ng \in \Gamma_{\mathbb{Z}}$. The positive monomial ω^g is an n -th root of $\omega^{ng} \in \mathbb{R}(\text{Ord})$, so belongs to F_{alg} .

For positive infinite x , one has $\text{lead}(x) > 0$. By Lemma 26.1 this exponent is either a positive rational or positive infinite. Choose n with $1/n < \text{lead}(x)$. The quotient $x/\omega^{1/n}$ has positive leading exponent and positive leading coefficient, so is infinite. $\square$

Proposition 26.9 (A strict distinction). *The inclusion $F_{\text{alg}} \subseteq \mathcal{F}_{\Gamma}$ is strict. An explicit witness is*

$$\mathcal{E} = \sum_{n=0}^{\infty} \frac{\omega^{-n}}{n!}, \quad (26.5)$$

which lies in $\mathcal{F}_{\Gamma}$ and is transcendental over $\mathbb{R}(\text{Ord})$.

Proof. The inclusion follows because $\mathcal{F}_{\Gamma}$ is real closed and contains $\mathbb{R}(\text{Ord})$. The displayed series is summable and supported on negative integers, so belongs to $\mathcal{F}_{\Gamma}$.

If it were algebraic over $\mathbb{R}(\text{Ord})$, clear denominators of a nonzero polynomial relation to obtain coefficients in $\mathbb{R}[\text{Ord}]$. Their combined support is a finite subset of $\Gamma_{\mathbb{Z}}$. Partition it into cosets of $\mathbb{Z}$ in $\Gamma_{\mathbb{Z}}$, with representatives h_c , to write

$$P(Y) = \sum_c \omega^{h_c} P_c(\omega, Y), \quad P_c(T, Y) \in \mathbb{R}[T, T^{-1}, Y],$$

where at least one $P_c \neq 0$. After substituting $\mathcal{E}$, each term has support in $h_c + \mathbb{Z}$. Disjointness of the cosets forces $P_c(\omega, \mathcal{E}) = 0$ for every c . A nonzero Laurent-polynomial coefficient stays nonzero upon substituting ω , by uniqueness of finite normal forms. Thus one nonzero P_c gives an algebraic relation over $\mathbb{R}(\omega)$.

Put $t = \omega^{-1}$. The integer-exponent subfield is the usual Laurent-series field $\mathbb{R}((t))$, and $\mathcal{E}$ becomes $f(t) = \sum_{n \geq 0} t^n/n!$, satisfying $f' = f$. For completeness, f is transcendental over $\mathbb{R}(t)$. If it had monic minimal polynomial

$$Q(t, Y) = Y^d + \sum_{i < d} a_i(t) Y^i, \quad a_i(t) \in \mathbb{R}(t),$$

differentiation would give $(\partial_t Q + Y \partial_Y Q)(t, f(t)) = 0$. Minimality makes Q divide $\partial_t Q + Y \partial_Y Q$. The latter has degree d and leading coefficient d , so equals dQ . Hence $a'_i = (d-i)a_i$ for $i < d$. A nonzero rational function $a = p/q$ cannot satisfy $a' = ca$ for a nonzero constant c : the equation $p'q - pq' = cpq$ has left-hand degree at most $\deg p + \deg q - 1$ and right-hand degree $\deg p + \deg q$. Thus every $a_i = 0$, giving $Q = Y^d$, impossible at $f \neq 0$. $\square$

This does not assert that $\mathcal{E}$ is undefinable after adding the omnific predicate. It separates pure-field definable closure from the strong fixed field, which are distinct invariants.

26.6 An independent family in one omnific interval

Theorem 26.10 (Bounded ordinal-definable independence). *For $0 < \alpha \in \text{Ord}$, put*

$$m_{\alpha} = \omega^{-\alpha}, \quad y_{\alpha} = \omega^{m_{\alpha}}, \quad z_{\alpha} = \omega + y_{\alpha}.$$

The families (y_α) and (z_α) are algebraically independent over $\mathcal{F}_\Gamma$. Every z_α is an ordinal-definable omnific integer with

$$\omega < z_\alpha < \omega + \omega^{1/2}, \quad \text{ct}(z_\alpha) = 0.$$

None is intrinsically definable in $\mathfrak{S}$ from real and ordinal parameters, even if every such parameter is permitted. These assertions hold in every universe of ZFC, including $V = \text{HOD}$.

Proof. Every m_α is a positive infinitesimal. For distinct $\alpha_1, \dots, \alpha_m$ and integers k_j , not all zero,

$$\sum_{j=1}^m k_j m_{\alpha_j} \neq 0$$

by uniqueness of normal form. Its exponents are negative, so it is infinitesimal and lies outside $\Gamma_\mathbb{Q}$.

For a nonzero polynomial in finitely many variables, combine equal monomials to write

$$P(Y_1, \dots, Y_m) = \sum_{\nu \in F} a_\nu Y^\nu, \quad 0 \neq a_\nu \in \mathcal{F}_\Gamma.$$

After substituting $Y_j = y_{\alpha_j}$, the term at ν has support in

$$\Gamma_\mathbb{Q} + \sum_{j=1}^m \nu_j m_{\alpha_j}.$$

Distinct multi-indices give disjoint cosets. No coefficient of one nonzero term can cancel with another term, so the sum is nonzero. This applies the repository's disjoint-support method (Part I) to the newly identified field $\mathcal{F}_\Gamma$. Since $\omega \in \mathcal{F}_\Gamma$, translating each variable by ω is an invertible polynomial change of variables, so the z_α are independent as well.

The inequalities $0 < m_\alpha < 1/2$ give $1 < y_\alpha < \omega^{1/2}$. The normal form of z_α has exactly the two positive exponents $1, m_\alpha$, coefficients one, and zero constant term. This proves the omnific and interval assertions.

Canonical arithmetic and Ω are parameter-free definable in set theory, so z_α is ambiently definable from α , and z_1 is ambiently parameter-free definable. But $m_\alpha \notin \Gamma_\mathbb{Q}$, so neither y_α nor z_α lies in $\mathcal{F}_\Gamma$. The intrinsic nondefinability follows from Corollary 26.6.

Only finite subfamilies and set-sized normal forms were used. Ordinal-indexed independence means that each finite subfamily is independent, not that a proper-class polynomial or sum is formed. $\square$

The first omitted monomial has a familiar analytic name:

$$y_1 = \omega^{1/\omega} = \log \omega$$

for the canonical Gonshor logarithm. This identity is classical: van den Dries–Ehrlich, proof of Lemma 5.3, p. 186 [3]. Thus $\log \omega$ is an ambiently parameter-free definable omnific integer, but is not definable in the field with omnific, real, and ordinal predicates from any finite tuple of real and ordinal parameters.

Neither the canonical omega map nor the canonical exponential is definable in this reduct from such parameters. A definition of the omega map would define its value at $1/\omega$; a definition of the exponential would define the unique solution of $\exp(x) = \omega$. Both would define y_1 , a contradiction. These statements concern the specified reduct, not structures where either function has already been named.

[write] Antecedent, 4 October 2026. Without the predicate for Ord, these nondefinability statements also follow from the omnific-preserving automorphisms report: $\omega^{1/\omega}$ has the exponent ω^{-1} outside the

rational span of the exponents of any tuple from $\mathbb{R} \cup \text{Ord}$, so `opa:sc:cor:innerbound` excludes it; and `opa:thm:parameters` and `opa:sc:thm:undef` (ii), (iv) show that the graph of $x \mapsto \omega^x$ and the Gonshor exponential are not definable over any set of parameters in the field with **Oz** and all real constants. Source 02 does not cite them; with the predicate for *Ord* the statements are new.

26.7 Naming the omega map

In the ordered-field language expanded by Ω , $\omega = \Omega(1)$ is a closed term, and

$$y_1 = \Omega(1/\Omega(1))$$

is intrinsically parameter-free definable. Every z_α is definable from α .

Proposition 26.11 (Intrinsic bounded coding). *In the ordered-field language with Ω , the function*

$$q(x) = \Omega\left(\frac{1}{2}\left(1 + \frac{x}{\sqrt{1+x^2}}\right)\right)$$

*is a parameter-free definable bijection from **No** onto*

$$\mathcal{Q} = \{\Omega(t) : 0 < t < 1\} \subseteq \mathbf{Oz} \cap (1, \omega),$$

with parameter-free definable inverse. For every parameter family A and every x ,

$$\text{dcl}_\Omega(A \cup \{x\}) = \text{dcl}_\Omega(A \cup \{q(x)\}).$$

Proof. In every real closed field,

$$t = \frac{1}{2}\left(1 + \frac{x}{\sqrt{1+x^2}}\right)$$

is a bijection onto $(0, 1)$, with inverse

$$x = \frac{2t - 1}{2\sqrt{t(1-t)}}.$$

Positive square roots are uniquely definable, and direct substitution verifies the inverse identities. The omega map is increasing and injective, so q has the stated inverse: recover the unique t with $\Omega(t) = q(x)$, then apply the algebraic formula. For $0 < t < 1$, the monomial ω^t is an omnific integer between 1 and ω . Mutual definability gives the closure identity. $\square$

The monomial code already occurs as an ambient code in the repository (Part I). This proposition answers a specific part of its question about intermediate languages: the code and its inverse become intrinsic when the omega map is named. It does not classify the entire definable closure in that expansion.

[write] 4 October 2026. With **Oz** named as well, Part III goes much further in this language: every coefficient and truncation is definable (theorem 36.3), a bounded lexicographic pairing is definable (lemma 36.5), and $(V, \in)$ has a parameter-free quotient interpretation with representatives in $\mathcal{Q}$ (theorem 37.1). The definable closure is still not classified there.

26.8 Remaining stabilizer and language questions

Let F_{all} be the common fixed field of all $\mathbf{Oz}$ -preserving field automorphisms fixing $\mathbb{R} \cup \text{Ord}$, with no strongness assumption, and put $D_{\mathbf{Oz}} = \text{dcl}_{(\mathbf{No}; +, \cdot, <, \mathbf{Oz})}(\mathbb{R} \cup \text{Ord})$. The proven bounds are

$$F_{\text{alg}} \subseteq D_{\mathbf{Oz}} \subseteq F_{\text{all}} \subseteq \mathcal{F}_{\Gamma}, \quad F_{\text{alg}} \subsetneq \mathcal{F}_{\Gamma}. \quad (26.6)$$

They follow respectively from pure-field definitions, automorphism invariance, and the inclusion of the explicit strong automorphisms among all automorphisms considered.

The unresolved part of the repository's stabilizer question is to identify F_{all} , and independently to identify $D_{\mathbf{Oz}}$. Definable closure need not automatically equal an automorphism fixed field in a proper-class structure. The series $\mathcal{E}$ in (26.5) is a concrete test: every strong automorphism fixes it, while pure-field definability has been ruled out. Can an $\mathbf{Oz}$ -preserving automorphism fixing all real and ordinal parameters move it?

Further questions include intrinsic definable closure in the omega-map expansion, comparisons between real constants and a predicate for the real subclass, and versions without set choice. The irrational-coefficient shear uses a $\mathbb{Q}$ -linear functional on $\mathbb{R}$, so eliminating choice requires a separate argument.

Proposition 26.12 (The fixed field need not consist of definable elements). *If $V \neq \text{HOD}_{\mathbb{R}}$, then $\mathcal{F}_{\Gamma}$ contains an element outside ROD. In particular the bound on intrinsic definable closure by $\mathcal{F}_{\Gamma}$ is strict in such a universe.*

Proof. The global coding theorem yields an ordinal κ and $A \subseteq \kappa$ with $A \notin \text{ROD}$. Form the normal form

$$X_A = \sum_{\alpha < \kappa} (1 + \chi_A(\alpha)) \omega^{-\alpha}.$$

Its support is reverse well ordered, and $-\alpha \in \Gamma_{\mathbb{Q}}$ for every ordinal α . Thus $X_A \in \mathcal{F}_{\Gamma}$. Its support order type recovers κ and its coefficients recover A . If X_A were ROD, so would be A . Finally every element intrinsically definable from finite real and ordinal parameters in the stated predicate languages is ambiently ROD: translate the defining formula into set theory using the definable carriers and predicates. Hence X_A is outside those intrinsic closures. $\square$

27 Arithmetic consequences and distinctions that remain

27.1 The omnific ring remembers the parameter hierarchy

The equality $K_{\mathbb{R}} = \text{Frac}(I_{\mathbb{R}})$ does not mean that omitting nondefinable surreals is an exclusively field-theoretic phenomenon. The entire real-parameter inclusion hierarchy is already visible in the omnific rings I_r . Likewise the countable closure criterion can be tested using only positive, purely infinite omnific summands. Neither fractional parts nor arbitrary real coefficients in the test series are necessary to detect the missing countable ordinal information.

There are nevertheless many arithmetic invariants that cannot detect this hierarchy. We record elementary instances, already established in the fixed-parameter predecessor and now applicable to $I_{\mathbb{R}}$ (Part I).

Proposition 27.1 (Constant term and finite arithmetic). *Let $\Pi_{\mathbb{R}}$ be the purely infinite part of $I_{\mathbb{R}}$. Then*

$$I_{\mathbb{R}} = \Pi_{\mathbb{R}} \oplus \mathbb{Z}, \quad \text{ct} : I_{\mathbb{R}} \longrightarrow \mathbb{Z}$$

is a split unital ring retraction, and $\Pi_{\mathbb{R}}$ is divisible as an additive group. For every positive ordinary integer n ,

$$nI_{\mathbb{R}} = \Pi_{\mathbb{R}} \oplus n\mathbb{Z}, \quad I_{\mathbb{R}}/nI_{\mathbb{R}} \cong \mathbb{Z}/n\mathbb{Z}.$$

Every unital ring homomorphism from $I_{\mathbb{R}}$ to a finite ring factors through ct . A finite system of polynomial equations with ordinary integer coefficients has a solution in $I_{\mathbb{R}}$ if and only if it has a solution in $\mathbb{Z}$.

Proof. Normal-form extraction preserves ROD, giving the decomposition. Products of positive-exponent terms still have positive exponents, so constant term is a ring homomorphism. Dividing a purely infinite omnific integer by n preserves its positive exponent support and its definability. This proves divisibility and the displayed residue-ring formula. The latter can be formalized using the ordinary integer representatives $0, \dots, n-1$, without treating a proper class as a set-valued equivalence class.

If the additive group of a finite target ring has exponent m , any $p \in \Pi_{\mathbb{R}}$ has the form mq and maps to zero. Thus the homomorphism factors through ct . Applying the retraction coordinatewise to a polynomial solution proves the forward implication in the last assertion; ordinary integers supply the reverse implication. $\square$

Nonvanishing constraints change this conclusion. For example, $X^2 = 2Y^2$ has the nonzero solution $(X, Y) = (\sqrt{2}\omega, \omega)$ in $I_{\mathbb{R}}$, but no nonzero ordinary integer solution. Also $I_{\mathbb{R}}$ is not Noetherian, since

$$(\omega) \subsetneq (\omega^{1/2}) \subsetneq (\omega^{1/3}) \subsetneq \dots$$

The inclusion follows by monomial division; the reverse inclusion would require a nonzero infinitesimal to be omnific. These examples show why agreement of finite quotients with $\mathbb{Z}$ should not be confused with ordinary integer arithmetic in all logical languages.

27.2 A real parameter can encode long countable signs

Every countable-birthday surreal is definable from one real alone, in every ambient ZFC universe. A real can encode a well-order on a subset of ω and a sign on each of its elements. The well-founded collapse supplies the actual ordinal domain, not just a numerical approximation to it. This gives an exact definition of the surreal.

This statement does not say that every ordinal sequence is real-only definable. A countable sequence of ordinals may use ordinals far above ω_1 ; its entries need not have countable codes. The principle A_{ω} concerns exactly this extra information. Prikry forcing exploits that difference: it changes a high ordinal's cofinality without adding a real.

27.3 Initiality and order topology

Proposition 27.2 (Topological behavior of the varying-real field). *If $K_{\mathbb{R}} \neq \mathbf{No}$, then $K_{\mathbb{R}}$ is closed and nowhere dense in the surreal order topology, understood through intervals and definable classes. It can nevertheless fill every external countable cut.*

Proof. If $x \notin K_{\mathbb{R}}$, let x^- and x^+ be the signs of x extended by one minus or one plus. Every $y \in (x^-, x^+)$ extends x as a prefix. If such a y belonged to $K_{\mathbb{R}}$, its prefix of length $\text{b}(x)$ would belong to $K_{\mathbb{R}}$, using the same real and that ordinal. This is impossible. Thus the complement is open.

No proper subfield of an ordered field contains an open interval. Indeed, if it did, choose $c < d$ in such an interval and in the subfield. For any positive u of the ambient field, $c + (d - c)/(u + 1)$ lies strictly between c and d , so belongs to the subfield; solving for u puts u there as well. Negation gives the whole field. Hence $K_{\mathbb{R}}$ has empty interior. The final assertion is realized by [theorems 24.1](#) and [24.4](#). $\square$

The prefix-gap mechanism is an antecedent of the repository and of Hamkins’s analysis of ordinal-definable dense maps [13, 21]. Its application here emphasizes that countable interpolation is much weaker than ambient density. The omitted gaps can require uncountable information to specify from inside the field.

27.4 Examples by language

The following table separates representative notions. The negative claims about the enriched pair allow all real and ordinal parameters, but do not add the omega map or simplicity.

Object or class	Ambient set-theoretic information	Intrinsic comparison
ω	Parameter-free	Not definable from real parameters in the pure ordered field
$\omega^{1/\omega} = \log \omega$	Parameter-free, purely infinite omnific	Not definable in the field/omnific/real/ordinal predicate language even from all real and ordinal constants
$\omega + \omega^{\omega^{-\alpha}}, 0 < \alpha \in \text{Ord}$	Ordinal-definable from α	Outside the computed strong fixed field, hence outside the same intrinsic definable closure
A countable-birthday sign sequence	Definable from one coding real	This does not imply field definability from that real
Z_f from an ordinal sequence f	Exactly the information of f	Each individual summand is OD; the whole sum need not be ROD
All of $K_{\mathbb{R}}$	Varying real plus finite ordinals	One real suffices exactly when $\text{HOD}_{\mathbb{R}}$ satisfies Choice

27.5 Consequences and common incorrect inferences [05]

Seven conclusions about definable omnific integers. The results can now be read directly as a theory of definable integers, not merely as a theory of surreal fields with an extra predicate. First, for each policy the integer part determines the field by fractions with a denominator chosen from the same parameters. Second, the real-parameter spectrum of an arbitrary surreal can be transported intact to a single omnific monomial below ω . Third, a countable sequence of ordinals can be transported intact to a coefficient-1 omnific normal form of length ω .

Fourth, the countable-summation behavior of $I_{\mathbb{R}}$ detects exactly the same assembly principle as the entire field $K_{\mathbb{R}}$. Fifth, replacing finite ordinal information by one countable ordinal parameter sequence gives I_{co} , a countably sum-closed integer part of the canonical ambient-stable completion. Sixth, in an uncountable Cohen product there are non-ROD omnific integers below ω , although every countable Hahn sum of ROD omnific integers is ROD. Finally, in the Prikry example even a countable Hahn sum of OD omnific monomials can fail to be ROD.

The last two conclusions are compatible. In the Cohen model, a missing bounded integer cannot have a Hahn-summable countable presentation whose terms all lie in $K_{\mathbb{R}}$. In the Prikry model, the formula (25.2) gives exactly such a presentation.

Definable entries, definable support, and definable graphs. These are different kinds of information. If a whole normal form is ROD, every indexed exponent and coefficient is ROD because its ordinal index may be used as an additional parameter. Conversely, even when all coefficients are fixed to 1 and every exponent is OD, the support may fail to be ROD: (25.2) proves this.

The separated bands make the support and the ordinal sequence interdefinable. Since the coefficients are constant, a ROD support would give a ROD normal form and hence a ROD sum. In the negative example the support itself therefore cannot be ROD. This pinpoints the failure instead of attributing it to an unspecified difficulty with infinite series.

A definable *set* of possible coefficients or possible exponents is likewise insufficient. One needs the actual assignment, including which values occur and where. A collection of individually available objects is not automatically an available sequence of those objects.

All reals and real closedness do not give all surreals. The Cohen model has

$$\mathbb{R} \subseteq K_{\mathbb{R}} \subsetneq \mathbf{No}, \quad K_{\mathbb{R}} \text{ initial, real closed, and countably Hahn-sum closed.}$$

It also contains every ordinal. Thus neither possessing every real and every ordinal, nor adjoining the natural countable summation operation, forces equality with the full surreal class. Information distributed across uncountably many generic coordinates is still absent. Every element of $\mathbf{No} \setminus K_{\mathbb{R}}$ is transcendental over $K_{\mathbb{R}}$, since a real closed field has no proper ordered algebraic extension. This is an ordinary algebraic observation; no claim about a proper-class transience basis is needed.

Why there is no definability paradox. Nothing here forms a set of all surreal numbers, a set of all classes, or a uniform truth predicate for V . The code Q maps a proper class into a bounded *proper-class* interval, so there is no conflict with cardinal arithmetic for sets. The code $\mathcal{Z}(f)$ sums only an ω -indexed set of terms. The assembly theorem manipulates rank-local certificates, not a list of all true global formulas. The completion theorem uses a scheme of fixed definable maps, not an operation evaluating arbitrary global formula codes.

The internal definitions of ROD and COD legitimately absorb rank ordinals into their permitted parameters. The parameter-free and real-only notions are instead treated externally or as schemes. This is why the least missing ROD birthday is meaningful while an unqualified “least parameter-free undefinable surreal” construction is not justified by the same reasoning.

Source 05’s conclusion. Definability is not an intrinsic adjective attached to a surreal independently of its language, parameters, and universe. With those choices fixed, the algebra is stable: ambient definable surreals form real closed fields, and the corresponding definable omnific integers recover those fields by fractions. Ordinal parameters make the hereditary sign-code principle available; arbitrary real parameters produce a countably directed union of the fixed-real fields.

The central additional phenomenon is assembly. Real parameters absorb countable real data, but not automatically countable ordinal data. The explicit code $\mathcal{Z}(f)$ translates that distinction into a countable coefficient-1 omnific normal form in $(0, \omega)$. It gives an exact criterion for countable Hahn-sum closure, rather than a vague warning about nonuniform definitions. Allowing a countable ordinal sequence as a parameter yields the corresponding minimal ambient-stable countably summable extension.

The Cohen and Prikry models then separate what the definitions can do. A field can be initial, real closed, contain all reals and all ordinals, and be countably Hahn-sum closed, while still omitting surreals as early as birthday ω_1 . Conversely, without adding any new reals, a countable sum of ordinal-definable omnific monomials can fail to be real-ordinal definable. Both conclusions are obtained with explicit parameter accounting and without a proper-class sum or a truth predicate for the universe.

28 A precise agenda for further research

The questions below distinguish what has been resolved from what is still missing. They are proposed problems arising from this paper and the cited repository, not claims that every formulation is new to the literature.

28.1 The remaining stabilizer and language questions

Question 28.1 (Remove the strong-sum hypothesis). Compute the common fixed field of *all* ordered-field automorphisms fixing $\mathbb{R} \cup \text{Ord}$ pointwise, both with and without preservation of $\mathbf{Oz}$.

The theorem in [section 26](#) computes the strong case exactly. Without strong summation the lower inclusion for an arbitrary normal form is unavailable. The relative real closure of $\mathbb{R}(\text{Ord})$ is still fixed, while the full common fixed field is contained in the strong one. Determine whether either bound is exact, or whether a proper intermediate field occurs. A proof must control automorphisms that do not commute with an infinite Hahn sum; the coefficient-by-coefficient argument cannot simply be reused.

Question 28.2 (Definable closure versus fixed elements). Determine the exact intrinsic definable closure of $\mathbb{R} \cup \text{Ord}$ in $(\mathbf{No}; +, -, \cdot, <, 0, 1, \mathbf{Oz})$ and in the expansion also naming the predicates $\mathbb{R}$ and Ord .

The strong fixed field is an upper bound, not an identification of definable closure. In a universe with $V \neq \text{HOD}_{\mathbb{R}}$, it contains elements outside ROD , so it is strictly larger than these intrinsic closures. Even when $V = \text{HOD}_{\mathbb{R}}$, invariance under automorphisms need not supply a finite isolating formula in this proper-class structure. The parameter-free real trace of the omnific expansion is a separate remaining question from (Part I).

Question 28.3 (Name the omega map). Classify definable closure in the omega-map expansion, and compare it with the exponential and simplicity expansions.

Naming Ω immediately defines $\omega = \Omega(1)$ and $\omega^{1/\omega} = \Omega(\Omega(-1))$. It also makes bounded monomial coding intrinsic. The previous automorphisms do not preserve this function and therefore give no nondefinability theorem in that larger language. Determine which additional set-theoretic coding is actually expressible, rather than inferring it from the ambient graph of the map.

28.2 The real-parameter hierarchy and countable information

Question 28.4 (Forcing preservation of countable coherence). Characterize forcing notions that preserve A_ω , and determine its behavior under iterations and under forcing that makes new objects ordinal definable.

The Namba example in [theorem 25.6](#) already proves failure without a large-cardinal assumption, so the basic independence question is settled here. The Cohen and Namba examples also show that A_ω and Choice in $\text{HOD}_{\mathbb{R}}$ imply neither one another. What remains is a structural preservation theorem: the property concerns definability in the final universe, not merely whether ordinal sequences were added. Coding forcing can change that definability, so distributivity alone is not a complete account of its behavior.

Question 28.5 (Higher cardinal versions). For an uncountable regular cardinal κ , identify the correct parameter class and coding principle corresponding to Hahn sums of size less than κ and to interpolation of cuts of size less than κ .

At the countable level, countably many reals can be joined into one real. That is the special fact powering the theorem. For uncountable families it fails, so merely replacing ω by κ in the statement is unjustified. Parameters from $\mathbb{R}^{<\kappa}$ or $\text{Ord}^{<\kappa}$ are natural alternatives, and their hereditary models should be compared explicitly.

Question 28.6 (Cofinality and order types of real-parameter fields). Which inclusion preorders can the fields K_r realize? Which uncountable cardinals can be their cofinality? How do these invariants interact with A_ω and with Choice or Dependent Choice in $\text{HOD}_\mathbb{R}$?

The Cohen computation gives the precise ground countable-support poset, and in particular cofinality λ for regular uncountable λ over L . More complicated forcing may produce degree patterns that are not coordinate-inclusion orders. Finite joins are known exactly; countable least upper bounds require separate analysis.

Question 28.7 (Possible first missing birthdays). If $K_\mathbb{R} \neq \mathbf{No}$, which ordinals can be the first birthday outside $K_\mathbb{R}$, and what restrictions follow from A_ω ?

The lower bound is always ω_1 , and the Cohen model attains it. For a general transitive inner class, the first missing sign sequence is equivalent to the first missing subset of an ordinal. Here the relevant class can fail Choice, so arguments based on an internal cardinal well-order must be reconsidered rather than imported without checking. The interaction with cofinality-changing forcing is a particularly concrete target.

Question 28.8 (Uniformity and complexity). Find sharp syntactic complexity bounds for the maps from definitions of ordinal sequences to definitions of their omnific codes, and for the reverse normal-form decoding.

The proof gives fixed formula substitutions but does not bound their optimal position in the Levy hierarchy. It also does not identify semantic definability with computability. A useful formalized result would be an explicit transformation on formula codes with a proved parameter and complexity budget.

28.3 Normal forms, parameter recovery, and integer arithmetic

Question 28.9 (Minimal parameter fields for individual numbers). For which $x \in K_\mathbb{R}$ does the collection $\{K_r : x \in K_r\}$ have a least member? Can its failure be witnessed by a purely infinite omnific integer in a fixed bounded interval?

The finite-join theorem does not give such a least field: it provides upper bounds for parameters already selected. Reversible omnific codes transfer any answer between general surreals and omnific integers, so the integer restriction does not remove the issue.

Question 28.10 (Exact birthday cost of countable ordinal codes). Estimate sharply $b(Z_f)$ from information about f , and compare the rational-slot code with nested-monomial codes.

The construction controls support order type and numerical size, not the birthday of its output. These are very different invariants in $\mathbf{No}$. Any optimality claim must specify the permitted coding maps and parameter policy.

Question 28.11 (Arithmetic recovery of countable coherence). Can A_ω be characterized by a natural first-order or infinitary property of the ring $I_\mathbb{R}$ alone, without explicitly naming canonical normal forms or the omega map?

The current characterization uses external countable families and Hahn summation. Finite residue rings and finite integer-coefficient equations cannot distinguish the relevant models by [proposition 27.1](#). A successful intrinsic characterization would need additional arithmetic structure or a controlled infinitary language.

[write] **Status of source 02’s questions, 4 October 2026, batch 93.** All questions above stay open as printed, with these re-scopings. [Question 28.1](#) is the nonstrong remainder of `sse:sr:q:stabilizers`. The question on naming the omega map continues [question 17.1](#) of Part I; Part III answers its “which additional set-theoretic coding is actually expressible” part for the omega map together with **Oz**, at the level of an interpretation ([theorem 37.1](#)), while the classification of definable closure stays open there (its [question 42.2](#)) and in the language with the omega map alone. The question on uniformity and complexity meets [question 17.5](#); Part III gives Δ_1 graphs for the canonical operations and level preservation for the bounded code ([theorem 34.7](#) and [corollary 40.3](#)), not for the codes Z_f . The question on the birthday cost of countable codes refines [question 17.2](#).

28.4 Source 05’s questions [05]

Source 05 proposes ten problems. Except where an earlier source is explicitly named, they are not asserted to be previously published open problems, and none is an unstated hypothesis of the proved theorems. Two are answered by source 02 and are printed with their answers; four are source 02’s questions above and are printed once, with source 05’s discussion here; four are new.

Question 28.12 (The consistency strength of failed assembly [05]; answered by [theorem 25.6](#)). What is the exact consistency strength of $\neg A_\omega$? Can a transitive model of ZFC alone support a no-new-reals, sufficiently homogeneous extension adding a non-ROD countable ordinal sequence?

The Prikry construction gives a measurable-cardinal upper bound, not a lower bound. A proposed replacement by Namba-type or other cofinality-changing forcing must separately verify no new reals and the relevant homogeneity or nondefinability argument. It is not enough merely to add an ω -sequence of ordinals.

[write] **Answered, 4 October 2026, batch 93.** [Theorem 25.6](#) answers both parts: a transitive model of ZFC alone supports such an extension (Namba forcing over its L), and the consistency strength of $\neg A_\omega$ is exactly that of ZFC; see the note at the end of [section 25.2](#). Source 02 carried out exactly the checks the discussion asks for: no new reals (imported, under CH; see [question 28.17](#)), and cone homogeneity with the resulting nondefinability (proved).

Question 28.13 (Weakening ambient stability in the completion theorem [05]). Which explicitly listed surreal operations, together with countable Hahn sums, generate K_{co} from $K_{\mathbb{R}}$? Is there a useful natural intermediate field obtained using only field operations, the Conway map and its inverse, normal-form extraction, and countable summation?

The current minimality theorem uses all fixed ambient definable maps. Removing that hypothesis is a substantive algebraic/model-theoretic problem, not a cosmetic sharpening. A negative answer should identify an invariant preserved by the restricted operation list.

Source 05’s Question 13.3 (longer parameter sequences and longer sums) is the question on higher cardinal versions above. Source 05 asks: for an infinite cardinal μ , compare definability from a μ -sequence of ordinals with closure under μ -indexed Hahn sums; can one obtain a reversible normal-form code with a prescribed support order type and an exact analogue of [theorem 23.9](#)? The present dyadic-band construction uses disjoint bands indexed by ω . Uncountably many real bands are unavailable. A longer code would need transfinite surreal scales, with careful control of reverse well ordering and of the parameters needed to identify those scales.

Source 05's Question 13.4 (real-parameter spectra and least information) contains the question on minimal parameter fields above. Source 05 asks: which upward-closed subsets of the real-ordinal reducibility preorder occur as $\text{Spec}(x)$ for a surreal x ; when does a nonempty spectrum have a least degree; and which spectra already occur for single monomials in $(0, \omega)$? The bounded code answers the last occurrence question relatively: every spectrum occurring for a surreal also occurs for such a monomial. The remaining task is the classification of spectra themselves, not another coding argument. The Cohen support theorem gives concrete necessary exclusions for coordinate generics.

Question 28.14 (Minimal coordinate supports in Cohen products [05]). Refine [theorem 24.1](#) from existence of a countable ground-model support to a classification of possible minimal supports and intersection behavior. Which surreal elements lie in every W_J from a given family of countable coordinate sets?

A useful first step is to distinguish pairwise intersections of mutually generic subextensions from arbitrary intersections of a countable decreasing family. The distinction should be proved before using it to assert existence of a least support.

Source 05's Question 13.6 (the first omitted real-ordinal birthday) is the question on possible first missing birthdays above, asked also for cofinalities: what values can $\lambda_{\mathbb{R}}$ take, must it be a cardinal of the ambient universe, what cofinalities are possible, and can one prescribe it while independently controlling A_ω ? The Cohen product gives exactly ω_1 . For OD, the repository has a stronger first-birthday analysis through a single ZFC inner model HOD. The union over all reals lacks the same immediate canonical well-order argument, so those conclusions should not be transferred without proof.

Question 28.15 (Global real parameters versus one fixed real [05]; answered negatively by [corollary 22.5](#)). Can $\text{ROD} = V$ hold while $V \neq \text{HOD}(r)$ for every real r ? How do any such examples appear through the bounded omnific test of [proposition 22.9](#) and [theorem 6.8](#)?

The Cohen example proves that a proper $K_{\mathbb{R}}$ need not equal any single K_r . It does not settle the stronger global situation posed here. The quantifiers $\forall x \exists r$ and $\exists r \forall x$ must remain distinct.

[write] **Answered negatively, 4 October 2026, batch 93.** No. Over ZFC, [corollary 22.5](#) (and source 06's [remark 22.7](#)) shows that $\text{ROD} = V$ implies $V = \text{HOD}(r)$ for one real r : by Choice the reals have a well-order, which is $\text{OD}(r)$ for some real r , and then every real, hence every set, is $\text{OD}(r)$. The quantifiers $\forall x \exists r$ and $\exists r \forall x$ are distinct in general (the Cohen example), but they coincide when the x range over all sets. Through the bounded omnific test the answer reads: if every omnific integer of $(0, \omega)$, or even every monomial ω^a with $0 < a < 1$, is ROD, then one real suffices for all of them.

Source 05's Question 13.8 (birthday and syntax cost of the multiplexer) is the question on the birthday cost of countable codes above, together with the question on uniformity and complexity. Source 05 asks for sharp bounds on $b(\mathcal{Z}(f))$ from the birthdays or ordinal sizes of the entries of f , a comparison with alternative reversible countable omnific codes, and effective translations of definition complexity through the encoder and decoder. The construction fixes the support order type and coefficient alphabet, not the exact birthday. This refines the earlier repository question about the birthday cost of bounded coding. Fixed-formula definability preservation does not automatically preserve a specific Levy-hierarchy level or an optimal description length.

Question 28.16 (Real-only definability without ordinal parameters [05]). Study the external first omitted ordinal and the maximal initial fragment of K_{rd} in specified transitive models. How do these depend on the supply of reals and on the model's height?

Every countable-birthday surreal is already in K_{rd} , but the rank-local-certificate argument cannot silently add rank ordinals under this policy. A solution should specify its external truth semantics and not confuse K_{rd} with $K_{\mathbb{R}}$.

Source 05’s Question 13.10 (intrinsic languages between the field and set theory) is the question on naming the omega map above and [question 17.1](#): which parts of the real-parameter hierarchy are captured by the ordered field expanded by the omnific predicate, the Conway map, or simplicity, and can any of the countable-assembly phenomena be detected in a precisely specified set-sized first-order structure? This continues the language question in (Part I). The pure field answer is [theorem 3.4](#), but the present ambient codes use normal forms and set-theoretic reconstruction. They do not solve the intrinsic definability problem merely by existing as ambient functions. See the status note above for what Part III adds.

28.5 Further questions and research added at the write [write]

The following item is recorded under Vladimir’s standing rule of 4 October 2026 that unproved claims are kept as open questions with their source, argument sketch and what is missing.

Question 28.17 (A reference for the Namba input [write]). Give a complete proof, or a textbook reference that states it for the presentation used, that Namba forcing in the Laver-style presentation of [section 25.1](#) (every node above the stem has ω_2 immediate successors) adds no reals over a ground model of CH.

Source 02 imports this from Lietz’s exposition [33], which distinguishes the two tree presentations and states that the classical no-new-reals proof applies to both, and from Miller’s lecture notes [34, Theorem 23 and the table on p. 4]. The standard textbook treatments of Namba forcing under CH are [10, Chapter 28] and [35, Chapter XI]; the latter also shows that the two presentations are not forcing equivalent (Magidor and Shelah, its Claim XI.4.2, as reported in [33]). The collection imports the theorem for the classical presentation from [36, Section 7.6] (`univ:gs:thm:nambainput`). The write consulted neither textbook and did not confirm for which presentation they prove it. The argument for either presentation is a fusion over $\aleph_2$ -splitting trees that uses CH to handle the $\aleph_1$ possible values of a name for a real; what is missing here is a written check of that fusion for trees in which every node above the stem splits. Everything else in [theorem 25.6](#), in particular cone homogeneity and the computation of $\text{HOD}_{\mathbb{R}}$, is proved in the text. Should the input fail for this presentation, the theorem would still hold for any presentation that adds no reals under CH, is cone homogeneous, and adds a cofinal ω -sequence in ω_2^W .

29 Formalization plan and reproducibility ledger

29.1 A safe sequence of formal developments

The following order keeps set-theoretic semantics separate from surreal calculations.

1. Define canonical signs, birthdays, and the translation from sets of ordinals to sign sequences. Prove that all hereditary components of a sign graph except the graph itself are OD.
2. Formalize satisfaction for set structures, rank-local $\text{OD}(r)$, real joining, and substitution of definitions. Prove the inner-model axioms for H_r and $\text{HOD}_{\mathbb{R}}$ with Choice kept as a separate conclusion for the latter.
3. Derive the parameter inclusion and capture theorems. This stage needs no sophisticated surreal arithmetic: real injection and the canonical sign representation are enough.

4. Formalize the rational-slot exponents and their inequalities, then normal-form decoding and the finite-prefix cut argument. These are the central algebraic lemmas of the countable theorem.
5. Formalize additive index relabeling and Hahn lifting, the reverse-well-ordered orbit lemma, and the rational-span description. For the coefficient shear, isolate the ordinary set-sized Hamel-basis choice as an explicit hypothesis or a ZFC construction.
6. Add the Cohen, Namba, and Prikry models through a set-theoretic forcing library. Until that is available, keep their exact forcing inputs as named assumptions and formally prove the conditional surreal consequences.

For an implementation in a type theory with universes, **No** at one universe level is a set or type only from a larger level. Moving to the larger level also changes which parameter families and sequences are available. The external countable closure statements must retain their ambient-universe annotation. Likewise, a finite formula with constants from a proper parameter class is represented by its finite list of constants, not by a set-sized language assumed to contain the entire class at the same level.

29.2 What was checked and what was imported

Item	Status and dependency
Canonical signs, real closure, omega map and normal forms	Classical surreal theory; sources [1, 2, 3].
Internal ordinal definability and real-plus-ordinal parameters	Classical rank-satisfaction framework; [28].
Fixed-parameter closure, floor, denominators and bounded coding	Already in the predecessor (Part I); short proofs included where needed.
Varying-real inclusion and Choice criterion	Proved here by real joining, hereditary coding, and a well-order of the actual real set.
Countable assembly and interpolation equivalence	Proved here with a uniform reversible ordinal-sequence code and explicit prefix inequalities.
Cohen field hierarchy	Standard forcing inputs, with the exact parameter-field computation and closure consequences proved here.
Negative countable-assembly models	Namba gives failure relative to ZFC consistency; Prikry gives the displayed measurable-cardinal example. The exact Namba no-new-reals theorem is imported; cone homogeneity and definability consequences are proved here.
Hahn lifts and support-orbit mechanisms	Antecedents in [24, 25]; rederived for the specific maps used.
Strong stabilizer of all real and ordinal constants	Exact rational-span fixed field proved here; strong version of the repository question resolved.
Full nonstrong stabilizer or full intrinsic definable closure	Not resolved.

Item	Status and dependency
Machine verification	No new theorem is claimed to be checked in Lean or Rocq. The finite code-inequality script is an auxiliary check only.

29.3 Package contents

The supplied source compiles with a standard LaTeX installation using `latexmk -pdf definable_surreals.tex`, or repeated runs of `pdflatex`. The package includes a self-contained TeX source, the compiled PDF, a short build guide, and an exact-arithmetic check of the coding inequalities. There are no downloaded third-party articles in the package. Bibliographic links identify the primary sources and the immutable repository snapshot instead.

[write] **On the package contents, 4 October 2026.** The preceding subsection describes source 02’s delivery. In this report only the checking script and the build script of source 02 are shipped, as `code/08-real-parameters-finite_code_check.py` and `code/08-real-parameters-build.sh`; the latter builds the undelivered `definable_surreals.tex` and is kept only as delivered. The manuscript source and PDF survive in the arrival commit’s archive. See [section 19.5](#) and the README.

29.4 Source 05’s formalization plan and proof ledger [05]

Separate the mathematical interfaces. A realistic formalization should not begin by treating the entire proper class **No** as a small type in its own universe. The first component is a universe-polymorphic sign-sequence representation, with the ordinal domain recorded in a larger universe when necessary. A set-model semantics can alternatively represent exactly the surreal codes belonging to a specified model.

The second component is a normal-form interface: uniqueness of the support/coefficient graph, definable construction of the Conway map, and the formal Hahn summation operation for set-indexed families. The separated-band theorem only needs ordered-field identities, monomial comparison, the normal-form existence theorem, and an ordinal-code decoder. It should be formalized independently of HOD or forcing.

The third component is first-order syntax and semantics for set structures, including satisfaction for V_θ and reflection as an explicitly stated theorem. It must distinguish external standard formulas from any internal nonstandard syntax. The proof of [theorem 23.9](#) then becomes a certificate-coding theorem plus the normal-form code, rather than an appeal to an informal truth oracle.

The fourth component is forcing: ccc countable supports, the cone-homogeneity lemma, and the Prikry property. These can initially be isolated as named hypotheses so that the new transfer theorem is checked without pretending that an existing library already contains a complete development of Prikry forcing.

What has actually been checked here. The accompanying program uses exact rational arithmetic to check the finite specialization of (23.3) and (23.7), including nonmonotone inputs and repetitions. A successful run is useful for catching an incorrect factor of 2 or band endpoint. It is not a formal proof of any statement involving infinite ordinals, real parameters, satisfaction, or generic extensions.

The LaTeX source is compiled to PDF and inspected for layout defects. No Lean source is presented as verified, and no Lean build was run. The repository’s earlier definability report is treated according to its own pending formalization status (Part I).

Table 7: Proof and provenance ledger.

Result	Mathematical input	Status in this manuscript
Parameter taxonomy and certificates	Set satisfaction, reflection, finite parameter coding	Standard foundations, made explicit
Field, floor, fractions, fixed-real HOD	Classical surreal theory and unique-output closure	Earlier repository results; proofs reproduced in a self-contained form
Countable-birthday real coding	Countable well-order codes	Elementary consequence, proved here
Bounded monomial code Q	Real closed field algebra and Ω	Explicitly credited to the repository
Countable code $\mathcal{Z}$	Separated rational bands and normal forms	Proposed tailored construction; full proof
Theorem 23.9	Certificates, Choice, $\mathcal{Z}$	Main proposed equivalence; full proof
Theorem 23.15	Classical countable-ordinal coding and $\mathcal{Z}$	Proposed surreal minimality bridge; closure policy explicit
Theorems 24.1 and 24.9	ccc, product factorization, homogeneity, $L[r]$	Proposed classification/application; detailed proof
Theorem 25.7	Classical normal-measure Prikry property; cone isomorphisms	Proposed omnific application; exact imported premise identified
Finite verification	Python standard library, exact fractions	Executed regression checks only
Historical priority	Selected repository and primary-source search	Not certified

Source 05's build notes (its Appendix B) describe a delivery of its source, its PDF, a README, a separate proof audit and the regression script with its recorded JSON output, built with `latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex` and checked with `python3 code/verify_code.py --output verification.json`. The regression script tests only the finite rational specializations of the countable code; the mathematical proofs, rather than that program, justify its transfinite use. No third-party papers, repository source copies or font files were redistributed, and the delivery modified nothing in the repository. [write] Of these, the script, its recorded output, the audit and the `Makefile` are shipped under the names given in [section 19.5](#); the audit still writes the band exponents $e_n(a)$ (here $b_n(a)$). Rerun only on a copy, as the README explains: the delivered `Makefile` and command line write over the recorded output.

Part III

Definable operations: parameter hierarchies, absolute normal forms, uniform coding, and an interpretation of set theory

Definable Surreals and Omnific Arithmetic

Research report prepared for Vladimir Reshetnikov, 4 October 2026

Abstract. We distinguish definability in the surrounding universe of sets from definability in a specified surreal structure, and develop separate notions using no parameters, fixed parameters, ordinal parameters, real parameters, and combinations of these. The principal construction is a parameter-free first-order interpretation of the set-theoretic universe in $\mathfrak{S}_\Omega = (\mathbf{No}; +, -, \cdot, <, 0, 1, \mathbf{Oz}, \Omega)$, where $\Omega(x) = \omega^x$ is Conway’s omega map. The interpretation uses equivalence classes of codes lying in a definable subclass of the bounded omnific monomial interval $\{\omega^a : 0 < a < 1\}$. Its mechanism is explicit: recover the real coefficients, define normal-form supports, encode all subsets and relations on a support, and collapse pointed well-founded extensional graphs. We also define Hahn summability and summation for uniformly coded set-indexed families in this same first-order language.

Further results concern parameters and operations. Allowing a different real and finitely many ordinals for each surreal yields every surreal exactly when one real r already satisfies $V = \text{HOD}(r)$. A canonical decoder of real codes for countable sign sequences has a parameter-free right inverse exactly when $\omega_1^{\text{HOD}} = \omega_1$, with a corresponding criterion at every infinite cardinal. We prove absoluteness of canonical normal forms between transitive inner models without requiring them to have the same reals, and analyze the syntactic cost of reversible coding. The article builds on, and explicitly separates its extensions from, the existing ProveIt reports. It establishes mathematical claims by written proofs; it does not claim a formal verification, a complete determination of intrinsic definable closure, or a certified historical priority.

[write] **Added 4 October 2026, batch 93.** This Part is manuscript 06 of batch 93 of ProveIt’s incoming-reports intake, *Definable Surreals and Omnific Arithmetic: Parameter hierarchies, absolute normal forms, uniform coding, and an interpretation of set theory*, printed in full after Part II (provenance, notation and the relation to Parts I–II: [sections 30.3 to 30.5](#)). Like the other Parts it is AI-assisted, unrefereed and not formalized. To keep the earlier numbers, its sections continue Part II’s: **the manuscript’s Section k is Section $k + 29$ here, and its statement or equation $k.m$ is $(k + 29).m$** . Thus its Theorem 5.1 is [theorem 34.1](#), its Theorem 8.1 is [theorem 37.1](#) and its Theorem 11.4 is [theorem 40.4](#). Its Theorem 4.2 is the same theorem as source 02’s Corollary 4.5 and is printed once, in Part II ([corollary 22.5](#) with [remark 22.7](#)); its Corollary 4.3 is printed once as source 05’s Theorem 5.2 ([theorem 22.8](#)); both keep their numbers here as pointer remarks. Apart from this, the notation changes of [section 30.4](#), the dated [write] notes and the editorial changes listed in [section 30.3](#), no statement, proof, number or non-claim of the manuscript was changed. “This article” in its text means the manuscript.

30 Scope, conclusions, and relation to the repository

The expression “a definable surreal number” has no unique meaning until three choices have been made: the ambient model, the language, and the allowed parameters. The first choice matters because forcing may add new sign sequences or change which old objects have definitions. The second matters because an ordered field can express much less than its canonical set-theoretic

construction. The third matters because an ordinal can specify a transfinite position, whereas a real can carry an arbitrary countable amount of information.

This distinction becomes especially consequential for omnific integers. An omnific integer is discrete in the numerical order, but it may have a transfinite normal form with arbitrary real coefficients at positive exponents. Discreteness therefore places almost no bound on its information content. A single bounded omnific monomial can already encode an arbitrary surreal when Conway’s omega map is available.

30.1 The existing starting point

The repository snapshot used here is commit `1404038dfb1cad09eb8a42e38b95f193e5fa54ba`, dated 4 October 2026. The exact-topic report (Part I) already proves real closedness and elementarity of ambient definability hulls, same-parameter fraction representations, the identification with HOD, bounded omnific coding, several independence results, and a maximal initial definable core. Its [question 17.1](#) leaves the omega-map expansion open; [question 17.3](#) asks for further support fields, and [question 17.5](#) asks about syntactic costs. The companion normal-form report supplies an inner-model instance under a same-reals hypothesis [26]. The reconstruction and automorphism reports distinguish the omnific predicate from a named omega map [22, 24].

Consequently, elementary closure statements and the bounded unary code are background in the present article. The main additions are the support-based interpretation, its operation calculus, the real-parameter uniformization and section criteria, and the removal of the same-reals restriction from normal-form absoluteness. The syntactic analysis gives explicit preservation statements rather than an assertion that interdefinability automatically preserves every complexity measure.

The novelty audit was targeted, not exhaustive. In particular, Chen–Hamkins–Yang have announced a bi-interpretation for a *birthday-enriched* surreal structure [45]; the general well-founded-graph technique is not new. Camacho’s work shows the logical strength of truncation in Hahn fields [42]. Here the specific signature, the coefficient-recovery route, and the full universe coded through arbitrary reverse-well-ordered supports are the point of the construction. We prove a mutual interpretation, with one composite identified in set theory. We do not infer a bi-interpretation of the omega-map structure from this.

30.2 Principal results in operational form

The following list also indicates how to read the article.

1. **Parameter classification.** Ambient definability hulls form real closed fields and their omnific parts are integer parts with the same fraction field. Ordinal-definable sign sequences are precisely the surreals belonging to HOD. Individual real parameters and the single set $\mathbb{R}$ are different permissions.
2. **Global real consolidation.** If each surreal is definable from some real and ordinals, then one real works for the entire universe: $V = \text{HOD}(r)$. This is stronger than a separate statement about each number, and uses Choice.
3. **Uniform code selection.** Every countable-birthday surreal has a real code. A uniform parameter-free selection of such codes exists exactly when HOD computes ω_1 correctly. The analogous obstruction at κ is correctness of κ^+ .
4. **Absolute operations.** Canonical arithmetic, the omega map, and normal-form evaluation agree in transitive inner models on common inputs. The real coefficients need not exhaust

the ambient reals.

5. **Intrinsic support calculus.** The omnific predicate recovers the canonical real field and the zero-exponent coefficient. Adding Ω recovers every coefficient and every support, and gives a definable pairing compatible with support order.
6. **Set theory in bounded integer codes.** Pointed well-founded extensional graphs have first-order codes in $\mathfrak{S}_\Omega$; their quotient is the ambient $(V, \in)$. Codes can be compressed into omnific monomials strictly between 1 and ω .
7. **Infinite operations with explicit inputs.** The same language defines summability and the Hahn sum of a coded set-indexed family. This does not introduce a proper-class sum or a universal truth evaluator.

30.3 Provenance of this Part [write]

This Part is one manuscript: batch-93 manuscript 06, *Definable Surreals and Omnific Arithmetic: Parameter hierarchies, absolute normal forms, uniform coding, and an interpretation of set theory*, “research report prepared for Vladimir Reshetnikov”, dated 4 October 2026, pinned to commit 1404038dfb1cad09eb8a42e38b95f193e5fa54ba (the batch-90 write of Part VIII of the birthday-cutoff report). It arrived as `definable_surreal_operations.zip` in commit `de37a66d1` and was placed in commit `47a77daba`; its labels carry the prefix `dsn:op:.` At the pin, Part I read as now, except for the batch-90 note before [theorem 8.2](#) and the dated notes of batch 93. It is a separate Part rather than a third member of Part II because its spine is different: intrinsic definability in $\mathfrak{S}_\Omega$, absoluteness without a same-reals hypothesis, and uniform code sections. It shares one theorem with Part II (its Theorem 4.2, source 02’s Corollary 4.5) and one elementary fact (its Corollary 4.3, source 05’s Theorem 5.2); each is printed once, in Part II.

The only delivered file shipped besides the text is `10-definable-operations-PROOF_STATU S.txt`, byte-identical; its theorem numbers are the manuscript’s (add 29 to the section number). The manuscript ships no code, data or checksum manifest; its source, PDF (41 pages) and delivery README are not shipped and survive in the arrival commit’s archive. Its preparation checks (a clean build, rendering of all pages, an independent AI review) are recorded in that file and could not be re-checked at intake.

Editorial changes: citations of the predecessor report became references into Part I, with its numbered questions and sections cited by label; the bibliography was merged into the report’s ([28] is the manuscript’s `DoraisHamkins2019`, [45] its `CHY`, [26], [22], [24] and [27] its `RepoLC`, `RepoDio`, `RepoAut` and `RepoVec`); its Theorem 4.2 with its proof and the paragraph after it, and the statement of its Corollary 4.3, were moved to Part II as described above; its `\sq (<)` for proper sign prefix is written $\prec_s$.

30.4 Notation in this Part [write]

Part I’s symbols keep their meaning, and so do Part II’s ROD , $\text{HOD}_\mathbb{R}$ and $\mathcal{Q}$. The manuscript’s $U = \{\Omega(a) : 0 < a < 1\}$ is written $\mathcal{Q}$, as in Part II, because $\mathcal{U}$ is Part I’s code interval $\mathcal{U}$; and its proper-prefix symbol $<$ is written $\prec_s$. Its $\mathfrak{S}_0$ is Part I’s $\mathfrak{S}$; its τ and Q are Part I’s τ and $\mathcal{Q}$; its $e_\alpha = \omega^{-(\alpha+1)}$ in [corollary 40.2](#) is Part I’s; and for a real p its $H_p = \text{HOD}(p)$ is Part II’s H_τ . The following letters have several local meanings in the manuscript and are kept, because each is introduced where it is used: D_A^M (definability hulls, as in Part I), D and D_κ (the real and κ -code decoders of [section 33](#)), D_b (the derivations of [theorem 39.1](#)) and D for a support set in [section 36](#); $B(a, b)$ and $p(a, b)$ (the pairing of (36.4), not Part I’s B_n nor a set of ordinals p); K_b ,

J_b (the operators of [proposition 39.2](#), not hulls K_A); H_D (a product ring); $T_{>a}$ (truncations, not Part I's T); and U in [section 40](#), a putative universal function in a diagonal argument.

30.5 Relation to Parts I and II and to neighbouring reports [write]

The manuscript credits Part I, the large-cardinal, omnific Diophantine, automorphism and real-vector-space reports. The following antecedents in the collection, checked at the write, are not cited by it.

- Parts (1)–(2) of [theorem 34.1](#) (old cuts, arithmetic and the omega map) are `sse:pf:lem: absolute` of `surreal/surreal-self-embeddings` [25], whose note already observes that the same-reals hypothesis is not needed for them, and items (i)–(iii) of `univ:prop: absolute` [21]; its [corollary 34.5](#) is `univ:prop: absolute` (iv). Parts (3)–(5), absoluteness of complete normal forms and sums without the same-reals hypothesis of `lce:lem: absolute` [26], are new; so is [corollary 34.4](#), which supplies the inner-model instance that [remark 17.6](#) (Part I) and `lce:rem: dsnsupport` record as not covered.
- [Proposition 35.1](#) largely re-proves `odg: def: thm: realrecovery` and `odg: def: cor: arithmetic` of `surreal/omnific-diophantine-geometry` [22] and [proposition 15.3](#) (i) of Part I, with the same dilation S_2 as `odg: def: ex: dilation`; its inclusion $\text{dcl}_{\mathfrak{S}_0}(\mathbb{R}) \subseteq \mathbb{R}$ also follows from `opa: sc: cor: innerbound` [24].
- The quotient interpretation [theorem 37.1](#) is for the signature $(\mathbf{No}; +, \cdot, <, \mathbf{Oz}, \Omega)$; the birthday-cutoff report proves uniform bi-interpretations for the birthday-enriched structures $(\text{hset: thm: biinterpretation}, \text{hset: thm: main-rounding})$ and records the transfer $M \equiv N \iff \mathfrak{B}^M \equiv \mathfrak{B}^N$ (`hset: rem: internal`, `hset: cor: equiv`) and the detectability of an outer universe, $\mathfrak{B}^M \prec \mathfrak{B}^N \iff M = N$ (`hset: thm: outer`) [20]. [Theorem 40.4](#) is the analogue for $\mathfrak{S}_\Omega$; see the note after it.
- [Section 41](#) re-proves for inner models the epsilon lemma of [section 10](#) and the cardinal threshold of `univ: prop: beta` [21] (general $M \subseteq N$), of which [theorem 11.2](#) is the case $M = \text{HOD}$.
- Part II's [proposition 26.11](#) (source 02) is the special case of the bounded code in the omega-map language; Part III's support calculus and interpretation go far beyond it.

No statement of this Part is formalized, and no Lean or Rocq declaration of ProveIt states any of its theorems.

31 Foundations and the objects being defined

31.1 Set theory, models, and class notation

Unless stated otherwise, the background is ZFC. A surreal is represented by a function $s : \alpha \rightarrow \{-, +\}$, where α is an ordinal. Thus every surreal is a set, while $\mathbf{No}$ is a definable proper class. Its birthday is $\text{b}(s) = \alpha$. We use standard set encodings of functions and ordered pairs and fixed finite tags for the two signs. The numerical order is lexicographic, treating an ended sequence as a temporary symbol lying strictly between minus and plus. Proper initial segment is the simplicity relation $s \prec_s t$.

Statements about a fixed formula on a definable class are translated into set theory by restricting its quantifiers to that class. For example, $\mathfrak{S}_\Omega \models \varphi$ means the corresponding translation of this particular formula into the ambient universe. It does not invoke a set-theoretic truth predicate that accepts arbitrary formula codes for V .

When discussing all definitions at once, we may instead fix a set model $M \models \text{ZFC}$ in an external metatheory. Then its satisfaction relation is an ordinary set externally, and its surreal class gives an external set structure $\mathbf{No}^M$. For absoluteness results we explicitly require transitivity. The interpretation theorem itself also has an internal reading in nonstandard models: subsets, well-foundedness, and Mostowski collapse are then those of the model. No assumption that ZFC proves the existence of a transitive set model is made.

No argument forms $\mathcal{P}(\mathbf{No})$, sums over all ordinals, or treats all class functions as a set. A quotient interpretation will be used through a domain formula and an equivalence-relation formula; its equivalence classes need not be sets.

31.2 Normal forms, monomials, and omnific integers

We use the classical Conway–Gonshor normal-form theorem [1, 2, 41]. Every surreal has a unique expression

$$x = \sum_{\xi < \lambda} r_\xi \omega^{a_\xi}, \quad r_\xi \in \mathbb{R} \setminus \{0\}, \quad a_\xi > a_\eta \quad (\xi < \eta < \lambda), \quad (31.1)$$

with a set ordinal λ . Conversely every such family has a surreal value. The support is the reverse-well-ordered set $\text{supp}(x) = \{a_\xi : \xi < \lambda\}$. Here and throughout, “reverse well ordered” means that every nonempty subset has a greatest element in the numerical order. The zero series has empty support.

The map $\Omega(a) = \omega^a$ is an increasing isomorphism from the additive group of $\mathbf{No}$ onto the multiplicative group of Conway monomials. In particular,

$$a < b \implies n\Omega(a) < \Omega(b) \quad \text{for every } n \in \mathbb{N}. \quad (31.2)$$

Its notation must not be confused with the Gonshor exponential $\exp$. In general $\Omega(x) \neq \exp(x \log \omega)$. The distinction between omega maps and ordinary exponentials is also central in [43].

Write $x = x_{>0} + r + x_{<0}$ by restricting its normal form to positive, zero, and negative exponents. The middle term is an ordinary real, possibly zero. The negative part is infinitesimal: its absolute value is less than every positive real. Define

$$\Pi = \{x : \text{supp}(x) \subseteq \mathbf{No}_{>0}\}, \quad \mathbf{Oz} = \Pi \oplus \mathbb{Z}. \quad (31.3)$$

These are the purely infinite additive subgroup and the omnific integer ring, respectively. A negative element of Π is allowed; “positive support” describes exponents, not the sign of the number.

The ring $\mathbf{Oz}$ is discretely ordered, with least positive element 1. If an omnific integer has a nonempty positive support, its leading term has absolute value larger than every ordinary integer; otherwise it is an ordinary integer. This proves the discreteness directly. The coefficients at positive exponents can be arbitrary reals: for example $\pi\omega \in \mathbf{Oz}$, while $\pi \notin \mathbf{Oz}$.

Proposition 31.1 (The omnific floor). *For every $x \in \mathbf{No}$ there is a unique $z \in \mathbf{Oz}$ such that $z \leq x < z + 1$. If $x = p + r + \varepsilon$ is the preceding decomposition, then*

$$\lfloor x \rfloor_{\mathbf{Oz}} = p + \begin{cases} \lfloor r \rfloor - 1, & r \in \mathbb{Z} \text{ and } \varepsilon < 0, \\ \lfloor r \rfloor, & \text{otherwise.} \end{cases}$$

Proof. When $r \notin \mathbb{Z}$, both distances to its adjacent integers are positive reals, so adding an infinitesimal changes neither inequality. When $r \in \mathbb{Z}$, a negative infinitesimal puts the number just below $p + r$, requiring the displayed correction. The remaining cases are immediate. Two different answers would be omnific integers at distance less than 1, contradicting discreteness. $\square$

For instance $\lfloor \omega - \omega^{-1} \rfloor_{\mathbf{Oz}} = \omega - 1$. Simply discarding negative exponents gives the wrong answer in this case. This floor formula and the surrounding facts are already developed in (Part I; [22]).

32 A precise hierarchy of definable numbers

32.1 Ambient parameters and language parameters

Fix externally a model M and an allowed collection $A \subseteq M$. An element $x \in M$ is *ambiently definable from A* if a standard finite first-order formula $\varphi(v, \bar{a})$, using a finite tuple $\bar{a}$ from A , uniquely defines x in $(M, \in)$. Write

$$D_A^M = \text{Def}^M(A), \quad K_A^M = D_A^M \cap \mathbf{No}^M, \quad I_A^M = D_A^M \cap \mathbf{Oz}^M.$$

The entire collection A is not itself a parameter unless explicitly allowed. In particular, permitting every real as an individual parameter differs from naming the single set $\mathbb{R}^M$.

By contrast, for a specified structure $\mathfrak{S}$ on surreals, $\text{dcl}_{\mathfrak{S}}(A)$ consists of unique solutions of formulas in the *language of that structure*. Such a formula has no access to arbitrary membership questions about a surreal's sign representation unless the structure can interpret them. Every intrinsic definition in one of the canonical structures considered here translates to an ambient definition, with the same parameters. The converse can fail radically.

Permission	Notation	Meaning
No parameters	$D_{\emptyset}$	Unique ambient definition using no named input.
A fixed real r	$D_{\{r\}}$	That real is available; arbitrary ordinals are not.
Arbitrary individual reals	$D_{\mathbb{R}}$	A finite real tuple, equivalently one real, may vary with the object.
Ordinal parameters	OD	A finite ordinal tuple may vary with the object.
One real and ordinals	$\text{OD}(r)$	The fixed real and any finite ordinal tuple.
Reals and ordinals	ROD	Some real and some finite ordinal tuple.
The single set $\mathbb{R}$	$D_{\{\mathbb{R}\}}$	Equals $D_{\emptyset}$ in the ambient language of sets.

32.2 Why ordinal definability is internally expressible

For a fixed set of ordinals p , define $x \in \text{OD}(p)$ by the existence of a rank segment V_{θ} , a formula code, and finitely many ordinal parameters such that x is uniquely defined there using p and those parameters. Satisfaction for a *set* structure is definable. The reflection theorem identifies this with the usual external notion of definability from p and ordinals. The extra rank θ can be included in the ordinal parameter tuple. The same argument cannot silently be used to define parameter-free definability: adding θ would change the permitted parameters. See [28, 12] for this distinction.

Set

$$\text{HOD}(p) = \{x : \text{TC}(\{x\}) \subseteq \text{OD}(p)\}, \quad \text{HOD} = \text{HOD}(\emptyset).$$

For a set of ordinals p , this is a transitive inner model of ZFC and has its usual canonical well-order definable from p . These statements are about $\text{HOD}(p)^V$. They do not replace it by $\text{HOD}^{\text{HOD}(p)}$, which is a separate computation.

Proposition 32.1 (Hereditary definability of sign codes). *For a surreal in its canonical sign representation and a set of ordinals p ,*

$$x \in \text{OD}(p) \iff x \in \text{HOD}(p).$$

Consequently $\mathbf{No} \cap \text{OD} = \mathbf{No} \cap \text{HOD}$, and the analogous statement holds for $\mathbf{Oz}$.

Proof. All hereditary components of the graph of $x : \alpha \rightarrow \{-, +\}$, except possibly the full graph itself, are ordinals or fixed finite constructions from ordinals and sign tags. They are ordinal definable. Thus definability of x from p and ordinals already gives hereditary such definability. The reverse implication is part of the definition of HOD . $\square$

This identification is representation-sensitive in its formulation: an arbitrary noncanonical game name could contain extraneous nondefinable sets. The canonical number and that particular game name need not have the same definability degree.

32.3 Elementary closure and the role of whole inputs

Lemma 32.2 (Closure under a uniquely specified operation). *If a class function F is ambiently definable from a finite tuple $\bar{b}$, and its input tuple $\bar{x}$ is definable from A , then $F(\bar{x})$ is definable from $A \cup \{\bar{b}\}$.*

Proof. Conjoin definitions of the inputs with the graph definition of F , and existentially quantify the input variables. Uniqueness of the inputs and of the function value proves uniqueness of the output. Only finitely many formulas and parameters are combined. $\square$

Theorem 32.3 (Background structure of ambient definability hulls). *Whenever the hulls above are interpreted in a fixed transitive model, K_A is a real closed field, $I_A = K_A \cap \mathbf{Oz}$ is its integer part, and*

$$K_A = \text{Frac}(I_A), \quad K_A \preceq \mathbf{No} \quad \text{in the ordered-field language.}$$

They are closed under every parameter-free, uniquely defined ambient operation whose inputs and outputs have the indicated types. In particular this includes Ω , its inverse on monomials, the Gonshor exponential and logarithm, and the omnific floor.

Proof. Arithmetic and the named canonical functions have parameter-free set-theoretic definitions, so lemma 32.2 applies. The positive square root is unique. For an odd-degree polynomial choose its least root in the real closed surreal field. Its finite root set is ordered, so this is a uniquely specified operation. These two root properties characterize real closedness. Quantifier elimination for real closed ordered fields gives elementarity. The floor provides the integer part.

For completeness, a denominator can be selected without adding a fresh ordinal parameter. If $S = \text{supp}(x)$, let $\delta(x)$ be the least ordinal such that $a + \delta(x) > 0$ for every $a \in S$, where addition is surreal addition. Such an ordinal exists because S is a set and the ordinal surreals are cofinal in $\mathbf{No}$. More explicitly, $|a| \leq b(a)$, and the ordinal supremum of the values $b(a) + 1$ suffices. For empty support take $\delta(0) = 0$. Both $q = \omega^{\delta(x)}$ and $p = xq$ are omnific integers, with $q > 0$, and both are definable from x . Thus $x = p/q$ belongs to $\text{Frac}(I_A)$. The other inclusion follows from field closure. $\square$

These are results of the existing reports (Part I; [22]); the proof is included to specify exactly which closure principles are used. The denominator is a definable *function of the input*. No new permission to use arbitrary ordinal parameters has been introduced.

The phrase “closed under normal forms” needs an input discipline. If an entire ordinal-indexed family is definable, its canonical sum is definable. Knowing that each term separately has some definition does not supply a single definable family or a definable collector of those definitions. The repository already supplies counterexamples to this mistake (section 9 of Part I). The intrinsic summation theorem later in this article therefore includes an explicit code for the entire family among its inputs.

Proposition 32.4 (Elementary arithmetic of the definable integer part). *Let $\Pi_A = K_A \cap \Pi$. Then*

$$I_A = \Pi_A \oplus \mathbb{Z}, \quad nI_A = \Pi_A \oplus n\mathbb{Z} \quad (1 \leq n \in \mathbb{N}), \quad I_A/nI_A \cong \mathbb{Z}/n\mathbb{Z}.$$

Proof. Constant-term extraction is a canonical operation, so the zero exponent coefficient of an element of I_A is an ordinary integer in the same hull. The difference lies in Π_A . Division of a purely infinite element by a positive integer preserves its support and belongs to the same hull, so $n\Pi_A = \Pi_A$. The formulas and quotient identification follow. In particular the small finite quotients do not measure the transfinite information in an omnific normal form. $\square$

32.4 A reusable bounded code

Define parameter-free maps

$$\tau(x) = \frac{1}{2} \left(1 + \frac{x}{\sqrt{1+x^2}} \right), \quad Q(x) = \Omega(\tau(x)). \quad (32.1)$$

The positive square root is understood. The map τ bijects **No** increasingly with $(0, 1)$, with inverse

$$\tau^{-1}(t) = \frac{2t-1}{\sqrt{1-(2t-1)^2}}.$$

Hence Q bijects **No** with

$$\mathcal{Q} = \{\Omega(a) : 0 < a < 1\} \subseteq \mathbf{Oz} \cap (1, \omega). \quad (32.2)$$

Both directions are definable in ambient set theory, and already in the ordered field expanded by Ω . Thus

$$\text{Def}(A, x) = \text{Def}(A, Q(x)) \quad (32.3)$$

for any ambient allowed parameters, with the corresponding intrinsic statement in a language containing Ω . This bounded code is prior repository work (theorem 6.8 of Part I). It will serve as a compression step for the new interpretation.

Corollary 32.5 (The ordinal-definability test, recalled). *Over ZFC, each of the following is equivalent to $V = \text{HOD}$: every surreal is OD; every omnific integer is OD; every member of $\mathcal{Q}$ is OD.*

Proof. The bounded code recovers every surreal from a member of $\mathcal{Q}$. Every subset of an ordinal has a canonical characteristic sign sequence. If all surreals are OD, every subset of every ordinal is in HOD. By Choice, any set has a pointed well-founded extensional relation code on an ordinal; that relation and its distinguished point can be encoded by a subset of an ordinal. Mostowski collapse then puts the original set in HOD. The converse is immediate. $\square$

33 Real parameters: individual descriptions and uniform coding

Throughout this section the ambient universe satisfies ZFC, and all definability is in its language of set theory. Real parameters may equivalently be taken from 2^ω : there are fixed definable coding maps between ordinary reals and subsets of ω , and finitely many such codes combine into one. We keep four notions separate:

$D_{\mathbb{R}}$	definability from one real, without ordinal parameters,
$OD(r)$	definability from the fixed real r and ordinal parameters,
ROD	$\bigcup_{r \in \mathbb{R}} OD(r)$,
$HOD_{\mathbb{R}}$	$\{x : TC(\{x\}) \subseteq ROD\}$.

The first line is an external semantic notion, or a scheme over a fixed model. The last three have first-order definitions by reflection and satisfaction in rank segments. Our $HOD_{\mathbb{R}}$ is the convention often written $HOD(\mathbb{R})$; that notation must not be confused with permitting only the single set $\mathbb{R}$ as a parameter. That single parameter adds no definability here, since the set of reals is itself parameter-free definable. Dorais and Hamkins discuss the reflection construction and the real-only distinction explicitly in [28, Section 4].

For a fixed set of ordinals p , write $H_p = HOD(p)$, with definitions allowed to use p and ordinal parameters. This is the usual transitive inner model of ZFC with its canonical well-order definable from p . The requirement that p be a set of ordinals ensures that it is itself hereditarily $OD(p)$. A real is a special case. All occurrences of H_p in what follows are computed in the same ambient universe V .

Proposition 33.1 (Exactly when arbitrary real parameters add information). *The following are equivalent:*

1. every real is ordinal definable;
2. $ROD = OD$;
3. $\mathbf{No} \cap ROD = \mathbf{No} \cap OD$;
4. $\mathbf{Oz} \cap ROD = \mathbf{Oz} \cap OD$.

Proof. If every real is OD, substitute an ordinal definition of the real parameter into any ROD definition; this proves (1) $\Rightarrow$ (2), which implies (3) and (4). Condition (3) implies (1), since the canonical embedding of a real in $\mathbf{No}$ and its inverse are definable. For (4), use the omnific integer $r\omega$ associated to a real r . It is ROD, and division by the parameter-free definable ω recovers r . Thus (4) also implies (1). $\square$

The hereditary sign-code argument gives

$$\mathbf{No} \cap ROD = \mathbf{No} \cap HOD_{\mathbb{R}}.$$

Indeed, every element in the transitive closure of a sign sequence other than the sequence itself is constructed from ordinals and fixed finite tags. This assertion concerns hereditary membership and does not assume that $HOD_{\mathbb{R}}$ satisfies Choice.

33.1 A single real suffices when all surreals are ROD

The next statement strengthens a pointwise permission to choose a new real for each object into one common real for the entire universe. It uses ambient Choice essentially.

Remark 33.2 (Global real-parameter consolidation; printed in Part II). The manuscript's Theorem 4.2, with its proof and the paragraph after it on relativization to a set of ordinals p , is printed once, in Part II: it is the same theorem as source 02's [corollary 22.5](#), and its formulation with six clauses, its proof and the relativization are [remark 22.7](#). Over ZFC it says: $V = \text{HOD}_{\mathbb{R}}$ if and only if every set, or every surreal, or every omnific integer, or every monomial ω^a with $0 < a < 1$ is ROD, if and only if $V = H_r$ for one real r . Its proof uses ambient Choice essentially: a well-order of $\mathbb{R}$ is $\text{OD}(r)$ for some real r .

33.2 A uniform decoder of countable sign sequences

Fix a definable coding of triples

$$(d, E, \varepsilon), \quad d \subseteq \omega, \quad E \subseteq d \times d, \quad \varepsilon : d \longrightarrow \{-, +\},$$

by subsets of ω . Such a code is *valid* when E is a strict well-order. Let α be its order type and let $\pi : (d, E) \cong (\alpha, \in)$ be the unique order isomorphism. The decoded surreal is the sign sequence s of length α given by $s(\pi(n)) = \varepsilon(n)$. Send invalid codes to 0. We obtain one parameter-free function

$$D : \mathcal{P}(\omega) \longrightarrow \mathbf{No}_{<\omega_1}. \quad (33.1)$$

Every countable-birthday surreal has a valid code, by taking any enumeration of its domain and pulling back the ordinal order and its signs. Hence D is onto. In particular:

Remark 33.3 (One-real descriptions below ω_1 ; printed in Part II). The manuscript's Corollary 4.3 says that every surreal of countable birthday, in particular every omnific integer of countable birthday, is definable from one real without ordinal parameters: $\mathbf{No}_{<\omega_1} \subseteq D_{\mathbb{R}} \cap \mathbf{No} \subseteq \text{ROD} \cap \mathbf{No}$. It is printed once, as source 05's [theorem 22.8](#); here it follows at once from the surjectivity of D .

This construction selects a code only existentially. Reversing its direction by one definable operation is a genuinely stronger demand.

Theorem 33.4 (Uniform real coding detects correctness of ω_1). *Let p be a set of ordinals. The following are equivalent:*

1. *there is an $\text{OD}(p)$ function*

$$C : \mathbf{No}_{<\omega_1^V} \longrightarrow \mathcal{P}(\omega) \quad \text{such that} \quad D(C(x)) = x;$$

2. *there is such a function definable from p alone;*

3. $(\omega_1)^{H_p} = (\omega_1)^V$.

Thus allowing ordinal parameters in the coding operation does not improve the criterion. When p is omitted, a parameter-free section exists exactly when HOD computes ω_1 correctly.

Proof. For (1) $\Rightarrow$ (3), let $\alpha < \omega_1^V$ and regard α as its all-plus surreal. If $\alpha > 0$, $C(\alpha)$ must be a valid real code for a well-order of type α , since invalid codes decode to 0. The case $\alpha = 0$ is already countable in every inner model. The code $C(\alpha)$ is $\text{OD}(p)$: a definition of C and the ordinal parameter α define its value. As a subset of ω , it belongs to H_p . Its well-order and order-type collapse are absolute between H_p and V . Therefore H_p regards α as countable. Every $\alpha < \omega_1^V$ is consequently below $\omega_1^{H_p}$. Together with the general inequality $\omega_1^{H_p} \leq \omega_1^V$, this gives equality.

Assume (3). For each $\alpha < \omega_1^V$, the inner model H_p has a real code for an enumeration of α . Select the least such enumeration under the canonical H_p well-order. More explicitly, select a

triple $(d_\alpha, E_\alpha, \pi_\alpha)$ with $(d_\alpha, E_\alpha) \cong (\alpha, \in)$ and $d_\alpha \subseteq \omega$; it is enough to select the least real coding (d_α, E_α) , as the collapse π_α is then unique. This selection is uniform in α and definable from p .

For an arbitrary input x , of birthday α , keep d_α, E_α and add the labels

$$\varepsilon_x(n) = x(\pi_\alpha(n)).$$

The resulting real is $C(x)$. The input x need not belong to H_p : only the chosen enumeration of its ordinal domain does. The labels are added in V , by a uniquely defined operation on p, x . By construction $D(C(x)) = x$. This proves (3) $\Rightarrow$ (2), and (2) $\Rightarrow$ (1) is immediate. $\square$

Remark 33.5 (Formula complexity and absoluteness). The decoder D can be given a Δ_1 graph over ZFC, in the Levy hierarchy of set theory. A valid code has a set-sized ordinal collapse witnessing its output; failure of the strict order axioms has a bounded witness, and failure of well-foundedness has a nonempty-subset witness with no least element. Both the graph and its complement therefore have existential set witnesses. For the encoder, use the standard $\Delta_2(p)$ surjection from Ord onto OD(p), as in [28, Section 4], and take the least ordinal code that produces a real well-order of the required type. The tested predicate is $\Delta_2(p)$ and leastness adds only bounded ordinal quantification. The resulting uniquely determined output has a $\Sigma_2(p)$ graph; on its total domain, a different uniquely determined output gives a $\Sigma_2(p)$ definition of the complement. Under the equivalent hypotheses of the theorem, extending the function by a fixed real off $\mathbf{No}_{<\omega_1}$ gives a total $\Delta_2(p)$ operation. The intended input domain is Σ_1 : the birthday admits an injection into ω , or into the fixed cardinal κ in the version below. Its complement is Π_1 , so adjoining the fixed default off that domain preserves the Σ_2 graph bound. These are upper bounds, not optimality claims.

33.3 The corresponding theorem at every infinite cardinal

The real case is part of an exact family of statements. Fix an infinite V -cardinal κ and use a definable pairing of κ to encode a signed well-order on a subset of κ by a member of $\mathcal{P}(\kappa)$. Decoding valid codes and sending invalid ones to 0 gives a function definable from κ ,

$$D_\kappa : \mathcal{P}(\kappa) \longrightarrow \mathbf{No}_{<(\kappa^+)^V}.$$

It is onto: the domain of a surreal on the right has cardinality at most κ . The ordinal pairing may be chosen by ordering pairs first by their maximum and then lexicographically; on an infinite initial ordinal its order type is κ . It is therefore available with the same interpretation in H_p and V .

Theorem 33.6 (Uniform cardinal-sized coding). *For a set of ordinals p and an infinite V -cardinal κ , the decoder D_κ has an OD(p) right inverse if and only if*

$$(\kappa^+)^{H_p} = (\kappa^+)^V. \quad (33.2)$$

When this holds, a right inverse is definable from p, κ alone, and has a $\Delta_2(p, \kappa)$ total extension.

Proof. The ordinal κ is also a cardinal in H_p , because a bijection with a smaller ordinal there would be one in V . Evaluate a proposed right inverse at each ordinal $0 < \alpha < (\kappa^+)^V$. Its OD(p) output is a subset of κ , hence belongs to H_p , and codes there a well-order of cardinality at most κ and order type α . Thus $\alpha < (\kappa^+)^{H_p}$. The reverse inequality between the two successor cardinals always holds, proving necessity. For sufficiency, choose in H_p the least κ -code of an enumeration of each such α , and add the input's signs just as in Theorem 33.4. The complexity argument is unchanged, with κ added as a parameter. $\square$

Corollary 33.7 (A uniform family detects all cardinal correctness). *For a fixed set of ordinals p , the following are equivalent:*

1. H_p and V have exactly the same cardinals;
2. for every infinite V -cardinal κ , D_κ has an $\text{OD}(p)$ right inverse;
3. one class operation definable from p assigns to each pair (κ, x) , where κ is an infinite V -cardinal and $b(x) < (\kappa^+)^V$, a κ -code for x .

Proof. Cardinal agreement implies successor-cardinal agreement, and the least-code construction in Theorem 33.6 is one formula uniform in κ , proving (1) $\Rightarrow$ (3). Clearly (3) implies (2). If (2) holds, every V -cardinal has the same successor in H_p and V . All V -cardinals are already H_p -cardinals. For any additional infinite H_p -cardinal λ , take $\kappa = |\lambda|^V$. Then $\kappa < \lambda < (\kappa^+)^V$, contradicting $(\kappa^+)^{H_p} = (\kappa^+)^V$. The finite cardinals and ω are absolute, so (1) follows. $\square$

Cardinal correctness is substantially weaker than $V = H_p$. For example, adjoining a Cohen real to L preserves cardinals while HOD remains L ; hence the parameter-free real section exists even though there are non- OD real surreals. In the other direction, a homogeneous collapse making ω_1^L countable has HOD equal to L and fails the parameter-free real-section criterion. These applications use the ordinary homogeneity argument: an ordinal-definable set of ordinals in an ordinal-definable homogeneous forcing extension comes from the ground model. The constructible sets remain OD .

Corollary 33.8 (An exact all-real-parameter test for a coding operation). *There is a ROD right inverse of the real decoder D if and only if there is a real r such that*

$$\omega_1^{\text{HOD}(r)} = \omega_1^V.$$

Proof. A ROD definition uses some single real r and ordinal parameters. Apply Theorem 33.4 with $p = r$. $\square$

The coding theorem is an operation-level obstruction, stronger than the observation that a particular set lacks a canonical real code. Every countable-birthday surreal has a real description, and ZFC supplies some set-theoretic right inverse of D . The theorem determines exactly when such a right inverse can be specified with a fixed permitted parameter budget. The cardinal-sized version gives the same answer at every successor cardinal.

34 Canonical surreal operations are absolute without a same-reals hypothesis

The repository's question about inner-model support fields leaves open the case in which the inner model has fewer reals than the ambient universe. The following argument removes that restriction. Its key point is that evaluating an *existing* real coefficient never requires enumerating all reals. At limit stages, a normal form can be evaluated using only the sign sequences in one existing, set-sized sequence.

Throughout this section, a surreal is represented canonically by a sign sequence, equivalently by a pair (α, A) where α is an ordinal and $A \subseteq \alpha$ records the positions with sign $+$. Write $u \preceq_s v$ when u is an initial sign segment of v , including equality, and write $u \prec_s v$ for a proper segment. “Simplest” means having the least birthday in a nonempty convex class; that element is unique and is an initial sign segment of every element of the class.

Theorem 34.1 (Absoluteness of the canonical structure). *Let $M \subseteq W$ be transitive models of ZFC. The models may be set models in a larger metatheory or transitive inner classes. For inner classes the assertions below are understood formula by formula. They need not have the same reals. Then:*

1. *The canonical surreals of M are exactly the canonical surreal sign sequences of W that belong to M . Their order, birthdays and simplicity relation agree in the two models.*
2. *The inclusion preserves every set cut whose two option sets belong to M , the ordered-field operations, and Conway's omega map $a \mapsto \omega^a$.*
3. *If $x \in \mathbf{No}^M$, its complete canonical normal-form graph computed in W belongs to M and is exactly the graph computed in M . In particular its coefficients lie in $\mathbb{R}^M$ and its exponents in $\mathbf{No}^M$.*
4. *Canonical normal-form evaluation agrees for every normal-form graph belonging to M . More generally, summability and the sum agree for every set-indexed family of surreals that belongs to M .*
5. *The canonical real and omnific-integer predicates agree on old surreals, and so does the omnific floor. Consequently*

$$\mathbb{R}^M = \mathbb{R}^W \cap \mathbf{No}^M, \quad \mathbf{Oz}^M = \mathbf{Oz}^W \cap \mathbf{No}^M.$$

For these local assertions, equality of the ordinal heights is not needed. In the usual inner-model application both models contain the same ordinals.

Lemma 34.2 (A bounded test for the simplest separator). *Let L, R be sets of canonical surreal sign sequences with $L < R$. A surreal z equals $\{L \mid R\}$ if and only if*

$$L < z < R \quad \text{and} \quad \text{no proper sign prefix of } z \text{ lies strictly between } L \text{ and } R.$$

The test uses only comparisons with the members of L, R and the ordinal-indexed prefixes of z .

Proof. The forward implication follows from simplicity. Conversely suppose z satisfies the displayed test and that y is another separator with a smaller birthday. If y is a prefix of z , it is a forbidden proper prefix. Otherwise y, z first differ at some position γ ; their common prefix of length γ lies strictly between them in the surreal order. Since the class of separators is convex, that common prefix is again a separator, also a contradiction. Thus z has the least birthday. The same common-prefix argument rules out two distinct least-birthday separators. $\square$

Proof of Theorem 34.1, parts (1) and (2). Being a function from an ordinal to the two fixed sign symbols is absolute for transitive models. The order is determined by the first differing sign, with an absent sign placed between $-$ and $+$, so it is absolute as well. Prefixes of an old sign sequence, and the sequence of all its prefixes, belong to M by Separation and Replacement.

If $L, R \in M$ and M computes $z = \{L \mid R\}$, every test in Lemma 34.2 is absolute. Hence z is also the simplest separator in W . This argument does not assume that all surreals with birthday below that of z belong to M .

For addition, let $x_\beta = x \upharpoonright \beta$ for $\beta \leq \mathbf{b}(x)$, and similarly define y_γ . All additions needed to compute $x + y$ occur in the set-sized table

$$T(\beta, \gamma) = x_\beta + y_\gamma.$$

The recursive definition of each table entry uses entries with at least one strictly smaller coordinate and the other coordinate unchanged. The coordinatewise predecessor relation is well

founded; for example the natural ordinal sum $\beta \oplus \gamma$ strictly decreases along it. Corresponding entries agree by well-founded induction, since the cut at each step agrees. Negation is just sign reversal.

The multiplication table on the same prefix pairs is treated by the same induction. Each of its cut options is a finite sum or difference of products with smaller prefix pairs. Addition has already been shown absolute, so the options and their simplest separator agree. Reciprocals then agree by uniqueness of y with $xy = 1$; existence is supplied in M by its field theorem. This avoids any assumption about an external class recursion or about the correctness of a choice of noncanonical game representatives.

For the omega map use its usual uniform cut in the form

$$\omega^a = \left\{ 0, n\omega^{a^L} \mid 2^{-n}\omega^{a^R} \right\}, \quad n = 1, 2, 3, \dots,$$

where a^L, a^R range over the canonical left and right prefixes of a . Integer multiples are cofinal among positive real multiples on the left, and the positive dyadic multiples are coinital on the right; the displayed cut therefore gives the standard omega map. Its entire dependency table is indexed by the prefixes of a . Induction on their lengths, using the arithmetic and cut results, proves agreement. There is no quantification over a possibly different set of positive reals in this recursion. $\square$

34.1 The limit step for complete normal forms

Here is the exact sign-limit operation needed in the proof. If $f = \langle x_\xi : \xi < \lambda \rangle$ is an ordinal-indexed sequence and λ is a nonzero limit ordinal, put

$$\text{slim}_{\xi < \lambda} x_\xi = \bigcup \left\{ u : \begin{array}{l} u \preceq_s x_\xi \text{ for some } \xi < \lambda, \\ (\exists \eta < \lambda)(\forall \xi \in [\eta, \lambda)) u \preceq_s x_\xi \end{array} \right\}.$$

Two eventually persistent prefixes are comparable, since a sufficiently late member of the sequence extends both. The union is therefore a sign sequence. Its length is at most $\sup_{\xi < \lambda} b(x_\xi)$, and its construction uses only a subset of the set of prefixes of the members of f .

Equivalently a sign at position γ survives if there is a *single* tail of the sequence on which the entire prefix through γ is defined and constant. Coordinatewise eventual agreement, with different tails and no common tail for a whole transfinite prefix, is insufficient. This distinction is essential at infinite positions. For a sequence of successor length its sign limit is its last member; for the empty sequence set its sign limit equal to 0.

Lemma 34.3 (Absoluteness of the sign limit). *If the sequence f belongs to M , its sign limit computed in M and in W is the same sign sequence.*

Proof. The collection of all prefixes of members of f is the same set in both models. Eventual persistence of any such prefix is tested by quantifiers bounded by λ and the existing sign graphs. The selected collection and its union thus agree. $\square$

We use the following standard normal-form fact in a particularly useful form. Lipparini and Mezo state it as Corollary 3.2 of *Surreal limits*, citing Conway's normal-form theory and Gonshor's Theorem 5.12 [40]. If $\langle (r_\xi, a_\xi) : \xi < \lambda \rangle$ is a legal Conway normal form, evaluate it by

$$\begin{aligned} S_0 &= 0, \\ S_{\xi+1} &= S_\xi + r_\xi \omega^{a_\xi}, \\ S_\delta &= \text{slim}_{\xi < \delta} S_\xi \end{aligned} \quad (\delta \text{ a nonzero limit}).$$

Then S_λ is its canonical Conway sum. The fact applies to arbitrary set ordinal lengths and to arbitrary real coefficients, provided the exponents occur in strictly decreasing order and zero coefficients have been omitted. It does *not* assert that sign limits or sign sums agree with arbitrary analytic limits or arbitrary rearrangements of series.

[write] **A check of the imported normal-form fact, 4 October 2026.** The fact is also a consequence of Gonshor’s sign-expansion theorem [2, Theorems 5.11–5.12], in the formulation of Bournez and Guilment [5, Definition 2.20 and Theorem 2.21], which the write consulted (it did not consult Gonshor’s book itself). That theorem gives the sign expansion of a legal normal form $\sum_{\xi < \lambda} r_\xi \omega^{a_\xi}$ as the juxtaposition, in normal-form order, of one block for each term: the sign expansion of $\omega^{a'_\xi} r_\xi$, where the reduced exponent a'_ξ is obtained from a_ξ by deleting the minus signs already treated by the earlier exponents a_η ($\eta < \xi$) and, at a successor ξ , one more minus when $r_{\xi-1}$ is not dyadic. The block of the ξ th term thus depends only on the terms of index at most ξ . Hence for $\xi \leq \eta$ the sign expansion of the partial sum S_ξ is an initial segment of that of S_η , and at a limit δ the sign expansion of S_δ is the union of those of the S_ξ , $\xi < \delta$. Along a $\preceq_s$ -increasing sequence a prefix is eventually persistent exactly when it is a prefix of some member, so that union is the sign limit $\text{slim}_{\xi < \delta} S_\xi$, as the displayed recursion requires. The large-cardinal report’s proof of `lce:lem:absolute` rests on the same sign-expansion theorem of Gonshor’s Chapter 5. Thus the dependence of parts (3)–(4) of [theorem 34.1](#), and of item (4) of [theorem 34.7](#), on the Lipparini–Mezo corollary reduces to Gonshor’s classical theorem.

Proof of Theorem 34.1, parts (3) and (4). An old canonical real remains a real in the outer model: equivalently, its rational Dedekind cut is unchanged. In sign notation a canonical real has finite length, or has length ω with both signs occurring arbitrarily far along the sequence. This characterization is absolute.

Let $N = \langle (r_\xi, a_\xi) : \xi < \lambda \rangle \in M$ be a normal-form graph computed in M . It is still a legal normal-form graph in W . Its coefficients are the same nonzero canonical reals; its exponents are the same surreals in the same decreasing order; and its ordinal indexing is unchanged. The ordinal indexing itself witnesses the well-order type of the support and cannot be destroyed by passing to W .

Construct the displayed evaluation table $\langle S_\xi : \xi \leq \lambda \rangle$ in each model. The tables agree by transfinite induction: successor steps use the already absolute field operations and omega map, while limit steps use Lemma 34.3. The normal-form fact identifies the last entry with the canonical sum in both models. This proves the evaluation assertion without presupposing the absoluteness of normal forms.

Now take $x \in \mathbf{No}^M$ and set $N = \text{NF}^M(x)$. The graph N belongs to M by the internal normal-form theorem. What we have just proved says that its canonical value in W is also x . Uniqueness of normal form in W yields

$$\text{NF}^W(x) = N = \text{NF}^M(x).$$

This step proves that $\text{NF}^W(x)$ belongs to M ; it does not assume that fact.

For the more general assertion let $f : I \rightarrow \mathbf{No}$ belong to M . Its family of support graphs and the union B of their monomials are sets in M . Summability says that B is reverse well ordered and that each monomial occurs in only finitely many members of the family. The latter condition is absolute for transitive models. If M judges B reverse well ordered, its order-isomorphism to an ordinal belongs to M and witnesses the same assertion in W . The converse follows by restricting the well-order assertion to subsets belonging to M . Thus summability agrees. The coefficients of the sum are finite sums of old coefficients, hence agree. The resulting normal-form graph belongs to M , and the evaluation assertion proves agreement of its sum. $\square$

Proof of Theorem 34.1, part (5). The real assertion was established above. A normal form represents an omnific integer exactly when all its exponents are nonnegative and its coefficient

at exponent zero, if present, is an ordinary integer. These conditions are absolute once the normal form is. If M computes $z = \lfloor x \rfloor_{\mathbf{Oz}}$, then in W the same z is an omnific integer and satisfies $z \leq x < z + 1$. Uniqueness of the integer part in W proves that the floors agree. $\square$

Corollary 34.4 (Support fields from arbitrary inner models). *For every transitive inner model M of ZFC, the class $K = \mathbf{No}^M$ is an initial real closed subfield of $\mathbf{No}^V$. It contains all ordinals, is closed under the omega map and its inverse on monomials, and contains every coefficient, exponent and truncation of the normal form of each of its elements. Its normal forms agree with the ambient normal forms. All sums of summable families coded in M are preserved.*

Thus the hypotheses in the repository's support-field construction hold for $\mathbf{No}^M$ without the additional assumption $\mathbb{R}^M = \mathbb{R}^V$. In particular they apply to L or to HOD even when those models omit reals.

Proof. Initiality follows because all prefixes of an element of M belong to M . The internal real-closed field theorem supplies roots of every finite old polynomial; its assertions remain true externally since the arithmetic and order agree. The omega-map assertion is the theorem. If an old element is an ambient monomial, its absolute normal form has a single term with coefficient one; its exponent is therefore old, which proves closure under the inverse omega map. The entire normal form belongs to M , hence so do its entries and its ordinal-indexed truncations and their values. $\square$

[write] **Effect on Part I, 4 October 2026.** This corollary answers the instance of [question 17.3](#) that [remark 17.6](#) and `lce:rem:dsnsupport` leave open: for every transitive inner model $M \neq V$ of ZFC, including L and HOD when they lack reals, $\mathbf{No}^M$ satisfies (8.3) and is proper (a transitive inner model containing every set of ordinals is V), so [theorem 8.2](#) applies to it; [corollary 40.2](#) carries this out. The classification asked in [question 17.3](#) stays open ([question 42.11](#)). Part II's [remark 22.10](#) adds the field $K_{\mathbb{R}}$, which need not be of the form $\mathbf{No}^M$.

Corollary 34.5 (Ordered-field elementarity). *The inclusion $\mathbf{No}^M \subseteq \mathbf{No}^W$ is elementary in the language of ordered fields. For class-sized fields this is a scheme, one assertion for each standard first-order formula.*

Proof. Both structures are real closed fields and the inclusion is an ordered-field embedding. Quantifier elimination for real closed fields reduces each formula to a quantifier-free formula, whose truth is preserved by the inclusion. No assertion of elementary inclusion in the language of set theory is involved. $\square$

34.2 A Levy-hierarchy refinement

Fix standard effective set codes for ordinals, sign sequences, ordered pairs and finite syntax. A class relation is called Δ_1 here when, over ZFC, it has equivalent Σ_1 and Π_1 definitions on its stated domain. Using the sign-pair code (α, A) , comparison, restriction tests and the test in [Lemma 34.2](#) are bounded formulas in the standard rudimentary coding expansion. This is sufficient for the Δ_1 conclusions in the pure membership language as well.

Lemma 34.6 (Set-sized computation certificates). *Suppose a total, uniquely specified operation is evaluated on a set-sized well-founded dependency diagram. Assume its dependency diagram is specified by bounded conditions from its inputs and that each local transition has a Σ_1 graph. If the required evaluation tables exist in ZFC, then its graph is Σ_1 . Totality and uniqueness make its graph Δ_1 on the stated domain.*

Proof. A certificate consists of the dependency diagram, an ordinal-valued rank function when one is needed to verify its well-foundedness, the table of values, and certificates for the local transitions. For the prefix and ordinal-indexed diagrams used here their well-foundedness already follows from their specified ordinal coordinates. Write the latter graphs as $\exists w \vartheta$, with ϑ bounded. There are only set many local transitions. Collection supplies one set of their witnesses, so all local checks become bounded checks on a single existentially quantified certificate. The output graph therefore has the form $\exists C \Psi(C, \text{input}, \text{output})$, with Ψ bounded. On an input in the domain, failure of the graph is equivalent to existence of a certificate for a different output, again a Σ_1 assertion. This gives the complementary Π_1 definition.

One must actually specify a set-sized dependency diagram. Merely saying that an operation is “defined by a class recursion” does not establish the certificate claim. $\square$

Theorem 34.7 (Δ_1 graphs of canonical operations). *On their natural domains, the graphs of the following canonical operations are Δ_1 over ZFC:*

1. *simplest set cuts, negation, addition, multiplication, nonzero reciprocals and nonnegative square roots;*
2. *the omega map, and its inverse graph on monomials;*
3. *the sign limit of an ordinal-indexed set sequence;*
4. *evaluation of a legal canonical normal-form graph and the map $x \mapsto \text{NF}(x)$;*
5. *the omnific-integer predicate and the omnific floor.*

Proof. For simplest cuts use the bounded prefix test. The addition and multiplication dependency diagrams are precisely the prefix-pair tables described above. Every use of an already constructed arithmetic operation is accompanied by its certificate, and Collection packages the set many certificates into one. This proves the Σ_1 assertions for the arithmetic graphs without a hidden proper-class object. Their standard existence and uniqueness theorems give Δ_1 graphs. Alternatively, the reciprocal graph is the already Δ_1 relation $xy = 1$, restricted to $x \neq 0$. The nonnegative square-root graph is the Δ_1 relation $y \geq 0$ and $y^2 = x$, restricted to $x \geq 0$; its existence and uniqueness follow from real closedness.

The omega-map dependency diagram is the sequence of prefixes of its input. At a stage β , the option sets have an explicit indexing by ω times a subset of β , and their values use earlier omega values and the previously certified field operations. The cut test is bounded. The certificate lemma applies. Transposing the graph gives the inverse graph; after the normal-form result below, its monomial domain is also Δ_1 .

For the sign limit, take the set of prefixes of the members of the input sequence and apply the bounded persistence test. A certificate for that prefix set and its union is a set-sized certificate of the required form. Equivalently one can directly test that every proper prefix through a position of the proposed output is eventually persistent, and that the output extends every eventually persistent prefix of an input member. These are bounded tests on the input sequence and its sign graphs.

Legality of a normal-form graph is a bounded condition in the sign-code expansion: its index is an ordinal, coefficients are nonzero canonical reals, and its exponents strictly decrease. The sign characterization of a canonical real uses only ω and its sign graph. The evaluation table has domain $\lambda + 1$. At successor stages it uses certified arithmetic and omega values, and at limit stages it uses the sign-limit test. The certificate lemma therefore proves that evaluation has a Δ_1 graph.

Finally the assertion that $N = \text{NF}(x)$ says that N is legal and its certified value is x . This is Σ_1 ; the existence and uniqueness of the canonical normal form make it Π_1 as well.

Testing nonnegative exponents and an integral constant coefficient in this graph gives the Δ_1 omnific-integer predicate. Its complement is verified using the same unique graph, so no unbounded search through all normal forms is needed. The floor graph is then the Δ_1 assertion $z \in \mathbf{Oz}$ and $z \leq x < z + 1$. $\square$

Remark 34.8 (What the certificate theorem does not say). The theorem is a complexity classification after a fixed standard coding and theory are chosen. It is not an optimal symbol-count or proof-length bound, and it does not claim that the operations are computable on finite descriptions of arbitrary surreal inputs. The normal-form theorem is an explicit mathematical dependency, not something silently replaced by an algorithm for all class-sized objects.

Proposition 34.9 (Transfer of definition complexity). *Let $n \geq 1$ be a fixed standard integer.*

1. *A total operation with Δ_1 graph sends a uniquely Σ_n -definable input tuple to a uniquely Σ_n -definable output, with the same permitted parameters. It also preserves uniquely Δ_n -definable objects.*
2. *Let $E : X \rightarrow Y$ be a fixed injective coding whose graph, domain, range and inverse graph are Δ_1 . Then x and $E(x)$ have the same Σ_n -, Π_n - and Δ_n -singleton definability status, with the same parameters.*

Proof. For (1), if $\varphi(u)$ is a unique Σ_n definition of the input, the output is defined by $\exists u [\varphi(u) \wedge F(u) = y]$. Substitution of a Δ_1 graph preserves the Σ_n class. When the input is Δ_n , the complementary output predicate is given by $\exists u [\varphi(u) \wedge F(u) \neq y]$ and is also Σ_n , by uniqueness and totality.

For (2), a Π_n definition $\varphi(x)$ transfers to

$$y \in \text{ran}(E) \wedge \forall x [E(x) = y \longrightarrow \varphi(x)].$$

Use the Σ_1 version of the graph inside the antecedent; its negation is Π_1 , so this remains Π_n . The Σ_n transfer uses the existential form, and the inverse gives the reverse implications. The Δ_n claim follows. $\square$

The injectivity and inverse/range conditions matter in part (2). The image of an arbitrary uniquely Π_n -definable input under a total Δ_1 operation is not asserted to retain a Π_n definition. This is the appropriate distinction when applying the result to a reversible omnific code instead of to an arbitrary definable operation.

34.3 Limits of the absoluteness theorem

The theorem concerns the canonical operations just specified. A parameter-free ambient set-theoretic definition need not be absolute: for example, the constant surreal-valued function that is zero if the continuum hypothesis holds and one otherwise can change in a forcing extension. Similarly, an old set of surreals whose members each belong to M , but whose *whole family* is not a set in M , is outside the summation assertion. No claim is made that $\mathbf{No}^M$ contains every ambient Hahn sum of old coefficients and old exponents. Those are precisely the distinctions needed for the termwise-versus-uniform-definability phenomenon.

35 What the choice of language permits

35.1 The ordered field alone

Let $\mathfrak{F} = (\mathbf{No}; +, -, \cdot, <, 0, 1)$. Quantifier elimination for real closed fields gives an exact answer in this language [7]:

$$\mathrm{dcl}_{\mathfrak{F}}(A) = \{x \in \mathbf{No} : x \text{ is algebraic over } \mathbb{Q}(A)\}. \quad (35.1)$$

The right side is the relative real closure inside $\mathbf{No}$. Indeed, a real algebraic root is singled out by its position in the ordered finite root set of a polynomial. Conversely, a formula reduces to a Boolean combination of polynomial sign conditions over a field generated by finitely many parameters. At a point transcendental over that field none of the finitely many nonzero polynomials vanishes. All signs remain unchanged on some interval about the point, so that formula cannot isolate a singleton.

In particular,

$$\mathrm{dcl}_{\mathfrak{F}}(\emptyset) = \mathbb{R}_{\mathrm{alg}}, \quad \mathrm{dcl}_{\mathfrak{F}}(\mathbb{R}) = \mathbb{R}.$$

Thus allowing every real as an intrinsic ordered-field parameter still does not define any infinite or nonzero infinitesimal surreal. The field defines only ordinary integers among its parameter-free definable omnific integers, because $\mathbb{R}_{\mathrm{alg}} \cap \mathbf{Oz} = \mathbb{Z}$.

The same elimination theorem makes every definable subset of the line a finite union of points and intervals. Consequently $\mathbf{Oz}$, an infinite discrete subclass, is not definable in this reduct with any finite parameters. The monomial class is also infinite and discrete: the interval $(m/2, 2m)$ contains no other monomial when m is a monomial. Hence the omega map is not definable there either. These arguments apply formula by formula to the class structure.

35.2 The ordinary exponential and the omnific predicate

The real exponential field is an elementary substructure of the Gonshor surreal exponential field; this is a classical result of van den Dries–Ehrlich, read with its erratum [3, 4]. It follows that the parameter-free definable elements of that reduct are real. If a formula has a unique solution in the large structure, elementarity supplies a solution in $\mathbb{R}$, and uniqueness identifies them. This does not calculate which reals are definable in the real exponential field. It does show that naming the ordinary exponential does not parameter-freely define ω .

The pair $\mathfrak{S}_0 = (\mathfrak{F}, \mathbf{Oz})$ behaves differently. The next section gives short formulas defining its canonical $\mathbb{R}$, coefficient at zero, and positive-support projection. Yet its parameter-free definable elements are still real. One elementary witness is the normal-form dilation

$$S_2 \left(\sum_a r_a \omega^a \right) = \sum_a r_a \omega^{2a}.$$

It is a field automorphism preserving $\mathbf{Oz}$. Its fixed field is $\mathbb{R}$: if a fixed series had a nonzero support exponent a , invariance under doubling and halving would give an infinite increasing sequence in its reverse-well-ordered support. For $a > 0$ use $a, 2a, 4a, \dots$; for $a < 0$ use $a, a/2, a/4, \dots$. Thus any uniquely parameter-free definable point, which must be fixed by S_2 , is real. This observation and much stronger parameterwise obstructions are in (Part I; [24]).

Because $\mathbb{R}$ and $\mathbb{Z} = \mathbb{R} \cap \mathbf{Oz}$ are definable in the pair, every real definable in $(\mathbb{R}; +, -, \cdot, <, \mathbb{Z})$ is definable there. The exact upper bound remains unresolved here. Fixedness under automorphisms alone cannot give an exact description of definable closure: even the ordinary ordered real field is rigid while its parameter-free definable elements are only the real algebraic numbers.

35.3 Definability in the omnific ring itself

In this subsection work in the canonical universe, or externally in a fixed transitive model. There is also a natural intrinsic definition that uses only the ring of omnific integers:

$$\mathfrak{I} = (\mathbf{Oz}; +, -, \cdot, 0, 1).$$

Its definable elements and operations should be distinguished from ambiently definable omnific integers. The elementary fraction-field construction makes the relationship with $\mathfrak{S}_0$ precise.

Proposition 35.1 (The native ring and its fraction interpretation). *The numerical order on $\mathbf{Oz}$ is definable in its pure unital ring language. The structure $\mathfrak{I}$ and the pair $\mathfrak{S}_0$ are bi-interpretable using the usual fraction quotient. Moreover,*

$$\text{dcl}_{\mathfrak{I}}(\emptyset) = \mathbb{Z}, \quad \text{dcl}_{\mathfrak{S}_0}(\mathbb{R}) = \mathbb{R}.$$

Proof. The same-denominator construction in [theorem 32.3](#) gives $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$. For $a, b \in \mathbf{Oz}$, use the formula

$$a < b \iff \exists p \exists q (p \neq 0 \wedge q \neq 0 \wedge (b - a)q^2 = p^2).$$

If $b - a > 0$, express its positive square root in $\mathbf{No}$ as a nonzero omnific fraction p/q . Conversely, the displayed identity makes $b - a$ a nonzero square in the ordered fraction field, hence positive.

Interpret $\mathbf{No}$ by pairs $(p, q) \in \mathbf{Oz}^2$ with $q \neq 0$, modulo $pq' = p'q$. The field operations and order have their usual fraction definitions, and the interpreted omnific predicate is $\exists z (p = zq)$. In the opposite direction the omnific domain is already a definable subset of $\mathfrak{S}_0$. The comparison map on the ring is $z \mapsto [(z, 1)]$; on the field it is the relation between x and the fraction class of (p, q) specified by $p = xq$. These are definable bijections onto the relevant quotients and preserve the operations. This proves the asserted bi-interpretation.

The dilation S_2 just constructed fixes exactly $\mathbb{R}$ in $\mathbf{No}$ and exactly $\mathbb{R} \cap \mathbf{Oz} = \mathbb{Z}$ in $\mathbf{Oz}$. Therefore every parameter-free definable element of $\mathfrak{I}$ is an ordinary integer. Conversely every ordinary integer is named by a finite ring term using $0, 1$. Since S_2 fixes every real parameter, the same argument gives $\text{dcl}_{\mathfrak{S}_0}(\mathbb{R}) \subseteq \mathbb{R}$, and the reverse inclusion follows from permission to name each real. $\square$

Thus naming the order adds no expressive power to the native ring. This fraction interpretation is elementary algebraic background, using the repository's fraction-field and dilation results (Part I; [\[22, 24\]](#)). It is a different comparison from the set-universe interpretation studied below.

[write] **Antecedents, 4 October 2026.** The order formula, the fraction bi-interpretation and $\text{dcl}_{\mathfrak{S}_0}(\mathbb{R}) = \mathbb{R}$ are largely in the omnific Diophantine report (`odg:def:thm:realrecovery`, `odg:def:cor:arithmetic`, with the same dilation as `odg:def:ex:dilation`) and in [proposition 15.3](#) (i) of Part I; the inclusion $\text{dcl}_{\mathfrak{S}_0}(\mathbb{R}) \subseteq \mathbb{R}$ also follows from `opa:sc:cor:innerbound`. The manuscript credits these reports in general terms.

A nonintegral real cannot literally be a parameter in a one-sorted $\mathbf{Oz}$ -structure, since it is not in its domain. It can be used as a parameter in the interpreted fraction sort, as an equivalence class. Naming a chosen numerator and denominator separately can add information: the pair $(r\omega, \omega)$ represents r but also names ω . The second definable-closure equality above shows that allowing real parameters in the fraction sort still does not define a nonordinary omnific integer.

Finally the native ring has definable quotient and remainder operations. For every $a, b \in \mathbf{Oz}$ with $b > 0$, there are unique $q, r \in \mathbf{Oz}$ such that

$$a = bq + r, \quad 0 \leq r < b.$$

Existence follows by taking $q = \lfloor a/b \rfloor_{\mathbf{Oz}}$ and $r = a - bq$. For uniqueness, a nonzero difference of two quotients has absolute value at least one, while the corresponding difference of remainders has absolute value less than b , a contradiction. These equations are first-order graphs, using the definable order. They do not assert termination of an iterated division procedure: the positive omnific order is not well founded. For example, starting with $(a, b) = (\sqrt{2}\omega, \omega)$, the first quotient is 1 and all later quotients are 2. The successive positive remainders are $(\sqrt{2} - 1)^n\omega$, for $n = 1, 2, \dots$; they never become zero at any finite step.

Language on No	Consequence relevant to definability
Ordered field	Exact closure (35.1); all unary definable sets are semialgebraic.
Ordered field and exp	No parameter-free infinite element; exact real definable closure is not computed here.
Ordered field and Oz	Defines canonical reals, zero coefficient, positive truncation and floor; parameter-free definable elements are real.
Ordered field and Ω	Defines $\omega = \Omega(1)$, a natural valuation, and bounded finite-tuple codes.
Ordered field, Oz , Ω	Defines all coefficients, coded Hahn sums and the quotient interpretation of set theory proved below.
Ordered field and birthday order	Bi-interpretation with set theory is the announced Chen–Hamkins–Yang result; a different signature.

Naming all ordinals or all reals in one of these intrinsic languages still means allowing finitely many named elements in each formula. The ambient classification in section 32 should not be transferred to a reduct merely by retaining the same informal word “definable.”

36 Definable support calculus

Write

$$\mathfrak{S}_0 = (\mathbf{No}; 0, 1, +, -, \cdot, <, \mathbf{Oz}), \quad \mathfrak{S}_\Omega = (\mathfrak{S}_0, \Omega), \quad \Omega(a) = \omega^a.$$

The distinction between the two structures is substantial. The omnific predicate suffices to recover the real coefficient field and the coefficient at exponent zero. Naming the omega map then permits a uniform shift to every exponent. This gives the entire support relation, and supports can carry arbitrary set-sized relations. We give the intermediate steps explicitly before drawing the set-theoretic conclusion.

36.1 Recovering the coefficient at zero

Let Π denote the additive subgroup of purely infinite surreals, including zero: its elements have normal forms supported on strictly positive exponents. Let $\mathcal{A}$ be the subring of normal forms supported on nonnegative exponents. Thus

$$\mathbf{Oz} = \mathbb{Z} + \Pi, \quad \mathcal{A} = \mathbb{R} + \Pi.$$

These equalities refer to the canonical normal form. The following formulas recover these objects inside $\mathfrak{S}_0$.

Lemma 36.1 (A quadratic detector and its multiplier ring). *The classes Π , $\mathcal{A}$, and $\mathbb{R}$ are parameter-free definable in $\mathfrak{S}_0$. Explicit definitions are*

$$\text{PI}(x) \longleftrightarrow x \in \mathbf{Oz} \wedge \exists y \in \mathbf{Oz} (y^2 = 2x^2), \quad (36.1)$$

$$\text{A}(t) \longleftrightarrow \forall p (\text{PI}(p) \longrightarrow \text{PI}(tp)), \quad (36.2)$$

$$\text{Real}(t) \longleftrightarrow t = 0 \vee (t \neq 0 \wedge \text{A}(t) \wedge \text{A}(t^{-1})). \quad (36.3)$$

Here an inverse is an ordinary definable field operation, and all displayed restricted quantifiers abbreviate first-order formulas.

Proof. The constant coefficient on $\mathbf{Oz}$ is a ring homomorphism into $\mathbb{Z}$: in a product of two nonnegative-support series, the only contribution to exponent zero comes from their constant terms. If $y^2 = 2x^2$ with $x, y \in \mathbf{Oz}$, their integer constant terms n, m satisfy $m^2 = 2n^2$, whence $m = n = 0$. Conversely, if $x \in \Pi$, then $\sqrt{2}x \in \Pi \subseteq \mathbf{Oz}$ witnesses the formula. This proves (36.1).

Every element of $\mathcal{A}$ multiplies Π into Π . Conversely, suppose the normal form of t contains a nonzero term at an exponent $g < 0$. Put $p = \Omega(-g/2) \in \Pi$. Multiplication by this monomial shifts every exponent by $-g/2$; in particular, tp has a nonzero term at $g/2 < 0$. Thus $tp \notin \Pi$. No cancellation at this exponent is possible because translation of exponents is injective. This proves (36.2).

Every nonzero real and its inverse belong to $\mathcal{A}$. If $0 \neq t \in \mathcal{A}$ is not real, its largest exponent is strictly positive. The largest exponent of t^{-1} is its negative, so $t^{-1} \notin \mathcal{A}$. This proves (36.3). $\square$

The use of the omega map in the proof of the multiplier identity does not put it into the defining formula. It constructs a witness showing that (36.2), a formula in the smaller language, has the stated extension. The quadratic detector and coefficient reconstruction are already part of the repository's omnific Diophantine work [22]; the proof is included to make the present interpretation independent of a lengthy imported reconstruction argument.

Proposition 36.2 (Global coefficient and positive-part operations). *In $\mathfrak{S}_0$, the infinitesimal class $\mathfrak{m}$, the finite-element valuation ring $\mathcal{O}$, and the functions*

$$x \longmapsto c_0(x), \quad x \longmapsto x_{>0}$$

are parameter-free definable. Here $c_0(x)$ is the real coefficient of ω^0 in the full Conway normal form, and $x_{>0}$ is its positive exponent truncation. The graph of c_0 is given by

$$\text{Real}(r) \wedge \exists p (\text{PI}(p) \wedge x - p - r \in \mathfrak{m}),$$

where

$$\epsilon \in \mathfrak{m} \longleftrightarrow \forall r (\text{Real}(r) \wedge r > 0 \longrightarrow |\epsilon| < r).$$

Proof. The formula for $\mathfrak{m}$ expresses precisely that $\epsilon = 0$ or its largest exponent is negative. Similarly,

$$x \in \mathcal{O} \longleftrightarrow \exists r (\text{Real}(r) \wedge r > 0 \wedge |x| < r).$$

Split the support of a normal form at exponent zero. This gives $x = p + r + \epsilon$ with $p \in \Pi$, $r \in \mathbb{R}$, and $\epsilon \in \mathfrak{m}$. The splitting is unique because $\Pi \cap \mathcal{O} = \{0\}$ and $\mathbb{R} \cap \mathfrak{m} = \{0\}$. Hence the displayed formula defines $r = c_0(x)$ uniquely, and the same decomposition defines $p = x_{>0}$. $\square$

This global coefficient projection is not a field homomorphism. For example, $c_0(\omega) = c_0(\omega^{-1}) = 0$ whereas $c_0(\omega\omega^{-1}) = 1$. On the finite valuation ring, its restriction is the ordinary standard-part map and is a ring homomorphism.

36.2 Every coefficient and every truncation

Theorem 36.3 (Uniform support calculus). *The following operations and relations are parameter-free definable in $\mathfrak{S}_\Omega$, uniformly in the variables shown:*

$$\begin{aligned} c_a(x) &= c_0(x/\Omega(a)), \\ T_{>a}(x) &= \Omega(a)(x/\Omega(a))_{>0}, \\ T_{\geq a}(x) &= T_{>a}(x) + c_a(x)\Omega(a), \\ \text{Supp}(a, x) &\longleftrightarrow c_a(x) \neq 0. \end{aligned}$$

Thus $T_{>a}(x)$ and $T_{\geq a}(x)$ are the canonical normal-form truncations at any surreal exponent a , and $\text{Supp}(a, x)$ means that a occurs in the Conway support of x . The complementary truncations are definable by subtraction.

Proof. The identity $\Omega(b)/\Omega(a) = \Omega(b - a)$ shifts all exponents by $-a$. The coefficient at zero after this shift is exactly the old coefficient at a , and the positive part after the shift is exactly the old part above a , scaled by $\Omega(-a)$. Apply Proposition 36.2 and compose definable functions. $\square$

The theorem is a statement about the richer language. The absence of a definable omega map in $\mathfrak{S}_0$ prevents transferring these uniform exponent-indexed formulas to that reduct. A monomial supplied as a parameter still allows coefficient extraction at that particular scale.

Remark 36.4 (Valuation already from the omega map). Even without the omnific predicate, $(\mathbf{No}; 0, 1, +, -, \cdot, <, \Omega)$ defines the leading exponent of a nonzero x . It is the unique a such that

$$\forall b [(b < a \longrightarrow \Omega(b) < |x|) \wedge (a < b \longrightarrow |x| < \Omega(b))].$$

The actual largest exponent satisfies the formula because distinct monomials have distinct Archimedean classes. If a different a did so, choose b strictly between a and the largest exponent to obtain a contradiction. Negating the leading exponent gives the natural valuation. In particular its valuation ring and infinitesimal ideal are definable already in this language. This observation does not by itself select the canonical real residue representatives or recover all coefficients.

36.3 A bounded lexicographic pairing operation

The parameter-free ordered-field map

$$\tau(x) = \frac{1}{2} \left(1 + \frac{x}{\sqrt{1+x^2}} \right)$$

is an increasing bijection from $\mathbf{No}$ onto $(0, 1)$, with inverse

$$\tau^{-1}(u) = \frac{2u-1}{\sqrt{1-(2u-1)^2}}.$$

Define

$$B(a, b) = \Omega \left(\frac{2 + \tau(a)}{4} \right) + \Omega \left(\frac{\tau(b)}{4} \right), \quad (36.4)$$

$$p(a, b) = \tau(B(a, b)). \quad (36.5)$$

Lemma 36.5 (Pairing and preservation of reverse well-orders). *The function B is a parameter-free definable injection $\mathbf{No}^2 \rightarrow \mathbf{Oz} \cap (1, \omega)$ in the omega-map expansion. It is an order embedding for the lexicographic order with first coordinate primary. Consequently p is a definable lexicographic order embedding $\mathbf{No}^2 \rightarrow (0, 1)$. If D, E are reverse well-ordered sets of surreals, then $p[D \times E]$, and each of its subsets, is reverse well-ordered.*

Proof. The first exponent in (36.4) belongs to $(1/2, 3/4)$ and the second to $(0, 1/4)$. Thus both exponents are positive, strictly separated, and below one. Their sum is an omnific integer strictly between one and ω . If $a < a'$, the leading monomial in $B(a', b')$ dominates both terms of $B(a, b)$, whatever b, b' are. Hence $B(a, b) < B(a', b')$. When $a = a'$, comparison reduces to the second term and therefore to comparison of b, b' . This proves the lexicographic embedding and in particular injectivity. Composition with τ preserves it.

For any nonempty subset of $D \times E$, first choose the greatest first coordinate appearing in it, and then the greatest second coordinate above that first coordinate. These exist by reverse well-ordering and give its lexicographically greatest pair. The same assertion therefore holds in its image under p . $\square$

The inverse of B on its image is definable by its graph and uniqueness. Iterating B consequently compresses any fixed finite tuple to one omnific integer below ω . In particular,

$$\text{dcl}_{\mathfrak{S}_\Omega}(A, a, b) = \text{dcl}_{\mathfrak{S}_\Omega}(A, B(a, b))$$

for every allowed parameter set A . This is an intrinsic version of the repository's bounded coding principle. No normal-form decoding symbol needs to be added to the language for the finite-tuple assertion.

36.4 Powersets of supports and arbitrary relations

Call a surreal u a *unit-coefficient series* if

$$\text{Unit}(u) : \quad \forall a (c_a(u) = 0 \vee c_a(u) = 1).$$

It may be zero. Its support determines it uniquely.

Lemma 36.6 (Exact powerset coding). *Let d be a unit-coefficient series and put $D = \{a : \text{Supp}(a, d)\}$. The first-order predicate*

$$\text{Sub}(u, d) : \quad \text{Unit}(u) \wedge \forall a (\text{Supp}(a, u) \longrightarrow \text{Supp}(a, d))$$

has exactly one solution u_X for each subset $X \subseteq D$, namely

$$u_X = \sum_{a \in X} \Omega(a).$$

If d, e have supports D, E , the analogous relation-code predicate

$$\begin{aligned} \text{Rel}(q; d, e) : \quad & \text{Unit}(q) \wedge \\ & \forall t (\text{Supp}(t, q) \longrightarrow \exists a \exists b [\text{Supp}(a, d) \wedge \text{Supp}(b, e) \wedge t = p(a, b)]) \end{aligned}$$

has exactly one solution for each relation $R \subseteq D \times E$:

$$q_R = \sum_{(a, b) \in R} \Omega(p(a, b)).$$

For this solution, $R(a, b)$ is expressed by $\text{Supp}(p(a, b), q_R)$.

Proof. A subset of a set-sized reverse well-ordered support is again a set-sized reverse well-ordered support. Hence every displayed unary series exists as a Conway normal form. Its nonzero coefficients are exactly one, and uniqueness of normal forms gives uniqueness of the code. Conversely, any solution of Sub has just this form for its support. For the binary assertion use Lemma 36.5: every subset of $p[D \times E]$ is reverse well-ordered, so every relation has a legal normal-form code. Injectivity of p and normal-form uniqueness give both decoding and uniqueness. All relation codes have support in $(0, 1)$ and therefore lie in $\mathbf{Oz} \cap [0, \omega)$. $\square$

This lemma covers *every* ambient subset of D and every relation on $D \times E$, not just subsets definable from the original parameters. Definability is used for the membership test and the family of codes. An individual code need not be parameter-free definable. This distinction is what allows the following first-order expression of genuine well-foundedness.

37 Interpreting set theory in bounded omnific monomials

The support-subset mechanism has precedents in interpretations of second-order arithmetic in Hahn fields [42]. Here it applies to supports of arbitrary set order type and therefore to arbitrary set-sized membership graphs.

37.1 Pointed well-founded extensional graphs

A candidate set code is a triple (d, q, r) with d a unit-coefficient series, $D = \text{supp}(d) \subseteq (0, 1)$ nonempty, $\text{Rel}(q; d, d)$, and $r \in D$. Write

$$a E_q b \iff \text{Supp}(p(a, b), q)$$

for its membership-oriented edge relation. The following are all first-order conditions in $\mathfrak{S}_\Omega$.

1. *Well-foundedness*: for every u with $\text{Sub}(u, d)$ and $u \neq 0$, there is $a \in \text{supp}(u)$ with no E_q -predecessor in $\text{supp}(u)$.
2. *Extensionality*: if $a, b \in D$ have the same E_q -predecessors in D , then $a = b$.
3. *Accessibility from the root*: every u with $\text{Sub}(u, d)$ whose support contains r and is closed under E_q -predecessors has support equal to D .

For instance, the first condition is literally

$$\forall u (\text{Sub}(u, d) \wedge u \neq 0 \longrightarrow \exists a [\text{Supp}(a, u) \wedge \forall b (\text{Supp}(b, u) \longrightarrow \neg(b E_q a))]).$$

There are no quantifiers over sets of surreals in these formulas: u is one surreal number. Lemma 36.6 proves that ranging over these numbers really ranges over all subsets needed in the displayed well-foundedness test.

Call such a triple *admissible*. The set-sized Mostowski collapse of (D, E_q) sends its root to a unique set, denoted here by $\text{val}(d, q, r)$. The collapse function is used in the proof in the ambient universe, not named in the surreal language.

Theorem 37.1 (Set theory in the omega-omnific structure). *In ZFC, $\mathfrak{S}_\Omega$ has a parameter-free first-order quotient interpretation of $(V, \in)$. Its representatives may all be taken to be single monomials in $\mathcal{Q} = \{\Omega(a) : 0 < a < 1\} \subseteq \mathbf{Oz} \cap (1, \omega)$. More precisely, there are fixed first-order formulas for a class C , an equivalence relation $\equiv$ on C , and a relation E^* invariant under $\equiv$, such that the Mostowski-collapse valuation gives*

$$(C/\equiv, E^*) \cong (V, \in).$$

The displayed quotient means a definable quotient presentation; it does not require proper-class equivalence classes to be sets.

Proof. First use admissible triples as representatives. The admissibility conditions were given above and are first-order. By exact powerset coding, the asserted well-foundedness is actual well-foundedness of the set relation E_q , so the set-sized Mostowski collapse applies. Extensionality

makes it an isomorphism onto a transitive set T . Accessibility says that T is exactly the transitive closure of the singleton containing the collapsed root. Indeed, the nodes reachable from the root by a finite sequence of predecessor steps form a predecessor-closed subset containing the root; by accessibility this is all of D .

Two representatives are declared equivalent when there is a root-preserving isomorphism between their graphs. This is first-order: for supports D, D' , quantify over a relation code h satisfying $\text{Rel}(h; d, d')$ and require its decoded relation to be the graph of a total bijection $D \rightarrow D'$ that takes root to root and preserves edges in both directions. These requirements are finite first-order formulas. The relation-coding lemma supplies every bijection required, so this condition is equivalent to equality of the collapsed roots.

For membership, declare a representative A to be E^* -related to a representative B if there are a node b in the graph of B and a coded map h with $\text{Rel}(h; d_A, d_B)$ satisfying the following conditions: b is an immediate E_B -predecessor of the root of B ; h is an injective map from the entire domain of A into that of B ; it sends the root of A to b ; it preserves and reflects edges; and its image is closed under E_B -predecessors. All these assertions are first-order statements about a relation code. The last requirement can be written

$$\forall a \in D_A \forall v \in D_B (v E_B h(a) \longrightarrow \exists a' \in D_A (h(a') = v)),$$

using the graph of h in place of function notation.

Such a map identifies A with the full predecessor closure rooted at b . To see this, its predecessor-closed image contains b , so it contains every node reachable downwards from b . Conversely, accessibility of A puts every image node in that closure. Applying collapse, its existence is therefore equivalent to $\text{val}(A) \in \text{val}(B)$. If this membership holds, the corresponding isomorphism of transitive closures supplies a map h , and exact relation coding supplies its code. Equality and membership are consequently independent of the chosen representatives.

For surjectivity, fix any set x . By Choice, the set $\text{TC}(\{x\})$ has an enumeration by an ordinal α . Label its elements bijectively by the reverse well-ordered set

$$D_\alpha = \left\{ \frac{1}{\underline{\beta} + 2} : \beta < \alpha \right\} \subseteq (0, 1),$$

where $\underline{\beta}$ is the canonical ordinal surreal. Use the unit-coefficient series supported on D_α for d , use Lemma 36.6 to code the membership relation as q , and take for r the label of x . This is an admissible triple whose collapsed root is x . The empty set causes no exception: its pointed transitive closure is the singleton $\{\emptyset\}$.

Finally replace the triple (d, q, r) by the single number

$$Q(B(d, B(q, r))) = \Omega(\tau(B(d, B(q, r)))).$$

Lemma 36.5 and (32.2) place it in $\mathcal{Q}$ and provide a definable inverse on its image. Pulling back the preceding domain, equality, and membership formulas gives the asserted one-coordinate presentation. $\square$

Example 37.2 (A code for the ordinal two). Put $a_0 = 1/2$, $a_1 = 1/3$, and $a_2 = 1/4$, and take

$$d = \Omega(a_0) + \Omega(a_1) + \Omega(a_2), \quad q = \Omega(p(a_0, a_1)) + \Omega(p(a_0, a_2)) + \Omega(p(a_1, a_2)).$$

The root is $r = a_2$. The graph has edges $a_0 E_q a_1$, $a_0 E_q a_2$, and $a_1 E_q a_2$. Collapse sends the three nodes to $\emptyset$, $\{\emptyset\}$, and $\{\emptyset, \{\emptyset\}\}$, respectively. Thus its value is the set-theoretic ordinal 2. The single representative $Q(B(d, B(q, a_2)))$ is an omnific monomial between 1 and ω . Different choices of labels give equivalent representatives; they need not give the same surreal number.

Corollary 37.3 (Uniform translation of set-theoretic formulas). *For each fixed first-order set-theoretic formula $\varphi(x_1, \dots, x_n)$ there is an effectively obtained formula $\varphi^*(z_1, \dots, z_n)$ in the language of $\mathfrak{S}_\Omega$ such that, for admissible codes,*

$$\mathfrak{S}_\Omega \models \varphi^*(z_1, \dots, z_n) \iff V \models \varphi(\text{val}(z_1), \dots, \text{val}(z_n)).$$

Proof. Replace atomic equality and membership by their formulas from Theorem 37.1; preserve Boolean connectives; and relativize each quantifier to the admissible-code domain. Induction on the fixed formula, using surjectivity of the valuation map, proves the claim. This is a syntactic translation, not a truth predicate for arbitrary formulas in V or in the class structure $\mathfrak{S}_\Omega$. $\square$

What this resolves. The repository asks how much ambient definability can be captured after adjoining the omega map to the field with the omnific predicate. The theorem supplies a strong answer at the level of interpretation: this expansion can express arbitrary first-order set theory through bounded omnific codes. It also gives explicit intrinsic definitions of all normal-form coefficients and exponent truncations. These conclusions do not identify the exact definable closure of the empty set, nor do they prove a bi-interpretation. In particular, the quotient interpretation does not supply a definable choice of one representative for each set, or a definable identification between an original surreal and the canonical surreal reconstructed inside the interpreted set universe. Those are additional mathematical requirements.

[write] **Effect on Part I, 4 October 2026.** This answers the omega-map half of [question 17.1](#) at the level of an interpretation: with **Oz** and Ω named, first-order set theory is expressible through bounded omnific codes. Source 02’s [proposition 26.11](#) is the special case of the bounded unary code. The exact definable closures remain open ([question 42.2](#)), as does a bi-interpretation ([question 42.1](#)). The birthday-enriched structures of the birthday-cutoff report are bi-interpretable with the corresponding H_κ (`hset:thm:biinterpretation`, `hset:thm:main-rounding`); the present signature names no birthday relation.

Choice and the interpretation domain. The proof of exact powerset and relation coding does not use Choice. Without Choice, every admissible graph still has a well-orderable domain, since its domain is a reverse well-ordered support. Its collapsed transitive closure is therefore well-orderable. Conversely, every set whose transitive closure is well-orderable has a code by the same argument. Thus over ZF the quotient presents exactly the hereditarily well-orderable sets, with their actual membership relation; Choice makes this all of V .

Class semantics. All normal-form supports, graph domains, and relations used in a single code are sets. Consequently the proof uses ordinary set recursion and the set-sized Mostowski collapse. No collapse of a proper-class graph is invoked. The theorem can be read schematically over ZFC or externally for the surreal structure of a transitive set model. In the latter reading, “every subset” means every subset in that model, and the interpreted structure is precisely its set universe.

38 Definable Hahn summation of coded families

The support calculus also makes an infinitary operation first-order definable once its set-sized input family is represented by one number. The coding of that family is part of the input: individual definability of all of its values is not a substitute for definability of the family.

For a unit-coefficient series d with support D , define

$$F_f(i, x) \iff \text{Supp}(p(i, x), f).$$

Let $\text{Fam}(f, d)$ be the conjunction of

1. $\text{Unit}(d)$ and $\text{Unit}(f)$;
2. every element of $\text{supp}(f)$ is $p(i, x)$ for some $i \in D$ and some surreal x ;
3. for every $i \in D$ there is exactly one x with $F_f(i, x)$.

These three conditions are a fixed first-order formula. They say that f is the unit-coefficient support code of the graph of a function $i \mapsto x_i$ on D .

Lemma 38.1 (Every support-indexed family has a code). *For every set function $i \mapsto x_i$ on a reverse well-ordered set D , the normal form*

$$f = \sum_{i \in D} \Omega(p(i, x_i))$$

exists and satisfies $\text{Fam}(f, d)$, where $d = \sum_{i \in D} \Omega(i)$. Conversely every such code defines a set-sized family.

Proof. Every nonempty subset of the graph has a largest first coordinate i , and the corresponding second coordinate is unique. Thus its image under the lexicographic embedding p is reverse well-ordered. The normal-form existence theorem and injectivity of p give the code. Conversely, decoding the set-sized support of f gives a set of pairs; the family conditions make it the graph of a set function on D . $\square$

Lemma 38.2 (First-order finiteness of a support). *For unit-coefficient t , the support of t is finite if and only if*

$$\text{Fin}(t) : \quad \forall u (\text{Sub}(u, t) \wedge u \neq 0 \longrightarrow \exists i [\text{Supp}(i, u) \wedge \forall j (\text{Supp}(j, u) \longrightarrow i \leq j)]).$$

Proof. Every nonempty subset of a finite linear order has a least element. Conversely, an infinite reverse well-ordered set has a strictly decreasing sequence consisting of its first ω elements in reverse order. The set of those elements is an allowed sub-support and has no numerical least element. Its unit-coefficient code therefore falsifies the displayed formula. This uses the fact that all subsets are represented, as proved in Lemma 36.6. $\square$

Theorem 38.3 (Definability of summability and the Hahn sum). *There are parameter-free first-order formulas $\text{Summable}(f, d)$ and $\text{Sum}(f, d, y)$ in $\mathfrak{S}_\Omega$ which, for family codes, express respectively that $(x_i)_{i \in D}$ is Hahn-summable and that*

$$y = \sum_{i \in D} x_i.$$

The sum formula defines a unique output exactly on the summable inputs.

Proof. Require $\text{Fam}(f, d)$. The union of the supports is expressed by the formula

$$U_f(a) : \quad \exists i \exists x (\text{Supp}(i, d) \wedge F_f(i, x) \wedge \text{Supp}(a, x)).$$

That this union is reverse well-ordered is equivalent to

$$\exists u (\text{Unit}(u) \wedge \forall a (\text{Supp}(a, u) \longleftrightarrow U_f(a))).$$

Indeed an existing support is reverse well-ordered, and every set-sized reverse well-ordered set has its unit-coefficient series. The union is a set because the decoded family is a set.

For each exponent a , require a code t for precisely the indices at which its coefficient is nonzero:

$$\text{Fib}(a, t; f, d) : \quad \text{Sub}(t, d) \wedge \forall i (\text{Supp}(i, t) \longleftrightarrow [\text{Supp}(i, d) \wedge \exists x (F_f(i, x) \wedge c_a(x) \neq 0)]).$$

Such a t always exists uniquely by exact subset coding. The condition

$$\forall a \exists t (\text{Fib}(a, t; f, d) \wedge \text{Fin}(t))$$

says exactly that every exponent occurs in only finitely many members of the family. This, together with the preceding union-support condition, is the usual Hahn-summability criterion, so defines $\text{Summable}(f, d)$.

For completeness, the finite coefficient sum in the output specification can itself be expressed by a fixed formula, without a variable-length term. For the finite fibre $I = \text{supp}(t)$, write

$$\text{Next}_t(j, i) : \quad i, j \in I \wedge i < j \wedge \neg \exists k \in I (i < k < j).$$

This is the next step in descending order. To say that c is the sum of the coefficients $c_a(x_i)$ for $i \in I$, use the following alternatives. If $t = 0$, require $c = 0$. Otherwise quantify a family code g on t and require all of the following finite list of first-order conditions:

1. $\text{Fam}(g, t)$;
2. for every $i \in I$, every x, v with $F_f(i, x)$ and $F_g(i, v)$, if i is greatest in I , then $v = c_a(x)$;
3. for every $i, j \in I$ and every x, v, w with $\text{Next}_t(j, i)$, $F_f(i, x)$, $F_g(i, v)$, and $F_g(j, w)$, require $v = w + c_a(x)$;
4. for the least $i \in I$, require $F_g(i, c)$.

Greatest and least are expressed by quantifying over I , a definable support. A finite linear order has unique descending partial sums, so these conditions determine c uniquely, with the intended value. Their witness family g has a code by Lemma 38.1. Denote this fixed formula, including the finite-fibre requirement, by $\text{CoeffSum}(a, f, d, c)$.

Now define

$$\text{Sum}(f, d, y) : \quad \text{Summable}(f, d) \wedge \forall a \text{ CoeffSum}(a, f, d, c_a(y)).$$

For summable inputs the standard Hahn construction forms these finite real coefficient sums, whose nonzero support is contained in the reverse well-ordered union already obtained. Thus a surreal y with exactly these coefficients exists. Normal-form uniqueness proves uniqueness. $\square$

The formula does not assign a Hahn sum to a family that fails the summability criterion even if some informal regrouping would cancel its terms. It defines precisely the usual unconditional Hahn operation. Since the operation has a unique output and a parameter-free graph, it preserves each intrinsic definable closure when both its domain code and family code belong to that closure. The corresponding ambient definability-preservation statement follows by the same unique-output argument.

39 An intrinsically definable commuting family of derivations

The support calculus provides explicit derivations without choosing a Hamel basis or an unspecified derivation. These are diagonal Euler derivations, rather than the Berarducci–Mantova derivation.

Theorem 39.1 (Coefficient-indexed derivations). *There is a parameter-free definable function $(b, x) \mapsto D_b(x)$ in $\mathfrak{S}_\Omega$ characterized by*

$$c_a(D_b(x)) = c_b(a)c_a(x) \quad (a \in \mathbf{No}).$$

For every surreal b , the function D_b is a strongly $\mathbb{R}$ -linear derivation. The derivations commute pairwise, satisfy $D_b(\mathbf{Oz}) \subseteq \Pi$, and have common constant field exactly $\mathbb{R}$:

$$\{x : \forall b \ D_b(x) = 0\} = \mathbb{R}.$$

In particular D_0 is a nonzero parameter-free definable derivation and $D_0(\omega) = \omega$.

Proof. For $x = \sum_a r_a \Omega(a)$, set

$$D_b(x) = \sum_a c_b(a) r_a \Omega(a).$$

The new nonzero support is a subset of the old support, so the series exists. Its coefficients characterize it uniquely and yield the displayed first-order definition. The map $a \mapsto c_b(a)$ is $\mathbb{R}$ -linear on the additive surreal group: addition and real scalar multiplication act coefficientwise on normal forms. Thus D_b is $\mathbb{R}$ -linear.

For multiplication, the contribution to any exponent g in xy is a finite sum over pairs $a + a' = g$ of exponents occurring in the two inputs. Since

$$c_b(g) = c_b(a) + c_b(a'),$$

its coefficient in $D_b(xy)$ equals the coefficient in $D_b(x)y + xD_b(y)$. Normal-form uniqueness proves the Leibniz rule. The same coefficient argument proves strong linearity: for a summable family (x_i) , each output support is contained in its original support, so $(D_b(x_i))$ is summable, and the finite coefficient sums commute with multiplication by $c_b(a)$.

The coefficient at a in either $D_b D_{b'}(x)$ or $D_{b'} D_b(x)$ is $c_b(a)c_{b'}(a)c_a(x)$, proving commutation. An omnific input has no negative exponents; the coefficient at zero in its derivative vanishes because $c_b(0) = 0$. Hence its derivative belongs to Π .

Every real is annihilated by every D_b . If x is not real, choose a nonzero exponent a with $c_a(x) \neq 0$. Since $a \neq 0$, its own normal form has some exponent b with $c_b(a) \neq 0$. Then $D_b(x)$ has nonzero coefficient at a , so x is not a common constant. Finally, $c_0(1) = 1$, and the only term of ω has exponent one, yielding $D_0(\omega) = \omega$. $\square$

Proposition 39.2 (Canonical partial integration). *For each b there are uniformly definable $\mathbb{R}$ -linear operators K_b, J_b such that*

$$\begin{aligned} c_a(K_b(x)) &= c_a(x), & c_b(a) &= 0, & c_a(J_b(x)) &= \begin{cases} c_a(x)/c_b(a), & c_b(a) \neq 0, \\ 0, & c_b(a) = 0. \end{cases} \\ c_a(K_b(x)) &= 0, & c_b(a) &\neq 0, \end{aligned}$$

They satisfy

$$D_b J_b = J_b D_b = \text{id} - K_b, \quad \text{im}(K_b) = \ker(D_b), \quad \text{im}(D_b) = \ker(K_b).$$

Consequently $D_b(y) = x$ has a solution exactly when $K_b(x) = 0$; then $J_b(x)$ is the unique solution with $K_b(y) = 0$.

Proof. All displayed series have support contained in the support of their input, so they exist and their coefficient specifications are first-order definitions with unique outputs. The identities follow coefficient by coefficient. If $D_b(y) = x$, the vanishing multiplier at every exponent with $c_b(a) = 0$ implies $K_b(x) = 0$. Conversely, for $K_b(x) = 0$, the identity $D_b J_b(x) = x - K_b(x)$ gives a solution. Two solutions differ by an element of $\ker D_b = \text{im } K_b$, so the stated normalization makes it unique. $\square$

For clarity, D_0 is not the standard surreal derivation normalized by $\partial\omega = 1$: here $D_0\omega = \omega$, and its kernel is much larger than $\mathbb{R}$. For example, $D_0(\Omega(\omega)) = 0$ because $c_0(\omega) = 0$. The advantage of this family is its explicit intrinsic definability in the stated language. The general classification of strong real-linear derivations diagonal on monomials is already discussed in the repository's real-vector-space report [27]; the present result isolates a uniformly definable coefficient family and supplies its definable partial inverses.

40 Further consequences for parameters, fields, and operations

40.1 Coefficientwise operations and full product rings

Once coefficients are intrinsically available, many useful operations have short definitions even when their normal forms are transfinite. Coordinatewise linear algebra and diagonal operators are developed in the repository's vector-space report [27]. The point here is the uniform intrinsic definability of their coefficient formulas. The following is a general existence and preservation principle.

Theorem 40.1 (Coefficientwise lifting). *Let $f : \mathbb{R}^k \rightarrow \mathbb{R}$ be definable in $\mathfrak{S}_\Omega$, with specified parameters, and assume $f(0, \dots, 0) = 0$. There is a unique operation $\tilde{f} : \mathbf{No}^k \rightarrow \mathbf{No}$ given by*

$$c_a(\tilde{f}(x_1, \dots, x_k)) = f(c_a(x_1), \dots, c_a(x_k)) \quad (a \in \mathbf{No}).$$

It is definable with the same parameters. If $f[\mathbb{Z}^k] \subseteq \mathbb{Z}$, then $\tilde{f}[\mathbf{Oz}^k] \subseteq \mathbf{Oz}$.

Proof. The new support is contained in the finite union of the input supports, because the all-zero coefficient tuple is sent to zero. A finite union of reverse-well-ordered sets is reverse well ordered: a nonempty subset has a greatest member in each nonempty component, and the greatest of those finitely many choices is its greatest member. The normal-form existence theorem therefore gives the output. The coefficient equations give uniqueness and a first-order graph, using theorem 36.3. For omnific inputs there are no negative-exponent coefficients, and the coefficient at zero is an integer tuple; both properties survive under the stated assumptions. $\square$

For example, define the Hadamard product and support indicator by

$$c_a(x \odot y) = c_a(x)c_a(y), \quad c_a(e_x) = \begin{cases} 1, & c_a(x) \neq 0, \\ 0, & c_a(x) = 0. \end{cases}$$

Both are definable operations on surreals and preserve omnific integers. The Hadamard product differs from ordinary field multiplication: $\omega \odot \omega = \omega$, whereas $\omega \cdot \omega = \omega^2$. Coefficientwise absolute value gives another example: it sends $\omega - 1$ to $\omega + 1$, whereas numerical absolute value leaves $\omega - 1$ unchanged.

For a unit-coefficient series d with support D , set

$$H_D = \{x : \text{supp}(x) \subseteq D\}.$$

This is a definable domain with parameter d . Every coefficient function $D \rightarrow \mathbb{R}$ is allowed, because its nonzero support is a subset of the reverse-well-ordered set D . Consequently

$$(H_D, +, \odot, d) \cong \mathbb{R}^D \tag{40.1}$$

as unital rings. If $D \subseteq \mathbf{No}_{>0}$, the entire domain H_D lies in $\mathbf{Oz}$. Thus omnific integers carry copies of arbitrary set products of the real field under this additional definable multiplication. There is no claim that H_D is a subring for ordinary multiplication.

The idempotents for $\odot$ have only coefficients zero and one, so they correspond exactly to $\mathcal{P}(D)$. Their Boolean operations are

$$e \wedge f = e \odot f, \quad e \vee f = e + f - e \odot f, \quad \neg_D e = d - e.$$

This is another way to see the exact powerset coding used in the interpretation. There is no global Hadamard identity on all of $\mathbf{No}$: it would need coefficient one at every surreal exponent, a proper-class support.

40.2 The inner-model transcendence consequence

The next corollary applies the repository's support-amplification argument ([theorem 8.2](#) of Part I) after the missing absoluteness hypothesis has been established. We include the short algebraic mechanism.

Corollary 40.2 (A proper inner model misses a class of independent integers). *Let $M \subsetneq V$ be a transitive inner model of ZFC with the same ordinals, and put $K = \mathbf{No}^M$. There is an ordinal-indexed family of omnific integers, all in $(\omega, \omega + \sqrt{\omega})$, which is algebraically independent over K in the sense that every finite subfamily is algebraically independent. No equality of the real fields is required.*

Proof. The field K is proper. Otherwise every ordinal subset would have its characteristic sign sequence in M , and hence belong to M ; ordinary pointed-relation coding would then give $M = V$. Choose $t \in (0, 1) \setminus K$ by applying τ to an element outside K . Its inverse is algebraic, so τ cannot send that element into K .

For each ordinal α , put

$$e_\alpha = \omega^{-(\alpha+1)}, \quad u_\alpha = te_\alpha, \quad z_\alpha = \omega + \omega^{u_\alpha}.$$

All e_α belong to K . The cosets $u_\alpha + K$ are linearly independent over $\mathbb{Q}$ in the additive quotient $\mathbf{No}/K$. Indeed, a nontrivial finite rational relation would give $t \sum q_\alpha e_\alpha \in K$. The sum is a nonzero member of K , by uniqueness of normal form, so this would put t in K .

Consider a nonzero polynomial over K in finitely many ω^{u_α} . A term indexed by an exponent vector n has its coefficient's normal-form support translated by $\sum n_\alpha u_\alpha$. By [theorem 34.1](#), every exponent of every coefficient from K itself belongs to K . Distinct exponent vectors therefore produce disjoint support cosets, by the preceding linear independence. Their nonzero series cannot cancel. Thus the monomials are algebraically independent, and translating each by $\omega \in K$ preserves this property. Finally $0 < u_\alpha < 1/2$, giving the claimed omnific interval. $\square$

The result states finite polynomial independence for a definable family from the parameter t . It does not assign a cardinality to a proper-class transcendence basis or require Global Choice.

40.3 Exact preservation of definability levels by the bounded code

Corollary 40.3 (The bounded code does not raise a fixed Levy level). *For every fixed standard $n \geq 1$, every fixed allowed parameter family, and every surreal x , the objects x and $Q(x)$ have the same Σ_n -, Π_n -, and Δ_n -singleton definability status in ambient set theory.*

Proof. By [theorem 34.7](#), the graph of $Q = \Omega \circ \tau$ is Δ_1 . Its range is Δ_1 : a normal form must have exactly one term, coefficient one, and exponent in $(0, 1)$. The inverse uses that exponent and the algebraic formula for τ^{-1} , so its graph is Δ_1 too. Apply [proposition 34.9](#) in both directions. $\square$

There is also a modest literal description-length conclusion once a syntax convention is fixed. For literal finite-alphabet length, fix a finite parameter family A with named constant references; broader parameter permissions are treated one finite tuple at a time. Use a fixed finite alphabet, allow either of two fixed symbols v_0, v_1 as the sole free object variable, and write permitted parameters as constant references with a fixed cost convention. Let $L_A(x)$ be the minimum length among definitions with either output variable. If $\varphi(v_0)$ defines x , the context

$$\exists v_0(\varphi(v_0) \wedge \text{Graph}_Q(v_0, v_1))$$

defines $Q(x)$, with φ inserted unchanged. All variables bound inside φ retain their scopes. Reverse the roles of the two free symbols when necessary, using two precompiled versions of the fixed graph formula. The inverse graph gives the reverse bound. Thus for every A -definable x ,

$$|L_A(x) - L_A(Q(x))| \leq c \quad (40.2)$$

for a constant depending only on the coding formulas and the syntax convention. This is a statement about minimum lengths of successful definitions, not an algorithm for recognizing success or minimizing length. It supplies an additive overhead bound for this specific reversible code; it is not an optimal proof-length or birthday bound.

40.4 Elementary theories and elementary inclusions

Theorem 40.4 (The expansion retains the full elementary theory of sets). *For canonical surreal structures of models $M, N \models \text{ZFC}$,*

$$M \equiv N \iff \mathfrak{S}_\Omega^M \equiv \mathfrak{S}_\Omega^N.$$

For nested transitive models $M \subseteq W \models \text{ZFC}$, with the canonical inclusions,

$$M \preceq W \iff \mathfrak{S}_\Omega^M \preceq \mathfrak{S}_\Omega^W. \quad (40.3)$$

For proper classes each elementarity assertion is a scheme over standard formulas.

Proof. The class **No**, all the field operations, **Oz**, and Ω have fixed set-theoretic definitions. Relativizing quantifiers therefore translates every sentence of the enriched surreal language into a set-theoretic sentence. This proves the forward implication for elementary equivalence. The interpretation of [theorem 37.1](#) gives the reverse translation, and hence the reverse implication. In a nonstandard model the graph codes and collapses are interpreted internally, so this argument remains the ordinary syntactic argument for interpretations.

The same translation proves the forward implication of (40.3), now with parameters. For the converse, given any finite tuple of parameters $\bar{x} \in M$, choose in M admissible surreal codes $\bar{c}$ whose collapsed values are $\bar{x}$. Normal-form absoluteness preserves these codes and their relations in W . Their old pointed graphs remain well founded and extensional. Accessibility is preserved because it is equivalent to every node having a finite predecessor path from the root, and such paths are absolute. Mostowski collapse is absolute, so the codes have the same values $\bar{x}$ there. Elementarity of the surreal inclusion applied to the translation of any formula $\varphi(\bar{x})$ now gives $M \models \varphi(\bar{x})$ iff $W \models \varphi(\bar{x})$. $\square$

This sharply separates preservation of operations from preservation of all formulas. Every canonical inclusion in the last theorem is elementary in the ordered-field reduct by [corollary 34.5](#). The enriched inclusion fails to be elementary whenever the set universes disagree on a first-order formula with parameters from the smaller model. For example, the translated continuum-hypothesis sentence distinguishes canonical structures built in universes disagreeing about

CH. More generally the translation detects any first-order set-theoretic difference, including differences above the reals.

The theorem still does not prove a bi-interpretation. A choice of codes for parameters in its proof is an existential choice for a finite tuple, not a definable assignment of canonical codes to all original surreals. Nor does elementary equivalence classify isomorphism types.

[write] **On the nested clause, 4 October 2026.** For nested models of the same height, (40.3) holds only in the trivial case: both of its sides are equivalent to $M = W$. This observation is the write's, and is proved as follows; it does not contradict the theorem.

Observation. Let $M \subseteq W$ be transitive models of ZFC (set models, or inner classes with elementarity read as a scheme) with $\text{Ord} \cap M = \text{Ord} \cap W$. If $M \preceq W$, or equivalently $\mathfrak{S}_\Omega^M \preceq \mathfrak{S}_\Omega^W$, then $M = W$.

Proof. Suppose $M \preceq W$ and take $x \in W \setminus M$. By Choice in W , x is the root of the Mostowski collapse of a well-founded extensional relation on an ordinal, which an ordinal pairing codes by a set $A \subseteq \alpha$ of ordinals. If A were in M , the collapse, which is absolute between transitive models, would put x in M ; so $A \in W \setminus M$, while $\alpha \in M$ because the ordinals agree. Let $P = \mathcal{P}(\alpha)^M \in M$. Then $M \models \forall y (y \subseteq \alpha \rightarrow y \in P)$, whereas $W \models \exists y (y \subseteq \alpha \wedge y \notin P)$, witnessed by A . One formula with the parameters $\alpha, P \in M$ thus refutes $M \preceq W$. $\square$

This is the argument of `hset:thm:outer` of the birthday-cutoff report (there for birthday structures, with codes for α and $\mathcal{P}(\alpha)^M$ as parameters). The nested clause of the theorem therefore has content only for models of different heights, for example $M = V_\kappa$ and $W = V_\lambda$ when $\kappa < \lambda$ are inaccessible cardinals with $V_\kappa \prec V_\lambda$. The clause on elementary equivalence is unaffected; it is the analogue for $\mathfrak{S}_\Omega$ of $M \equiv N \iff \mathfrak{B}^M \equiv \mathfrak{B}^N$ in `hset:rem:internal` and `hset:cor:equiv`.

40.5 An operation calculus has no universal self-evaluator

There are two different meanings of “all definable operations.” One can prove a theorem scheme that each specified formula gives a parameter-preserving operation, or supply a uniform translation of finite syntax. One cannot conclude that there is an internal evaluator for truth of arbitrary formulas in the same universe.

Here is an elementary diagonal version in the enriched surreal language. Its ordinary naturals are definable by $\mathbb{N} = \{n \in \mathbf{Oz} \cap \mathbb{R} : n \geq 0\}$. Suppose a parameter-free definable total function $U : \mathbb{N} \times \mathbf{No} \rightarrow \mathbf{No}$ enumerated all parameter-free definable total unary surreal functions, in the sense that each such function was $x \mapsto U(n, x)$ for some natural n . Then the parameter-free function

$$g(x) = \begin{cases} U(x, x) + 1, & x \in \mathbb{N}, \\ 0, & x \notin \mathbb{N} \end{cases}$$

would have an index k . Evaluation at k would give $U(k, k) = U(k, k) + 1$, a contradiction. The support and summation formulas evade this obstruction because their inputs are explicitly coded mathematical families, not an alleged enumeration of every operation definable in their own language.

41 Size, first omissions, and the limits of collection

The parameter distinctions also control what a counting argument can show. For an externally fixed set model M , a set of parameters $A \subseteq M$ gives

$$|D_A^M|_{\text{external}} \leq \max(\aleph_0, |A|_{\text{external}}).$$

There are only countably many finite formulas and only the indicated number of finite parameter tuples. In particular a fixed finite parameter family A produces an externally countable definability hull. This does not prove that the hull is a countable set *inside* M , or that M

contains a sequence enumerating its successful definitions. Pointwise definable models illustrate why the distinction matters [12]: every element may have an external parameter-free definition while the model internally satisfies the usual uncountability theorems.

For the full ambient universe V , allowing all ordinals gives a proper-class parameter supply and the preceding set bound does not apply. For a fixed set model M , its ordinals are still an external set and the bound continues to apply. With individual real parameters, every real is trivially definable from itself, and [remark 33.3](#) gives much more. Nevertheless a particular formula uses only one finite tuple. Neither permission is an infinitary conjunction of names.

41.1 Two different first-omission ordinals

The repository distinguishes a first ordinal without an allowed definition from the first birthday at which a new sign sequence appears ([sections 10 and 11](#) of Part I). They measure different phenomena.

First fix a transitive model M and an external parameter family A . Suppose there is a least ordinal $\rho \in \text{Ord}^M$ not in D_A^M . Suppressing the superscript, the largest initial subclass of K_A , for sign simplicity, is

$$K_A \cap \mathbf{No}_{<\rho}. \quad (41.1)$$

Indeed, every prefix of a definable x of birthday below ρ is definable from x and its length, and that length is A -definable. Conversely, if $x \in K_A$ has birthday at least ρ , its prefix of length ρ cannot belong to K_A , since its definable birthday operation would then define ρ . No initial subclass of K_A can contain such an x .

The ordinal ρ , when it exists, is an epsilon number. It is not finite. If it were not a fixed point of ordinal $\alpha \mapsto \omega^\alpha$, its finite Cantor normal form would use only exponents strictly below ρ . Each such exponent and each finite coefficient is A -definable, so the normal form would define ρ , a contradiction. The stronger field and exponential properties of (41.1) are proved in the existing report using the classical birthday-cutoff theorems. This discussion does not assert that such a ρ exists for every parameter budget; with all ordinals permitted there is none.

Second let $M \subsetneq V$ be a transitive inner model of ZFC with all ordinals. Define

$$\lambda_M = \min\{\alpha : \mathcal{P}(\alpha)^M \neq \mathcal{P}(\alpha)^V\}.$$

This exists: otherwise all sets of ordinals, and then all sets by well-founded relation coding, would belong to M . Characteristic sign sequences show immediately that λ_M is the first birthday of a surreal not in $\mathbf{No}^M$.

For clarity, the standard cardinal argument is included. If λ_M were not a V -cardinal, let $\kappa = |\lambda_M|^V < \lambda_M$, and choose a bijection $f : \kappa \rightarrow \lambda_M$. Pull ordinal order back along f to a well-order E on κ . The usual ordinal pairing codes E by a subset of κ ; by minimality this code belongs to M . The unique ordinal collapse of E , computed in M , is absolute. It therefore recovers the original f and its range λ_M . Now every $A \subseteq \lambda_M$ has pullback $f^{-1}[A] \subseteq \kappa$ in M , and hence belongs to M , a contradiction. Thus λ_M is a V -cardinal, and therefore also an M -cardinal. The case $M = \text{HOD}$ is the repository's first non-OD birthday theorem.

[write] **Antecedents, 4 October 2026.** The epsilon-number argument is the lemma behind [theorem 10.3](#) ([section 10](#) of Part I), and the cardinal threshold is `univ:prop:beta` of the universes report, proved there for arbitrary transitive $M \subseteq N$; [theorem 11.2](#) is its case $M = \text{HOD}$.

These observations organize different questions without identifying them. The ordinal ρ concerns definitions using a fixed external budget; λ_{HOD} concerns hereditary ordinal definability; and $\omega_1^{\text{HOD}} = \omega_1$ concerns the ability to choose real codes uniformly. Cardinal correctness can hold in a proper inner model, as the Cohen example in [section 33](#) shows.

42 Further research questions and proposed program

The following questions are deliberately stated beyond what has been proved. The first three concern expressive power; the next group concerns parameters and complexity; the final group concerns operations and formalization. Their order reflects dependencies rather than a claim about difficulty.

Question 42.1 (The missing comparison map). Is $\mathfrak{S}_\Omega$ bi-interpretable with $(V, \in)$, or can one rule this out? Specifically, is there a first-order definable relation associating an original surreal x with exactly the equivalence class coding its canonical set-theoretic sign sequence in the interpretation?

Theorem 37.1 gives a quotient universe, and the ordinary set construction gives surreals within that universe. A bi-interpretation requires the second composite to be definably isomorphic to the original surreal structure. An abstract external isomorphism does not meet this requirement. An explicit sign-prefix relation, or a sufficiently strong intrinsic birthday relation, would be a promising route. The birthday-enriched Chen–Hamkins–Yang announcement [45] identifies a useful comparison target but does not supply the missing definition in the present signature.

Question 42.2 (Exact intrinsic definable closures). Determine

$$\text{dcl}_{\mathfrak{S}_0}(\emptyset), \quad \text{dcl}_{\mathfrak{S}_\Omega}(\emptyset), \quad \text{dcl}_{\mathfrak{S}_\Omega}(\mathbb{R}),$$

and their ordinal-parameter analogues, in specified ambient models.

For the smaller pair, the parameter-free closure lies in $\mathbb{R}$ and contains the reals definable in $(\mathbb{R}; +, \cdot, <, \mathbb{Z})$. For the richer structure, ω , all the operations proved here, and bounded tuple codes are available. Neither statement identifies the closure. An answer to question 42.1 may connect this problem to ambient definability; without that comparison map, the interpretation theorem cannot simply be substituted for an exact closure theorem. This is the remaining part of repository question 17.1 (Part I).

Question 42.3 (How much of the signature is necessary?). Can the named omega map be replaced by its range predicate, or by a weaker normal-form operation? Can the omnific predicate be removed while preserving an interpretation of set theory?

The proof uses **Oz** to recover canonical real coefficients and Ω to index them by arbitrary exponents and to build an order-compatible pairing. A monomial predicate names scales but does not automatically identify their surreal exponents. Conversely, Ω already defines a natural valuation (remark 36.4), but a valuation alone does not provide a canonical residue-field section. A useful first target is an exact replacement for one of these two missing ingredients, keeping the other unchanged.

Question 42.4 (Automorphisms compatible with the coding structure). Classify automorphisms of $\mathfrak{S}_\Omega$, and determine how they can act on representatives of the interpreted quotient universe.

Kaplan–Krapp–Serra ask about nontrivial omega-preserving automorphisms in Question 5.6 and a related extension problem in Question 5.7 [44]. Their pointwise-valuation rigidity result must be distinguished from preserving a valuation equivariantly while also acting on its value group. The present interpretation places an additional constraint on automorphisms when **Oz** is also named. Even a rigid interpreted quotient would leave the possibility of movement within individual code classes. Studying that action is a concrete way to test whether the missing comparison map in question 42.1 can exist.

Question 42.5 (The parameter degree of a uniform section). For a fixed universe, classify the sets of ordinals p , or the reals r , for which

$$\omega_1^{\text{HOD}(p)} = \omega_1, \quad (\kappa^+)^{\text{HOD}(p)} = \kappa^+.$$

What ordering of parameter strengths is induced by the existence of the corresponding definable coding sections?

Theorems 33.4 and 33.6 turn this into an exact inner-model question. Natural variants restrict the input birthdays, require sections to commute with embeddings, or require one parameter to work at every cardinal. The all-cardinal criterion is agreement on cardinals, not equality with V . Further work should distinguish changes in that criterion from changes in which individual surreals are OD.

Question 42.6 (Optimal logical complexity). When a real-code section exists, is the $\Delta_2(p)$ upper bound optimal? Which additional hypotheses permit a $\Delta_1(p)$ section? What are the sharp complexities for variants with a prescribed enumeration convention?

The conditional totality in the existing Δ_2 bound matters. If an eligible ordinal has no successful OD enumeration, adding a default for that additional failure is an unbounded negative search; the current argument does not establish the same bound for such an unconditional extension. This is a concrete place where a complexity lower bound could be informative, rather than merely shortening a formula.

Question 42.7 (Birthday and description costs). Find sharp birthday bounds for $Q(x)$, the lexicographic pair $B(a, b)$, and the pointed-graph code of a set. How do the bounds change when the leading term, coefficient alphabet, or target interval is prescribed?

The fixed-level preservation in corollary 40.3 and the additive literal length bound (40.2) concern logical syntax. They do not bound the ordinal length of the output sign sequence. For graph codes, the chosen enumeration of the transitive closure may also affect the cost. An initial project is to combine corrected classical birthday estimates [3, 4] with the explicit formulas for τ, B, Q , before asking for optimal constructions. This develops repository questions 17.2 and 17.5 (Part I).

Question 42.8 (Interpretations in bounded birthday structures). For which λ does an appropriate expansion of $\mathbf{No}_{<\lambda}$ interpret a substantial set-theoretic structure such as H_κ or V_α ? What is the exact interpreted domain?

The unrestricted proof quantifies over codes for *all* subsets of a support. At a birthday cutoff, those subset codes or the paired relation codes may lie outside the structure. In that case the well-foundedness formula need not express ambient well-foundedness. Determining precisely which closure and size hypotheses restore the proof is the first necessary step; merely restricting the quantifiers of the present formulas is not a solution.

Question 42.9 (Strong definable operations beyond diagonal calculus). Classify the intrinsically definable strong real-linear maps and derivations in $\mathfrak{S}_\Omega$. Which are determined by a definable action on monomials, and which require more general support changes?

The family D_b and its partial integration operators are support-preserving, which makes existence transparent. Support-changing operators require a summability criterion for their monomial images. The coded-family theorem supplies a way to express that criterion for a given family. One can then ask which such operators are definable uniformly, and how their kernels, images and commutants are described. The Berarducci–Mantova derivation is a separate target: its compatibility with exponential structure and its normalization cannot be inferred from the diagonal construction.

Question 42.10 (Coefficientwise algebra and definable product structures). Which definable real operations have coefficientwise lifts compatible with ordinary surreal multiplication? What substructures of the definable product rings $H_D \cong \mathbb{R}^D$ can be characterized uniformly?

Hadamard multiplication gives all Boolean support algebra directly. Ordinary multiplication instead convolves exponents. Requiring compatibility with both operations should impose strong constraints, but the present coefficientwise theorem assumes no such compatibility. Finite-support ideals, support filters, and the interactions between coordinate projections and convolution provide explicit starting examples.

Question 42.11 (First omissions and inner support fields). Which first-omission ordinals ρ and first new birthdays λ_M can occur under specified hypotheses on M and the parameter family? Which intermediate support fields arise as $\mathbf{No}^M$, and which do not?

The absoluteness theorem proves that every transitive inner model of ZFC gives a support field, including models with fewer reals. It does not classify all support fields, or all possible first-omission spectra. One can now compare combinatorial closure of a field with set-theoretic closure of a candidate inner model, without carrying the obsolete same-reals restriction. This extends the still-open classification parts of repository [questions 17.3](#) and [17.4](#) (Part I).

Question 42.12 (Axiom strength and formal verification). Which fragments of set theory suffice for the canonical-operation certificates, the powerset coding, and the interpretation? Can their formula translations and correctness proofs be formalized with an explicit dependency ledger?

There are several distinct obligations: existence of the relevant normal forms, Collection for evaluation certificates, the set-sized Mostowski collapse, and Choice for coding every set by a well-orderable domain. The ZF version identifies the hereditarily well-orderable quotient, but does not by itself calibrate weaker theories. Existing formal work on Conway normal forms [\[41\]](#) is useful background. Formalizing a finite syntactic translator alone would verify the translation algorithm, not the semantic adequacy of its graph codes.

[write] **Status of the questions, 4 October 2026, batch 93.** All twelve questions stay open as printed. [Question 42.2](#) is the remainder of [question 17.1](#) of Part I and contains source 02’s question on definable closure versus fixed elements in [section 28](#). The question on automorphisms meets `sse:sr:q:omegafield` of the self-embeddings report, which also starts from Kaplan–Krapp–Serra’s Questions 5.6–5.7. The question on birthday and description costs ([question 42.7](#)) develops [questions 17.2](#) and [17.5](#), as do source 02’s question on the birthday cost of countable codes and source 05’s Question 13.8 in [section 28](#). [Question 42.11](#) extends [questions 17.3](#) and [17.4](#); [remark 22.10](#) supplies one more support field.

42.1 Further questions and research added at the write [write]

The following items are recorded under Vladimir’s standing rule of 4 October 2026: claims made by the manuscript with an argument that it itself presents as a sketch or an upper bound are kept as open questions, with the argument and what is missing.

Question 42.13 (The complexity of the decoder and of its sections [write]). Prove in full the bounds of the remark after [theorem 33.4](#): that the real decoder D has a Δ_1 graph over ZFC, and that under the hypotheses of [theorems 33.4](#) and [33.6](#) the canonical section has a $\Sigma_2(p)$ graph and a total $\Delta_2(p)$ (or $\Delta_2(p, \kappa)$) extension.

The manuscript argues this in a remark, not a proof: existential set witnesses for the graph of D and its complement (an ordinal collapse; a bounded failure of the order axioms; a nonempty subset without least element); and, for the section, the standard $\Delta_2(p)$ surjection from Ord

onto $\text{OD}(p)$ of [28, Section 4] with a least-code search. What is missing is a written verification that the least-code search, the choice of a real well-order of the required type and the default off the intended domain stay within the stated classes; the manuscript itself calls these upper bounds and asks for optimality in [question 42.6](#).

Question 42.14 (The description-length bound [write]). With the syntax convention stated before (40.2), write out the two precompiled graph formulas of Q and Q^{-1} and the resulting constant c , and prove (40.2).

The manuscript’s argument inserts a definition $\varphi(v_0)$ unchanged into a fixed context $\exists v_0(\varphi(v_0) \wedge \text{Graph}_Q(v_0, v_1))$, renaming the free variable when necessary. What is missing is the explicit cost accounting for variable capture and the constant-reference convention; the bound depends on that convention, as the manuscript states.

43 Proof dependencies, contribution ledger, and verification status

43.1 What is proved here and what is imported

The following ledger records the scope of the claims. An extension means an addition to the audited repository treatment with a proof in this article. It is not a certification that no earlier source contains an equivalent result.

Claim or construction	Status	Dependency and precise scope
Surreal field and normal forms	Classical	Conway–Gonshor theory; legal supports are set-sized and reverse well ordered.
Real closed ambient hulls, floor, fraction representation	Repository background	Unique-output closure and classical field facts; rederived here to fix the parameter budget.
Bounded unary monomial code Q	Repository background	Parameter-free algebraic compression followed by Ω .
Canonical real field and coefficient at zero from Oz	Repository background	Quadratic detector and multiplier ring; no omega symbol is needed in the defining formulas.
Global real consolidation	Extension	Choice of a well-order of $\mathbb{R}$, then substitution of definitions; equivalent to $V = \text{HOD}(r)$ for one real.
Uniform real and κ -code sections	Extension	Exact HOD successor-cardinal criteria; Δ_2 encoder bound is conditional on the agreement hypothesis.
Normal-form absoluteness without common reals	Extension	Absolute cut tables and persistent sign prefixes, plus the classical normal-form sign-sum identity.
Set theory in bounded omnific monomials	Extension	Coefficient recovery, lexicographic pairing, exact support subsets and relations, and set-sized Mostowski collapse.
Hahn summability and sum for coded families	Extension	One fixed formula uses support-union existence, finite fibres and finite partial-sum codes.
Coefficient-indexed derivations and partial integration	Definable instance and extension	General diagonal Euler derivations have precedents; this family and its normalized partial inverses are given by intrinsic coefficient formulas.
Fixed Levy levels and literal description length	Extension	Δ_1 graph, range and inverse for Q ; a stated finite-alphabet syntax convention for the additive bound.

Claim or construction	Status	Dependency and precise scope
Elementary-theory and inclusion equivalences	Consequence	Two syntactic translations, and absoluteness of old graph codes for the inclusion statement.
Exact intrinsic definable closure; bi-interpretation	Not proved	The missing comparison map and exact closure problems remain research questions.

The main proof chain can be read in four layers. Classical surreal theory supplies legal series and uniqueness of coefficients. The omnific predicate and omega map expose these coefficients to a fixed first-order language. Ordered support coding then exposes all subsets and relations on each set support. Finally, ordinary well-founded collapse identifies the quotient with set theory. The real-code section criterion is independent of this intrinsic interpretation: it is an ambient definability theorem about HOD and ordinal enumerations.

43.2 A concrete formalization sequence

A useful formalization effort would proceed in the following order.

1. Fix canonical sign representations and prove the bounded simplest-separator test. Construct the finite-input, set-sized dependency tables for arithmetic and Ω .
2. Connect the sign and normal-form representations. Prove that the persistent-prefix limit evaluates legal normal forms, with explicit set witnesses at limit stages. Establish the Δ_1 graph results in the chosen set-theoretic coding.
3. Formalize the three reconstruction formulas for Π , $\mathcal{A}$ and $\mathbb{R}$, then coefficient extraction and truncation. These are small interfaces on which the later interpretation depends.
4. Verify the lexicographic pairing, the exact subset and relation coding lemmas, and the three admissibility conditions. Define the isomorphism and membership formulas without using collapse as a primitive of the surreal language.
5. Prove the semantic correspondence by the set-sized Mostowski collapse theorem. Only after that, verify the recursive translation of formulas and its preservation of truth, one standard formula at a time.
6. Add family codes, finite coefficient folds, Hahn sums and derivations. Develop the HOD and cardinal-correctness section in a separate set-theory layer, because those proofs have different foundational dependencies.

This article contains written proofs and was subjected to independent mathematical checks during preparation, particularly at the support coding, inner-model absoluteness, cardinal selection and syntax-cost steps. It is not a Lean, Mizar or other proof-assistant development. No finite numerical experiment is offered as evidence for a transfinite existence theorem. The supplied source compiles without an external bibliography database; the accompanying archive contains the PDF and the precise source used to generate it.

[write] **4 October 2026**. In this report the manuscript's source, its PDF and its delivery README are not shipped; they survive in the arrival commit's archive. Only `10-definable-operations-PROOF_STATUS.txt` is shipped ([section 30.3](#)).

References

- [1] J. H. Conway, *On Numbers and Games*, second edition, A K Peters, 2001; first edition, Academic Press, 1976.
- [2] H. Gonshor, *An Introduction to the Theory of Surreal Numbers*, London Mathematical Society Lecture Note Series 110, Cambridge University Press, 1986.
- [3] L. van den Dries and P. Ehrlich, Fields of surreal numbers and exponentiation, *Fundamenta Mathematicae* **167** (2001), 173–188. doi:10.4064/fm167-2-3. In particular Theorem 2.1, Lemmas 4.1–4.2, Proposition 4.7, and Corollary 5.5. To be read with [4].
- [4] L. van den Dries and P. Ehrlich, Erratum to “Fields of surreal numbers and exponentiation”, *Fundamenta Mathematicae* **168** (2001), 295–297. doi:10.4064/fm168-3-5.
- [5] O. Bournez and Q. Guilmant, Surreal fields stable under exponential and logarithmic functions, arXiv:2201.08199, 2022. <https://arxiv.org/abs/2201.08199>.
- [6] S. L’Innocente and V. Mantova, *A factorisation theory for generalised power series and omnific integers*, arXiv:1710.07304, version 5, January 22, 2024. <https://arxiv.org/abs/1710.07304>. Journal version: doi:10.1016/j.aim.2024.109513.
- [7] D. Marker, *Model Theory: An Introduction*, Graduate Texts in Mathematics 217, Springer, 2002. For the quantifier-elimination argument, see also the author’s *Introductory Lectures on Model Theory II: Quantifier Elimination*, October 14, 2020, <https://homepages.math.uic.edu/~marker/MSRI-2.pdf>.
- [8] L. van den Dries, *Tame Topology and o-Minimal Structures*, London Mathematical Society Lecture Note Series 248, Cambridge University Press, 1998.
- [9] P. Ehrlich and E. Kaplan, *Number systems with simplicity hierarchies: a generalization of Conway’s theory of surreal numbers II*, preprint, arXiv:1512.04001, version 1, 2015. <https://arxiv.org/abs/1512.04001v1>.
- [10] T. Jech, *Set Theory: The Third Millennium Edition, Revised and Expanded*, Springer Monographs in Mathematics, Springer, 2003. doi:10.1007/3-540-44761-X.
- [11] J. C. Shepherdson, A non-standard model for a free variable fragment of number theory, *Bulletin de l’Académie Polonaise des Sciences, Série des sciences mathématiques, astronomiques et physiques* **12** (1964), 79–86. The integer-part application to **Oz** is also stated explicitly in [6, Section 1.1].
- [12] J. D. Hamkins, D. Linetsky, and J. Reitz, Pointwise definable models of set theory, *Journal of Symbolic Logic* **78** (2013), no. 1, 139–156. <https://arxiv.org/abs/1105.4597>. Author’s discussion: <https://jdh.hamkins.org/pointwisedefinablemodelssettheory/>.
- [13] J. D. Hamkins, answer to “Definable map from all the ordinals to the surreal numbers with a dense image”, *MathOverflow*, 8 April 2012. <https://mathoverflow.net/questions/93468/definable-map-from-all-the-ordinals-to-the-surreal-numbers-with-a-dense-image>.
- [14] J. D. Hamkins and C. Leahy, Algebraicity and implicit definability in set theory, *Notre Dame Journal of Formal Logic* **57** (2016), no. 3, 431–439. <https://arxiv.org/abs/1305.5953>; doi:10.1215/00294527-3542326.

- [15] G. Goldberg and A. Poveda, Compactness phenomena in HOD, *Forum of Mathematics, Sigma* **13** (2025), e118. doi:10.1017/fms.2025.10070. Author's version: <https://math.berkeley.edu/~goldberg/Papers/CompactnessPhenomenaInHOD.pdf>.
- [16] T. Benhamou, J. Cummings, G. Goldberg, Y. Hayut, and A. Poveda, *Compactness phenomena in HOD and the Optimality of Magidor's Covering theorem*, arXiv:2608.24190, version 1, August 25, 2026. <https://arxiv.org/abs/2608.24190>. Theorems 1.1–1.2 supply the cited first-disagreement results; these are external inputs, not newly proved claims of this report.
- [17] J. D. Hamkins, *Surreal arithmetic is bi-interpretable with set theory*, CUNY Logic Workshop announcement, posted 4 February 2026 for a talk on 13 March 2026; joint work with Junhong Chen and Ruizhi Yang, described as work in progress. <https://jdh.hamkins.org/surreal-arithmetic-cuny-logic-workshop-march-2026/>. Cited as an announcement, not a refereed proof source.
- [18] Wikipedia contributors, *Definable real number*, user-specified starting reference, accessed September 23, 2026. https://en.wikipedia.org/wiki/Definable_real_number. No theorem depends on this secondary reference.
- [19] V. Reshetnikov and contributors, *Surreal*, GitHub repository; the sources inspected it at commit 9a385d3957bdfe3d9ea79f9a524751c90bd2c894, 23 September 2026. <https://github.com/VladimirReshetnikov/Surreal>. The repository describes its manuscripts as AI-assisted research drafts and tracks formalization separately.
- [20] *Birthday Cutoffs and Hereditary Sets*, merged research report in [19], docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets/ (label prefix hset:).
- [21] *Surreal Fields Across Set-Theoretic Universes*, merged research report in [19], docs/foundations-and-computation/surreal-fields-across-universes/ (label prefix univ:).
- [22] *Omnific Integers and Omnific–Diophantine Geometry*, merged research report in [19], docs/surreal/omnific-diophantine-geometry/ (label prefix odg:).
- [23] *Set-Sized Quotients of Omnific Integers*, merged research report in [19], docs/surreal/set-sized-quotients-of-omnific-integers/ (label prefix osq:).
- [24] *Omnific-Preserving Automorphisms*, merged research report in [19], docs/surreal/omnific-preserving-automorphisms/ (label prefix opa:).
- [25] *Surreal Self-Embeddings*, merged research report in [19], docs/surreal/surreal-self-embeddings/ (label prefix sse:). Its question sse:sr:q:stabilizers asks for the common fixed field with or without omnific or strong-sum constraints.
- [26] *Large-Cardinal Embeddings and Normal Forms*, research report in [19], docs/foundations-and-computation/large-cardinal-embeddings-and-normal-forms/ (label prefix lce:).
- [27] *The Surreal Numbers as a Real Vector Space: Conway–Hahn coordinates, Hamel bases, strong duality, canonical operators, and the Euler quotient*, research report in [19], docs/surreal/real-vector-space-structure/ (label prefix rvs:). Cited in Part III only for the general diagonal-derivation framework.
- [28] F. G. Dorais and J. D. Hamkins, When does every definable nonempty set have a definable element?, *Mathematical Logic Quarterly* **65** (2019), no. 4, 407–411; arXiv:1706.07285; doi:10.1002/malq.201700035. Sections 2 and 4: the reflected definition of OD(X) and the

Δ_2 ordinal enumeration. Used for the rank-local framework, not for an assertion that $\text{HOD}(\mathbb{R})$ satisfies Choice.

- [29] K. Kunen, *Set Theory*, Studies in Logic 34, College Publications, 2011. Used for standard forcing, homogeneity, and hereditary definability.
- [30] A. Karagila, *Lecture Notes: Forcing & Symmetric Extensions*, version dated 24 August 2026, especially Section 9.2 and Theorem 9.10. <https://karagila.org/files/Forcing-2023.pdf>. The filename predates the version consulted by source 02.
- [31] K. L. Prikry, Changing measurable into accessible cardinals, *Dissertationes Mathematicae* **68** (1970), 5–52.
- [32] T. Benhamou, Prikry forcing and tree Prikry forcing of various filters, *Archive for Mathematical Logic* **58** (2019), 787–817. <https://doi.org/10.1007/s00153-019-00660-3>; <https://arxiv.org/abs/1801.04424>. The classical normal-measure Prikry property and its no-new-bounded-subsets consequence are the inputs source 05 uses.
- [33] A. Lietz, *The Basics of Namba Forcing*, author-written exposition, 22 March 2024. <https://andreas-lietz.github.io/blog/NambaBlog.html>. The opening distinguishes the two tree presentations; the CH no-new-reals proof is stated to apply to both.
- [34] A. W. Miller, *Forcing Tidbits*, Math 873, Fall 2009, version dated 21 December 2009, Theorem 23 and the comparison table on p. 4. <https://people.math.wisc.edu/~awmil1e1/old/m873-09/forcing.pdf>.
- [35] S. Shelah, *Proper and Improper Forcing*, second edition, Perspectives in Mathematical Logic, Springer, 1998. Chapter XI treats Namba forcing. Added at the write as a standard reference; not consulted (see [question 28.17](#)).
- [36] V. Fischer, M. Koelbing and W. Wohofsky, Fresh function spectra, *Annals of Pure and Applied Logic* **174** (2023), no. 9, 103300. <https://doi.org/10.1016/j.apal.2023.103300>. Section 7.6 (Namba forcing under CH), as imported by the universes report.
- [37] P. Ehrlich and E. Kaplan, Surreal ordered exponential fields, *Journal of Symbolic Logic* **86** (2021), no. 3, 1066–1115. <https://arxiv.org/abs/2002.07739>. Background on canonical surreal structures, not a proof source for the assembly theorems.
- [38] R. M. Solovay, A model of set-theory in which every set of reals is Lebesgue measurable, *Annals of Mathematics* (2) **92** (1970), no. 1, 1–56. <https://www.jstor.org/stable/1970696>. The countable-ordinal definability construction is in III.2.2–III.2.6, pp. 51–52.
- [39] V. Kanovei and V. Lyubetsky, *A countable equivalence relation of second projective class not generated by projective functions*, arXiv:2605.03126v1, 2026. <https://arxiv.org/html/2605.03126v1>. Proposition 2.2(i) records countable ROD assembly in the Levy–Solovay extension, citing Solovay. The unversioned record has since used the title *Locally countable graphs of second projective class not generated by countably many projective functions*; the cited proposition is from version 1.
- [40] P. Lipparini and I. Mező, *Surreal limits*, arXiv:1603.09289v3, 8 June 2026. <https://arxiv.org/abs/1603.09289>. Definition 2.1 and Corollary 3.2: the persistent-prefix sign limit and legal normal-form sums.
- [41] K. Pąk and C. Kaliszyk, *Conway Normal Form: Bridging Approaches for Comprehensive Formalization of Surreal Numbers*, arXiv:2410.00065, 2024. <https://arxiv.org/abs/2410.00065>.

- [42] S. Camacho, Truncation in Hahn fields is undecidable and wild, *South American Journal of Logic* **5** (2019), no. 1, 39–47. <https://arxiv.org/abs/1706.03722>.
- [43] A. Berarducci, S. Kuhlmann, V. Mantova, and M. Matusinski, Exponential fields and Conway’s omega-map, *Proceedings of the American Mathematical Society* **151** (2023), no. 6, 2655–2669. doi:10.1090/proc/14577; <https://arxiv.org/abs/1810.03029>.
- [44] E. Kaplan, L. S. Krapp, and M. Serra, *Decomposing the automorphism group of the surreal numbers*, author-hosted manuscript, accessed 4 October 2026 by source 06. https://elliotakaplan.github.io/Decomposing_the_automorphism_group_of_the_surreal_numbers.pdf. See Proposition 5.5 and Questions 5.6–5.7.
- [45] J. D. Hamkins, *The elementary theory of surreal arithmetic is bi-interpretable with set theory*, Kobe conference announcement, posted 20 August 2025 for a September 2025 talk; joint work with Junhong Chen and Ruizhi Yang, described as work in progress. <https://jdh.hamkins.org/theory-of-surreal-arithmetic-bi-interpretable-with-set-theory-japan-september-2025/>. Source 06 also consulted the November 2025 Notre Dame slides. The signature includes birthday order; no theorem of Part III depends on the announcement. Compare [17].